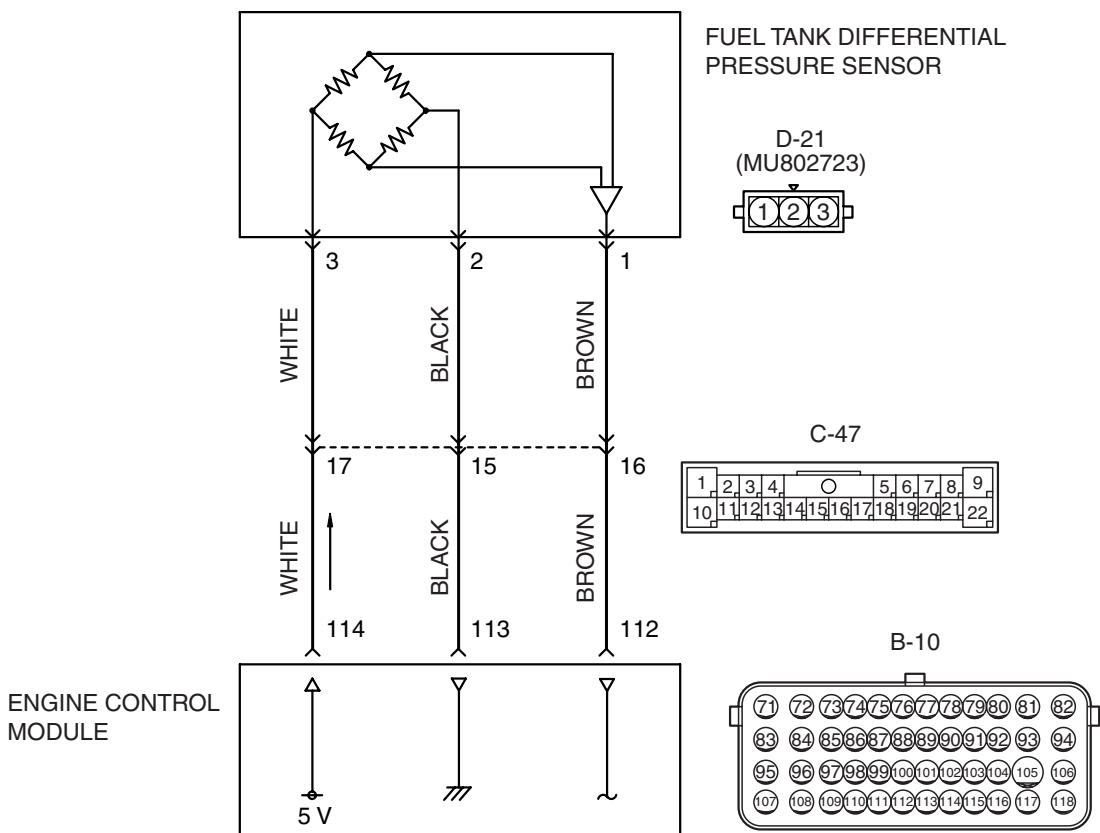
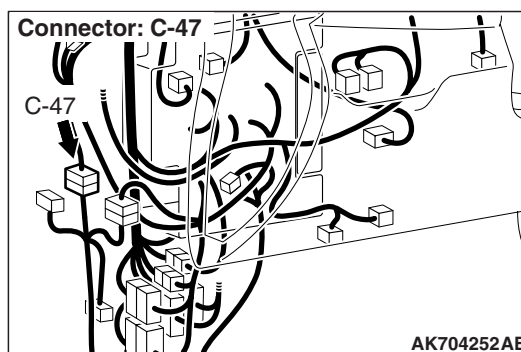
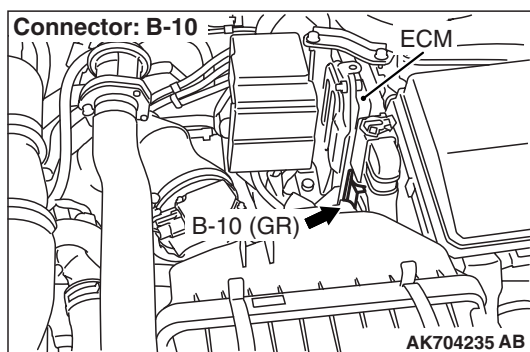


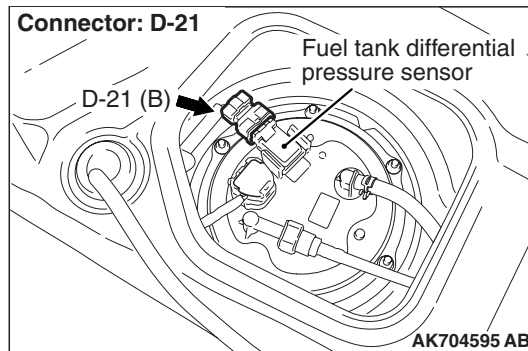
AK704684AB

FUEL TANK DIFFERENTIAL PRESSURE SENSOR CIRCUIT



AK604513 AE





CIRCUIT OPERATION

- The ECM (terminal No. 114) supplies a 5 volts reference signal to the fuel tank differential pressure sensor (terminal No. 3). The fuel tank differential pressure sensor (terminal No. 2) is grounded through the ECM (terminal No. 113).
- The fuel tank differential pressure sensor (terminal No. 1) returns a voltage signal to the ECM (terminal No. 112) that is proportional to the pressure in the fuel tank.

TECHNICAL DESCRIPTION

- The ECM monitors the fuel tank differential pressure sensor output voltage.
- The ECM determines whether the fuel tank differential pressure sensor signal voltage is within normal operating parameters.

DESCRIPTIONS OF MONITOR METHODS

Compare evaporative emission purge solenoid status with fuel tank differential pressure sensor output voltage.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (OTHER MONITOR AND SENSOR)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

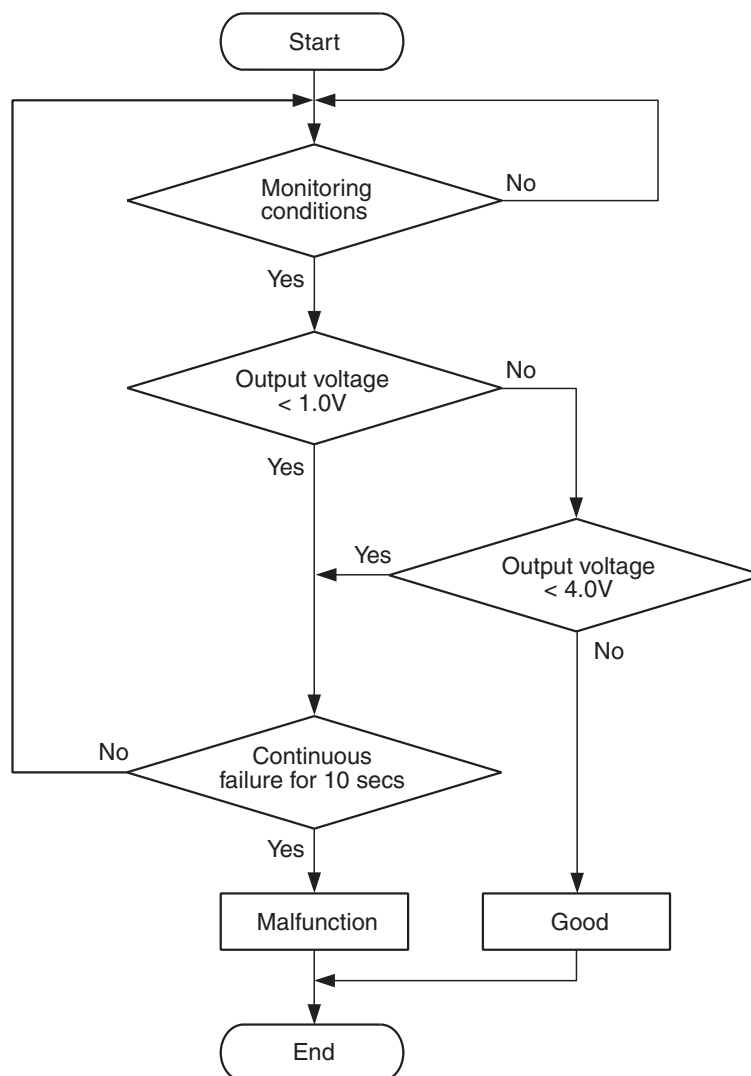
- Evaporative emission purge solenoid monitor
- Evaporative emission ventilation solenoid monitor
- Fuel tank temperature sensor monitor
- Fuel level sensor monitor

Sensor (The sensors below are determined to be normal)

- Mass airflow sensor
- Barometric pressure sensor
- Intake air temperature sensor
- Engine coolant temperature sensor
- Accelerator pedal position sensor

DTC SET CONDITIONS

Logic Flow Chart (Monitor Sequence)



AK704741

Check Conditions

- Intake air temperature is greater than 5° C (41° F).
- Engine speed is 1,594 r/min or greater.
- Volumetric efficiency is between 20 and 80 percent.

Judgement Criteria

- When the evaporative emission purge solenoid is off, the fuel differential pressure sensor output voltage remains 1.0 volt or less for ten seconds.

Check Conditions

- Intake air temperature is between 5° C (41° F) and 45° C (113° F) or greater.
- Engine speed is 1,594 r/min or greater.

- Volumetric efficiency is between 20 and 80 percent.

Judgement Criteria

- When the evaporative emission purge solenoid valve is fully operational (100 percent ratio), the fuel differential pressure sensor output voltage remains at 4.0 volts or greater for ten seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 4 [P.13A-9](#).

TROUBLESHOOTING HINTS (THE MOST LIKELY CAUSES FOR THIS CODE TO BE SET ARE:)

- Fuel tank differential pressure sensor failed.

- Fuel tank differential pressure sensor circuit harness damage, or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

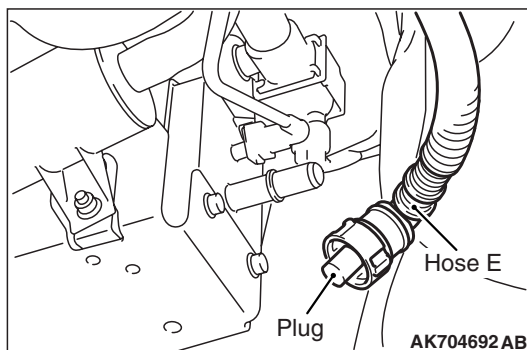
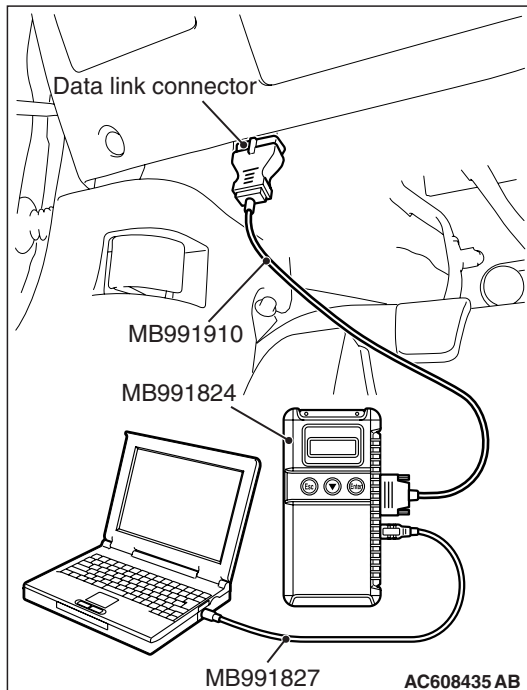
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Using scan tool MB991958, check data list item 52: Fuel Tank Differential Pressure Sensor.

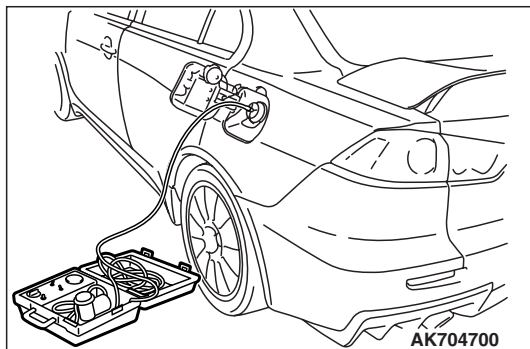
CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

(1) Connect scan tool MB991958 to the data link connector.



- Plug the disconnected end of hose E.
- Turn the ignition switch to the "ON" position.
- Remove the fuel cap.
- Set scan tool MB991958 to the data reading mode for item 52, Fuel Tank Differential Pressure Sensor.
 - Output voltage should be between 1,500 to 3,500 mV.



- (6) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382) and pressurize the fuel tank.
 - The fuel tank pressure reading should increase.
- (7) Turn the ignition switch to the "LOCK" (OFF) position. Then disconnect scan tool MB991958.
- (8) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.
- (9) Connect hose E to the evaporative emission canister.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting / Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2 .

STEP 2. Check harness connector D-21 at fuel tank differential pressure sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

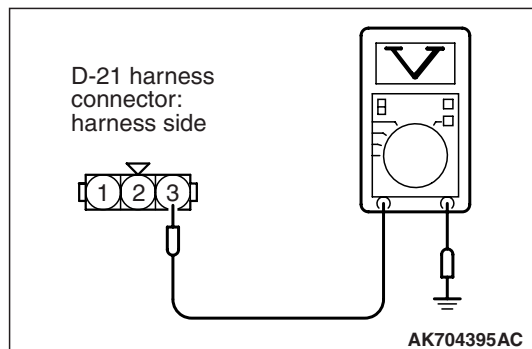
STEP 3. Measure the sensor supply voltage at fuel tank differential pressure sensor connector D-21 by backprobing.

- (1) Do not disconnect the connector D-21.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 3 and ground by backprobing.
 - Voltage should be between 4.9 and 5.1 volts.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage between 4.9 and 5.1 volts?

YES : Go to Step 6 .

NO : Go to Step 4 .



STEP 4. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to step 5 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

STEP 5. Check for harness damage between fuel tank differential pressure sensor connector D-21 (terminal No. 3) and ECM connector B-10 (terminal No. 114).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection P.00E-2. Then go to Step 12.

Q: Is the harness wire in good condition?

YES : Go to Step 11 .

NO : Repair it. Then go to Step 12 .

STEP 6. Measure the ground voltage at fuel tank differential pressure sensor connector D-21 by backprobing.

(1) Do not disconnect the connector D-21.

(2) Turn the ignition switch to the "ON" position.

(3) Measure the voltage between terminal No. 2 and ground by backprobing.

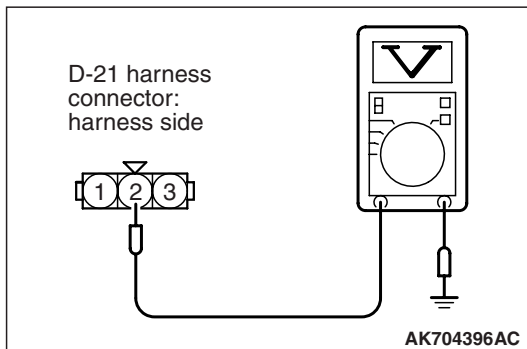
- Voltage should be 0.5 volt or less.

(4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage 0.5 volt or less?

YES : Go to Step 9 .

NO : Go to Step 7 .



STEP 7. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 8 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection P.00E-2. Then go to Step 12 .

STEP 8. Check for harness damage between fuel tank differential pressure sensor connector D-21 (terminal No. 2) and ECM connector B-10 (terminal No. 113).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection P.00E-2. Then go to Step 12.

Q: Is the harness wire in good condition?

YES : Go to Step 11 .

NO : Repair it. Then go to Step 12 .

STEP 9. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 10 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

STEP 10. Check for harness damage between fuel tank differential pressure sensor connector D-21 (terminal No. 1) and ECM connector B-10 (terminal No. 112).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

Q: Is the harness wire in good condition?

YES : Go to Step 11 .

NO : Repair it. Then go to Step 12 .

STEP 11. Replace the fuel tank differential pressure sensor.

(1) Replace the fuel tank differential pressure sensor.

(2) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 4 [P.13A-9](#).

(3) Check the diagnostic trouble code (DTC).

Q: Is DTC P0450 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, Diagnosis –ID Code Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, Diagnosis –ID Code Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 12 .

NO : The inspection is complete.

STEP 12. Test the OBD-II drive cycle.

(1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 4 [P.13A-9](#).

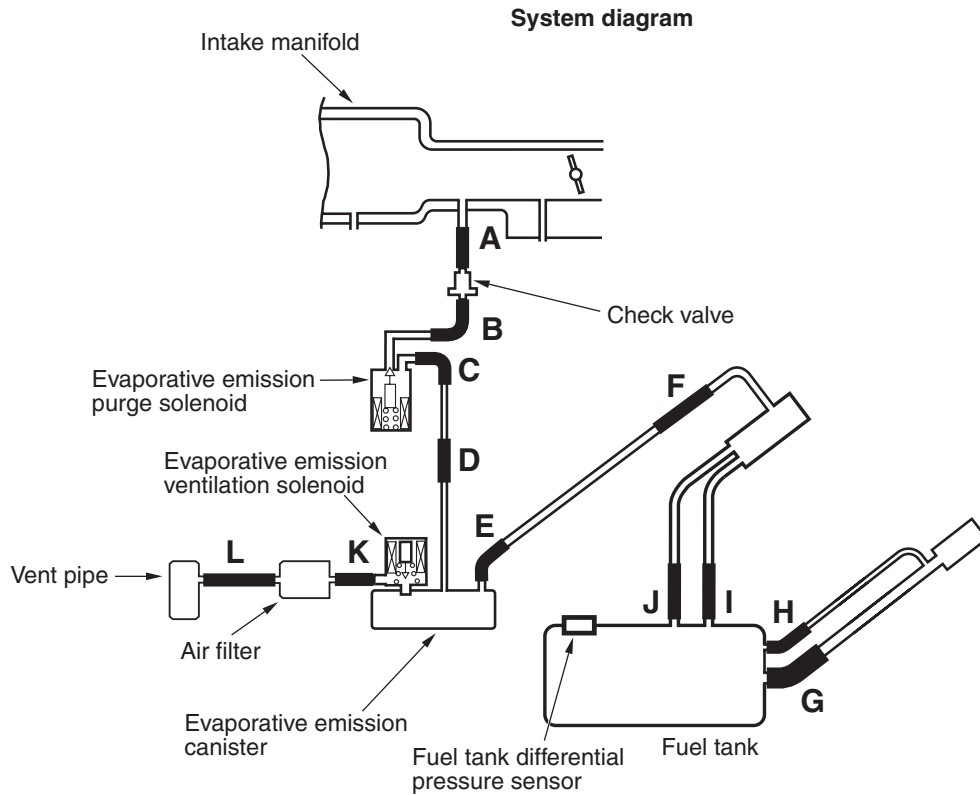
(2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0450 set?

YES : Repeat the troubleshooting.

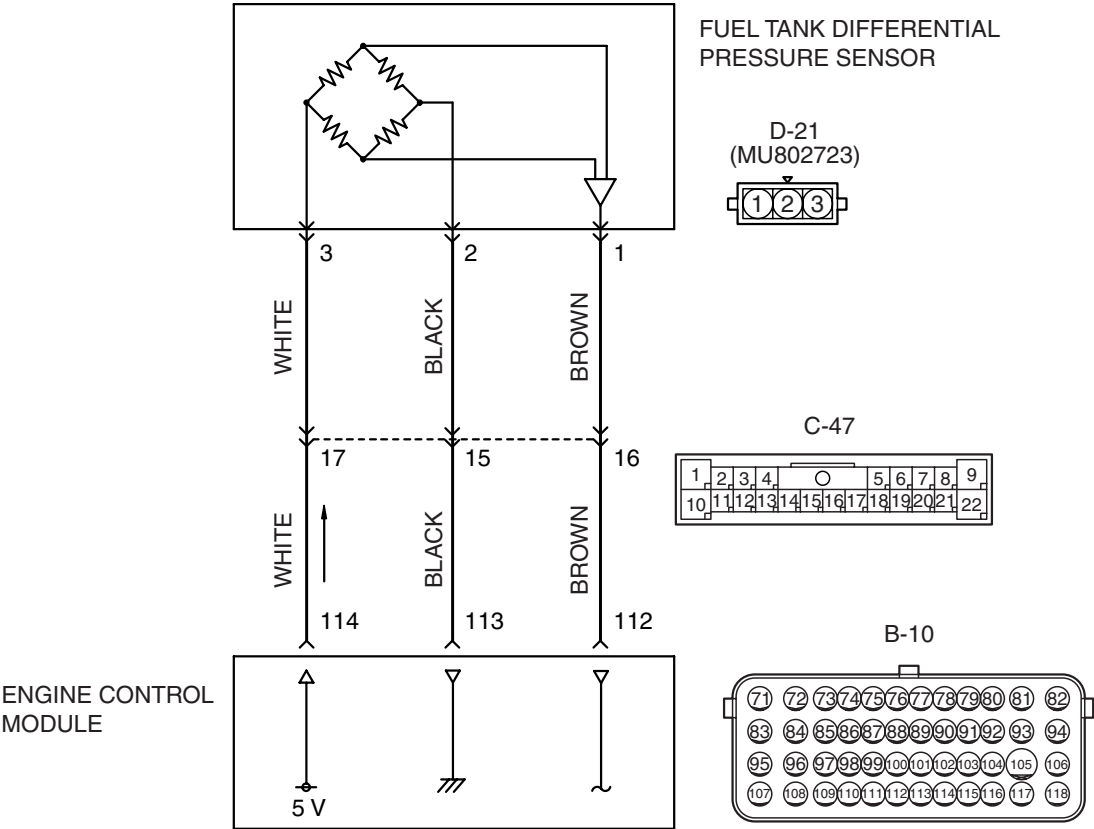
NO : The inspection is complete.

DTC P0451: Fuel tank Differential Pressure Circuit Range/Performance Problem

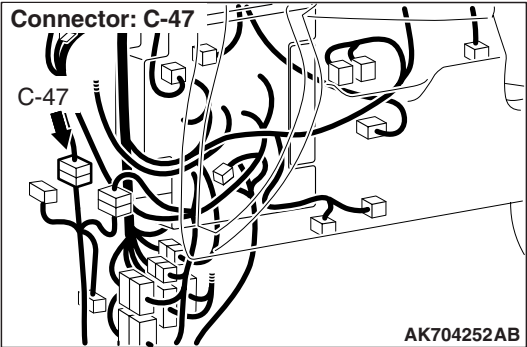
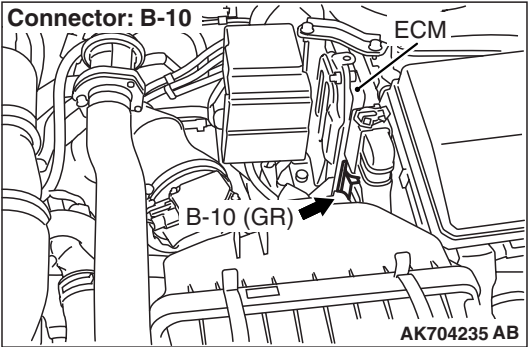


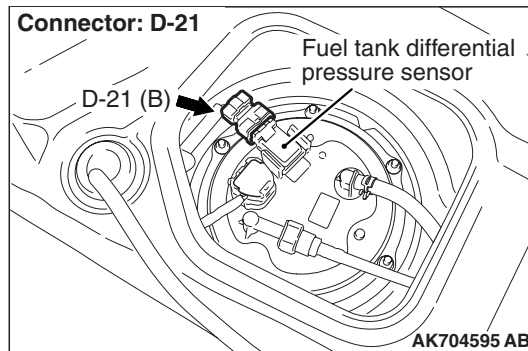
AK704684AB

FUEL TANK DIFFERENTIAL PRESSURE SENSOR CIRCUIT



AK604513 AE





CIRCUIT OPERATION

- The ECM (terminal No. 114) supplies a 5 volts reference signal to the fuel tank differential pressure sensor (terminal No. 3). The fuel tank differential pressure sensor (terminal No. 2) is grounded through the ECM (terminal No. 113).
- The fuel tank differential pressure sensor (terminal No. 1) returns a voltage signal to the ECM (terminal No. 112) that is proportional to the pressure in the fuel tank.

TECHNICAL DESCRIPTION

- The ECM monitors the fuel tank differential pressure sensor signal voltage.
- The ECM determines whether the fuel tank differential pressure sensor signal voltage is within normal operating parameters.

DESCRIPTIONS OF MONITOR METHODS

Detect malfunction if change of fuel tank differential pressure sensor output voltage during idling stays large during specified go/stop operations.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (OTHER MONITOR AND SENSOR)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

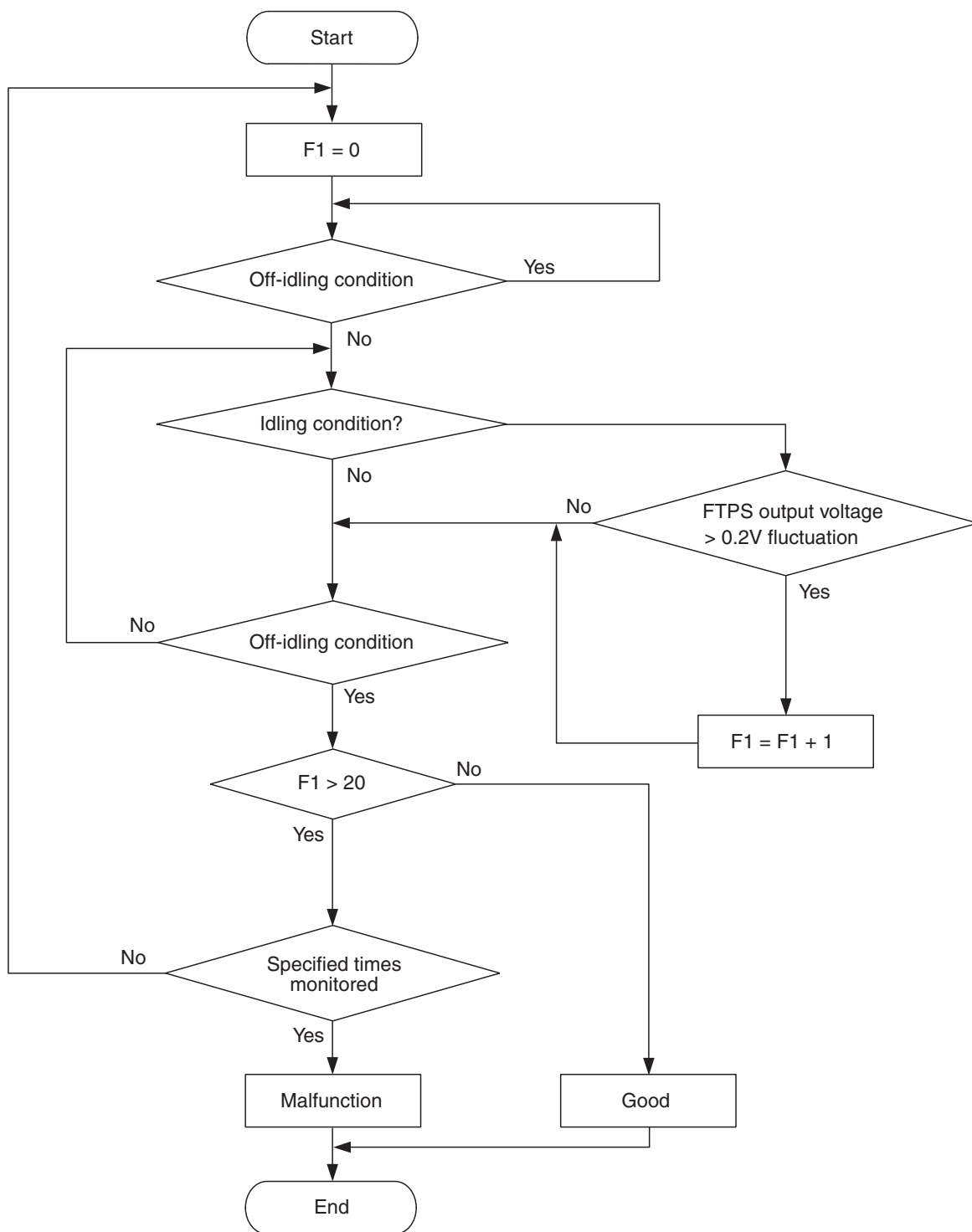
- Evaporative emission purge solenoid monitor
- Evaporative emission ventilation solenoid monitor
- Fuel tank temperature sensor monitor
- Fuel level sensor monitor

Sensor (The sensors below are determined to be normal)

- Mass airflow sensor
- Barometric pressure sensor
- Intake air temperature sensor
- Engine coolant temperature sensor
- Accelerator pedal position sensor

DTC SET CONDITIONS

Logic Flow Chart (Monitor Sequence)



AK704744

Check Conditions

- Throttle valve is closed.

- Vehicle speed is 1.5 km/h (0.93 mph) or less.

Judgement Criteria

- If the voltage signal from the fuel tank differential pressure sensor changes by 0.2 volt or more, DTC P0451 will set. The code may also set if a sudden pressure fluctuation occurs twenty times while the engine is idling, and then four consecutive times during normal driving.

NOTE: If the number of sudden pressure fluctuations does not reach twenty during any one period of engine idling, or if the ignition switch is turned OFF, the counter will reset to zero.

NOTE: The conditions for deviating from idling operation are as follows:

- Vehicle speed is 50 km/h (31 mph) or greater.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 15 [P.13A-9](#).

TROUBLESHOOTING HINTS (THE MOST LIKELY CAUSES FOR THIS CODE TO BE SET ARE:)

- Fuel tank differential pressure sensor failed.
- Fuel tank differential pressure sensor circuit harness damage, or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

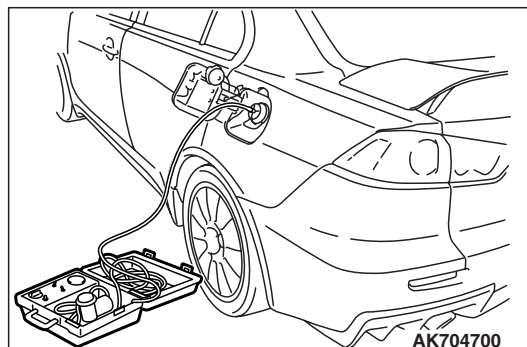
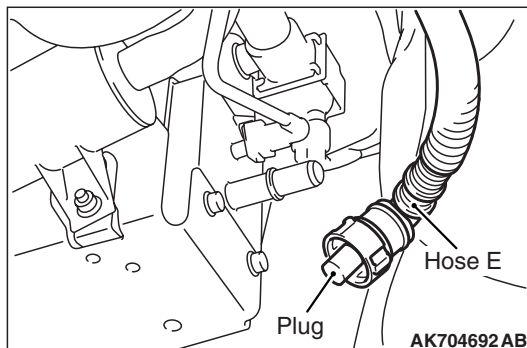
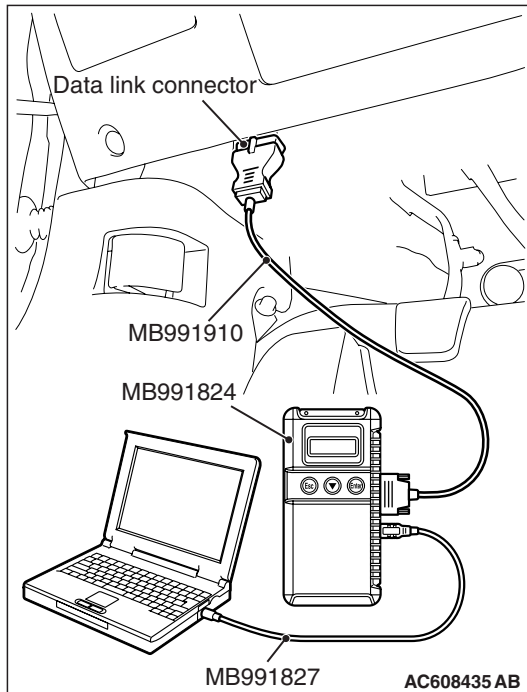
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Using scan tool MB991958, check data list item 52: Fuel Tank Differential Pressure Sensor.

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

(1) Connect scan tool MB991958 to the data link connector.



- (2) Plug the disconnected end of hose E.
- (3) Turn the ignition switch to the "ON" position.
- (4) Remove the fuel cap.
- (5) Set scan tool MB991958 to the data reading mode for item 52, Fuel Tank Differential Pressure Sensor.
 - Output voltage should be between 1,500 to 3,500 mV.

- (6) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382) and pressurize the fuel tank.
 - The fuel tank pressure reading should increase.
- (7) Turn the ignition switch to the "LOCK" (OFF) position. Then disconnect scan tool MB991958.
- (8) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.
- (9) Connect hose E to the evaporative emission canister.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting / Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2 .

STEP 2. Check harness connector D-21 at fuel tank differential pressure sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

STEP 3. Measure the sensor supply voltage at fuel tank differential pressure sensor connector D-21 by backprobing.

(1) Do not disconnect the connector D-21.

(2) Turn the ignition switch to the "ON" position.

(3) Measure the voltage between terminal No. 3 and ground by backprobing.

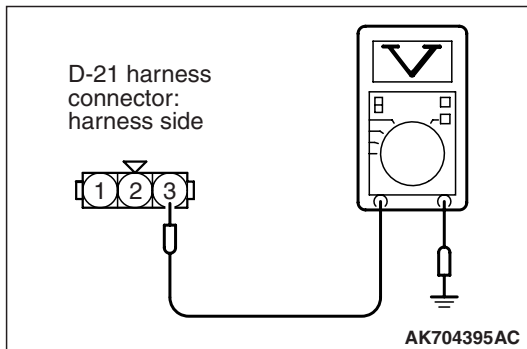
- Voltage should be between 4.9 and 5.1 volts.

(4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage between 4.9 and 5.1 volts?

YES : Go to Step 6 .

NO : Go to Step 4 .



STEP 4. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to step 5 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

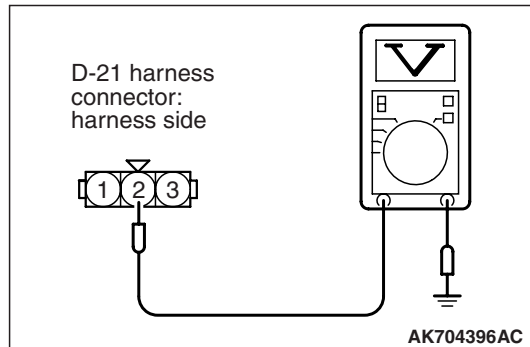
STEP 5. Check for harness damage between fuel tank differential pressure sensor connector D-21 (terminal No. 3) and ECM connector B-10 (terminal No. 114).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

Q: Is the harness wire in good condition?

YES : Go to Step 11 .

NO : Repair it. Then go to Step 12 .

**STEP 6. Measure the ground voltage at fuel tank differential pressure sensor connector D-21 by backprobing.**

- (1) Do not disconnect the connector D-21.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 2 and ground by backprobing.
 - Voltage should be 0.5 volt or less.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage 0.5 volt or less?**YES :** Go to Step 9 .**NO :** Go to Step 7 .**STEP 7. Check harness connector B-10 at ECM for damage.****Q: Is the harness connector in good condition?****YES :** Go to Step 8 .**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .**STEP 8. Check for harness damage between fuel tank differential pressure sensor connector D-21 (terminal No. 2) and ECM connector B-10 (terminal No. 113).**

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

Q: Is the harness wire in good condition?**YES :** Go to Step 11 .**NO :** Repair it. Then go to Step 12 .**STEP 9. Check harness connector B-10 at ECM for damage.****Q: Is the harness connector in good condition?****YES :** Go to Step 10 .**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

STEP 10. Check for harness damage between fuel tank differential pressure sensor connector D-21 (terminal No. 1) and ECM connector B-10 (terminal No. 112).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

Q: Is the harness wire in good condition?

YES : Go to Step 11 .

NO : Repair it. Then go to Step 12 .

STEP 11. Replace the fuel tank differential pressure sensor.

- (1) Replace the fuel tank differential pressure sensor.
- (2) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 15 [P.13A-9](#).
- (3) Check the diagnostic trouble code (DTC).

Q: Is DTC P0451 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, Diagnosis –ID Code Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, Diagnosis –ID Code Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 12 .

NO : The inspection is complete.

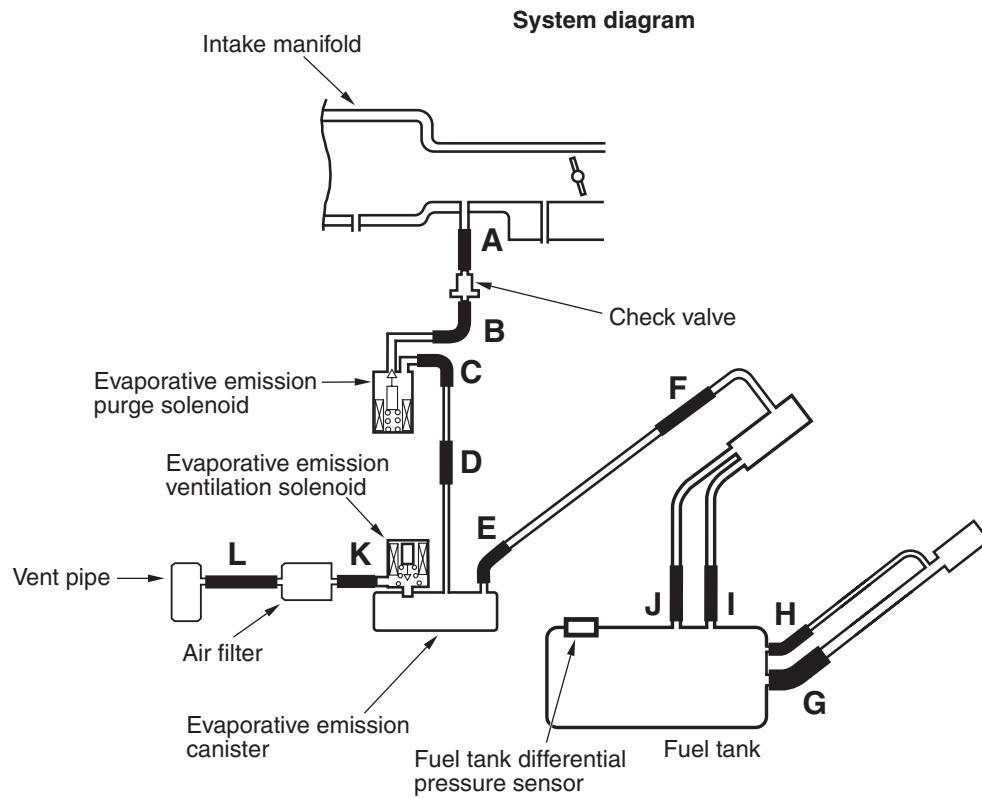
STEP 12. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 15 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0451 set?

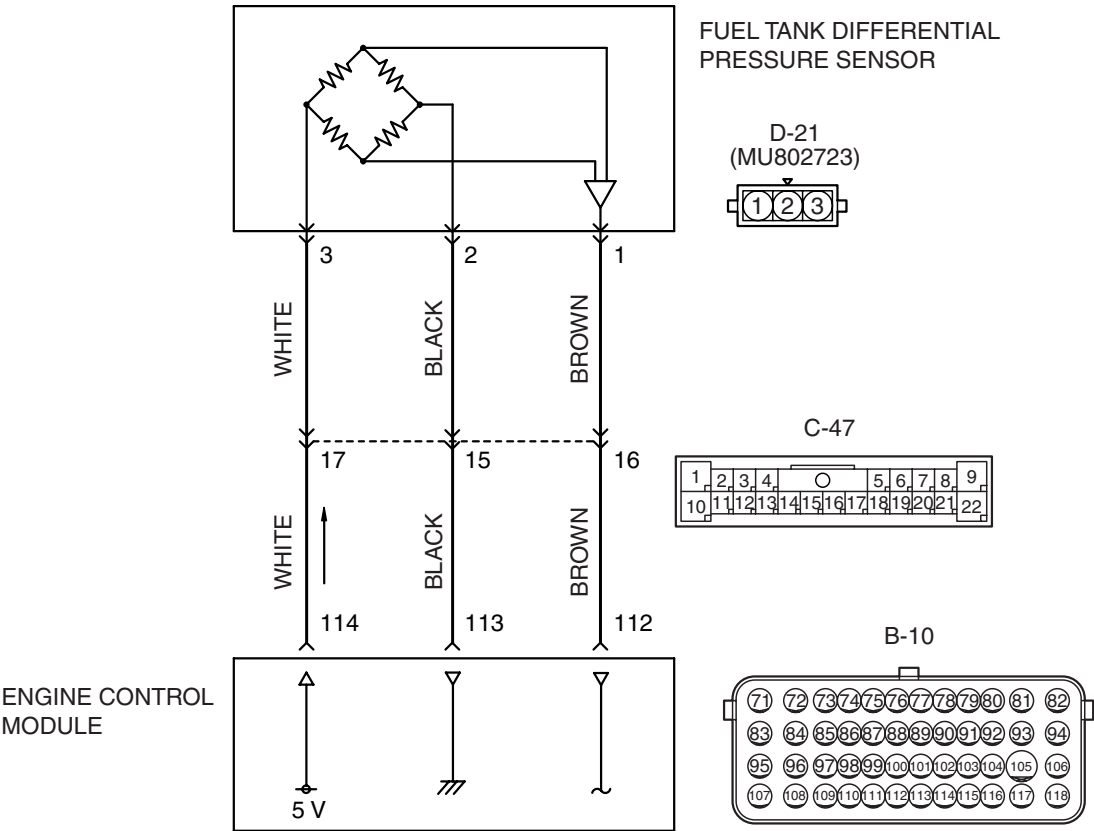
YES : Repeat the troubleshooting.

NO : The inspection is complete.

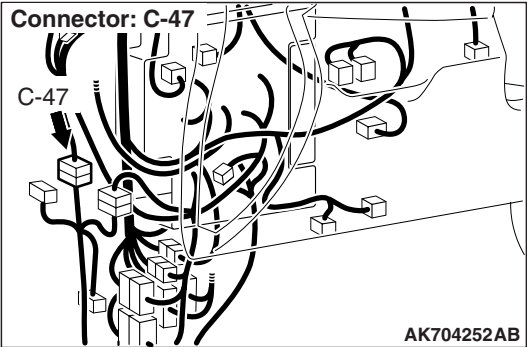
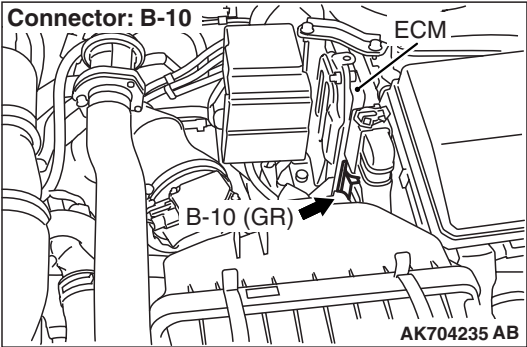
DTC P0452: Fuel Tank Differential Pressure Circuit Low Input

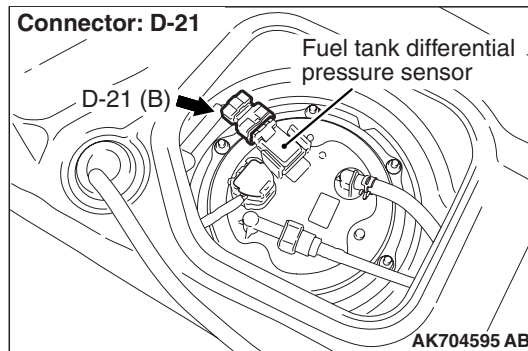
AK704684AB

FUEL TANK DIFFERENTIAL PRESSURE SENSOR CIRCUIT



AK604513 AE





CIRCUIT OPERATION

- The ECM (terminal No. 114) supplies a 5 volts reference signal to the fuel tank differential pressure sensor (terminal No. 3). The fuel tank differential pressure sensor (terminal No. 2) is grounded through the ECM (terminal No. 113).
- The fuel tank differential pressure sensor (terminal No. 1) returns a voltage signal to the ECM (terminal No. 112) that is proportional to the pressure in the fuel tank.

TECHNICAL DESCRIPTION

- The ECM monitors the fuel tank differential pressure sensor output voltage.
- The ECM determines whether the fuel tank differential pressure sensor signal voltage is within normal operating parameters.

DESCRIPTIONS OF MONITOR METHODS

Fuel tank differential pressure sensor output voltage is out of specified range.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (OTHER MONITOR AND SENSOR)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

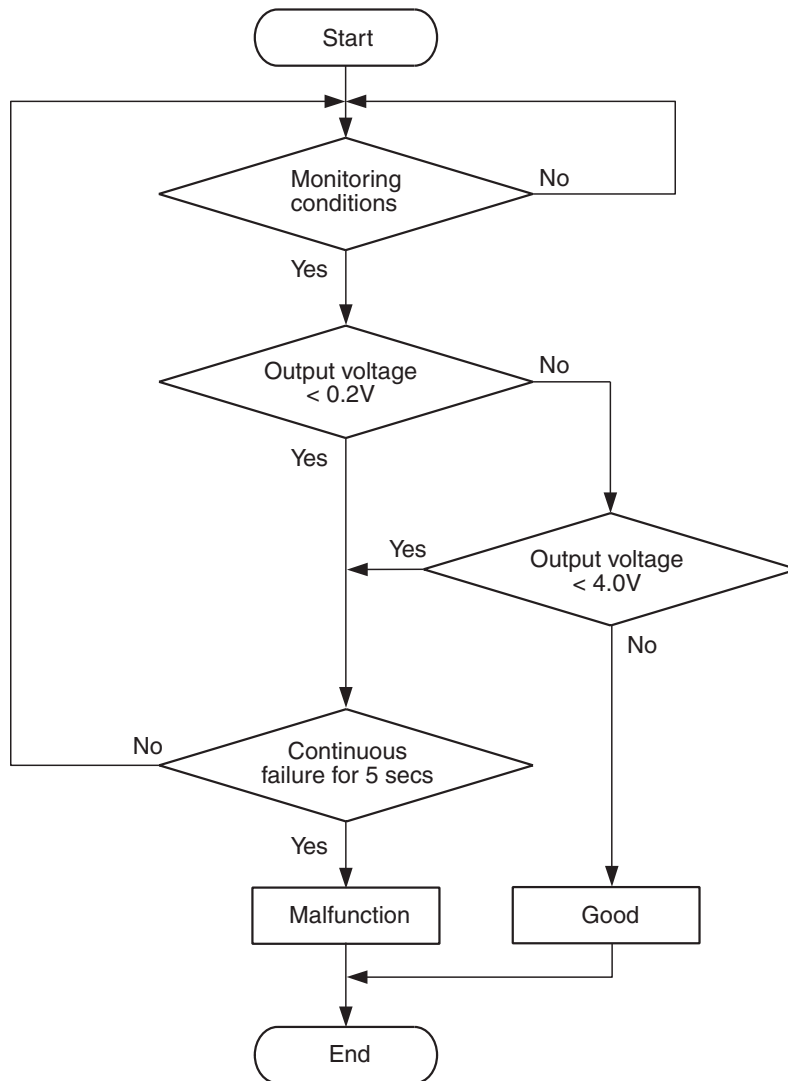
- Evaporative emission purge solenoid monitor
- Evaporative emission ventilation solenoid monitor
- Fuel tank temperature sensor monitor
- Fuel level sensor monitor

Sensor (The sensors below are determined to be normal)

- Mass airflow sensor
- Barometric pressure sensor
- Intake air temperature sensor
- Engine coolant temperature sensor
- Accelerator pedal position sensor

DTC SET CONDITIONS

Logic Flow Chart (Monitor Sequence)



AK704742

Check Conditions

- 2 seconds or more have passed since the starting sequence was completed.

Judgement Criteria

- The fuel tank differential pressure sensor output voltage remains 0.2 volt or less for 5 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (THE MOST LIKELY CAUSES FOR THIS CODE TO BE SET ARE:)

- Fuel tank differential pressure sensor failed.
- Open or shorted fuel tank differential pressure sensor circuit, connector damage.
- ECM failed.

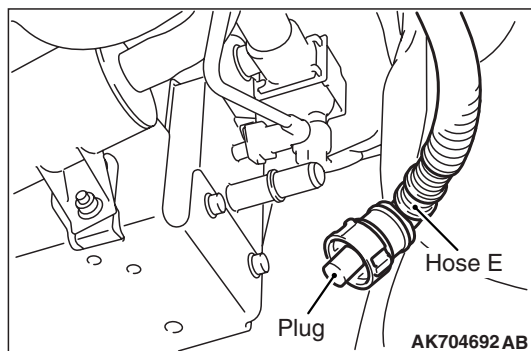
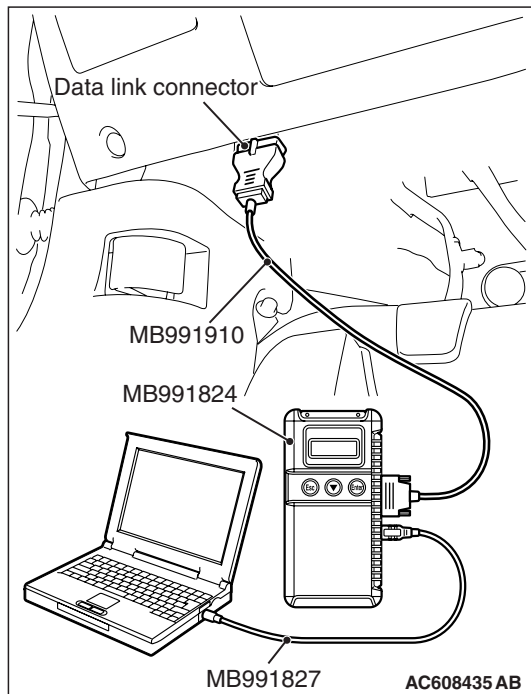
DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A
- MB992110: Power Plant ECU Check Harness

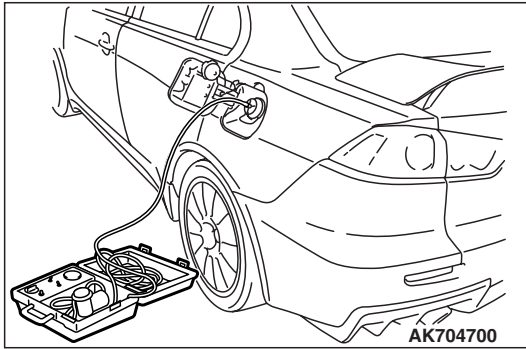
STEP 1. Using scan tool MB991958, check data list item 52: Fuel Tank Differential Pressure Sensor.**⚠ CAUTION**

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

(1) Connect scan tool MB991958 to the data link connector.



- (2) Disconnect hose E from the evaporative emission canister, and plug the hose.
- (3) Turn the ignition switch to the "ON" position.
- (4) Remove the fuel cap.
- (5) Set scan tool MB991958 to the data reading mode for item 52, Fuel Tank Differential Pressure Sensor.
 - Output voltage should be between 1,500 to 3,500 mV.



- (6) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382) and pressurize the fuel tank.
 - The fuel tank pressure reading should increase.
- (7) Turn the ignition switch to the "LOCK" (OFF) position. Then disconnect scan tool MB991958.
- (8) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.
- (9) Connect hose E to the evaporative emission canister.

Q: Is the sensor operating properly?

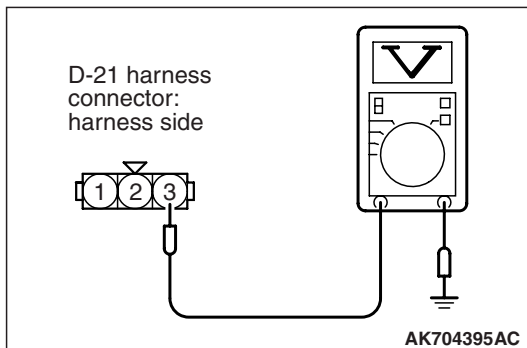
- YES :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).
- NO :** Go to Step 2 .

STEP 2. Measure the sensor supply voltage at fuel tank differential pressure sensor connector D-21 by backprobing.

- (1) Do not disconnect the connector D-21.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 3 and ground by backprobing.
 - Voltage should be between 4.9 and 5.1 volts.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage between 4.9 and 5.1 volts?

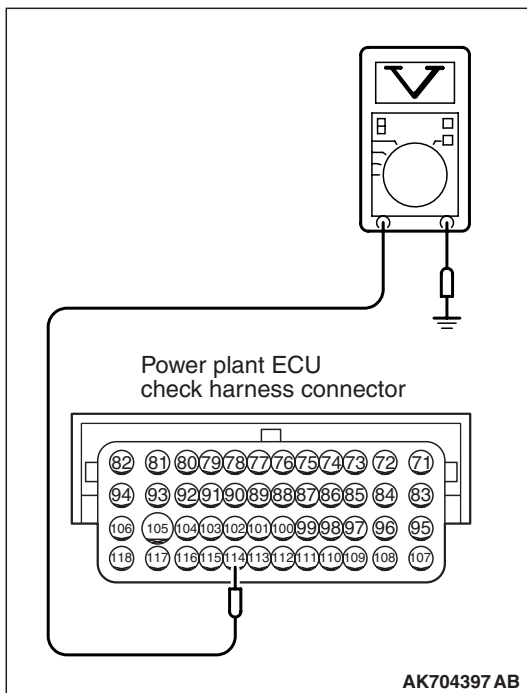
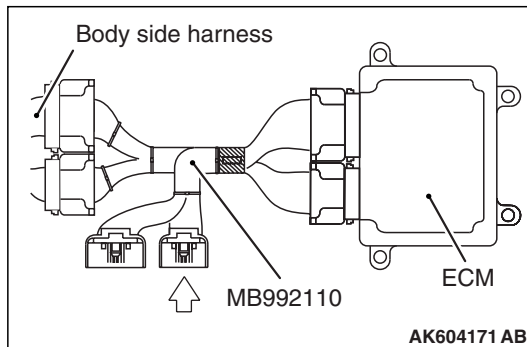
- YES :** Go to Step 8 .
- NO :** Go to Step 3 .



STEP 3. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

- YES :** Go to Step 4 .
- NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .



STEP 4. Measure the sensor supply voltage at ECM connector B-10 by using power plant ECU check harness special tool MB992110.

- (1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.
- (2) Turn the ignition switch to the "ON" position.

- (3) Measure the voltage between terminal No. 114 and ground.
 - Voltage should be between 4.9 and 5.1 volts.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage between 4.9 and 5.1 volts?

YES : Go to Step 7 .

NO : Go to Step 5 .

STEP 5. Check harness connector D-21 at the fuel tank differential pressure sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 6 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

STEP 6. Check for short circuit to ground between fuel tank differential pressure sensor connector D-21 (terminal No. 3) and ECM connector B-10 (terminal No. 114).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12.

Q: Is the harness wire in good condition?

YES : Go to Step 11 .

NO : Repair it. Then go to Step 12 .

STEP 7. Check harness connector D-21 at the fuel tank differential pressure sensor for damage.

Q: Is the harness connector in good condition?

YES : Check harness connector C-47 at intermediate connector for damage, and repair or replace as required. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). If intermediate connector is in good condition, repair harness wire between fuel tank differential pressure sensor connector D-21 (terminal No. 3) and ECM connector B-10 (terminal No. 114) because of open circuit. Then go to Step 12 .

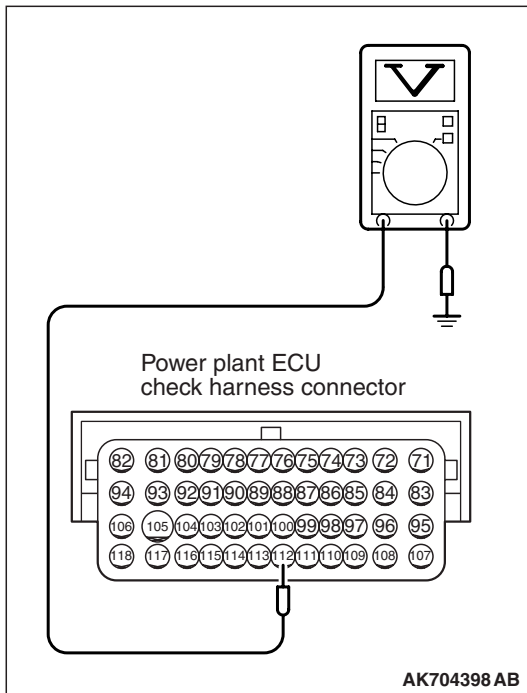
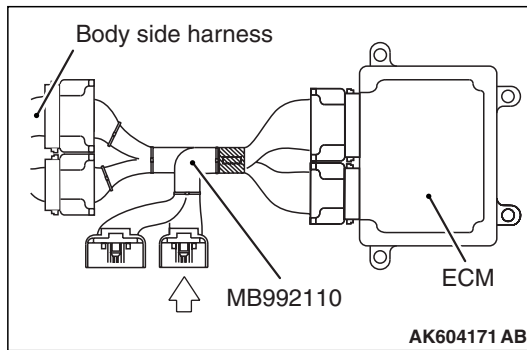
NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .

STEP 8. Check harness connector D-21 at the fuel tank differential pressure sensor and connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 9 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 12 .



STEP 9. Measure the sensor output voltage at ECM connector B-10 by using power plant ECU check harness special tool MB992110.

- (1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.
- (2) Turn the ignition switch to the "ON" position.
- (3) Remove the fuel cap

- (4) Measure the voltage between terminal No. 112 and ground.
 - Voltage should be between 1.5 and 3.5 volts.
- (5) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage normal?

YES : Go to Step 11 .

NO : Go to Step 10 .

STEP 10. Check for open circuit and short circuit to ground between fuel tank differential pressure sensor connector D-21 (terminal No. 1) and ECM connector B-10 (terminal No. 112).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection P.00E-2. Then go to Step 12.

Q: Is the harness wire in good condition?

YES : Replace the fuel tank differential pressure sensor.
Then go to Step 12 .

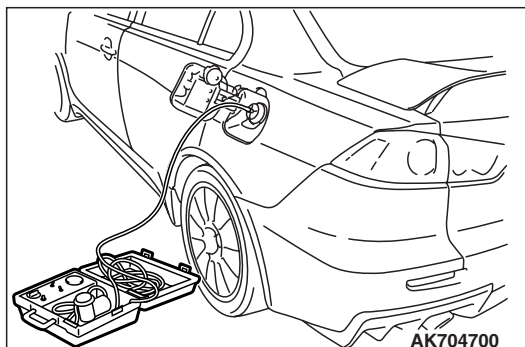
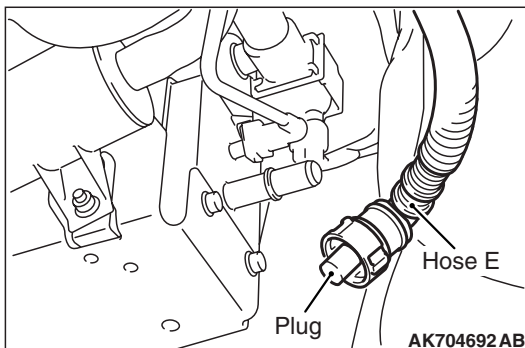
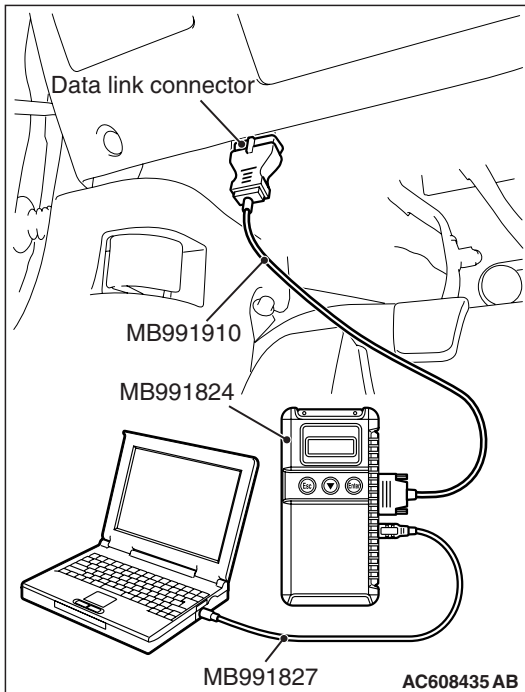
NO : Repair it. Then go to Step 12 .

STEP 11. Using scan tool MB991958, check data list item 52: Fuel Tank Differential Pressure Sensor.

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

(1) Connect scan tool MB991958 to the data link connector.



(2) Disconnect hose E from the evaporative emission canister, and plug the hose.

(3) Turn the ignition switch to the "ON" position.

(4) Remove the fuel cap.

(5) Set scan tool MB991958 to the data reading mode for item 52, Fuel Tank Differential Pressure Sensor.

- Output voltage should be between 1,500 to 3,500 mV.

(6) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382) and pressurize the fuel tank.

- The fuel tank pressure reading should increase.

(7) Turn the ignition switch to the "LOCK" (OFF) position. Then disconnect scan tool MB991958.

(8) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.

(9) Connect hose E to the evaporative emission canister.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent.
 Refer to GROUP 00, How to Use
 Troubleshooting/Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced,
 register the ID code. Refer to GROUP 42B, Diagnosis
 –ID Code Registration Judgment Table <Vehicles with
 KOS> [P.42B-12](#) or GROUP 42C, Diagnosis –ID Code
 Registration Judgment Table <Vehicles with WCM>
[P.42C-9](#). Then go to Step 12 .

STEP 12. Test the OBD-II drive cycle.

(1) Carry out a test drive with the drive cycle pattern. Refer to
 Diagnostic Function –OBD-II Drive Cycle –Pattern 23
[P.13A-9](#).

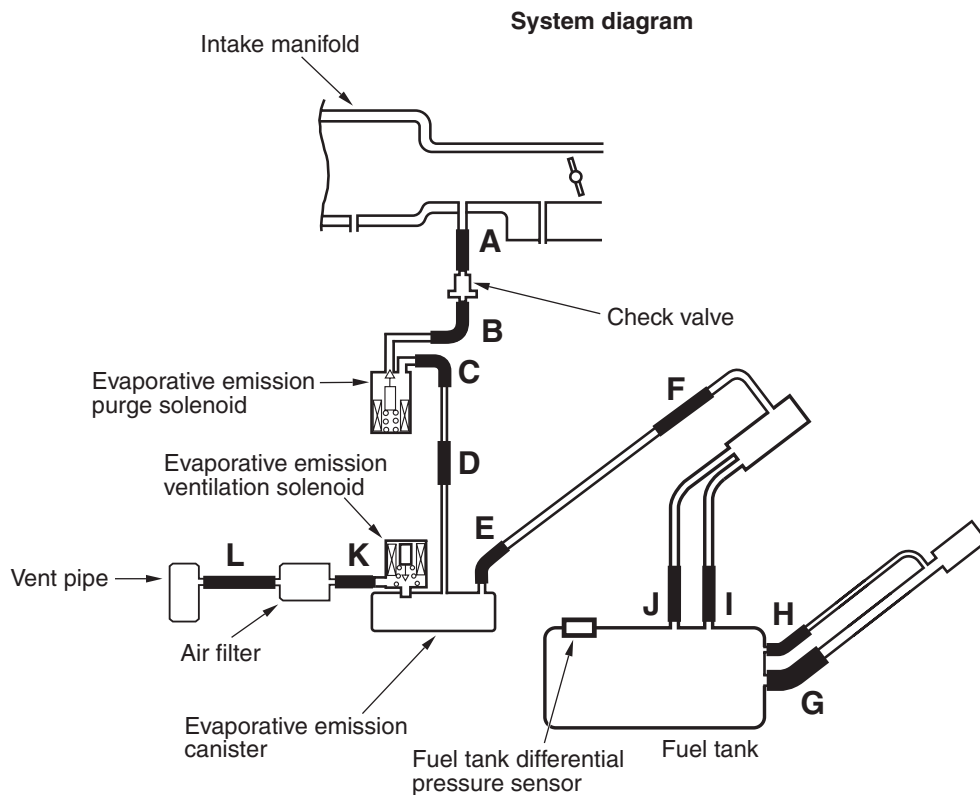
(2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0452 set?

YES : Retry the troubleshooting.

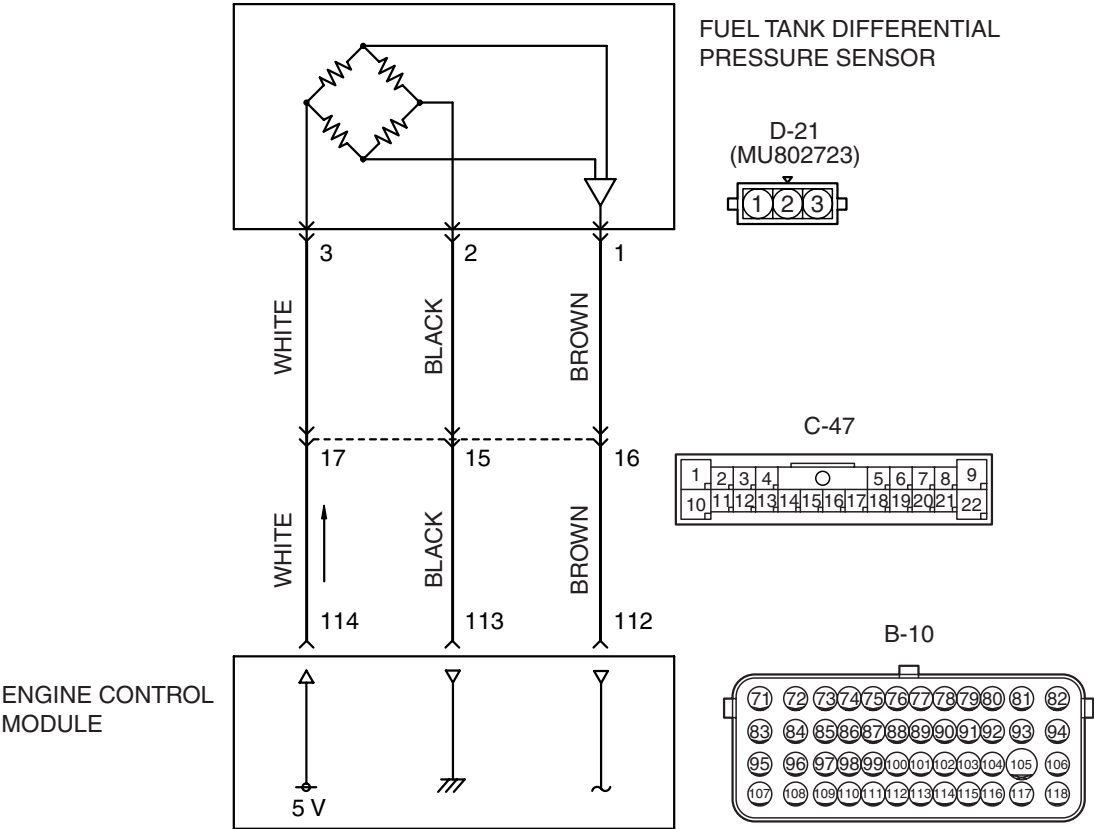
NO : The inspection is complete.

DTC P0453: Fuel tank Differential Pressure Circuit High Input

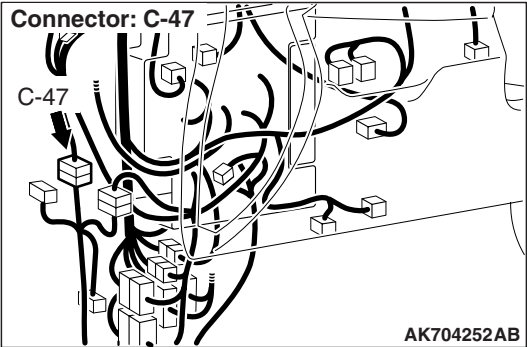
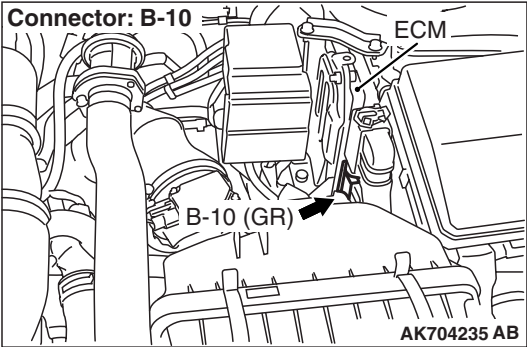


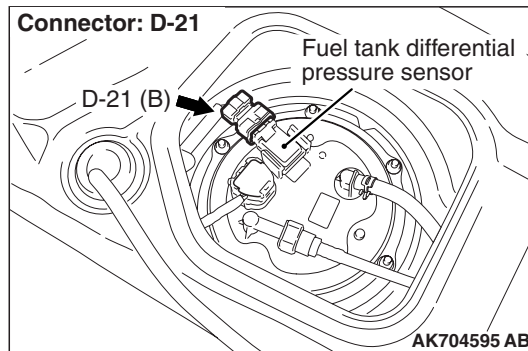
AK704684AB

FUEL TANK DIFFERENTIAL PRESSURE SENSOR CIRCUIT



AK604513 AE





CIRCUIT OPERATION

- The ECM (terminal No. 114) supplies a 5 volts reference voltage to the fuel tank differential pressure sensor (terminal No. 3). The ECM (terminal No. 113) supplies a ground to the fuel tank differential pressure sensor (terminal No. 2).
- The ECM (terminal No. 112) receives a voltage signal proportional to the pressure in the fuel tank from the fuel tank differential pressure sensor (terminal No. 1).

TECHNICAL DESCRIPTION

- To determine whether the fuel tank differential pressure sensor is defective, the ECM monitors the fuel tank differential pressure sensor output voltage.
- The ECM judges if the fuel tank differential pressure sensor output voltage is normal.

NOTE: In rare cases, this DTC may be also set under some fuel and driving conditions regardless of the fuel pressure sensor output voltage when the fuel system is clogged.

DESCRIPTIONS OF MONITOR METHODS

Fuel tank differential pressure sensor output voltage is out of specified range.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (OTHER MONITOR AND SENSOR)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

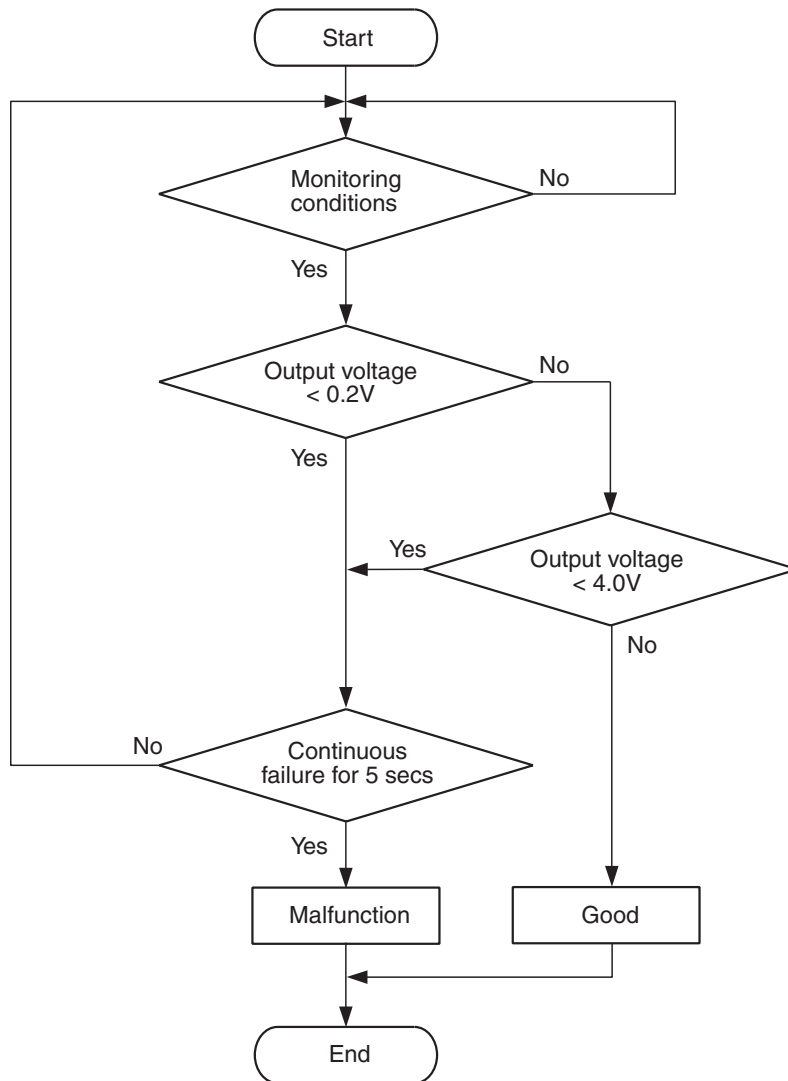
- Evaporative emission purge solenoid monitor
- Evaporative emission ventilation solenoid monitor
- Fuel tank temperature sensor monitor
- Fuel level sensor monitor

Sensor (The sensors below are determined to be normal)

- Mass airflow sensor
- Barometric pressure sensor
- Intake air temperature sensor
- Engine coolant temperature sensor
- Accelerator pedal position sensor

DTC SET CONDITIONS

Logic Flow Chart (Monitor Sequence)



AK704742

Check Conditions

- 2 seconds or more have passed since the starting sequence was completed.
- The fuel temperature is 36° C (97° F) or less.
- Remaining fuel level is 85 percent or less when the engine is started.

Judgement Criteria

- The fuel tank differential pressure sensor output voltage remains 4.8 volts or greater for 5 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (THE MOST LIKELY CAUSES FOR THIS CODE TO BE SET ARE:)

- Fuel tank differential pressure sensor failed.
- Open fuel tank differential pressure sensor circuit, or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

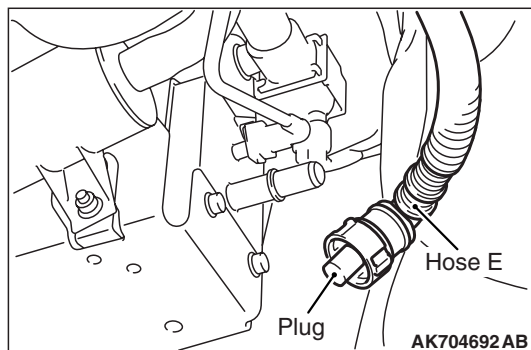
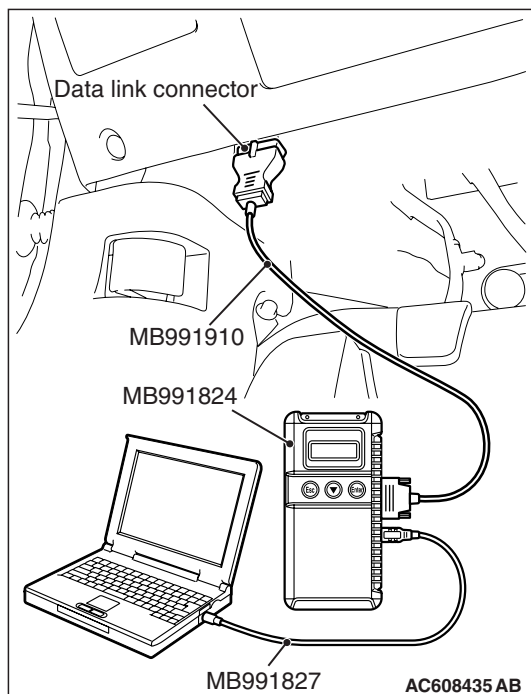
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A

STEP 1. Using scan tool MB991958, check data list item 52: Fuel Tank Differential Pressure Sensor.

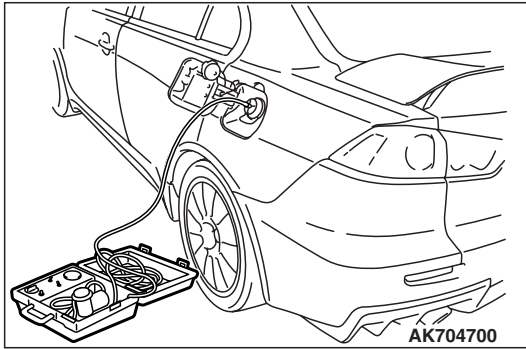
⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

(1) Connect scan tool MB991958 to the data link connector.



- (2) Disconnect hose E from the evaporative emission canister, and plug the hose.
- (3) Turn the ignition switch to the "ON" position.
- (4) Remove the fuel cap.
- (5) Set scan tool MB991958 to the data reading mode for item 52, Fuel Tank Differential Pressure Sensor.
 - Output voltage should be between 1,500 to 3,500 mV.



- (6) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382) and pressurize the fuel tank.
 - The fuel tank pressure reading should increase.
- (7) Turn the ignition switch to the "LOCK" (OFF) position. Then disconnect scan tool MB991958.
- (8) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.
- (9) Connect hose E to the evaporative emission canister.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. (Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#)).

NO : Go to Step 2 .

STEP 2. Measure the ground voltage at fuel tank differential pressure sensor connector D-21 by backprobing.

- (1) Do not disconnect the connector D-21.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 2 and ground by backprobing.

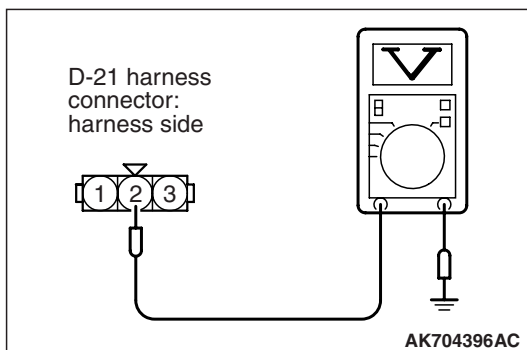
- Voltage should be 0.5 volt or less.

- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage 0.5 volt or less?

YES : Go to Step 6 .

NO : Go to Step 3 .



STEP 3. Check harness connector D-21 at the fuel tank differential pressure sensor and harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 4 .

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7 .

STEP 4. Check for open circuit between fuel tank differential pressure sensor connector D-21 (terminal No. 2) and ECM connector B-10 (terminal No. 113).

NOTE: Check harness after checking intermediate connector C-47. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7 .

Q: Is the harness wire in good condition?

YES : Go to Step 5 .

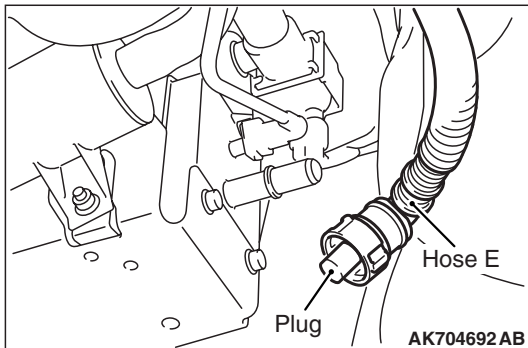
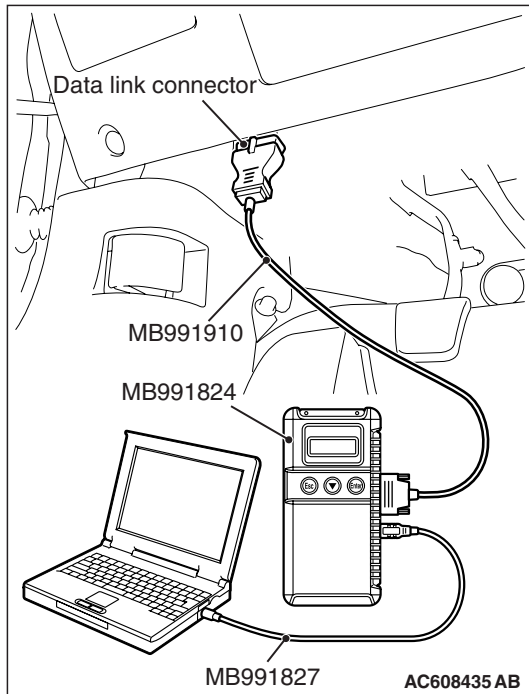
NO : Repair it. Then go to Step 7 .

STEP 5. Using scan tool MB991958, check data list item 52: Fuel Tank Differential Pressure Sensor.

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

(1) Connect scan tool MB991958 to the data link connector.



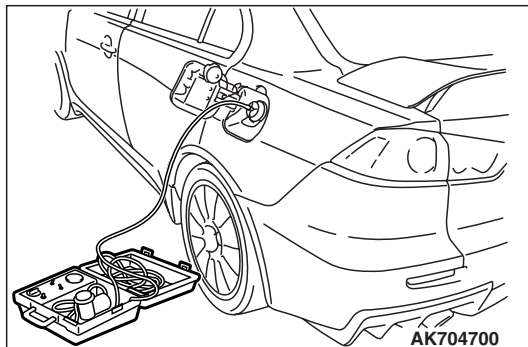
(2) Disconnect hose E from the evaporative emission canister, and plug the hose.

(3) Turn the ignition switch to the "ON" position.

(4) Remove the fuel cap.

(5) Set scan tool MB991958 to the data reading mode for item 52, Fuel Tank Differential Pressure Sensor.

- Output voltage should be between 1,500 to 3,500 mV.



(6) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382) and pressurize the fuel tank.

- The fuel tank pressure reading should increase.

(7) Turn the ignition switch to the "LOCK" (OFF) position. Then disconnect scan tool MB991958.

(8) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.

(9) Connect hose E to the evaporative emission canister.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent.
Refer to GROUP 00, How to Use
Troubleshooting/Inspection Service Points –How to
Cope with Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced,
register the ID code. Refer to GROUP 42B, Diagnosis
–ID Code Registration Judgment Table <Vehicles with
KOS> [P.42B-12](#) or GROUP 42C, Diagnosis –ID
Codes Registration Judgment Table <Vehicles with
WCM> [P.42C-9](#). Then go to Step 7 .

**STEP 6. Check harness connector D-21 at the fuel tank
differential pressure sensor and harness connector B-10
at ECM for damage.**

Q: Is the harness connector in good condition?

YES : Replace the fuel tank differential pressure sensor.
Then go to Step 7 .

NO : Repair or replace it. Refer to GROUP 00E, Harness
Connector Inspection [P.00E-2](#). Then go to Step 7 .

STEP 7. Test the OBD-II drive cycle.

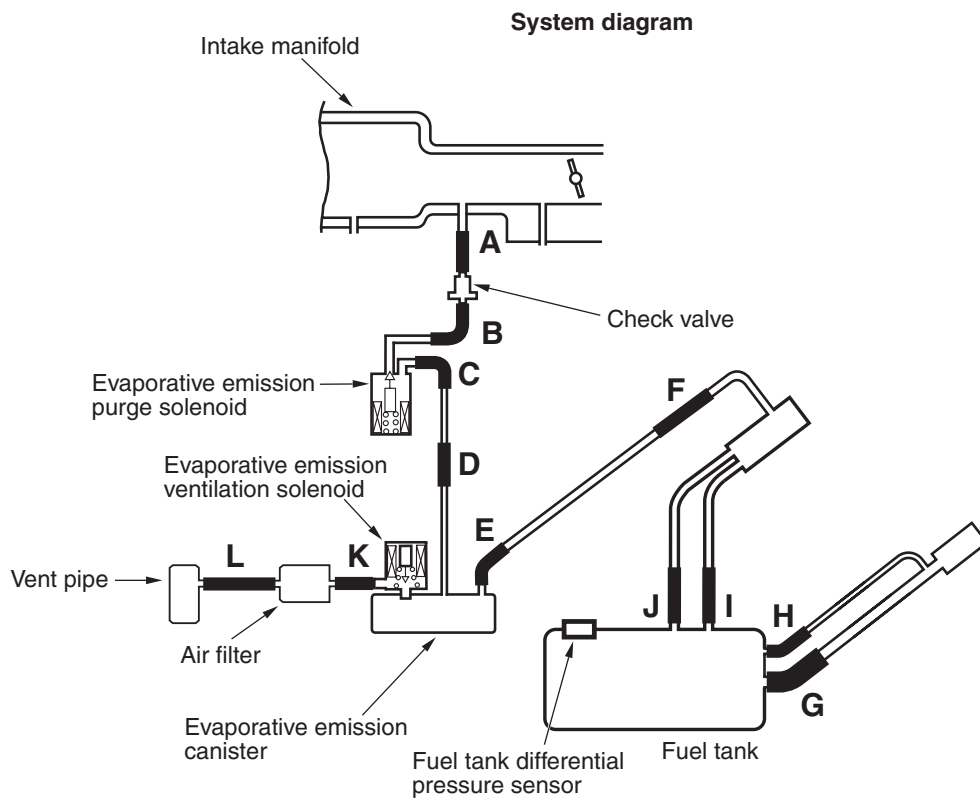
(1) Carry out a test drive with the drive cycle pattern. Refer to
Diagnostic Function –OBD-II Drive Cycle –Pattern 23
[P.13A-9](#).

(2) Check the diagnostic trouble code (DTC).

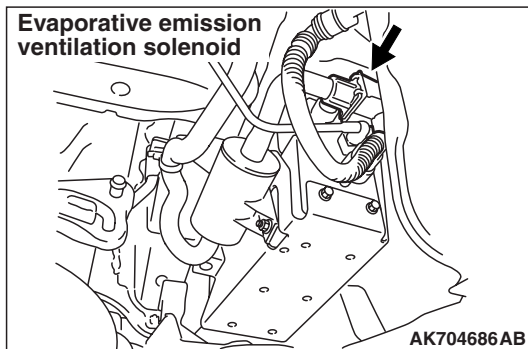
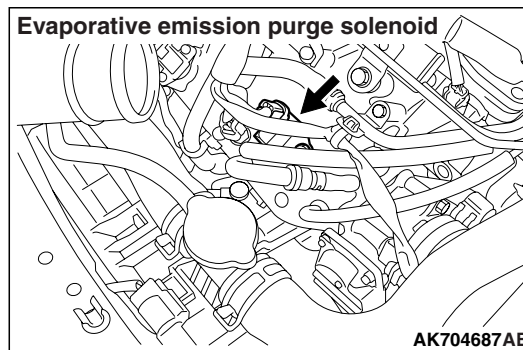
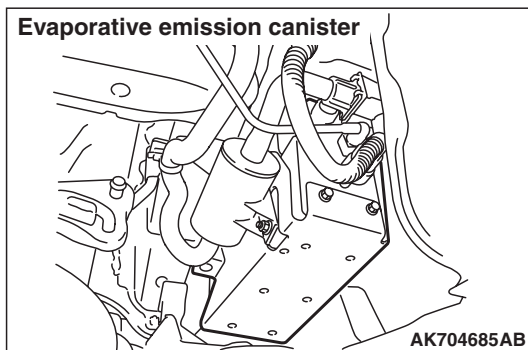
Q: Is DTC P0453 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0455: Evaporative Emission System Leak Detected (Gross Leak)

AK704684AB



TECHNICAL DESCRIPTION

- The fuel tank may be under a slight pressure or vacuum depending on the state of the Evaporative Emission (EVAP) System. The ECM monitors and responds to these pressure/vacuum changes. If the pressure/vacuum varies from the specified range, the ECM will set DTC P0455.
- The ECM energizes the evaporative emission ventilation solenoid to shut off the evaporative emission canister outlet port.
- The evaporative emission purge solenoid is activated to apply engine manifold vacuum to the EVAP system.
- When the fuel system develops a vacuum of 2 kPa (0.29 psi), the evaporative emission purge solenoid is turned "off" and the fuel system vacuum is maintained at 2 kPa (0.29 psi).
- The ECM determines whether there is a leak or clog in the fuel system by measuring the change in vacuum inside the fuel tank.
- The test is stopped when fuel vapor pressure is determined to be too high.

DESCRIPTIONS OF MONITOR METHODS

Depressurizing EVAP system by intake manifold negative pressure is impossible within specified period.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (OTHER MONITOR AND SENSOR)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

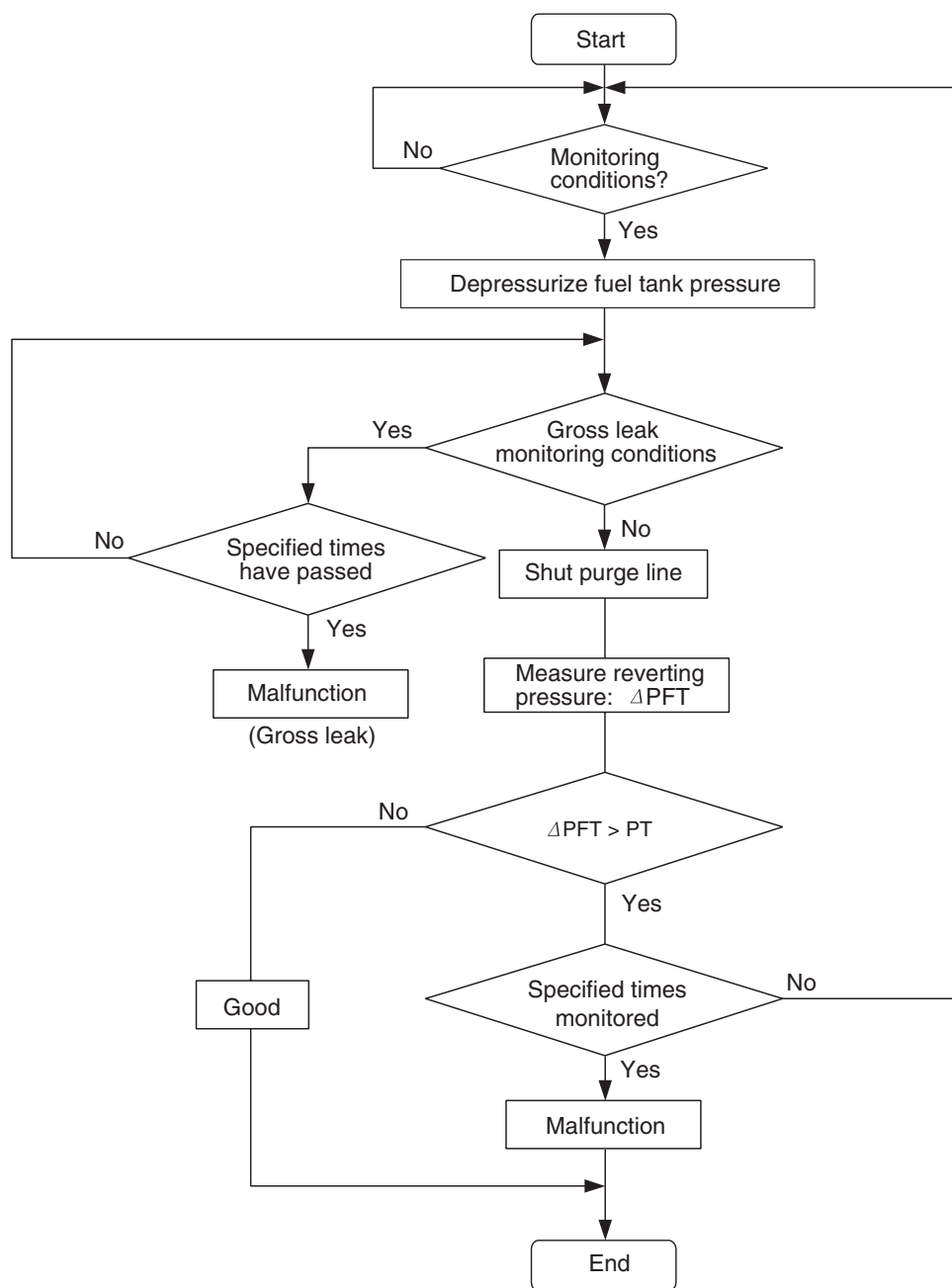
- Not applicable

Sensor (The sensors below are determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart (Monitor Sequence)



AK704743

DTC SET CONDITIONS

Check Conditions A: At Start up

- Intake air temperature is 36° C (97° F) or less upon engine start up.
- The engine coolant temperature is 36° C (97° F) or less upon engine start up.

Check Conditions B: For Test to Run

- The engine coolant temperature is 60° C (140° F) or greater and the fuel tank is 15 –40 percent full.
- The engine coolant temperature is 20° C (68° F) or greater and the fuel tank is 40 –85 percent full.
- The engine speed is greater than 1,594 r/min.
- Barometric pressure is greater than 76 kPa (11 psi).
- Volumetric efficiency is between 20 and 80 percent.

- The fuel temperature is 36° C (97° F) or less.
- The fuel tank differential pressure sensor output voltage is 1 –4 volts.

Check Conditions C: For Test to Stop

- The intake air temperature is greater than 5° C (41° F).
- When the evaporative emission purge solenoid and evaporative emission ventilation solenoid are closed, the pressure in the fuel tank rises to 451 Pa (0.065 psi) or less and the amount of remaining fuel is 15 –40 percent of capacity upon engine start-up.
- When the evaporative emission purge solenoid and evaporative emission ventilation solenoid are closed, the pressure in the fuel tank rises to 324 Pa (0.047 psi) or less and the amount of remaining fuel is 40 –85 percent of capacity upon engine start-up.
- 10 seconds have elapsed from the start of the previous monitoring.
- Monitoring time: 150 seconds.

Judgment Criteria

- The fuel tank internal pressure is -1961 Pa (-0.284 psi) or more after the evaporative emission purge solenoid valve has been driven when the fuel tank and vapor line were closed.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 4 [P.13A-9](#).

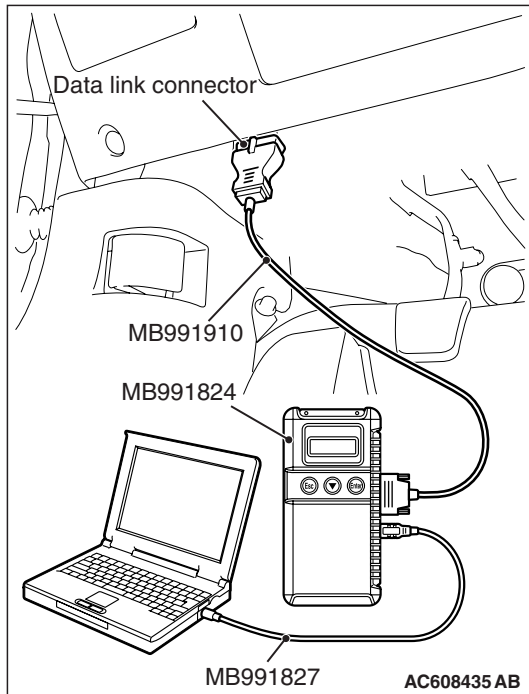
TROUBLESHOOTING HINTS (THE MOST LIKELY CAUSES FOR THIS CODE TO BE SET ARE:)

- Loose fuel cap.
- Fuel cap relief pressure is incorrect.
- Fuel overflow limiter valve failed.
- Purge line or vapor line is clogged.
- Fuel tank, purge line or vapor line seal failed.
- Evaporative emission purge solenoid valve failed.
- Evaporative emission ventilation solenoid valve failed.
- Fuel tank differential pressure sensor failed.
- Evaporative emission canister seal is faulty.
- Evaporative emission canister is clogged.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A



STEP 1. Using scan tool MB991958, check the evaporative emission system monitor test.

⚠ CAUTION

- To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.
- During this test, the ECM will automatically increase the engine speed to 1,600 r/min or greater. Check that the transaxle is set to "P" position.

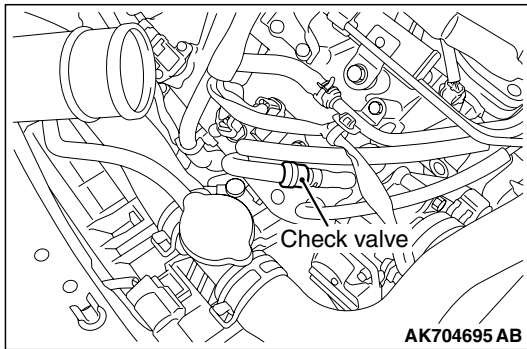
- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTCs using scan tool MB991958.
- (4) Check that the fuel cap is securely closed (Tighten until three clicks are heard).
- (5) Start the engine.
- (6) Select "Special Function"
- (7) Select "Evap Leak Monitor"
- (8) During this test, keep the accelerator pedal at the idle position.
- (9) Keep the engine speed and engine load within the specified range. When the monitor test starts, the "In Progress" item on scan tool MB991958 will change from "NO" to "YES".
- (10) Turn the ignition switch to the "LOCK" (OFF) position, and disconnect scan tool MB991958.

Q: Is "Evap Leak Mon. Completed. Test Failed and DTCs Set" displayed on scan tool MB991958?

YES : A malfunction has been detected during the monitor test. Refer to the Diagnostic Trouble Code Chart and diagnose any other DTCs that are set [P.13A-44](#). If no other DTC's have been set, go to Step 2 .

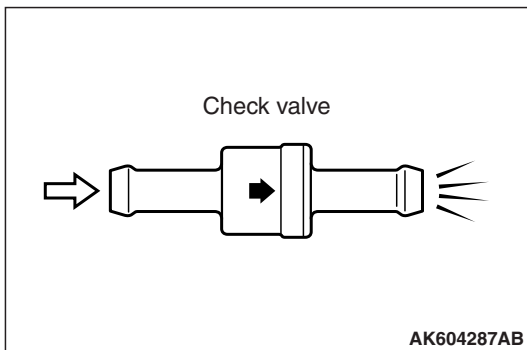
NO <"Evap Leak Mon. Completed. Test Passed" is displayed on scan tool MB991958.> : The evaporative emission system is working properly at this time. Explain to the customer that an improperly tightened fuel cap can cause the MIL to illuminate. Return the vehicle to the customer.

NO <"Evap Leak Mon. Discontinued. Retest again from the first" is displayed on scan tool MB991958.> : The EVAP monitor has been interrupted during the test. Turn the ignition switch to the "LOCK" (OFF) position once, and repeat the monitoring.



STEP 2. Test check valve.

(1) Check valve is a one-way check valve.

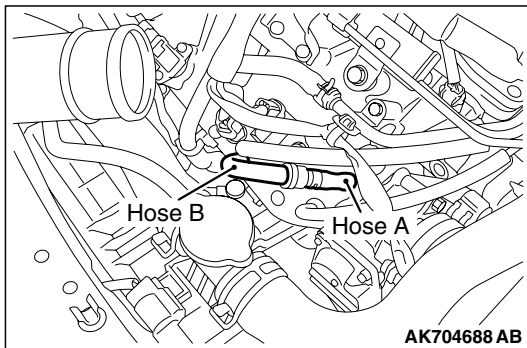


(2) Check valve should allow air to flow in only one direction.

Q: Does check valve allow air to pass in one direction only?

YES : Go to Step 3 .

NO : Replace check valve. Then go to Step 14 .



STEP 3. Check for leaks in evaporative emission hoses A and B.

Use a hand vacuum pump to test each hose A and B.

Q: Do the hoses hold vacuum?

YES : Go to Step 4 .

NO : Replace any damaged hose. Then go to Step 14 .

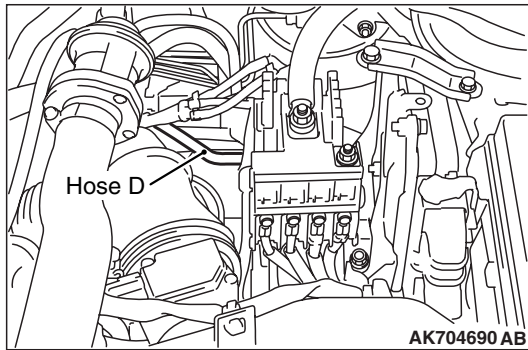
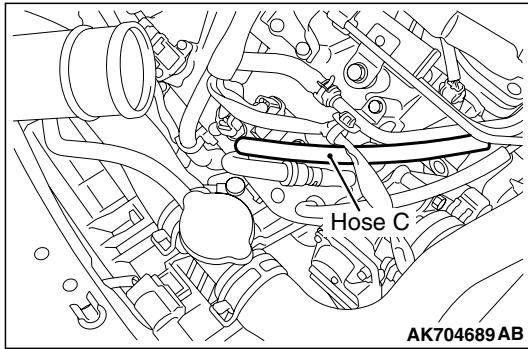
STEP 4. Check the evaporative emission purge solenoid for leaks.

Refer to GROUP 17, Emission Control –Evaporative Emission Control System –Evaporative Emission Purge Solenoid Check [P.17-77](#).

Q: Is the evaporative emission purge solenoid operating properly?

YES : Go to Step 5 .

NO : Replace the evaporative emission purge solenoid.
Then go to Step 14 .



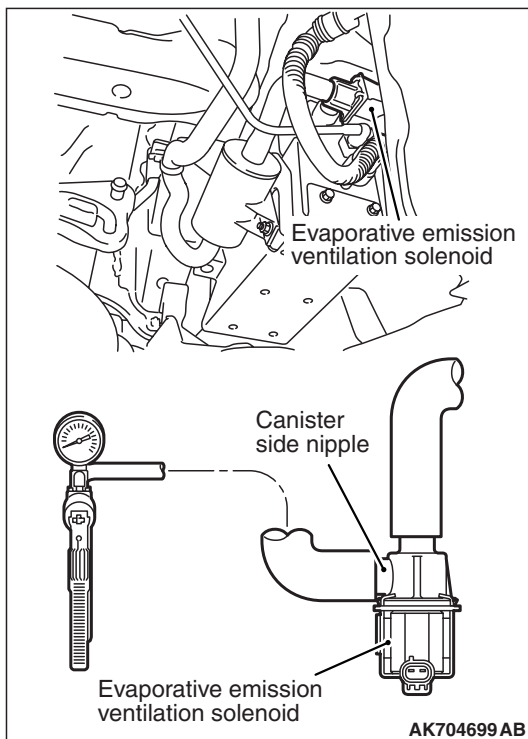
STEP 5. Check for leaks in evaporative emission hoses C and D.

Use a hand vacuum pump to test each hose C and D.

Q: Do the hoses hold vacuum?

YES : Go to Step 6 .

NO : Replace any damaged hose. Then go to Step 14 .



STEP 6. Using scan tool MB991958, check actuator test item 15: Evaporative emission ventilation solenoid.

- (1) Remove the canister cover.
- (2) Remove the evaporative emission ventilation solenoid. Do not disconnect the connector.
- (3) Connect the hose of the hand vacuum pump to the canister side nipple of the evaporative emission ventilation solenoid.
- (4) Turn the ignition switch to the "ON" position.
- (5) Set scan tool MB991958 to actuator test mode.
 - Item 15: Evaporative Emission Ventilation Solenoid.
 - While the evaporative emission ventilation solenoid is energized, operate the hand vacuum pump and confirm that the solenoid holds vacuum.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.
- (7) Disconnect the hand vacuum pump, and reinstall the evaporative emission ventilation solenoid.
- (8) Reinstall the canister cover.

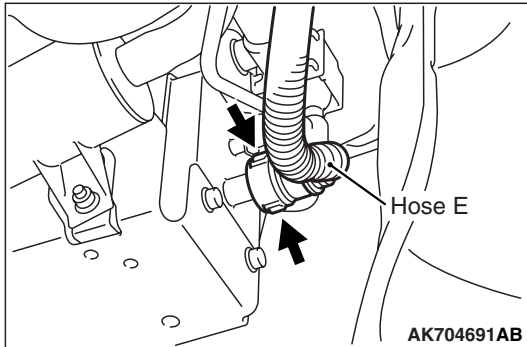
Q: Did the evaporative emission ventilation solenoid hold vacuum?

YES : Go to Step 7 .

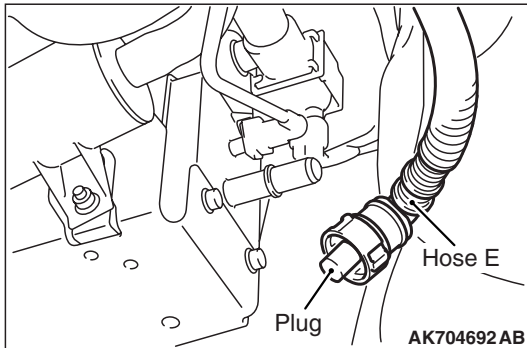
NO : Replace the evaporative emission ventilation solenoid. Then go to Step 14 .

STEP 7. Perform the pressure test on the evaporative emission system.

- (1) Disconnect hose E from the canister while holding the release buttons indicated in the illustration pressed by fingers.



- (2) Plug the disconnected end of hose E.
(3) Confirm that the evaporative emission system pressure pump (Miller number 6872A) is operating properly. Perform the self-test as described in the pump manufacturer's instructions.
(4) Remove the fuel cap.



- (5) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382).
(6) Pressure test the system to determine whether any leaks are present.

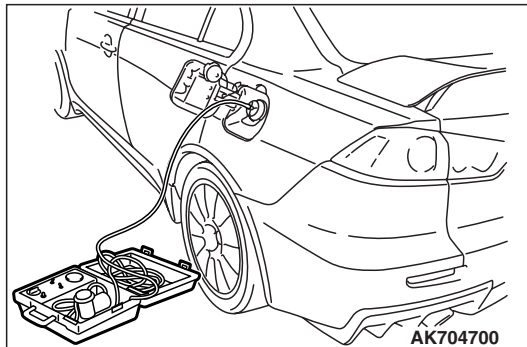
NOTE: The "Pressure test" in this procedure refers to the I/M240 Simulation Test. The eight steps of this test are described in the manufacturer's instructions for the evaporative emission system pressure pump, Miller number 6872A.

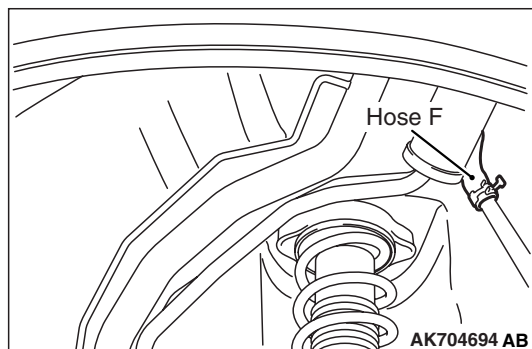
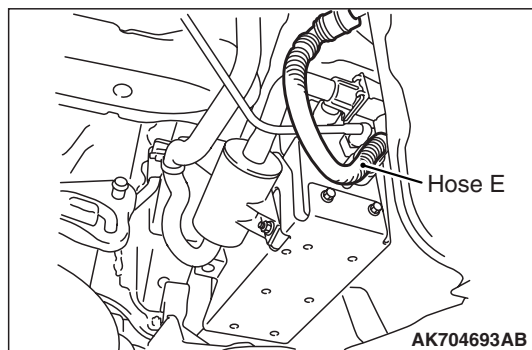
- (7) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.
(8) Connect hose E to the evaporative emission canister.

Q: Is the evaporative emission system line free of leaks?

YES : Go to Step 12 .

NO : Go to Step 8 .



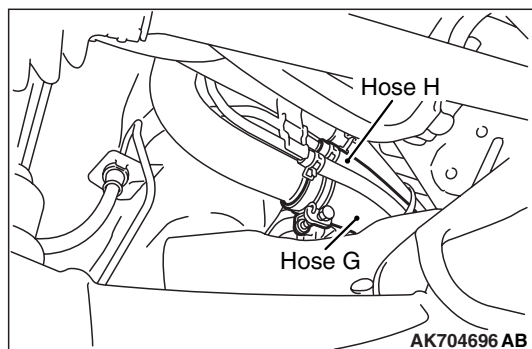
**STEP 8. Check for leaks in evaporative emission hoses E and F.**

Use a hand vacuum pump to test each hose E and F.

Q: Do the hoses hold vacuum?

YES : Go to Step 9 .

NO : Replace any damaged hose. Then go to Step 14 .

**STEP 9. Check for leaks in evaporative emission hoses G and H.**

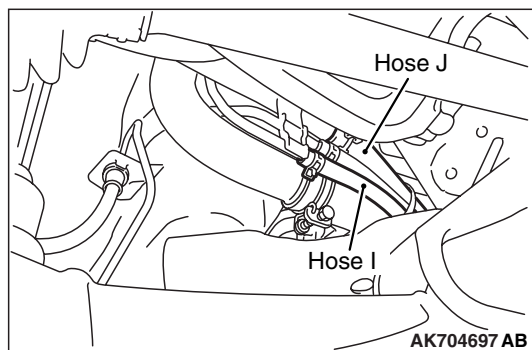
(1) Remove the fuel tank assembly.

(2) Use a hand vacuum pump to test each hose G and H.

Q: Do the hoses hold vacuum?

YES : Go to Step 10.

NO : Replace any damaged hose and reinstall the fuel tank assembly. Then go to Step 14 .

**STEP 10. Check for leaks in evaporative emission hoses I and J.**

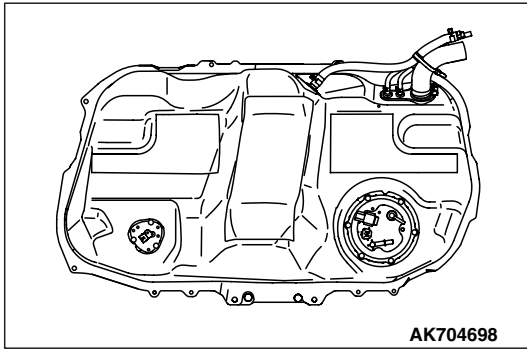
(1) Remove the fuel tank assembly.

(2) Use a hand vacuum pump to test each hose I and J.

Q: Do the hoses hold vacuum?

YES : Go to Step 11 .

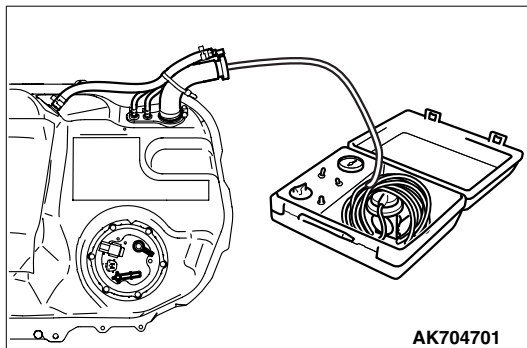
NO : Replace any damaged hose and reinstall the fuel tank assembly. Then go to Step 14 .



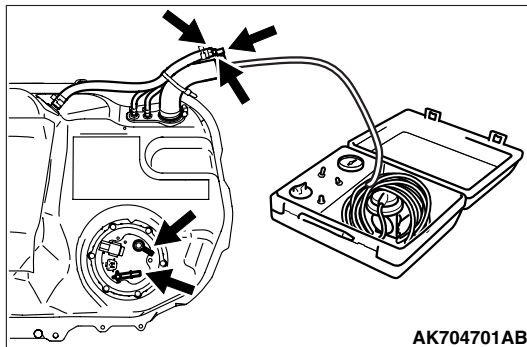
STEP 11. Check for leaks in the fuel tank.

- (1) Visually check for cracks or other leaks in the fuel tank.

NOTE: Carefully check the fuel pump module and the fuel tank differential pressure sensor installation in the fuel tank.



- (2) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel filler hose.



- (3) Plug the hose and the nipple shown in the illustration.

NOTE: If these items are not securely plugged now, the fuel could leak in the next step.

- (4) Pressurize the fuel tank with the evaporative emission system pressure pump.
(5) In the pressurized state, check for leaks by applying a soapy water solution to each section and look for bubbles.

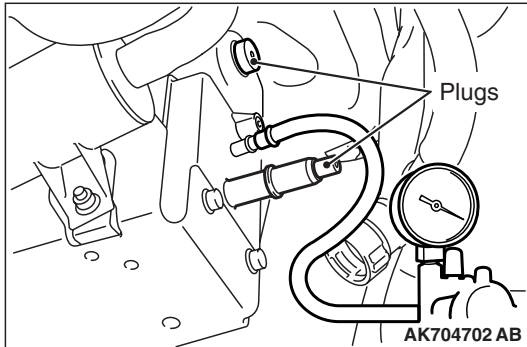
Q: Are any leaks found?

YES <When there is a leak from the attachment points of the fuel pump module, fuel tank differential pressure sensor, fuel level sensor or leveling valve.> :

Reassemble the leaked parts and check again that there are no leaks. Then reinstall the fuel tank. Then go to Step 14 .

YES <When there is a leak from the fuel tank.> : Replace the fuel tank. Then go to Step 14 .

NO : When there is no leak, reinstall the fuel tank. Then go to Step 13 .

**STEP 12. Check the evaporative emission canister for vacuum leaks.**

- (1) Connect a hand vacuum pump to the evaporative emission canister and plug the other nipples.
- (2) Apply a pressure on the hand vacuum pump, and confirm that air is maintained.
- (3) Disconnect the hand vacuum pump and remove the plugs.

Q: Is the evaporative emission canister in good condition?**YES :** Go to Step 13 .**NO :** Replace the evaporative emission canister. Then go to Step 14 .**STEP 13. Using scan tool MB991958, check the evaporative emission system monitor test.****⚠ CAUTION**

- During this test, the ECM automatically increases the engine speed to 1,600 r/min or greater. Check that the transaxle is set to "P" position.

- (1) Turn the ignition switch to the "ON" position.
- (2) Erase the DTCs using scan tool MB991958.
- (3) Check that the fuel cap is securely closed (Tighten until three clicks are heard).
- (4) Start the engine.
- (5) Select "Special Function"
- (6) Select "Evap Leak Monitor"
- (7) During the test, keep the accelerator pedal at the idle position.
- (8) Keep the engine speed and engine load within the specified range. When the monitor test starts, the "In Progress" item on scan tool MB991958 will change from "NO" to "YES".
- (9) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is "Evap Leak Mon. Completed. Test Failed and DTCs Set" displayed on scan tool MB991958?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Code Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 14 .

NO <"Evap Leak Mon. Completed. Test Passed" is displayed on scan tool MB991958.> : The evaporative emission system is working properly at this time. Go to Step 14 .

NO <"Evap Leak Mon. Discontinued. Retest again from the first" is displayed on scan tool MB991958.> : The EVAP monitor has been interrupted during the test. Turn the ignition switch to the "LOCK" (OFF) position once, and repeat the monitoring.

STEP 14. Test the OBD-II drive cycle.

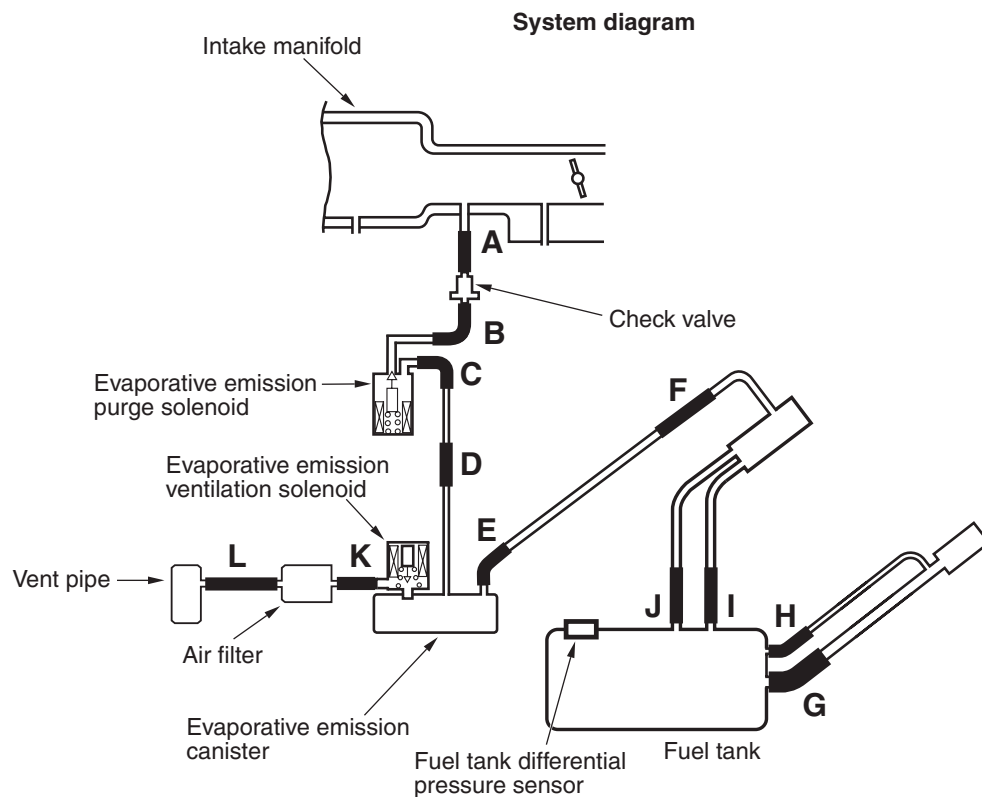
- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 4 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0455 set?

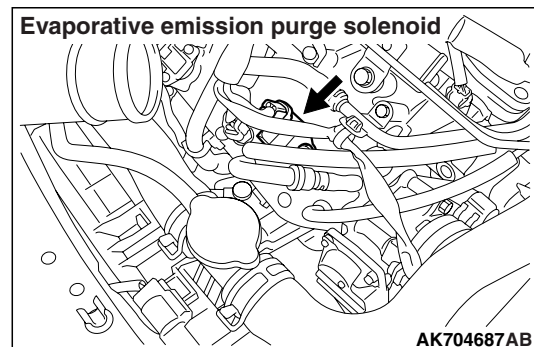
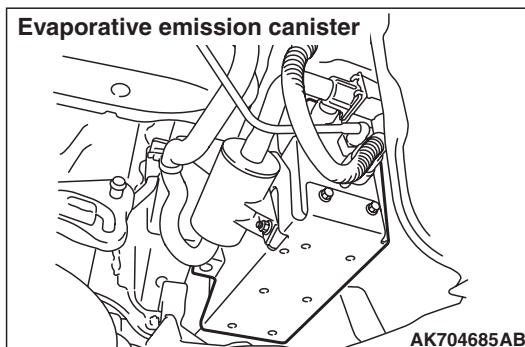
YES : Retry the troubleshooting.

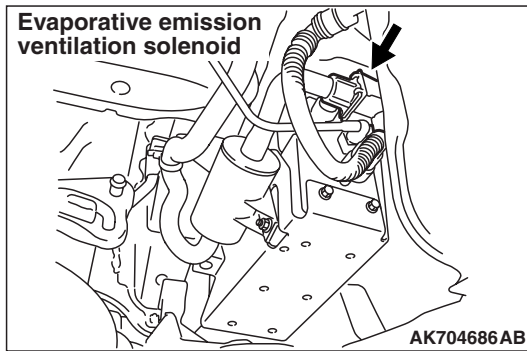
NO : The inspection is complete.

DTC P0456: Evaporative Emission System Leak Detected (Very Small Leak)



AK704684 AB





TECHNICAL DESCRIPTION

- The ECM monitors the Evaporative Emission (EVAP) System pressure.
- The ECM controls the evaporative emission ventilation solenoid. It closes the evaporative emission ventilation solenoid to seal the evaporative emission canister side of the system.
- The evaporative emission purge solenoid is opened to allow manifold vacuum to create low pressure (vacuum) in the EVAP system.
- When the EVAP system develops a vacuum of 2 kPa (0.29 psi), the evaporative emission purge solenoid is closed and the fuel system vacuum is maintained at 2 kPa (0.29 psi).
- The ECM determines whether there is a leak in the EVAP system by monitoring the vacuum inside the fuel tank.
- The test is stopped when fuel vapor pressure exceeds predetermined limits.

DESCRIPTIONS OF MONITOR METHODS

Measure reverting pressure after depressurizing by intake manifold negative pressure and detect malfunction if reverting pressure rises largely.

MONITOR EXECUTION

Once per driving cycle

MONITOR EXECUTION CONDITIONS (OTHER MONITOR AND SENSOR)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

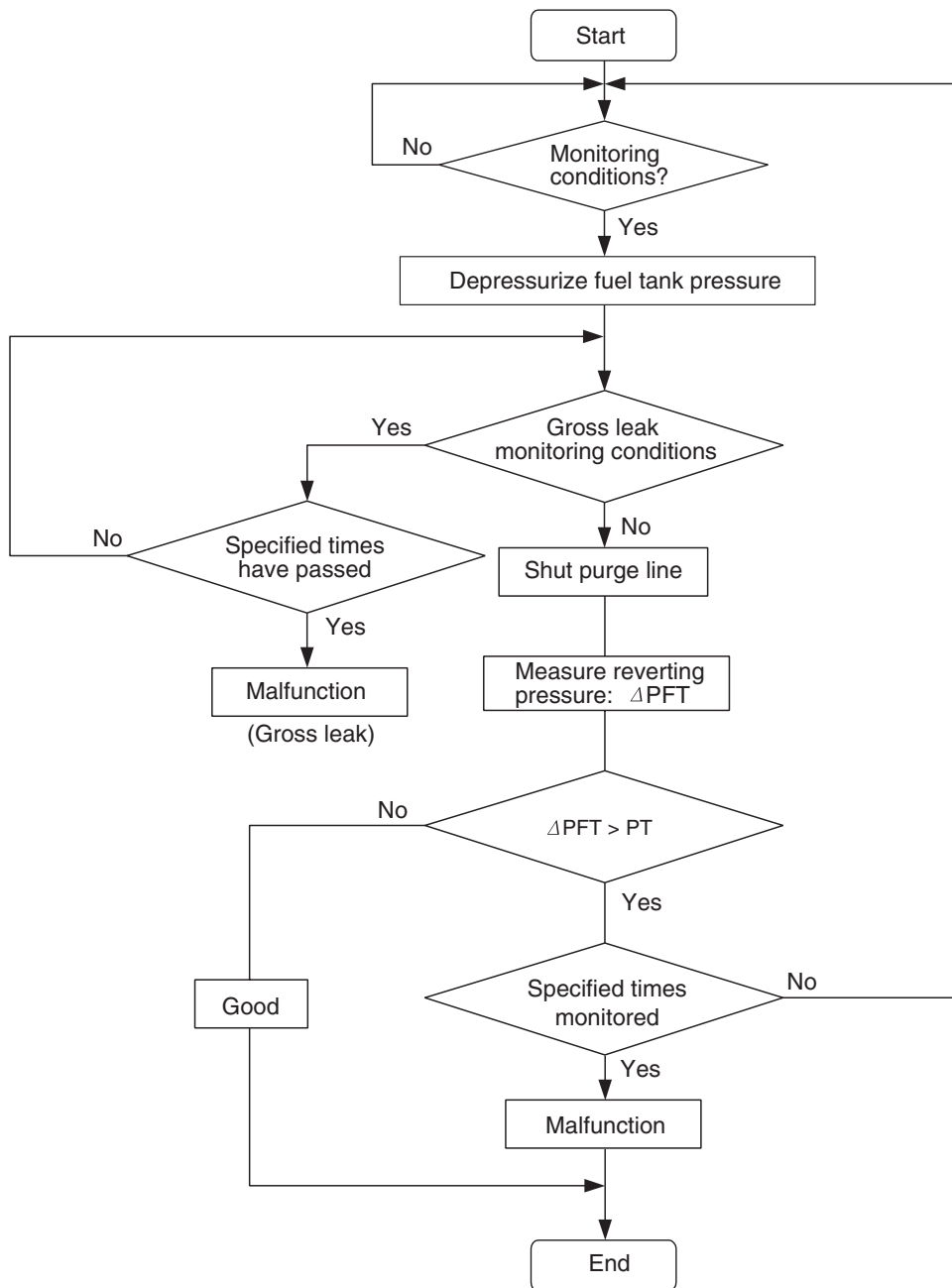
- Evaporative emission purge solenoid monitor
- Evaporative emission purge system monitor
- Fuel tank differential pressure sensor monitor
- Evaporative emission ventilation solenoid monitor
- Fuel level sensor monitor
- Fuel tank temperature sensor monitor

Sensor (The sensors below are determined to be normal)

- Mass airflow sensor
- Barometric pressure sensor
- Intake air temperature sensor
- Engine coolant temperature sensor

DTC SET CONDITIONS

Logic Flow Chart (Monitor Sequence)



AK704743

Check Conditions

Conditions A: At Start up

- Intake air temperature is 36° C (97° F) or less when the engine is started.
- The engine coolant temperature is 36° C (97° F) or less when the engine is started.
- Remaining fuel amount is 40 –85 percent of capacity.

Conditions B: For Test to Run

- Barometric pressure is greater than 76 kPa (11 psi).
- The fuel temperature is 33° C (91° F) or less.
- Fuel tank differential pressure sensor output voltage is 1 to 4 volts.

Conditions C: For Test to Stop

- Engine coolant temperature is greater than 20° C (68° F).

- Intake air temperature is greater than -10° C (14° F).
- When the evaporative emission purge solenoid and evaporative emission ventilation solenoid are closed, the pressure rises in the fuel tank is less than 324 Pa (0.047 psi).
- 10 seconds have elapsed from the start of the previous monitoring.
- Monitoring time: 10 –14 minutes.

Judgement Criteria

- Internal pressure of the fuel tank has changed more than 1,172 –1,433 Pa (0.170 –0.208 psi) in 128 seconds after the tank and vapor line were closed.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

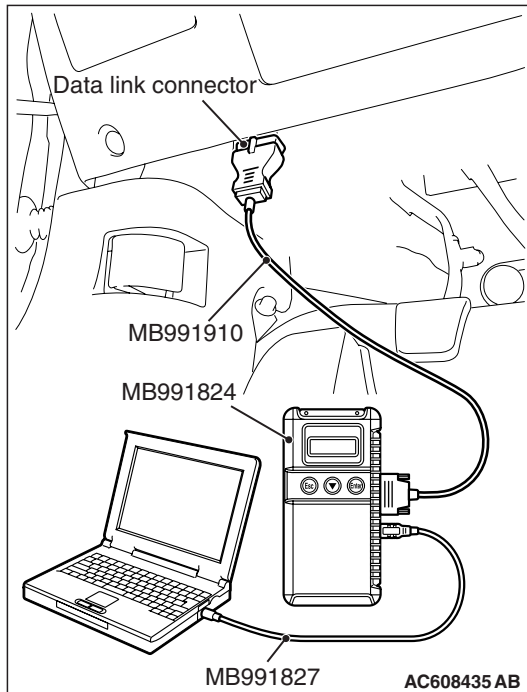
Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 5 [P.13A-9](#).

TROUBLESHOOTING HINTS (THE MOST LIKELY CAUSES FOR THIS CODE TO BE SET ARE:)

- Loose fuel cap.
- Fuel cap relief pressure is incorrect.
- Malfunction of the evaporative emission canister seal.
- Malfunction of the fuel tank, purge line or vapor line seal.
- Malfunction of the evaporative emission ventilation solenoid.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: M.U.T.-III USB Cable
 - MB991910: M.U.T.-III Main Harness A



STEP 1. Using scan tool MB991958, check the evaporative emission system monitor test.

⚠ CAUTION

- To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.
- During this test, the ECM will automatically increase the engine speed to 1,600 r/min or greater. Check that the transaxle is set to "P" position.

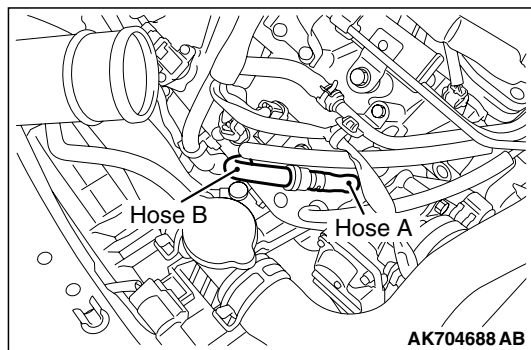
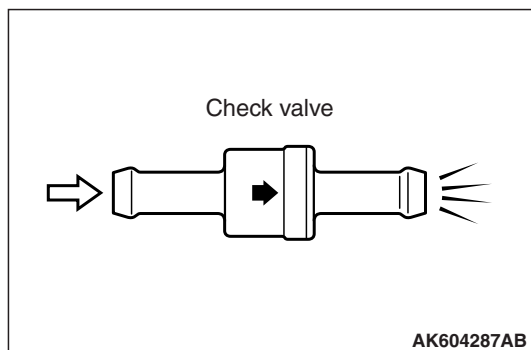
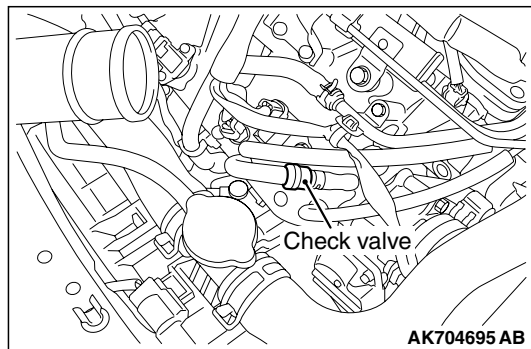
- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTCs using scan tool MB991958.
- (4) Check that the fuel cap is securely closed (Tighten until three clicks are heard).
- (5) Start the engine.
- (6) Select "Special Function"
- (7) Select "Evap Leak Monitor"
- (8) During this test, keep the accelerator pedal at the idle position.
- (9) Keep the engine speed and engine load within the specified range. When the monitor test starts, the "In Progress" item on scan tool MB991958 will change from "NO" to "YES".
- (10) Turn the ignition switch to the "LOCK" (OFF) position, and disconnect scan tool MB991958.

Q: Is "Evap Leak Mon. Completed. Test Failed and DTCs Set" displayed on scan tool MB991958?

YES : A malfunction has been detected during the monitor test. Refer to the Diagnostic Trouble Code Chart and diagnose any other DTCs that are set [P.13A-44](#). If no other DTC's have been set, go to Step 2 .

NO <"Evap Leak Mon. Completed. Test Passed" is displayed on scan tool MB991958.> : The evaporative emission system is working properly at this time. Explain to the customer that an improperly tightened fuel cap can cause the MIL to illuminate. Return the vehicle to the customer.

NO <"Evap Leak Mon. Discontinued. Retest again from the first" is displayed on scan tool MB991958.> : The EVAP monitor has been interrupted during the test. Turn the ignition switch to the "LOCK" (OFF) position once, and repeat the monitoring.

**STEP 2. Test check valve.**

(1) Check valve is a one-way check valve.

(2) Check valve should allow air to flow in only one direction.

Q: Does check valve allow air to pass in one direction only?

YES : Go to Step 3 .

NO : Replace check valve. Then go to Step 14 .

STEP 3. Check for leaks in evaporative emission hoses A and B.

Use a hand vacuum pump to test each hose A and B.

Q: Do the hoses hold vacuum?

YES : Go to Step 4 .

NO : Replace any damaged hose. Then go to Step 14 .

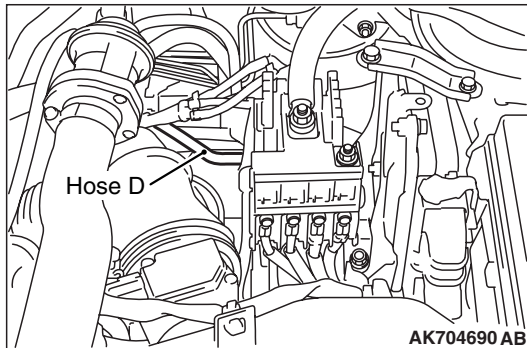
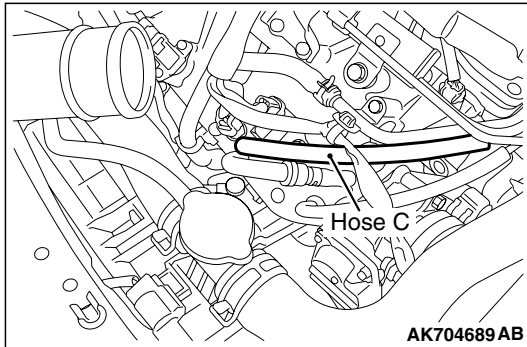
STEP 4. Check the evaporative emission purge solenoid for leaks.

Refer to GROUP 17, Emission Control –Evaporative Emission Control System –Evaporative Emission Purge Solenoid Check [P.17-77](#).

Q: Is the evaporative emission purge solenoid operating properly?

YES : Go to Step 5 .

NO : Replace the evaporative emission purge solenoid.
Then go to Step 14 .



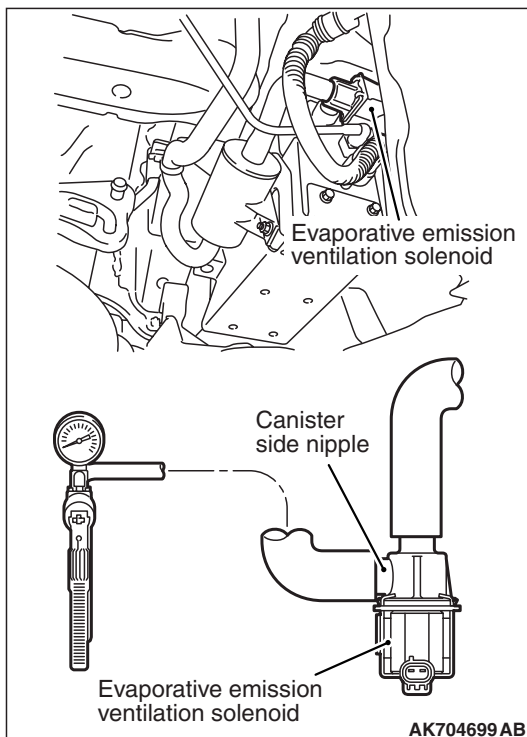
STEP 5. Check for leaks in evaporative emission hoses C and D.

Use a hand vacuum pump to test each hose C and D.

Q: Do the hoses hold vacuum?

YES : Go to Step 6 .

NO : Replace any damaged hose. Then go to Step 14 .



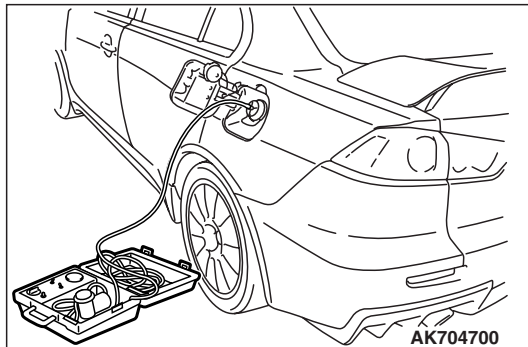
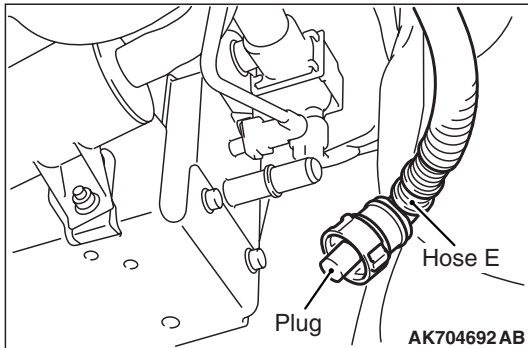
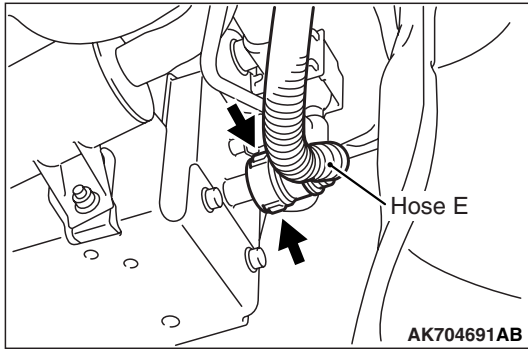
STEP 6. Using scan tool MB991958, check actuator test item 15: Evaporative emission ventilation solenoid.

- (1) Remove the canister cover.
- (2) Remove the evaporative emission ventilation solenoid. Do not disconnect the connector.
- (3) Connect the hose of the hand vacuum pump to the canister side nipple of the evaporative emission ventilation solenoid.
- (4) Turn the ignition switch to the "ON" position.
- (5) Set scan tool MB991958 to actuator test mode.
 - Item 15: Evaporative Emission Ventilation Solenoid.
 - While the evaporative emission ventilation solenoid is energized, operate the hand vacuum pump and confirm that the solenoid holds vacuum.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.
- (7) Disconnect the hand vacuum pump, and reinstall the evaporative emission ventilation solenoid.
- (8) Reinstall the canister cover.

Q: Did the evaporative emission ventilation solenoid hold vacuum?

YES : Go to Step 7 .

NO : Replace the evaporative emission ventilation solenoid. Then go to Step 14 .

**STEP 7. Perform the pressure test on the evaporative emission system.**

(1) Disconnect hose E from the canister while holding the release buttons indicated in the illustration pressed by fingers.

(2) Plug the disconnected end of hose E.

(3) Confirm that the evaporative emission system pressure pump (Miller number 6872A) is operating properly. Perform the self-test as described in the pump manufacturer's instructions.

(4) Remove the fuel cap.

(5) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel tank filler tube by using fuel tank adapter (MLR-8382).

(6) Pressure test the system to determine whether any leaks are present.

NOTE: The "Pressure test" in this procedure refers to the I/M240 Simulation Test. The eight steps of this test are described in the manufacturer's instructions for the evaporative emission system pressure pump, Miller number 6872A.

(7) Remove the evaporative emission system pressure pump (Miller number 6872A) and the fuel tank adapter (MLR-8382), and reinstall the fuel cap.

(8) Connect hose E to the evaporative emission canister.

Q: Is the evaporative emission system line free of leaks?

YES : Go to Step 12 .

NO : Go to Step 8 .

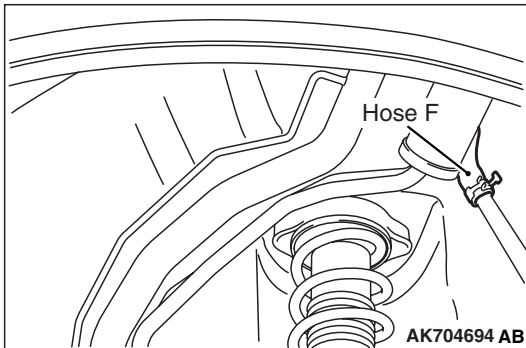
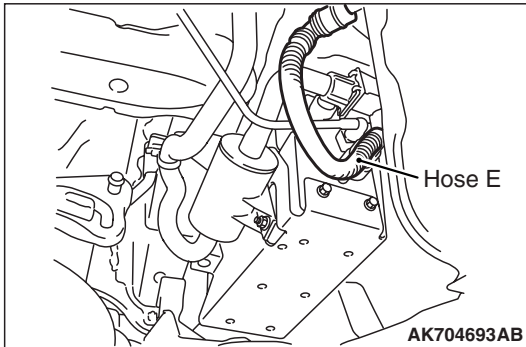
STEP 8. Check for leaks in evaporative emission hoses E and F.

Use a hand vacuum pump to test each hose E and F.

Q: Do the hoses hold vacuum?

YES : Go to Step 9 .

NO : Replace any damaged hose. Then go to Step 14 .



STEP 9. Check for leaks in evaporative emission hoses G and H.

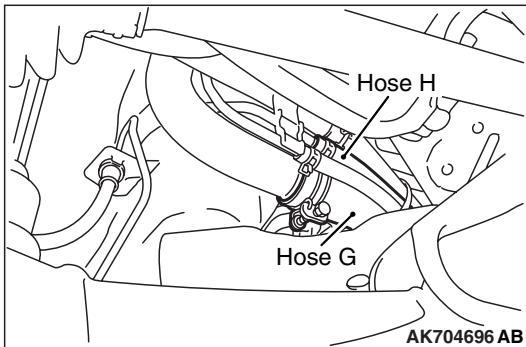
(1) Remove the fuel tank assembly.

(2) Use a hand vacuum pump to test each hose G and H.

Q: Do the hoses hold vacuum?

YES : Go to Step 10.

NO : Replace any damaged hose and reinstall the fuel tank assembly. Then go to Step 14 .



STEP 10. Check for leaks in evaporative emission hoses I and J.

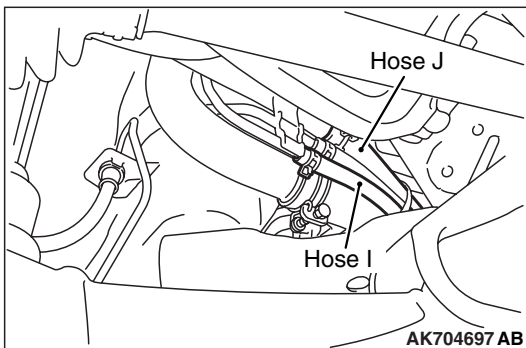
(1) Remove the fuel tank assembly.

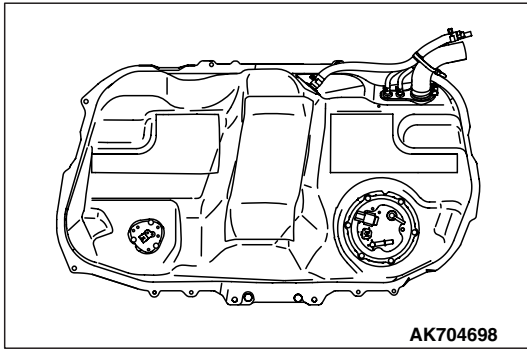
(2) Use a hand vacuum pump to test each hose I and J.

Q: Do the hoses hold vacuum?

YES : Go to Step 11 .

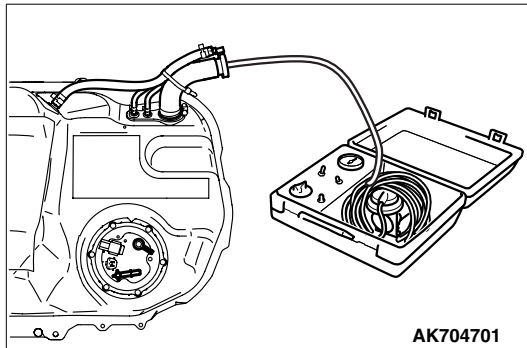
NO : Replace any damaged hose and reinstall the fuel tank assembly. Then go to Step 14 .



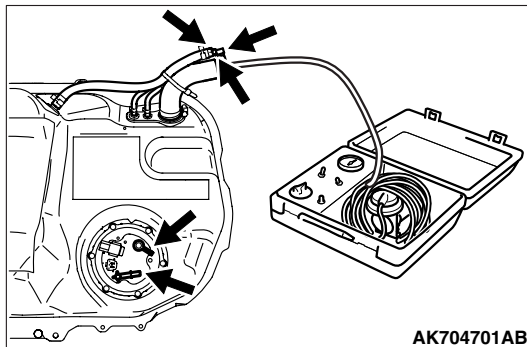
**STEP 11. Check for leaks in the fuel tank.**

- (1) Visually check for cracks or other leaks in the fuel tank.

NOTE: Carefully check the fuel pump module and the fuel tank differential pressure sensor installation in the fuel tank.



- (2) Connect the evaporative emission system pressure pump (Miller number 6872A) to the fuel filler hose.



- (3) Plug the hose and the nipple shown in the illustration.

NOTE: If these items are not securely plugged now, the fuel could leak in the next step.

- (4) Pressurize the fuel tank with the evaporative emission system pressure pump.
(5) In the pressurized state, check for leaks by applying a soapy water solution to each section and look for bubbles.

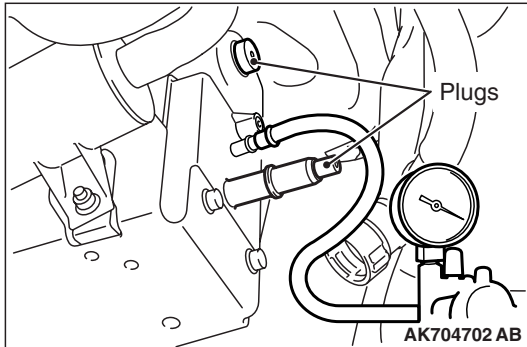
Q: Are any leaks found?

YES <When there is a leak from the attachment points of the fuel pump module, fuel tank differential pressure sensor, fuel level sensor or leveling valve.> :

Reassemble the leaked parts and check again that there are no leaks. Then reinstall the fuel tank. Then go to Step 14 .

YES <When there is a leak from the fuel tank.> : Replace the fuel tank. Then go to Step 14 .

NO : When there is no leak, reinstall the fuel tank. Then go to Step 13 .



STEP 12. Check the evaporative emission canister for vacuum leaks.

- (1) Connect a hand vacuum pump to the evaporative emission canister and plug the other nipples.
- (2) Apply a pressure on the hand vacuum pump, and confirm that air is maintained.
- (3) Disconnect the hand vacuum pump and remove the plugs.

Q: Is the evaporative emission canister in good condition?

YES : Go to Step 13 .

NO : Replace the evaporative emission canister. Then go to Step 14 .

STEP 13. Using scan tool MB991958, check the evaporative emission system monitor test.

⚠ CAUTION

- During this test, the ECM automatically increases the engine speed to 1,600 r/min or greater. Check that the transaxle is set to "P" position.

- (1) Turn the ignition switch to the "ON" position.
- (2) Erase the DTCs using scan tool MB991958.
- (3) Check that the fuel cap is securely closed (Tighten until three clicks are heard).
- (4) Start the engine.
- (5) Select "Special Function"
- (6) Select "Evap Leak Monitor"
- (7) During the test, keep the accelerator pedal at the idle position.
- (8) Keep the engine speed and engine load within the specified range. When the monitor test starts, the "In Progress" item on scan tool MB991958 will change from "NO" to "YES".
- (9) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is "Evap Leak Mon. Completed. Test Failed and DTCs Set" displayed on scan tool MB991958?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Code Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 14 .

NO <"Evap Leak Mon. Completed. Test Passed" is displayed on scan tool MB991958.> : The evaporative emission system is working properly at this time. Go to Step 14 .

NO <"Evap Leak Mon. Discontinued. Retest again from the first" is displayed on scan tool MB991958.> : The EVAP monitor has been interrupted during the test. Turn the ignition switch to the "LOCK" (OFF) position once, and repeat the monitoring.

STEP 14. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 5 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0456 set?**YES** : Retry the troubleshooting.**NO** : The inspection is complete.

DTC P0461: Fuel Level Sensor (main) Circuit Range/Performance

TECHNICAL DESCRIPTION

- The fuel level sensor converts the rest of the fuel to a voltage and sends it to the combination meter.
- The combination meter sends the data regarding the rest of the fuel to the ECM.
- The ECM checks whether this data is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Detect malfunction if change of fuel level sensor output voltage is small when sum of fuel injection is large.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

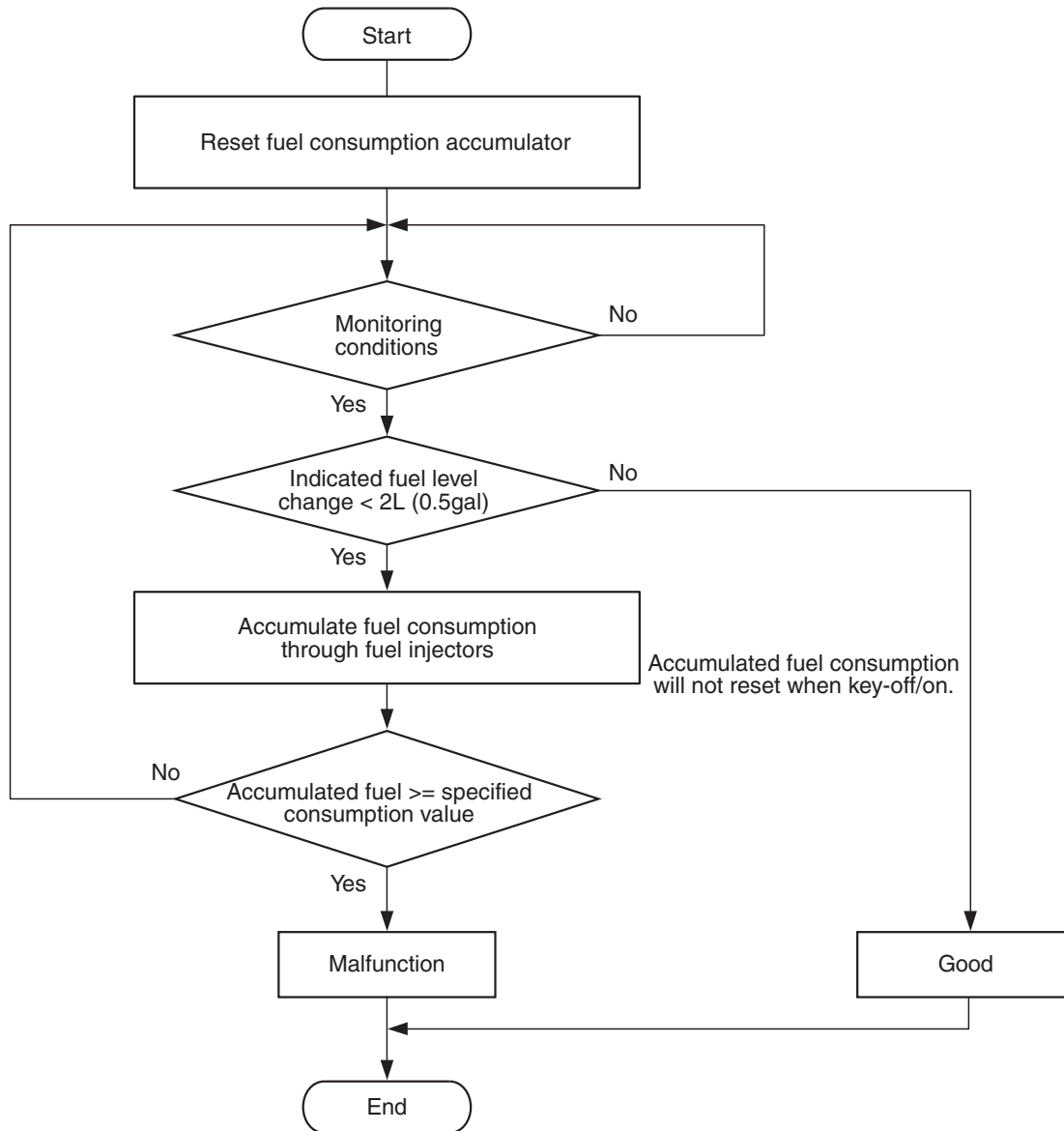
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604341

Check Condition, Judgement Criterion

- When the fuel consumption calculated from the operation time of the injector amounts to 30 liters (7.9 gal), the diversity of the amount of fuel in tank calculated from the fuel level sensor is 2 liters (0.5 gal) or less.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Combination meters assembly failed.
- Fuel level sensor failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the combination meter diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

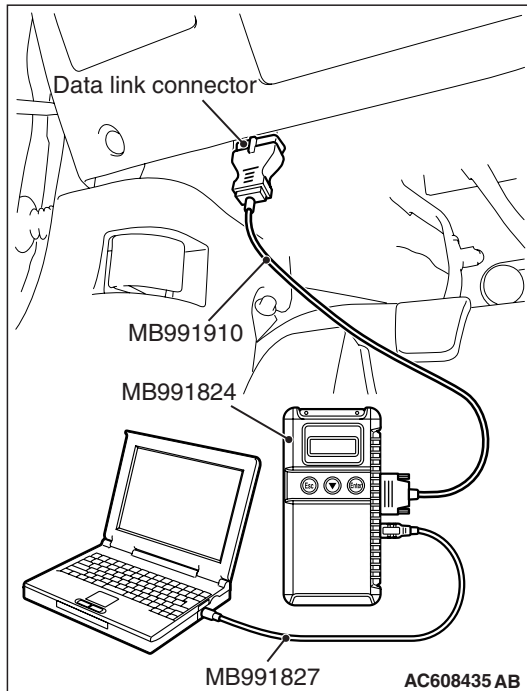
- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the combination meter DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the combination meter-DTC set?

YES : Refer to GROUP 54A, Chassis Electrical – Combination Meter –Diagnosis Trouble Code Chart

[P.54A-28](#)

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC)

1. Turn the ignition switch to the "ON" position.
2. After the DTC has been deleted, read the DTC again.
3. Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0461 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC)

1. Turn the ignition switch to the "ON" position.
2. After the DTC has been deleted, read the DTC again.
3. Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0461 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0462: Fuel Level Sensor Circuit Low Input

TECHNICAL DESCRIPTION

- The fuel level sensor converts the rest of the fuel to a voltage and sends it to the combination meter.
- The combination meter sends the data regarding the rest of the fuel to the ECM.
- The ECM checks whether this data is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

A short circuit is detected while monitoring the fuel level sensor output.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

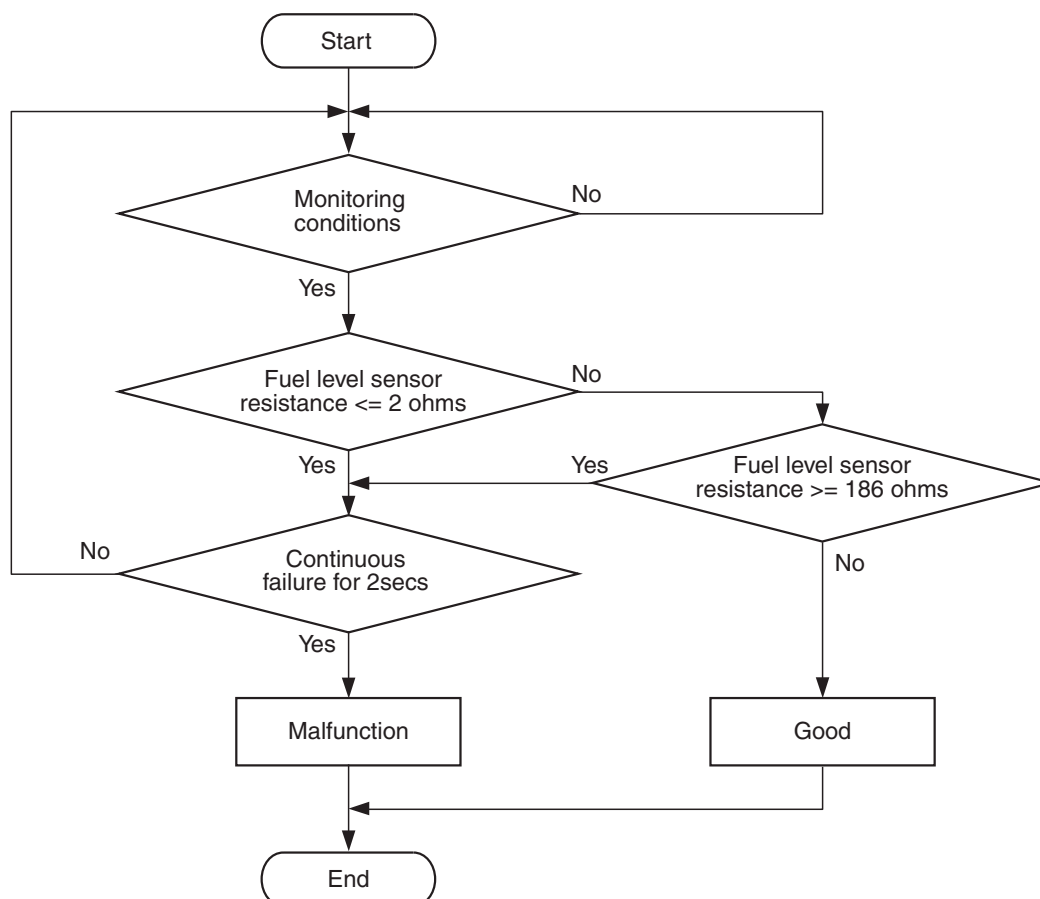
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604342

Check Conditions

- Battery positive voltage is between 11 and 16.5 volts.
- 2 seconds or more have passed since the engine starting sequence was completed.

Judgement Criterion

- Fuel level sensor resistance has continued to be 2 ohms or lower for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

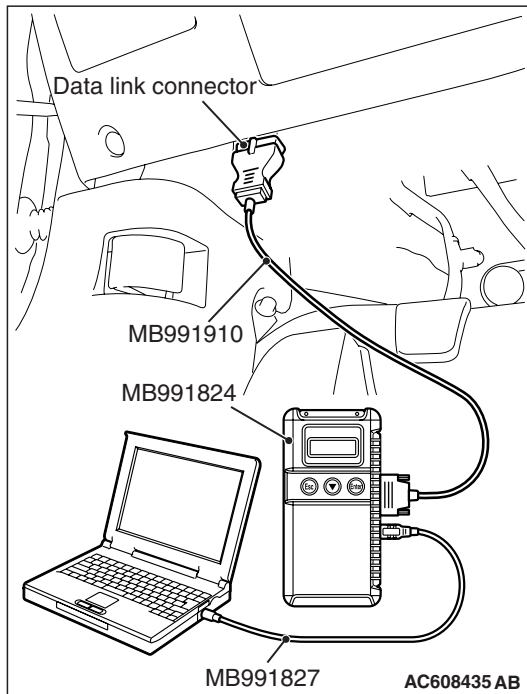
Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Fuel level sensor failed.
- Combination meters assembly failed.
- Connector damage
- Harness damage
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A



STEP 1. Using scan tool MB991958, read the combination meter diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the combination meter DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the combination meter DTC set?

YES : Refer to GROUP 54A, Chassis Electrical – Combination Meter –Diagnosis Trouble Code Chart

[P.54A-28](#)

NO : Go to Step 2.

STEP 2. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0462 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0462 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0463: Fuel Level Sensor Circuit High Input**TECHNICAL DESCRIPTION**

- The fuel level sensor converts the rest of the fuel to a voltage and sends it to the combination meter.
- The combination meter sends the data regarding the rest of the fuel to the ECM.
- The ECM checks whether this data is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

An open circuit is detected while monitoring the fuel level sensor output.

MONITOR EXECUTION

Continuous

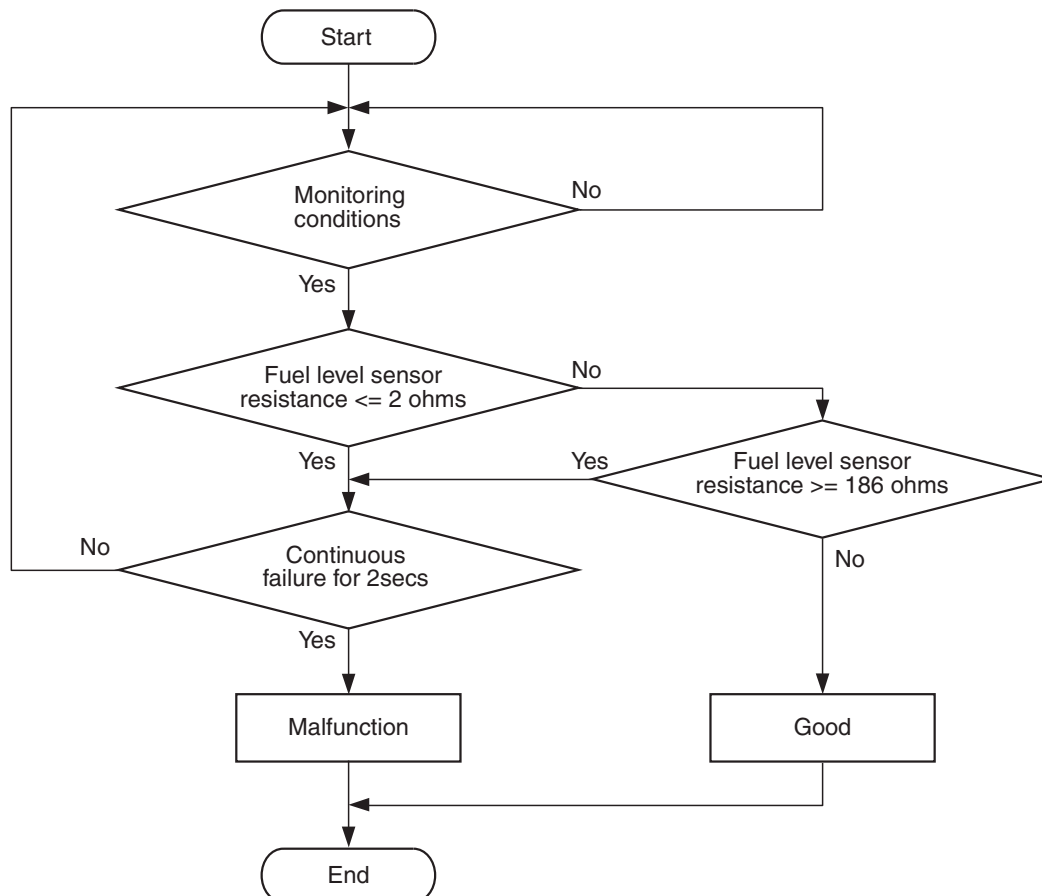
**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Logic Flow Chart**

AK604342

Check Conditions

- Battery positive voltage is between 11 and 16.5 volts.

- 2 seconds or more have passed since the engine starting sequence was completed.

Judgement Criterion

- Fuel level sensor resistance has continued to be 186 ohms or higher for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Fuel level sensor failed.
- Combination meters assembly failed.
- Connector damage
- Harness damage
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the combination meter diagnostic trouble code (DTC).

CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

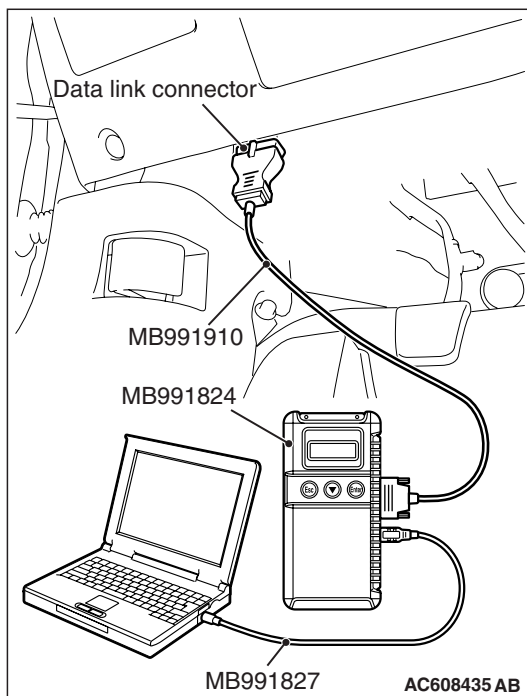
- Connect scan tool MB991958 to the data link connector.
- Turn the ignition switch to the "ON" position.
- Read the combination meter DTC.
- Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the combination meter DTC set?

YES : Refer to GROUP 54A, Chassis Electrical – Combination Meter –Diagnosis Trouble Code Chart

[P.54A-28](#)

NO : Go to Step 2.



STEP 2. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0463 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Code Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0463 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0500: Vehicle Speed Signal Malfunction

TECHNICAL DESCRIPTION

- The ECM receives the vehicle speed signal from the ASC-ECU by the CAN bus line.

DESCRIPTIONS OF MONITOR METHODS

When the signal not available as the vehicle speed information send from ASC-ECU, a malfunction is determined to have occurred.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

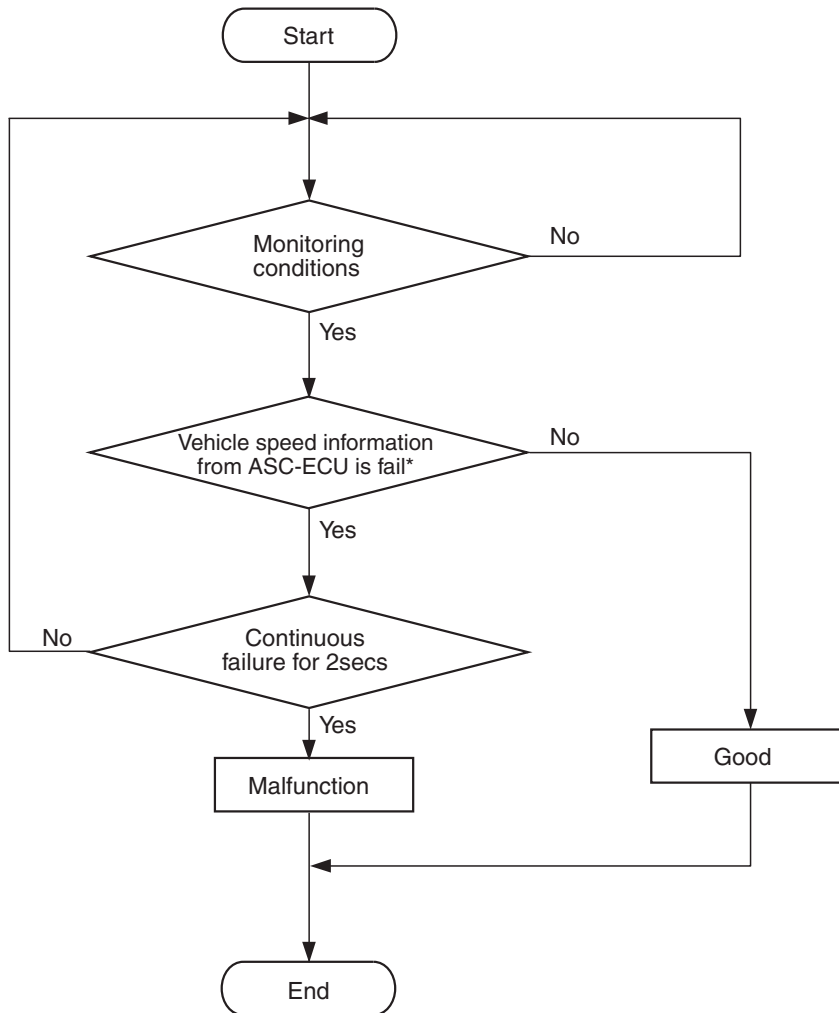
- Not applicable

Sensor (The sensor below is determined to be normal)

- Mass airflow sensor

DTC SET CONDITIONS (Entry Conditions and Malfunction Thresholds)

Logic Flow Chart



*The fail signal is sent when ASC-ECU detects all four wheel speed sensors failures.

AK704731

Check Conditions

- 2 seconds or more have passed since the engine starting sequence was completed.
- Battery positive voltage is between 10 and 16.5 volts.

Judgement Criterion

- Vehicle speed information from ASC-ECU is fail for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Wheel speed sensor of ASC system failed.
- ASC-ECU failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

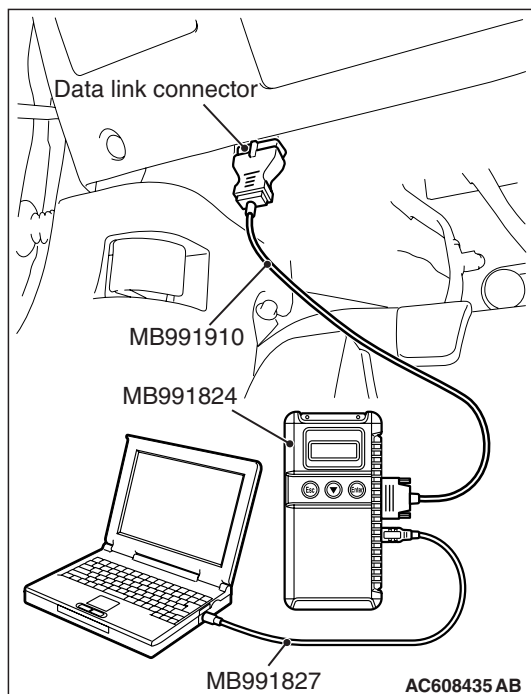
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958, read the ASC-DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the ASC-DTC set?

YES : Refer to GROUP 35C, Active Stability Control System
–Diagnostic Trouble Code Chart [P.35C-20](#).

NO : Go to Step 2.



STEP 2. Check the speedometer.

Q: Is the indication of speedometer normal?

YES : Go to Step 3.

NO : Replace the ASC-ECU.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0500 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/ Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

DTC P0506: Idle Control Sytem RPM Lower Than Expected

TECHNICAL DESCRIPTION

- The amount of air taken in during idling is regulated by the opening and closing of the throttle valve.
- The ECM checks the difference between the actual engine speed and the target engine speed.

DESCRIPTIONS OF MONITOR METHODS

Difference between actual and target idle speed is over the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

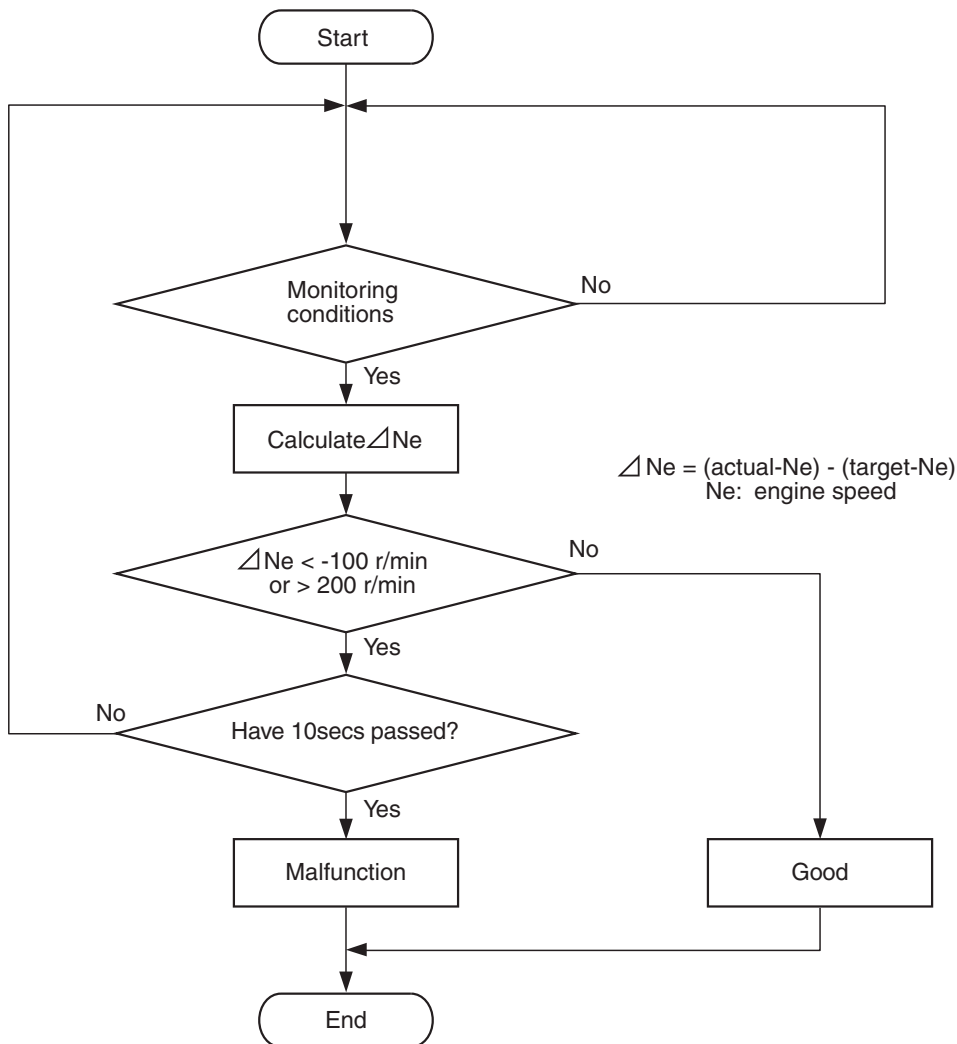
- Misfire monitor
- Fuel system monitor
- Vehicle speed signal monitor

Sensor (The sensor below is determined to be normal)

- Mass airflow sensor
- Engine coolant temperature sensor
- Intake air temperature sensor
- Barometric pressure sensor
- Throttle position sensor

DTC SET CONDITIONS

Logic Flow Chart



AK704413

Check Conditions

- Under the closed loop idle speed control.
- The engine coolant temperature is 41°C (105°F) or more.
- Battery positive voltage is higher than 10 volts.
- Power steering pressure switch: OFF
- Volumetric efficiency is lower than 40 percent.
- Barometric pressure is higher than 76 kPa (22.4 in.Hg).
- Intake air temperature is higher than -10°C (14°F).
- 3 seconds have elapsed from the start of the previous monitoring.
- Target airflow rate is 24 L/sec or more.

Judgement Criterion

- The actual idle speed is more than 100 r/min lower than the target idle speed for 10 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 18 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle valve area is dirty.
- ECM failed.

DIAGNOSIS

Required Special Tools

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

CAUTION

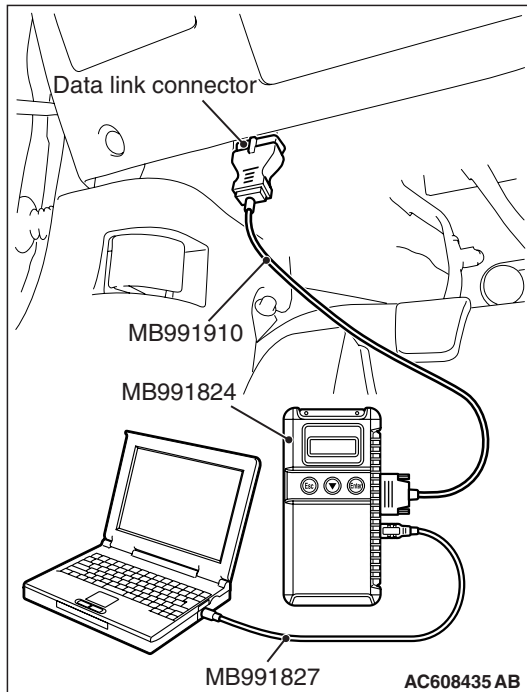
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958, read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P0506 set?

YES : Refer to Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Check the throttle body. (throttle valve area)

Q: Is the throttle valve area dirty?

YES : Perform cleaning. Refer to Throttle body cleaning [P.13A-897](#). Then go to Step 4.

NO : Go to Step 3.

STEP 3. Replace the throttle body assembly.

- (1) Replace the throttle body assembly
- (2) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 18 [P.13A-9](#).
- (3) Check the diagnostic trouble code (DTC).

Q: Is DTC P0506 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.

NO : The inspection is complete.

STEP 4. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 18 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0506 set?**YES** : Retry the troubleshooting.**NO** : The inspection is complete.

DTC P0507: Idle Control Sytem RPM Higher Than Expected

TECHNICAL DESCRIPTION

- The amount of air taken in during idling is regulated by the opening and closing of the throttle valve.
- The ECM checks the difference between the actual engine speed and the target engine speed.

DESCRIPTIONS OF MONITOR METHODS

Difference between actual and target idle speed is over the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)****Other Monitor (There is no temporary DTC stored in memory for the item monitored below)**

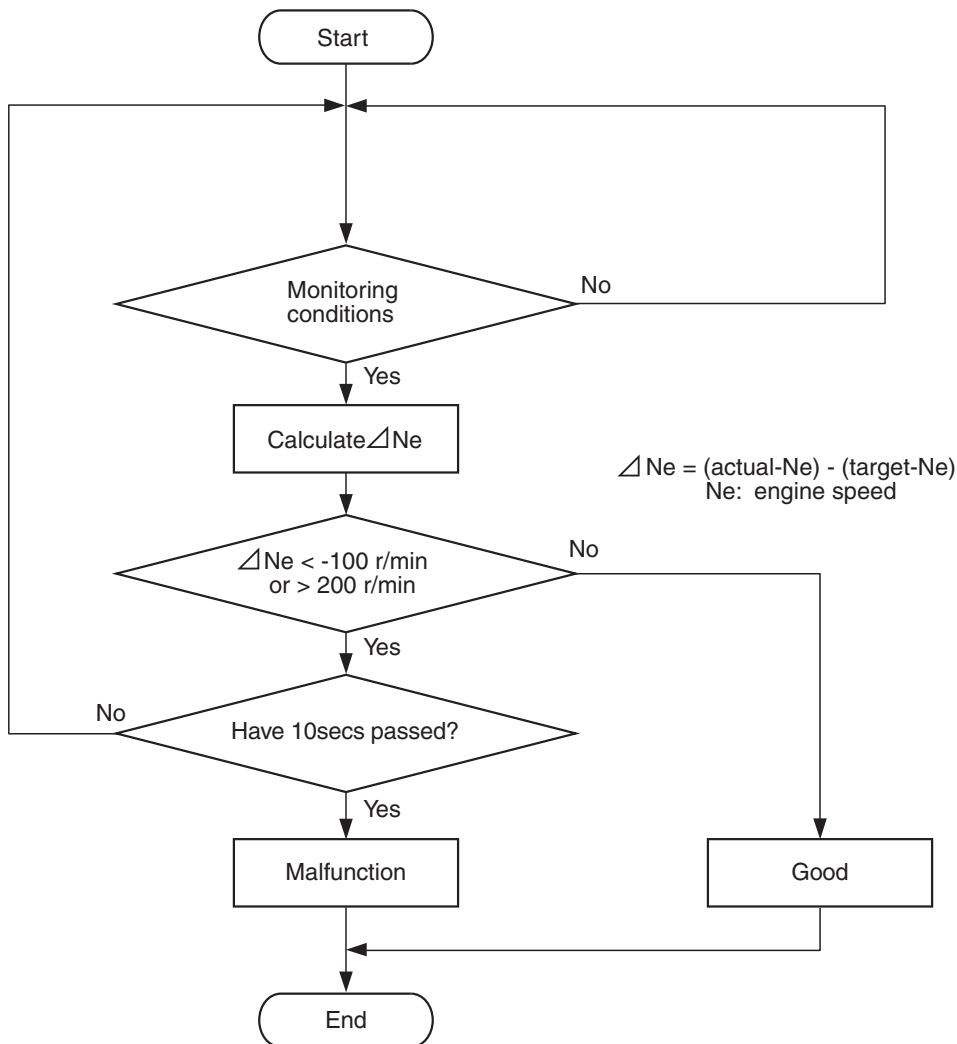
- Misfire monitor
- Fuel system monitor
- Vehicle speed signal monitor

Sensor (The sensor below is determined to be normal)

- Mass airflow sensor
- Engine coolant temperature sensor
- Intake air temperature sensor
- Barometric pressure sensor
- Throttle position sensor

DTC SET CONDITIONS

Logic Flow Chart



AK704413

Check Conditions

- Vehicle speed has reached 1.5 km/h (1.0 mph) or more at least once.
- Under the closed loop idle speed control.
- Engine coolant temperature is 41°C (105°F) or more.
- Battery positive voltage is higher than 10 volts.
- Barometric pressure is higher than 76 kPa (22.4 in.Hg).
- Intake air temperature is higher than -10°C (14°F).
- 3 seconds have elapsed from the start of the previous monitoring.
- Target airflow rate is 0 L/sec.

Judgement Criterion

- Actual idle speed has continued to be higher than the target idle speed by 200 r/min (300 r/min*) or more for 10 seconds.
- *: Specs in parentheses are applicable if the maximum air temperature during the previous operation was more than 45°C (113°F).

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 18 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Intake system vacuum leak.

- ECM failed.

DIAGNOSIS**Required Special Tools**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).**⚠ CAUTION**

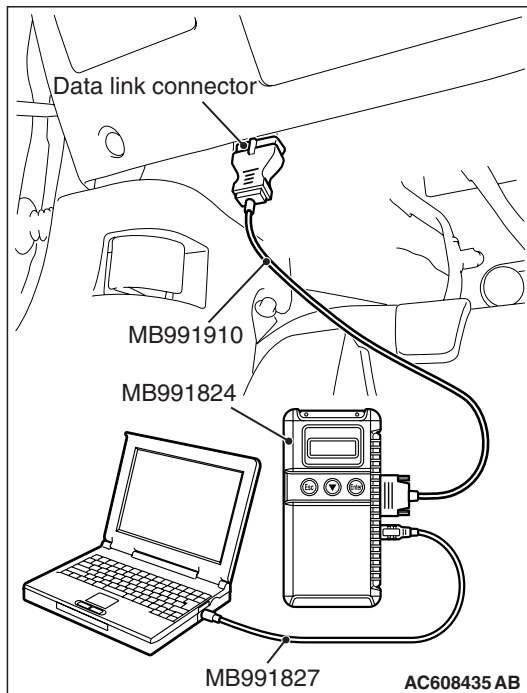
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958, read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P0507 set?

YES : Refer to Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.

**STEP 2. Check the intake system vacuum leak.****Q: Are there any abnormalities?**

YES : Repair or replace it. Then go to Step 4.

NO : Go to Step 3.

STEP 3. Replace the throttle body assembly.

- (1) Replace the throttle body assembly
- (2) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 18 [P.13A-9](#).
- (3) Check the diagnostic trouble code (DTC).

Q: Is DTC P0507 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.

NO : The inspection is complete.

STEP 4. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 18 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0507 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0513: Immobilizer Malfunction

TECHNICAL DESCRIPTION

- ECM monitors the communication condition with the immobilizer-ECU. When an abnormality in communication is found, ECM prevents engine start.

DTC SET CONDITIONS**Check Condition**

- Ignition switch is "ON" position.

Judgment Criterion

- When it was different in the encrypted code sent from immobilizer-ECU and operation result by ECM.

FAIL-SAFE AND BACKUP FUNCTION

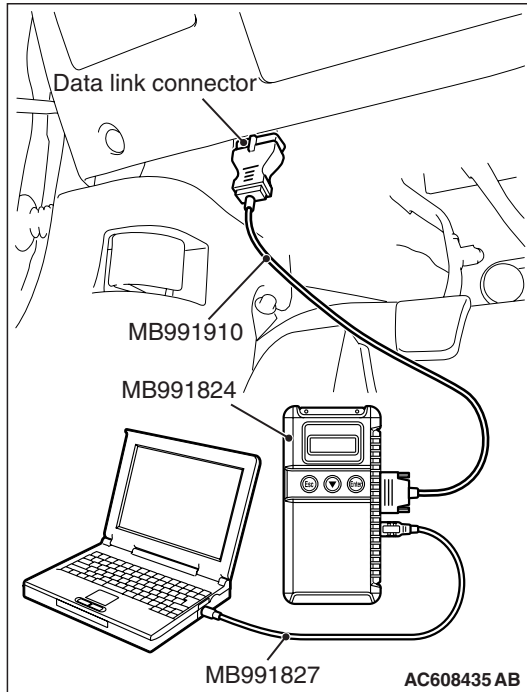
- Engine start is prohibited.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN system failed.
- Immobilizer system failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

**STEP 1. Using scan tool MB991958, check VIN Information.****⚠ CAUTION**

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Set scan tool MB991958 to the coding mode for VIN Information.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Has VIN (current) been written?

YES : Go to Step 2.

NO : Write VIN. Then go to Step 2.

STEP 2. Using scan tool MB991958, read the immobilizer diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the immobilizer-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the immobilizer-DTC set?

YES : Refer to GROUP 42B, Diagnosis –Diagnostic Trouble Code Chart <Vehicles with KOS> [P.42B-20](#) or GROUP 42C, diagnosis –Diagnostic Trouble Code Chart <Vehicles with WCM> [P.42C-14](#).

NO : Go to Step 3.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC)

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

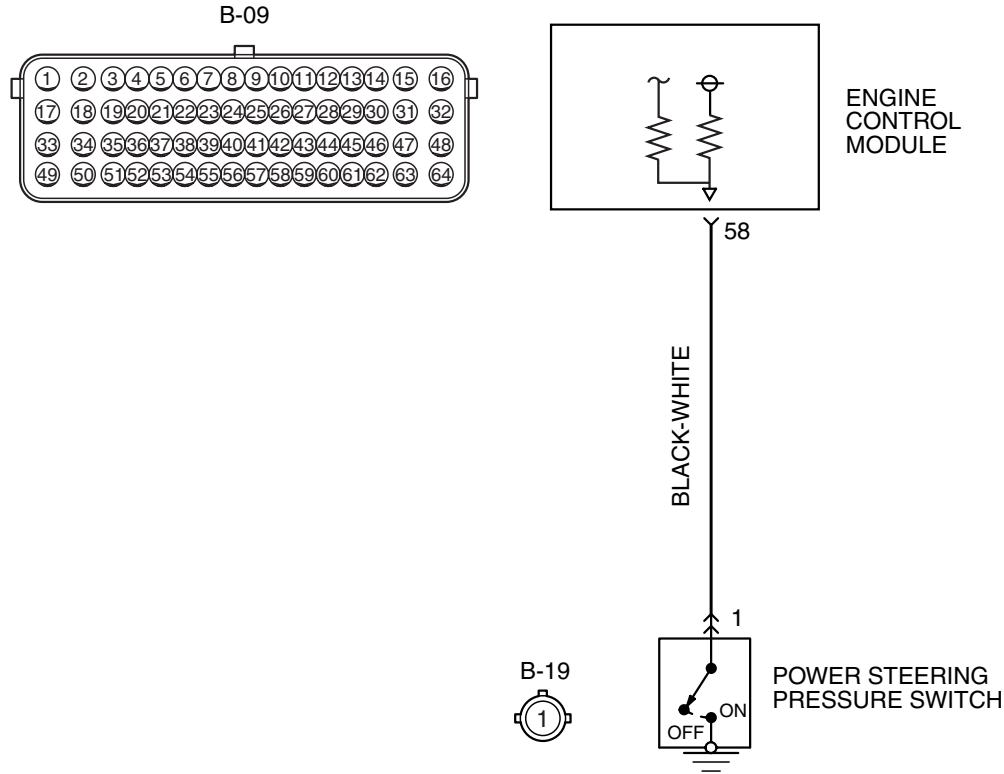
Q: Is DTC P0606 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

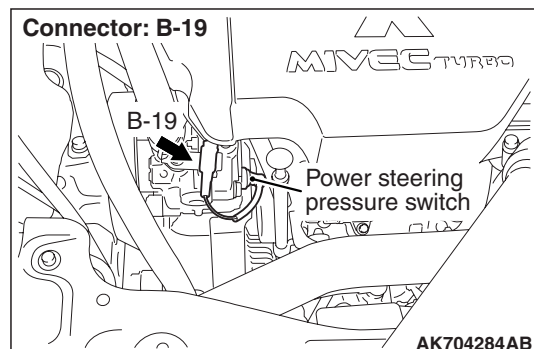
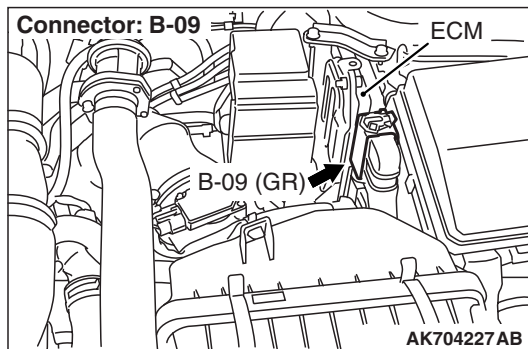
NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

DTC P0551: Power Steering Pressure Switch Circuit Range/Performance

POWER STEERING PRESSURE SWITCH CIRCUIT



AK704283AB



CIRCUIT OPERATION

- A battery positive voltage is applied to the power steering pressure switch output terminal (terminal No. 1) from the ECM (terminal No. 58) via the resistor in the ECM.

TECHNICAL DESCRIPTION

- The power steering pressure switch converts the existence of a power steering load into a high/low voltage, and inputs it into the ECM.

- When the steering wheel is turned, hydraulic pressure rises. The power steering pressure switch closes, and the applied battery positive voltage will be grounded. With this, the power steering pressure switch output voltage will fluctuate between 0 and 12 volts.
- While driving with the steering wheel held straight, the power steering pressure switch turns "OFF".
- The ECM checks whether the power steering pressure switch turns "OFF" or "ON" during driving.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)****Other Monitor (There is no temporary DTC stored in memory for the item monitored below)**

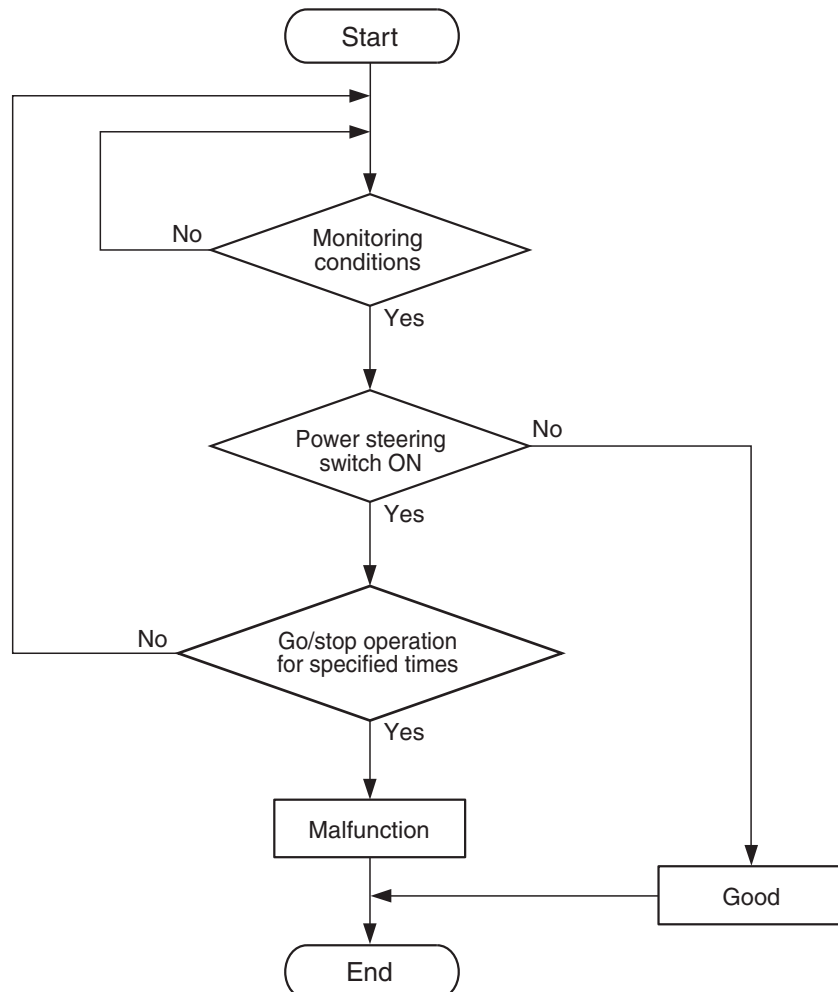
- Not applicable

Sensor (The sensor below is determined to be normal)

- Engine coolant temperature sensor

DESCRIPTIONS OF MONITOR METHODS

Power steering pressure switch stays on during specified go/stop operations.

DTC SET CONDITIONS**Logic Flow Chart**

Check Conditions

- Engine coolant temperature is higher than 65° C (149° F).
- Drive for 4 seconds or more with the vehicle speed is 50 km/h (31 mph) or more. Stop the vehicle [vehicle speed is 1.5 km/h (1.0 mph) or less]. Repeat 10 times or more.

Judgement Criterion

- Power steering pressure switch continues to be "ON".

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Power steering pressure switch failed.
- Open or shorted power steering pressure switch circuit, harness damage, or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A
- MB992110: Power Plant ECU Check Harness

STEP 1. Using scan tool MB991958, check data list item 83: Power Steering Pressure Switch.

CAUTION

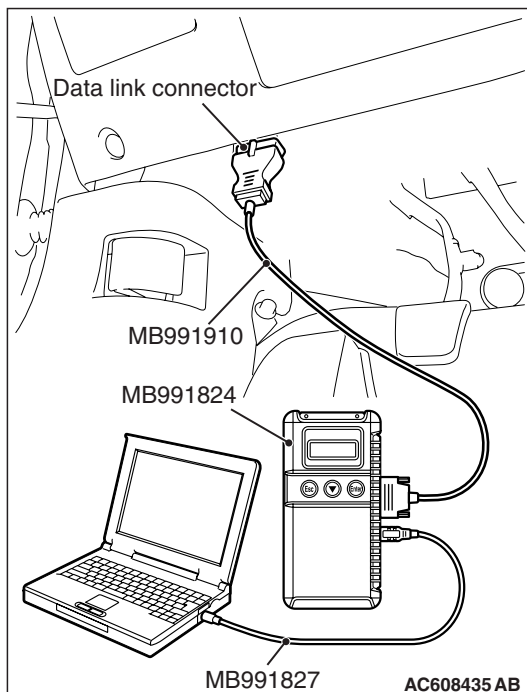
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

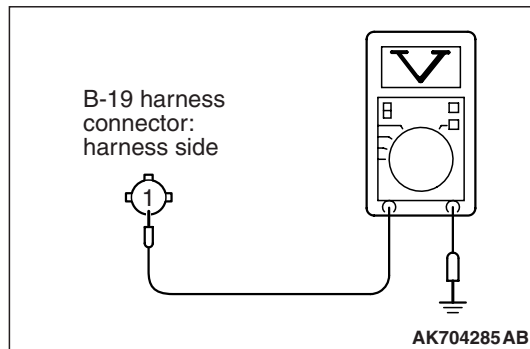
- (1) Connect scan tool MB991958 to the data link connector.
- (2) Start the engine and run at idle.
- (3) Set scan tool MB991958 to the data reading mode for item 83, Power Steering Pressure Switch.
 - If the steering wheel is not turned while idling, "OFF" will be displayed.
 - If the steering wheel is turned while idling, "ON" will be displayed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the switch operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2.




STEP 2. Measure the power supply voltage at power steering pressure switch connector B-19 by backprobing.

- (1) Do not disconnect the connector B-19.
- (2) Start the engine and run at idle.
- (3) Measure the voltage between terminal No. 1 and ground by backprobing.
 - When steering wheel is not turned, voltage should be battery positive voltage.
 - When steering wheel is turned, voltage should be 1 volt or less.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage within the specified range?

YES : Go to Step 3.

NO : Go to Step 5.

STEP 3. Check harness connector B-19 at power steering pressure switch for damage.
Q: Is the harness connector in good condition?

YES : Go to Step 4.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 13.

STEP 4. Using scan tool MB991958, check data list item 83: Power Steering Pressure Switch.

- (1) Start the engine and run at idle.
- (2) Set scan tool MB991958 to the data reading mode for item 83, Power Steering Pressure Switch.
 - If the steering wheel is not turned while idling, "OFF" will be displayed.
 - If the steering wheel is turned while idling, "ON" will be displayed.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the switch operating properly?

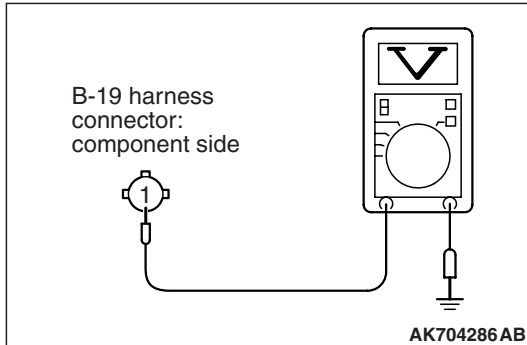
YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 13.

STEP 5. Check harness connector B-19 at power steering pressure switch for damage.
Q: Is the harness connector in good condition?

YES : Go to Step 6.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 13.



STEP 6. Measure the power supply voltage at power steering pressure switch harness side connector B-19.

- (1) Disconnect the connector B-19 and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 1 and ground.
 - Voltage should be battery positive voltage.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 10.

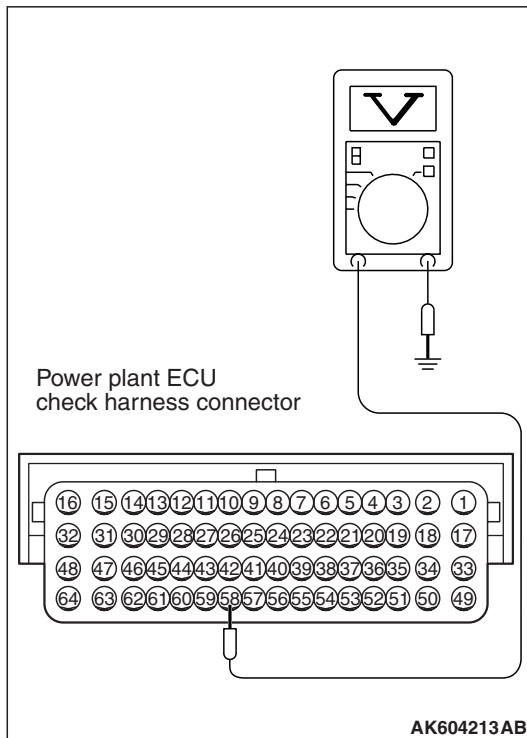
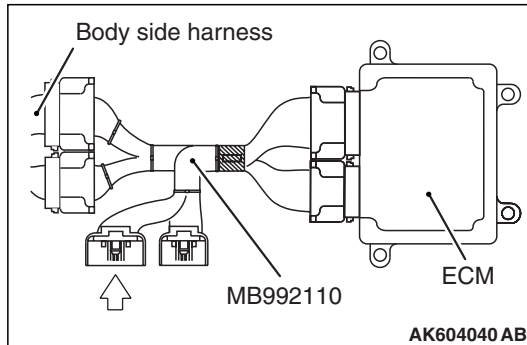
NO : Go to Step 7.

STEP 7. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 8.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 13.



STEP 8. Measure the power supply voltage at ECM connector B-09 by using power plant ECU check harness special tool MB992110.

- (1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.
- (2) Disconnect the power steering pressure switch connector B-19.
- (3) Turn the ignition switch to the "ON" position.

- (4) Measure the voltage between terminal No. 58 and ground.
 - Voltage should be between battery positive voltage.
- (5) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Repair harness wire between power steering pressure switch connector B-19 (terminal No. 1) and ECM connector B-09 (terminal No. 58) because of open circuit. Then go to Step 13.

NO : Go to Step 9.

STEP 9. Check for short circuit to ground between power steering pressure switch connector B-19 (terminal No. 1) and ECM connector B-09 (terminal No. 58).

Q: Is the harness wire in good condition?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 13.

NO : Repair it. Then go to Step 13.

STEP 10. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 11.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 13.

STEP 11. Check for harness damage between power steering pressure switch connector B-19 (terminal No. 1) and ECM connector B-09 (terminal No. 58).

Q: Is the harness wire in good condition?

YES : Go to Step 12.

NO : Repair it. Then go to Step 13.

STEP 12. Replace the power steering pressure switch.

(1) Replace the power steering pressure switch.

(2) Check the trouble symptoms.

(3) Read in the diagnostic trouble code (DTC).

Q: Is DTC P0551 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 13.

NO : Go to Step 13.

STEP 13. Using scan tool MB991958, check data list item 83: Power Steering Pressure Switch.

(1) Start the engine and run at idle.

(2) Set scan tool MB991958 to the data reading mode for item 83, Power Steering Pressure Switch.

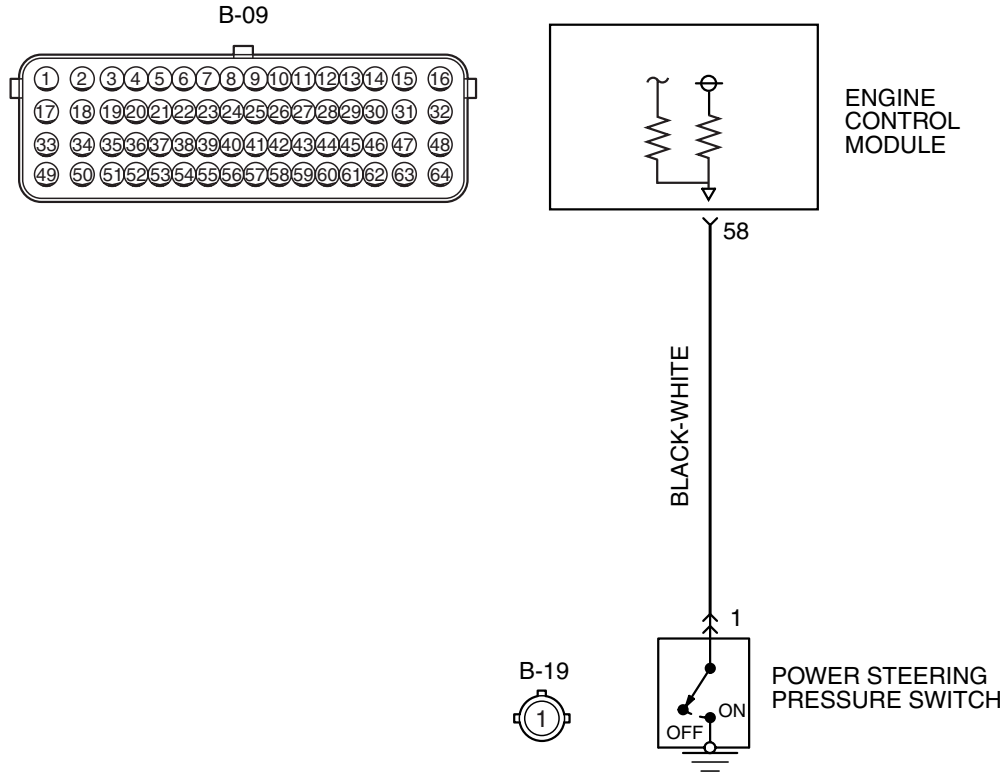
- If the steering wheel is not turned while idling, "OFF" will be displayed.
- If the steering wheel is turned while idling, "ON" will be displayed.

(3) Turn the ignition switch to the "LOCK" (OFF) position.

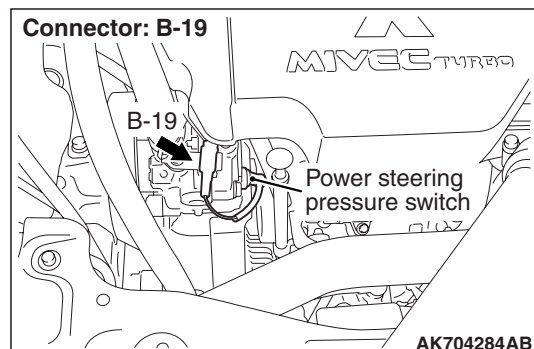
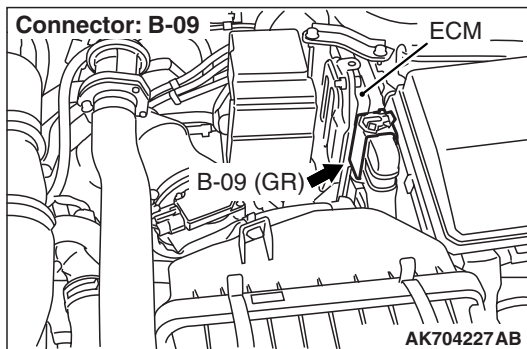
Q: Is the switch operating properly?

YES : The inspection is complete.

NO : Retry the troubleshooting.

DTC P0554: Power Steering Pressure Switch Circuit Intermittent**POWER STEERING PRESSURE SWITCH CIRCUIT**

AK704283AB

**CIRCUIT OPERATION**

- A battery positive voltage is applied to the power steering pressure switch output terminal (terminal No. 1) from the ECM (terminal No. 58) via the resistor in the ECM.

TECHNICAL DESCRIPTION

- The power steering pressure switch converts the existence of a power steering load into a high/low voltage, and inputs it into the ECM.

- When the steering wheel is turned, hydraulic pressure rises. The power steering pressure switch closes, and the applied battery positive voltage will be grounded. With this, the power steering pressure switch output voltage will fluctuate between 0 and 12 volts.
- While driving with the steering wheel held straight, the power steering pressure switch turns "OFF".
- The ECM checks whether the power steering pressure switch turns "OFF" or "ON" during driving.

DESCRIPTIONS OF MONITOR METHODS

Power steering pressure switch changes from off to on more than 10 times for 1 second.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

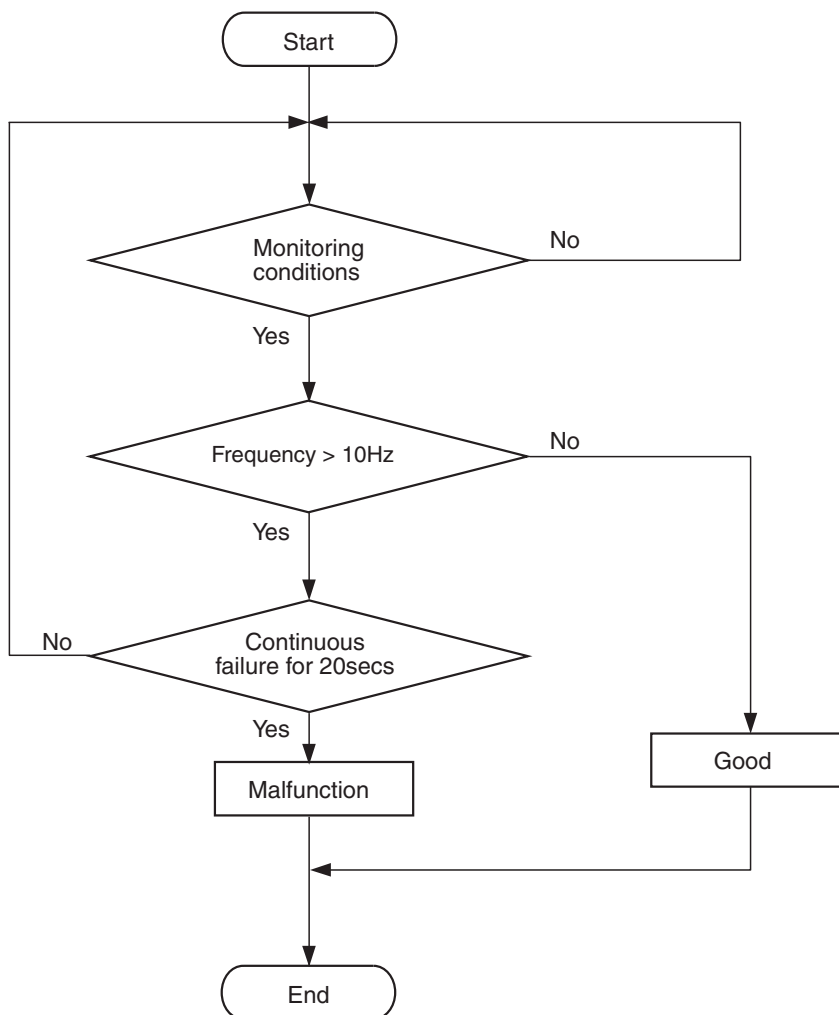
- Not applicable

Sensor (The sensor below is determined to be normal)

- Engine coolant temperature sensor

DTC SET CONDITIONS

Logic Flow Chart



AK703260

Check Conditions

- Engine coolant temperature is higher than 65° C (149° F).
- Vehicle speed is 50 km/h (31 mph) or higher.

Judgement Criterion

- The ON/OFF frequency of a power steering pressure switch is 10 Hz or more for 20 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 16 [P.13A-9](#)

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Power steering pressure switch failed.
- Incorrect power steering fluid level.
- Incorrect oil pump pressure.
- Harness damage in power steering pressure switch circuit, or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, check data list item 83: Power Steering Pressure Switch.

⚠ CAUTION

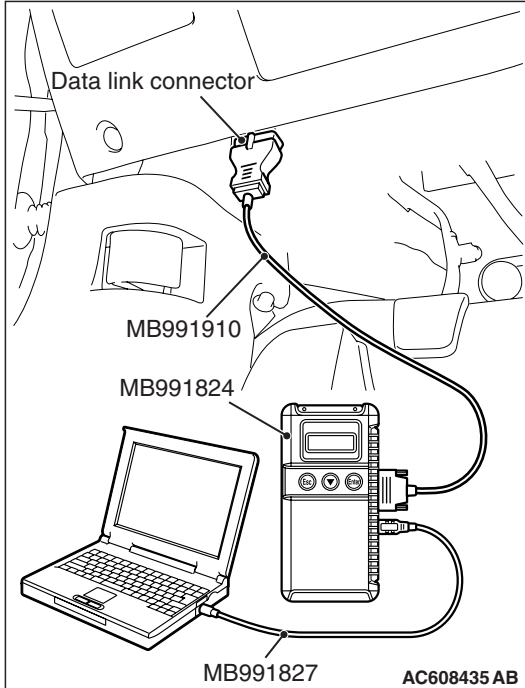
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Start the engine and run at idle.
- (3) Set scan tool MB991958 to the data reading mode for item 83, Power Steering Pressure Switch.
 - If the steering wheel is not turned while idling, "OFF" will be displayed.
 - If the steering wheel is turned while idling, "ON" will be displayed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2.



STEP 2. Check the power steering fluid level.

Refer to GROUP 37, On-vehicle Service –Fluid Level Check [P.37-17](#).

Q: Are there any abnormalities?

YES : Repair it. Then go to Step 7.

NO : Go to Step 3.

STEP 3. Check the power steering pressure switch.

Refer to GROUP 37, On-vehicle Service –Power Steering Pressure Switch Check [P.37-21](#).

Q: Are there any abnormalities?

YES : Replace the power steering pressure switch. Then go to Step 7.

NO : Go to Step 4.

STEP 4. Check the oil pump pressure.

Refer to GROUP 37, On-vehicle Service –Oil Pump Pressure Test [P.37-20](#).

Q: Are there any abnormalities?

YES : Repair it. Then go to Step 7.

NO : Go to Step 5.

STEP 5. Check harness connector B-19 at the power steering pressure switch and harness connector B-09 at ECM for damage.**Q: Is the harness connector in good condition?**

YES : Go to Step 6.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7.

STEP 6. Check for harness damage between power steering pressure switch connector B-19 (terminal No. 1) and ECM connector B-09 (terminal No. 58).**Q: Is the harness wire in good condition?**

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 7.

NO : Repair it. Then go to Step 7.

STEP 7. Test the OBD-II drive cycle.

(1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).

(2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0554 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0603: EEPROM Malfunction

TECHNICAL DESCRIPTION

- ECM stored the information such as the idle learned value and so on in the memory of ECM.

DESCRIPTIONS OF MONITOR METHODS

To check whether the information such as the idle learned value and so on is stored in the memory of ECM.

MONITOR EXECUTION

Once per driving cycle

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

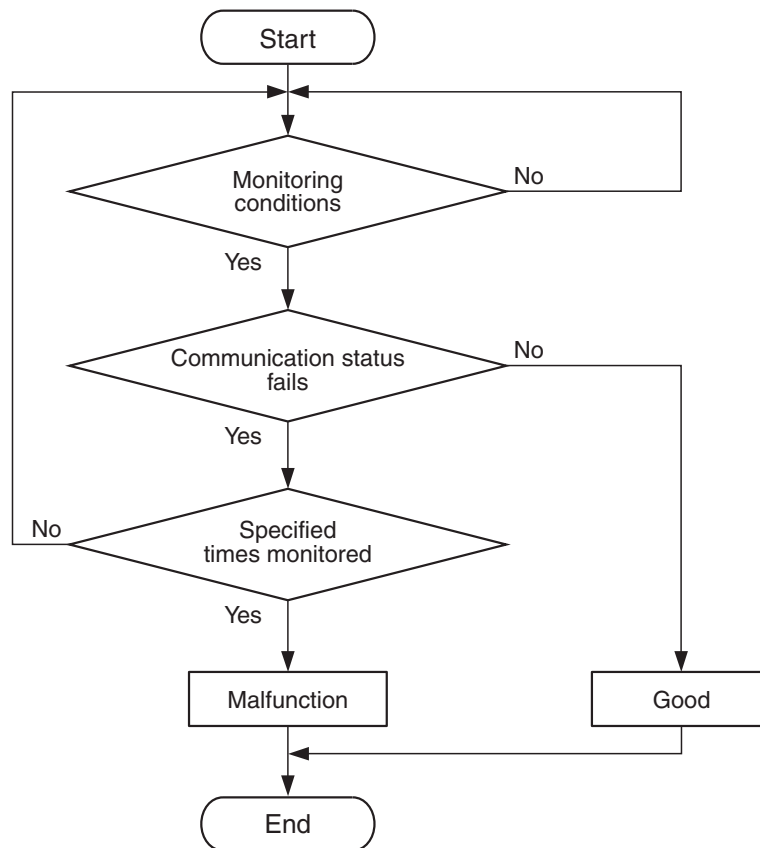
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK704415

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- The latest data that was flashed while the ignition switch was in "LOCK" (OFF) position are not stored correctly.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

FAIL-SAFE AND BACKUP FUNCTION

- None

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable

- MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC)

⚠ CAUTION

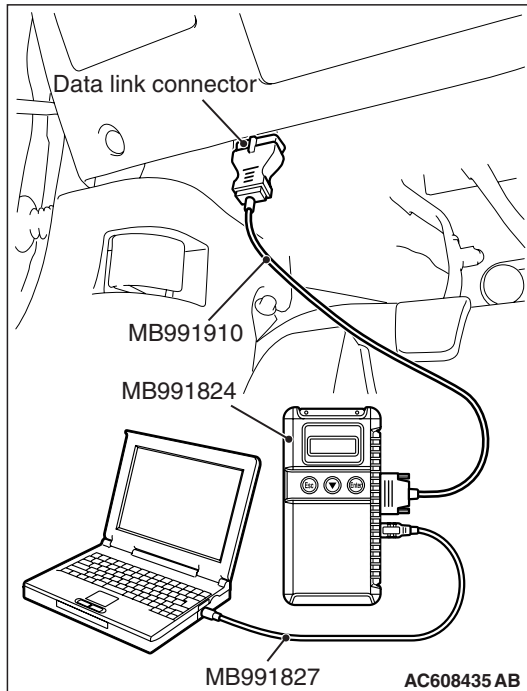
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0603 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P0606: Engine Control Module Main Processor Malfunction

TECHNICAL DESCRIPTION

- Throttle actuator control processor checks the ECM for abnormal conditions.

DESCRIPTIONS OF MONITOR METHODS

No watch dog pulse is detected.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS

(Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- No surveillance pulse signals should be input for 0.5 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC)

CAUTION

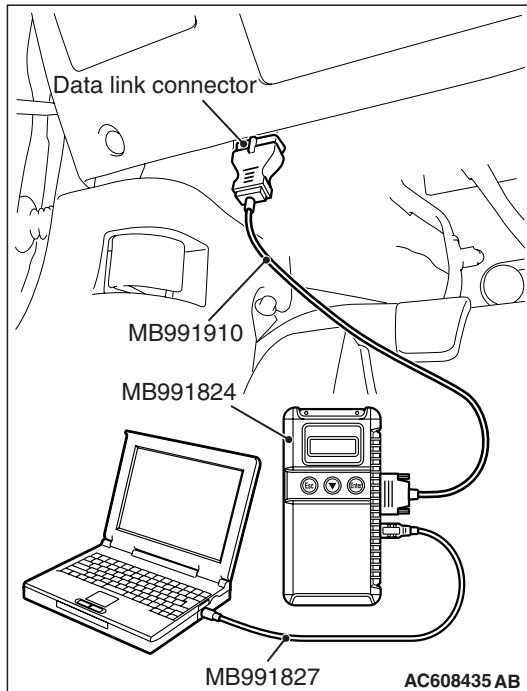
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0606 set?

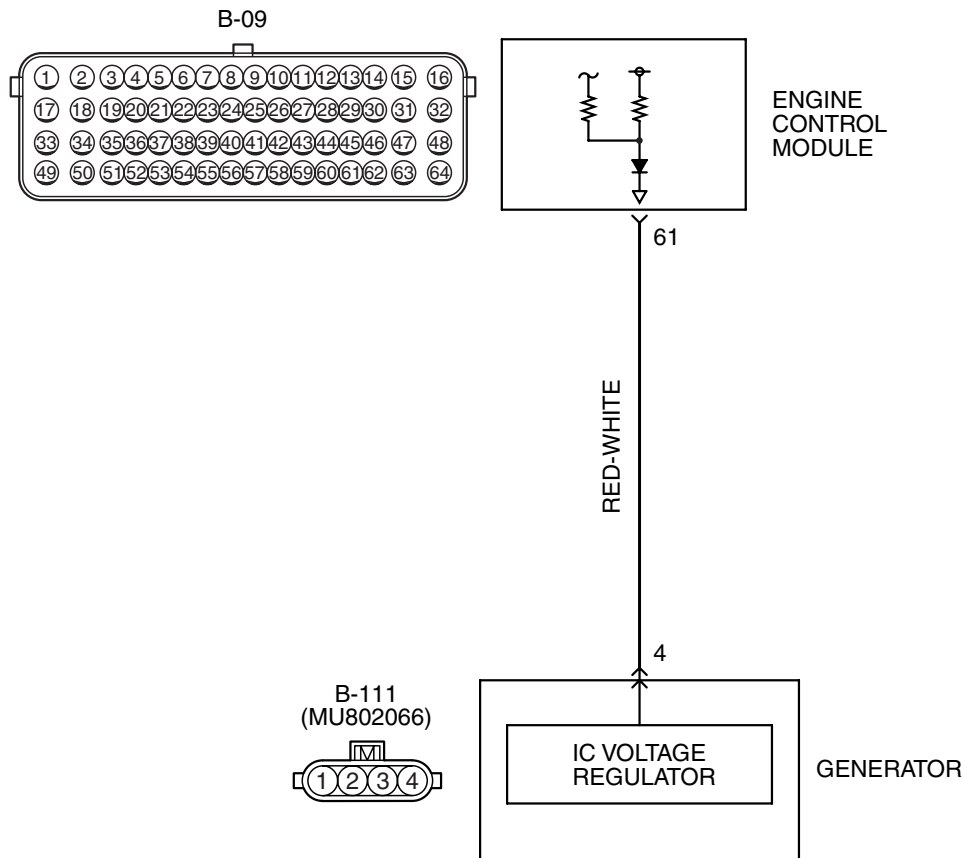
YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

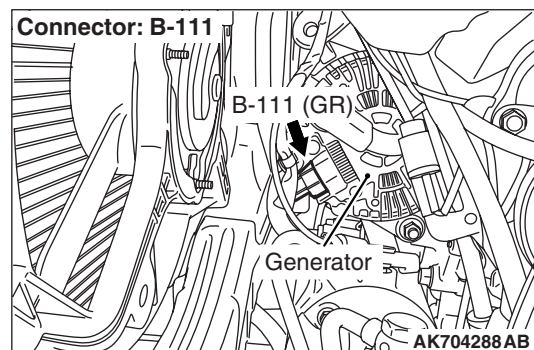
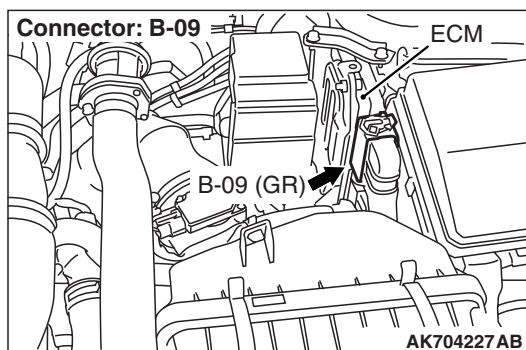


DTC P0622: Generator FR Terminal Circuit Malfunction

GENERATOR CIRCUIT



AK604253AD



CIRCUIT OPERATION

- The ECM (terminal No. 61) apply a battery voltage into the generator FR terminal No. 4 via resistance inside the unit.

TECHNICAL DESCRIPTION

- When the generator field coils are controlled, the generator FR terminal inputs signal to the ECM.
- The ECM detects the generator output with the input signal, and controls the idle air control motor according to the generator output.

DTC SET CONDITIONS

Check Condition

- Engine speed is higher than 50 r/min.

Judgement Criterion

- Input voltage from the generator FR terminal has continued to be battery positive voltage for 20 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- Prohibits generator output suppression control against current consumers. (Operates as a normal generator.)

TROUBLESHOOTING HINTS (The most likely causes for this code to be set area:)

- Generator failed.
- Open circuit in generator FR terminal circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

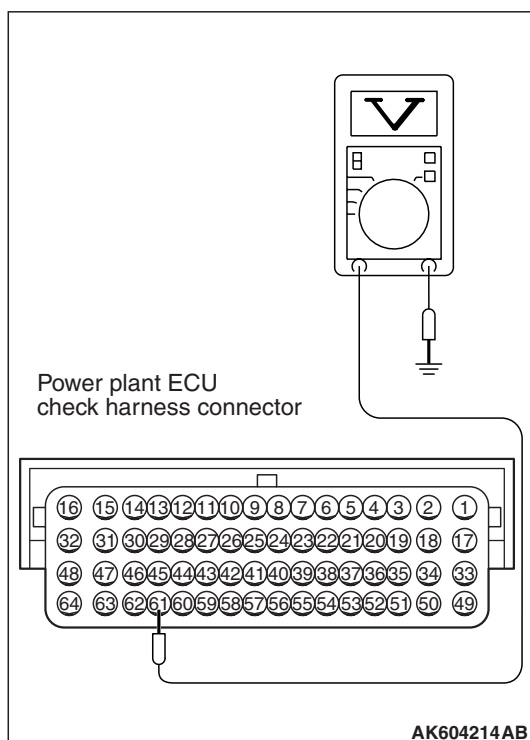
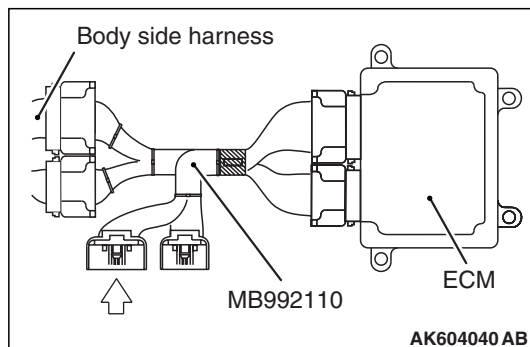
- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A
- MB992110: Power Plant ECU Check Harness

STEP 1. Check harness connector B-09 at ECM connector for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 8.



STEP 2. Measure the voltage at ECM connector B-09 by using power plant ECU check harness special tool MB992110.

- (1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.
- (2) Start the engine and run at idle.

- (3) Measure the voltage between terminal No. 61 and ground.

NOTE: Vehicle for Canada, the headlight, taillight, etc. remain lit even when the lighting switch is in "OFF" position but this is no problem for checks.

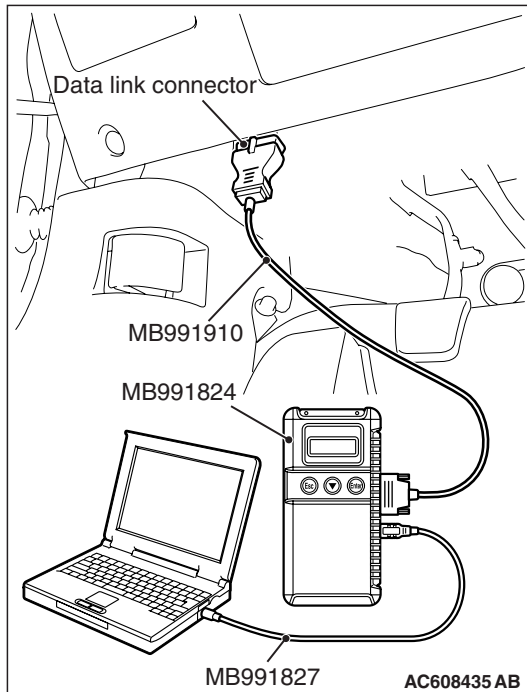
- a. Engine: warming up
- b. Radiator fan: stopped
- c. Headlight switch: OFF to ON
- d. Rear defogger switch: OFF to ON
- e. Stoplight switch: OFF to ON
 - Voltage should be drop.

- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Did the measured voltage drop?

YES : Go to Step 3.

NO : Go to Step 4.



STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Start the engine and run at idle.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0622 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 8.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 4. Check harness connector B-111 at generator connector for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 5.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 8.

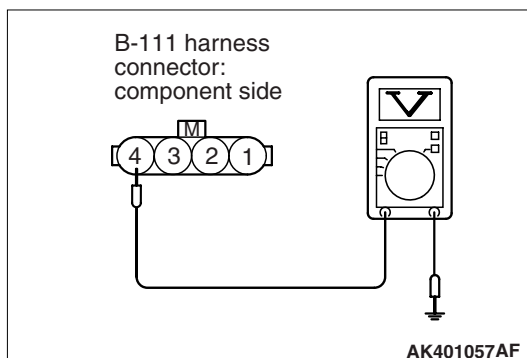
STEP 5. Measure the voltage at generator harness side connector B-111.

- (1) Disconnect the connector B-111 and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 4 and ground.
 - Voltage should measure battery positive voltage.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 7.

NO : Go to Step 6.



STEP 6. Check for open circuit and short circuit to ground between generator connector B-111 (terminal No. 4) and ECM connector B-09 (terminal No. 61).

Q: Is the harness wire in good condition?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 8.

NO : Repair it. Then go to Step 8.

STEP 7. Check for harness damage between generator connector B-111 (terminal No. 4) and ECM connector B-09 (terminal No. 61).

Q: Is the harness wire in good condition?

YES : Replace the generator. Then go to Step 8.

NO : Repair it. Then go to Step 8.

STEP 8. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

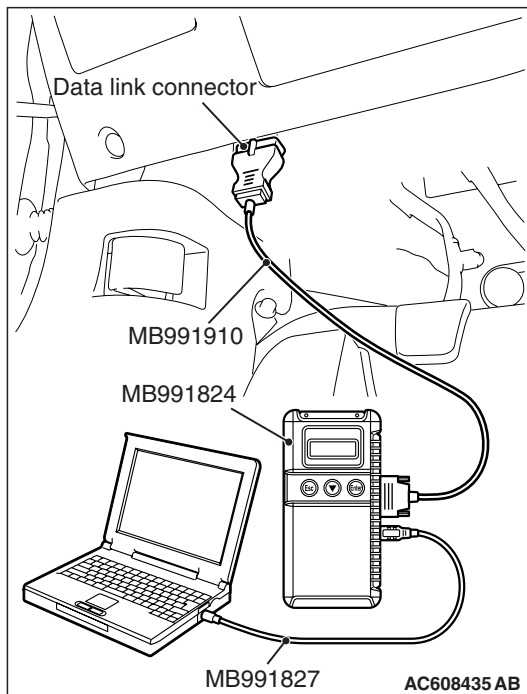
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Start the engine and run at idle.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0622 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.



DTC P0630: Vehicle Identification Number (VIN) Malfunction

TECHNICAL DESCRIPTION

- The Vehicle Identification Number (VIN) is stored in the ECM by the vehicle manufacturer.

DESCRIPTIONS OF MONITOR METHODS

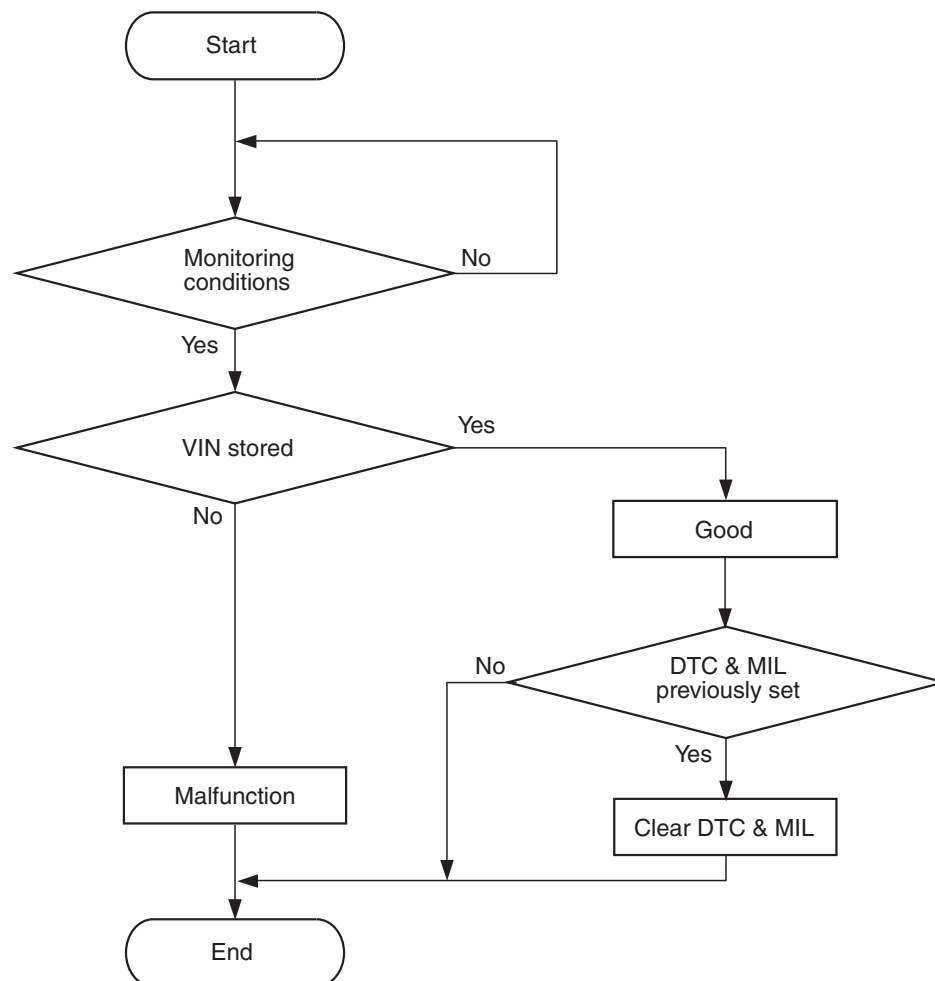
The ECM checks whether the VIN is being entered or not.

MONITOR EXECUTION

Continuous

DTC SET CONDITIONS

Logic Flow Chart



AK604349

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- VIN (current) has not been written.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, check VIN Information.**⚠ CAUTION**

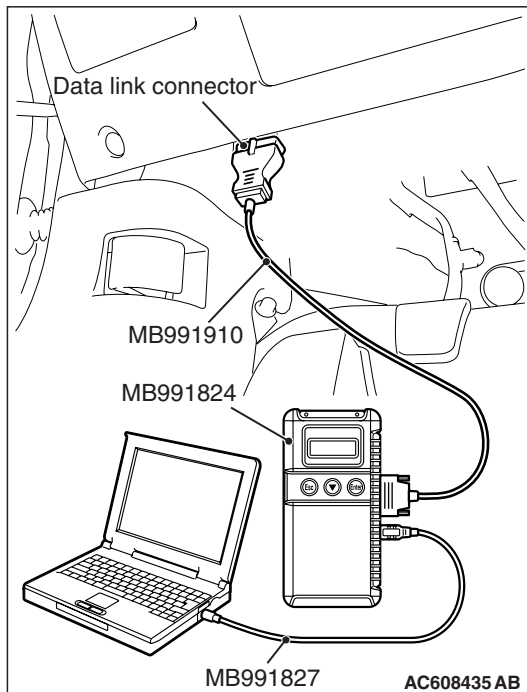
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Set scan tool MB991958 to the coding mode for VIN Information.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Has VIN (current) been written?

YES : Go to Step 2.

NO : Write VIN. Then go to Step 3.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC)

- (1) Turn the ignition switch to the "ON" position.
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0630 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).
- NO :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC)

- (1) Turn the ignition switch to the "ON" position.
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P0630 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).
- NO :** The inspection is complete.

DTC P0638: Throttle Actuator Control Motor Circuit Range/Performance

**THROTTLE ACTUATOR CONTROL
MOTOR CIRCUIT
RANGE/PERFORMANCE PROBLEM
CIRCUIT**

- Refer to DTC P2101 –Throttle Actuator Control Motor Magneto Malfunction [P.13A-668](#).

CIRCUIT OPERATION

- Refer to DTC P2101 –Throttle Actuator Control Motor Magneto Malfunction [P.13A-668](#).

TECHNICAL DESCRIPTION

- ECM checks the electronic controlled throttle system for abnormal conditions.

DESCRIPTIONS OF MONITOR METHODS

Difference between throttle position sensor (main) output and target opening is greater than the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Check Conditions**

- Battery positive voltage is higher than 8.3 volts.
- Throttle position sensor (main) output voltage is between 0.2 and 4.8 volts.
- Drop of throttle position sensor (main) output voltage per 100 milliseconds is 0.04 volt or more.

Judgement Criterion

- Throttle position sensor (main) output voltage has continued to be higher than the target throttle position sensor (main) voltage by 0.5 volt or more for 0.4 second.

Check Conditions

- Battery positive voltage is higher than 8.3 volts.
- Throttle position sensor (main) output voltage is between 0.2 and 4.8 volts.
- Target throttle position sensor (main) output voltage is 1.5 volts or lower.

Judgement Criterion

- Difference between throttle position sensor (main) output voltage and target throttle position sensor (main) voltage is 0.6 volt or higher for 0.4 second.

Check Conditions

- Battery positive voltage is higher than 8.3 volts.
- Throttle position sensor (main) output voltage is between 0.2 and 4.8 volts.

Judgement Criterion

- Difference between throttle position sensor (main) output voltage and target throttle position sensor (main) voltage is 1 volt or higher for 0.4 second.

Check Condition

- Battery positive voltage is higher than 8.3 volts.

Judgement Criterion

- Throttle position sensor (main) default opening learnt value is 4 volts or higher.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle valve return spring failed.
- Throttle valve operation failed.
- Throttle actuator control motor failed.
- Harness damage in throttle actuator control motor circuit, or connector damage.
- ECM failed.

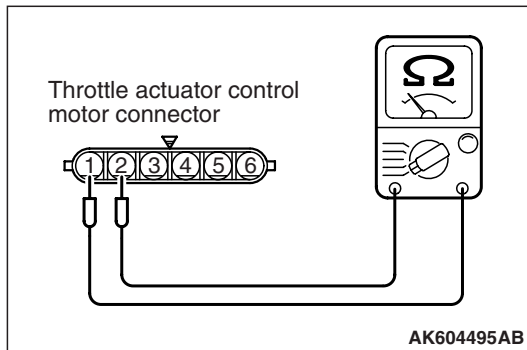
DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector B-11 at throttle actuator control motor for damage.**Q: Is the harness connector in good condition?**

YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7..



STEP 2. Check the throttle actuator control motor.

- (1) Disconnect the connector B-11.
- (2) Measure the resistance between throttle actuator control motor side connector terminal No. 1 and No. 2.

Standard value: 0.3 –80 Ω [at 20° C (68° F)]

Q: Is the measured resistance between 0.3 and 80 Ω [at 20° C (68° F)]?

YES : Go to Step 3.

NO : Replace the throttle body assembly. Then go to Step 7.

STEP 3. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 4.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7.

STEP 4. Check for harness damage between throttle actuator control motor connector B-11 (terminal No. 1) and ECM connector B-09 (terminal No. 15).

Q: Is the harness wire in good condition?

YES : Go to Step 5.

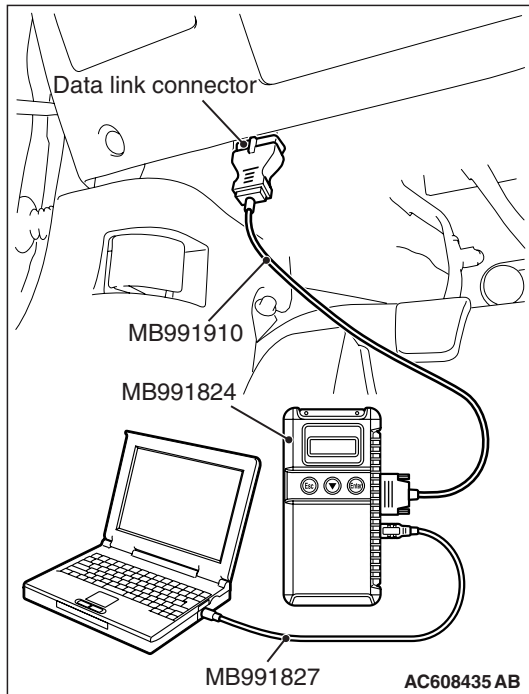
NO : Repair it. Then go to Step 7.

STEP 5. Check for harness damage between throttle actuator control motor connector B-11 (terminal No. 2) and ECM connector B-09 (terminal No. 16).

Q: Is the harness wire in good condition?

YES : Go to Step 6.

NO : Repair it. Then go to Step 7.



STEP 6. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

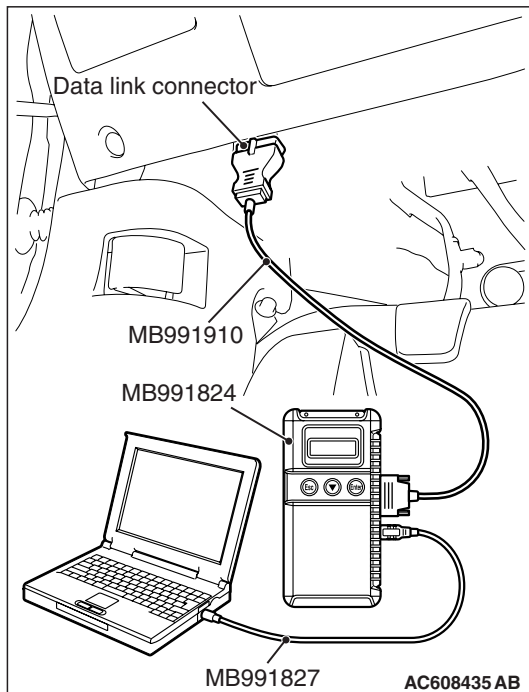
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0638 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 7.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 7. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0638 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P0642: Throttle Position Sensor Power Supply

TECHNICAL DESCRIPTION

- ECM checks the throttle position sensor power voltage for abnormal conditions.

DESCRIPTIONS OF MONITOR METHODS

Throttle position sensor source voltage is smaller than the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Check Conditions

- Battery positive voltage is higher than 6.3 volts.

Judgement Criteria

- Throttle position sensor power voltage should be 4.1 volts or less for 0.3 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

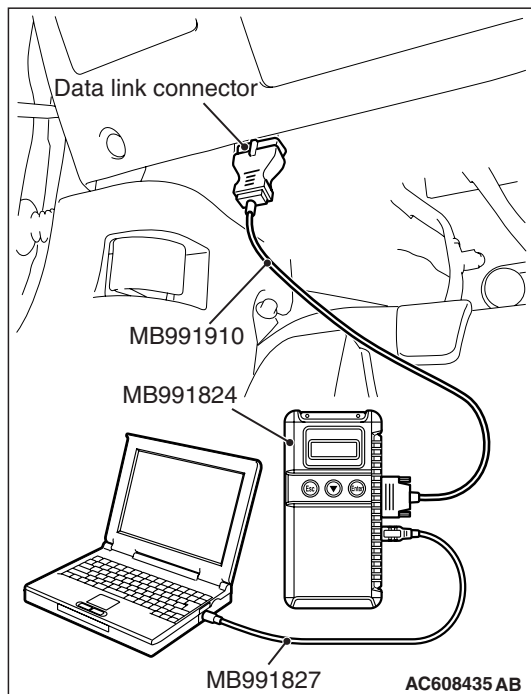
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK"(OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK"(OFF) position.

Q: Is DTC P0642 set?

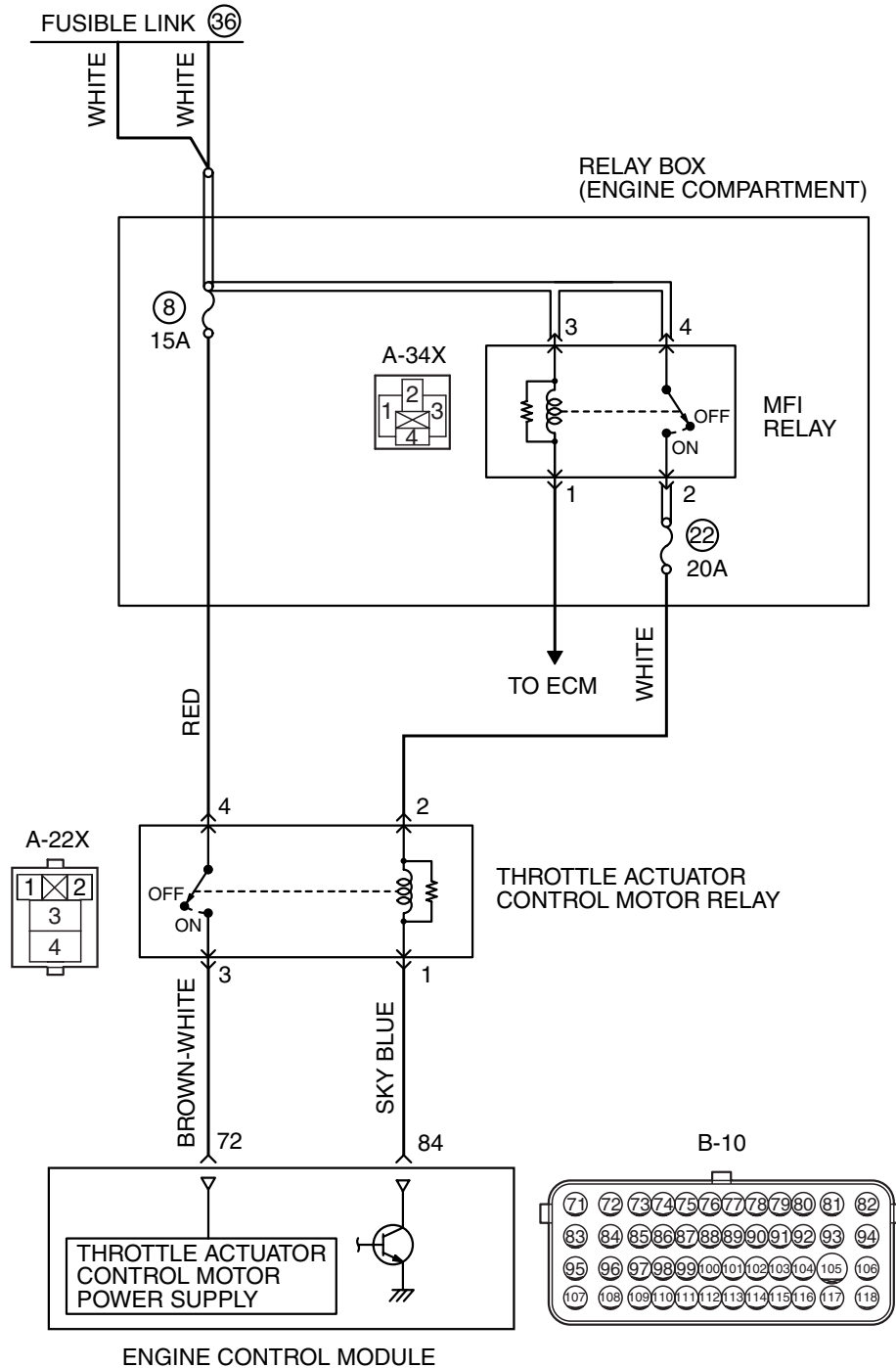
YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

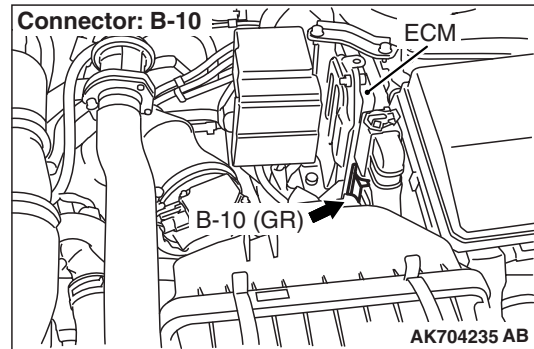
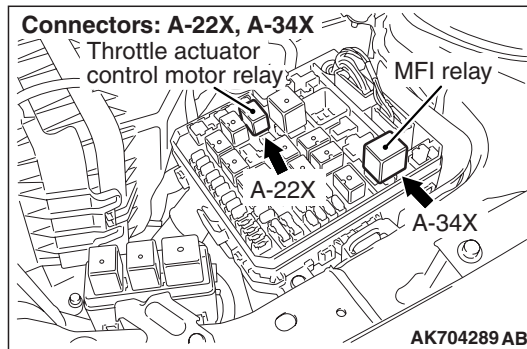


DTC P0657: Throttle Actuator Control Motor Relay Circuit Malfunction

THROTTLE ACTUATOR CONTROL MOTOR RELAY CIRCUIT



AK604254AD



CIRCUIT OPERATION

- Battery positive voltage is applied to the throttle actuator control motor relay terminal (terminal No. 4).
- Battery positive voltage is applied to the throttle actuator control motor relay terminal (terminal No. 2) from the MFI relay (terminal No. 2).
- ECM (terminal No. 84) applies current to the throttle actuator control motor relay coil by turning ON the power transistor in the unit in order to turn the relay ON.
- When the throttle actuator control motor relay turns ON, battery positive voltage is supplied by the throttle actuator control motor relay (terminal No. 3) to the ECM (terminal No. 72).

TECHNICAL DESCRIPTION

- When the ignition switch ON signal is input into the ECM, the ECM turns ON the throttle actuator control motor relay.

DESCRIPTIONS OF MONITOR METHODS

Throttle actuator control motor relay circuit voltage is smaller than the specified value.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Check Condition

- Battery positive voltage is higher than 8.3 volts.

Judgement Criterion

- The power line voltage of the electronic controlled throttle valve system should be 6.0 volts or less for 0.35 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle actuator control motor relay failed.
- Open or shorted throttle actuator control motor relay circuit, or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector A-22X at throttle actuator control motor relay for damage.

Q: Is the harness connector in good condition?

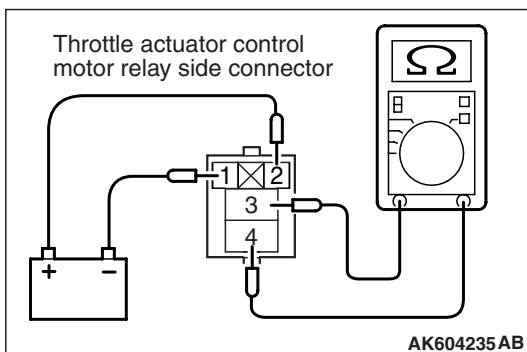
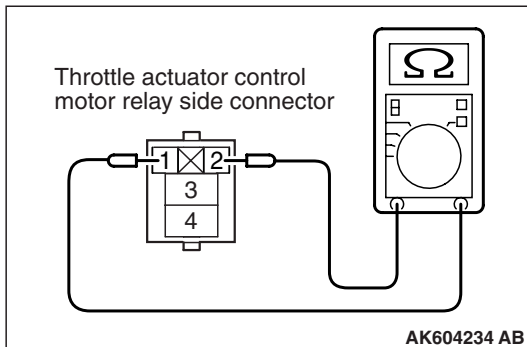
YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 14.

STEP 2. Check the throttle actuator control motor relay.

- (1) Remove the throttle actuator control motor relay.
- (2) Check for continuity between the throttle actuator control motor relay terminal No. 1 and No. 2.

- There should be continuity.



- (3) Use jumper wires to connect throttle actuator control motor relay terminal No. 2 to the positive battery terminal and terminal No. 1 to the negative battery terminal.

- (4) Check the continuity between the throttle actuator control motor relay terminal No. 3 and No. 4 while connecting and disconnecting the jumper wire at the negative battery terminal.

- Continuity (2 ohms or less) <Negative battery terminal connected.>
- Should be open loop. <Negative battery terminal disconnected.>

- (5) Install the throttle actuator control motor relay.

Q: Is the measured resistance normal?

YES : Go to Step 3.

NO : Replace the throttle actuator control motor relay. Then go to Step 14.

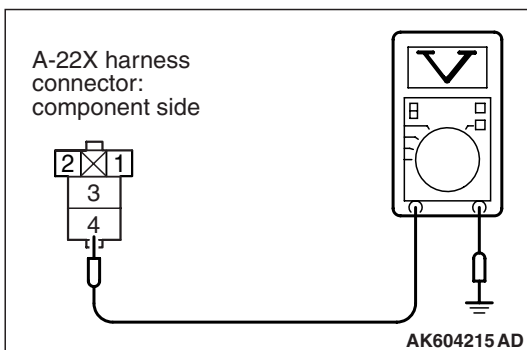
STEP 3. Measure the power supply voltage at throttle actuator control motor relay harness side connector A-22X

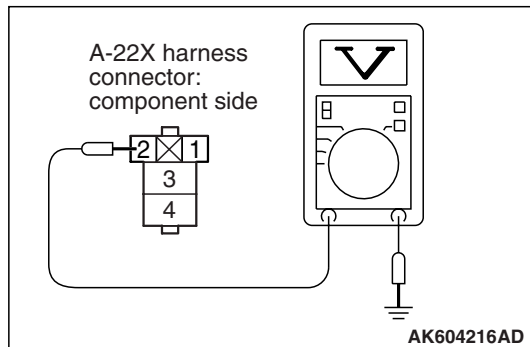
- (1) Disconnect the connector A-22X and measure at the harness side.
 - (2) Measure the voltage between terminal No. 4 and ground.
- Voltage should be battery positive voltage.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 4.

NO : Repair it harness wire between battery and throttle actuator control motor relay connector A-22X (terminal No. 4) because of open circuit or short circuit to ground. Then go to Step 14.



**STEP 4. Measure the power supply voltage at throttle actuator control motor relay harness side connector A-22X.**

- (1) Disconnect the connector A-22X and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 2 and ground.
 - Voltage should be battery positive voltage.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 6.

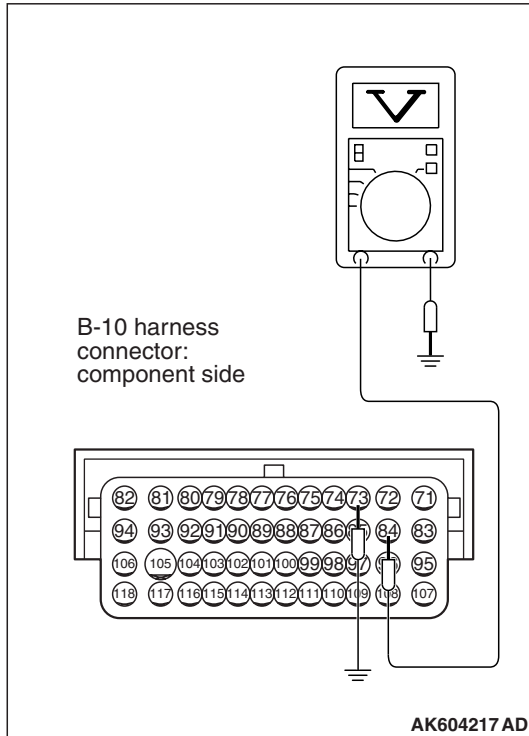
NO : Go to Step 5.

STEP 5. Check harness connector A-34X at MFI relay for damage.

Q: Is the harness connector in good condition?

YES : Repair harness wire between MFI relay connector A-34X (terminal No. 2) and throttle actuator control motor relay connector A-22X (terminal No. 2) because of open circuit or short circuit to ground. Then go to Step 14.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 14.



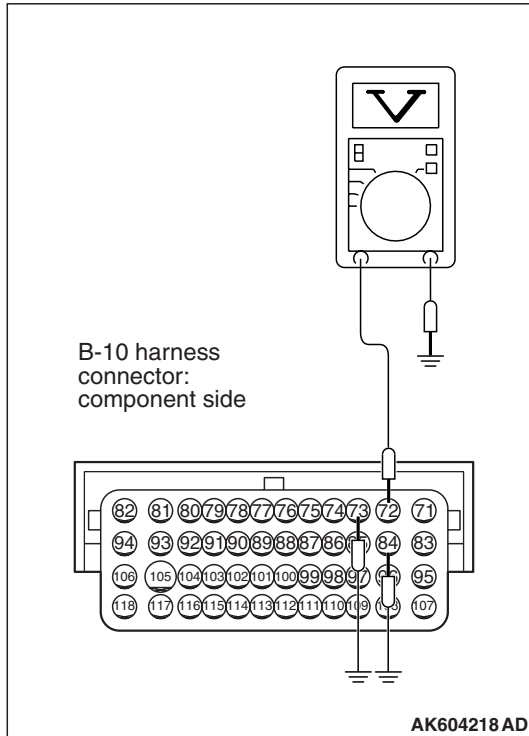
STEP 6. Measure the power supply voltage at ECM harness side connector B-10.

- (1) Disconnect the connector B-10 and measure at the harness side.
- (2) Short-circuit the terminal No. 73 of the ECM harness connector to the ground.
- (3) Measure the voltage between terminal No. 84 and ground.
 - Voltage should be battery positive voltage.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 7.

NO : Repair harness wire between throttle actuator control motor relay connector A-22X (terminal No. 1) and ECM connector B-10 (terminal No. 84) because of open circuit or short circuit to ground. Then go to Step 14.

**STEP 7. Measure the power supply voltage at ECM harness side connector B-10.**

- (1) Disconnect the connector B-10 and measure at the harness side.
- (2) Short-circuit the terminals No. 73 and No. 84 of the ECM harness connector to the ground.
- (3) Measure the voltage between terminal No. 72 and ground.
 - Voltage should be battery positive voltage.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 10.

NO : Go to Step 8.

STEP 8. Check for open circuit and short circuit to ground between throttle actuator control motor relay connector A-22X (terminal No. 3) and ECM connector B-10 (terminal No. 72).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 14.

STEP 9. Check for harness damage between MFI relay connector A-34X (terminal No. 2) and throttle actuator control motor relay connector A-22X (terminal No. 2).

Q: Is the harness wire in good condition?

YES : Repair harness wire between throttle actuator control motor relay connector A-22X (terminal No. 1) and ECM connector B-10 (terminal No. 84) because of harness damage. Then go to Step 14.

NO : Repair it. Then go to Step 14.

STEP 10. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 11.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 14.

STEP 11. Check for harness damage between battery and throttle actuator control motor relay connector A-22X (terminal No. 4).

Q: Is the harness wire in good condition?

YES : Go to Step 12.

NO : Repair it. Then go to Step 14.

STEP 12. Check for harness damage between throttle actuator control motor relay connector A-22X (terminal No. 3) and ECM connector B-10 (terminal No. 72).

Q: Is the harness wire in good condition?

YES : Go to Step 13.

NO : Repair it. Then go to Step 14.

STEP 13. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

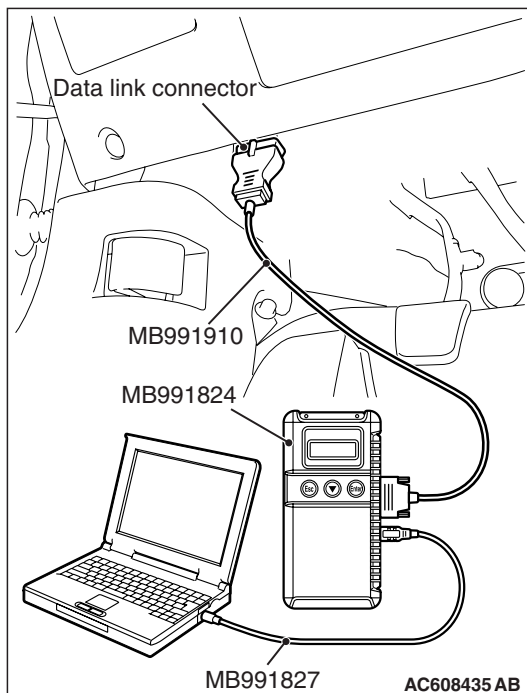
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

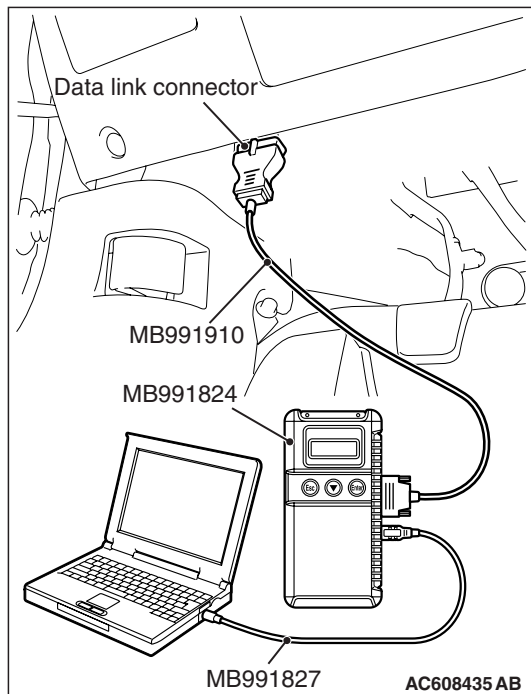
- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0657 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 14.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).





STEP 14. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

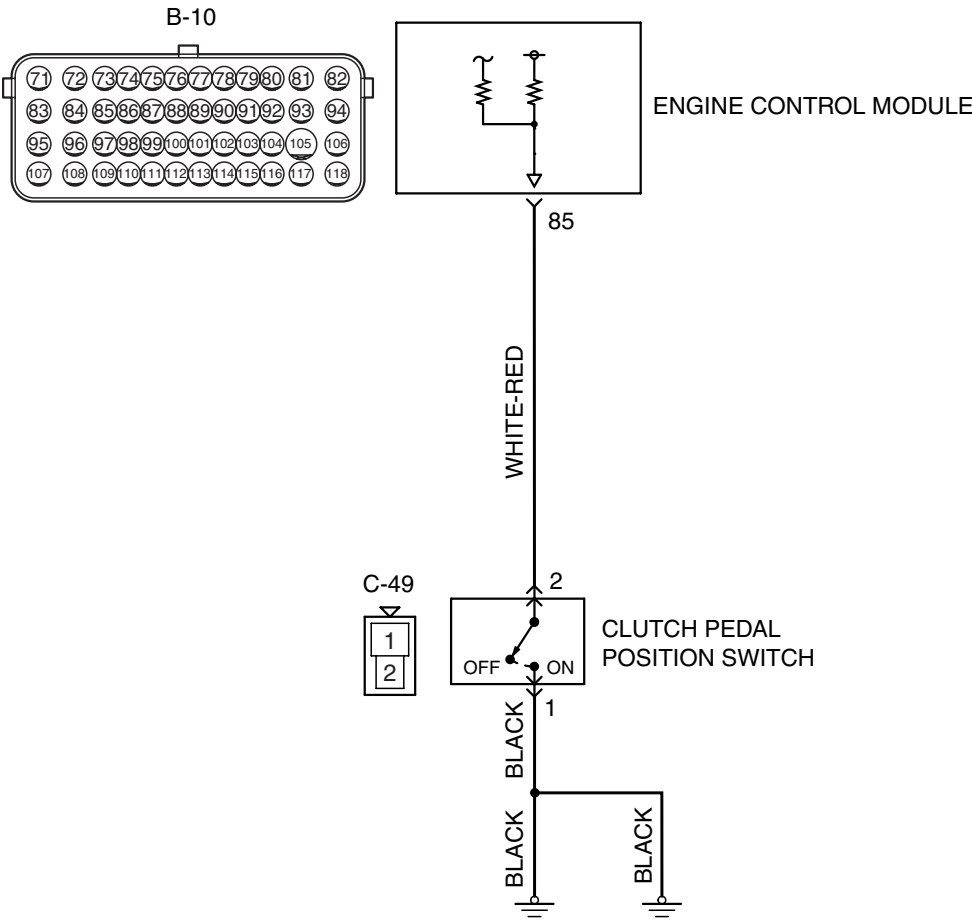
Q: Is DTC P0657 set?

YES : Retry the troubleshooting.

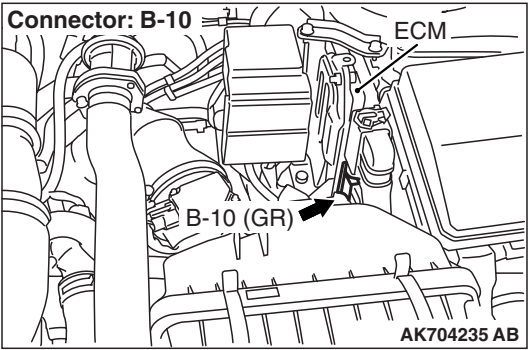
NO : The inspection is complete.

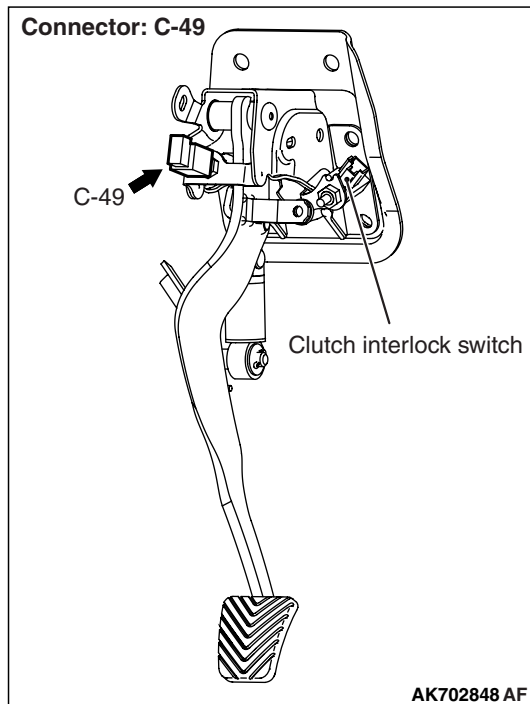
DTC P0830: Clutch Pedal Position Switch Circuit Range/Performance

CLUTCH PEDAL POSITION SWITCH CIRCUIT



AK704432 AB





CIRCUIT OPERATION

- A battery positive voltage is applied to the clutch pedal position switch output terminal (terminal No. 2) from the ECM (terminal No. 85) via the resistor in the ECM.

TECHNICAL DESCRIPTION

- The clutch pedal position switch inputs the high/low voltage into the ECM in accordance with whether the clutch pedal is being pressed.
- When the driver presses the clutch pedal while shifting gears, the contact of the clutch pedal position switch closes, causing the low signal to be input into the ECM. At this time, the ECM effects a correction to reduce the fuel injection volume.

DTC SET CONDITIONS

Check Condition, Judgment Criterion

- After the engine is started and the vehicle attains a minimum speed of 30 km/h (19 mph), the inversion of the high-low signals of the clutch pedal position switch is not detected even once.

FAIL-SAFE AND BACKUP FUNCTION

- None

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Clutch pedal position switch failed.
- Open or shorted clutch pedal position switch circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

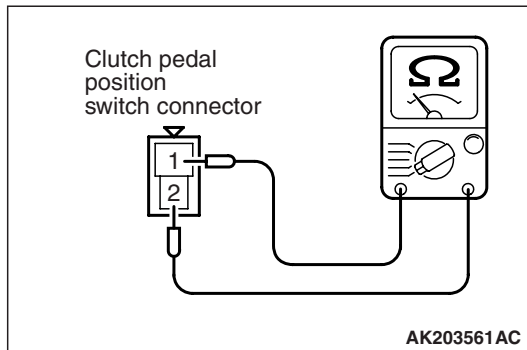
- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector C-49 at clutch pedal position switch for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.



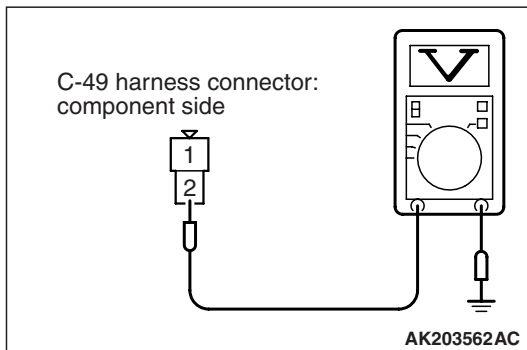
STEP 2. Check the clutch pedal position switch.

- (1) Disconnect the clutch pedal position switch connector C-49.
- (2) Check for continuity between the clutch pedal position switch side connector terminal No. 1 and No. 2.
 - If the clutch pedal is not pressed, the measurement should indicate the current is not going.
 - If the clutch pedal is pressed, the measurement should indicate the current is going.

Q: Are there any abnormalities?

YES : Repair it. Then go to Step 9.

NO : Go to Step 3.



STEP 3. Measure the power supply voltage at clutch pedal position switch harness side connector C-49.

- (1) Disconnect the connector C-49 and measure the harness side.
- (2) Turn the ignition switch to the "ON" position. Measure the voltage between terminal No. 2 and ground.
 - Voltage should be battery positive voltage.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 6.

NO : Go to Step 4.

STEP 4. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 5.

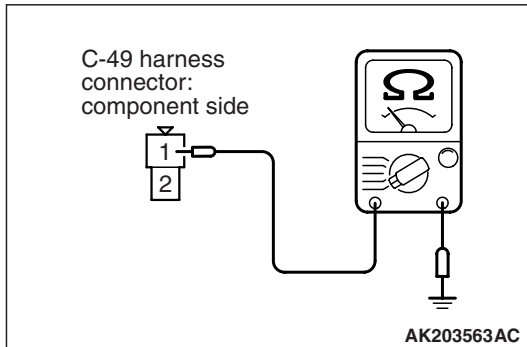
NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.

STEP 5. Check for open circuit and short circuit to ground between clutch pedal position switch connector C-49 (terminal No. 2) and ECM connector B-10 (terminal No. 85).

Q: Is the harness wire in good condition?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 9.

NO : Repair it. Then go to Step 9.



STEP 6. Check for continuity at clutch pedal position switch harness side connector C-49.

- (1) Disconnect the connector C-49 and measure the harness side.
- (2) Check for the continuity between terminal No. 1 and ground.
 - Should be less than 2 ohms.

Q: Does continuity exist?**YES :** Go to Step 7.**NO :** Repair harness wire between clutch pedal position switch connector C-49 terminal No. 1 and ground because of harness damage. Then go to Step 9.

STEP 7. Check harness connector B-10 at ECM for damage.
Q: Is the harness connector in good condition?**YES :** Go to Step 8.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.

STEP 8. Check for harness damage between clutch pedal position switch connector C-49 (terminal No. 2) and ECM connector B-10 (terminal No. 85).
Q: Is the harness wire in good condition?**YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 9.**NO :** Repair it. Then go to Step 9.

STEP 9. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

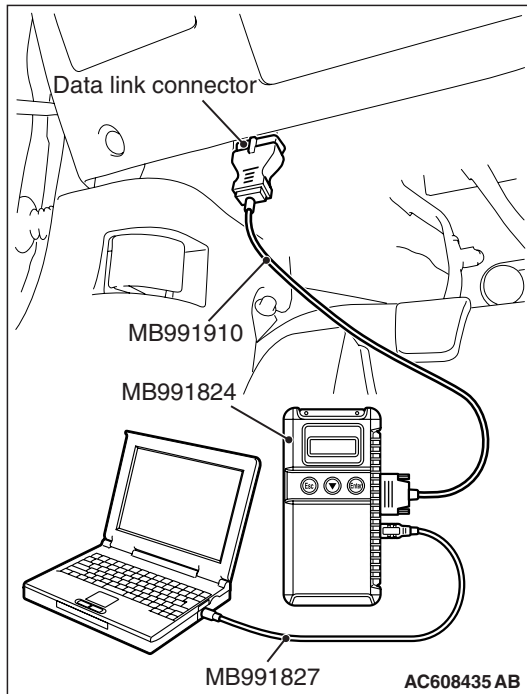
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTC.
- (4) Drive the vehicle at a minimum speed of 30 km/h (19 mph).
- (5) After completing the test drive, read the DTC. Turn the ignition switch to the "LOCK" (OFF) position.

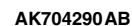
Q: Is DTC P0830 set?

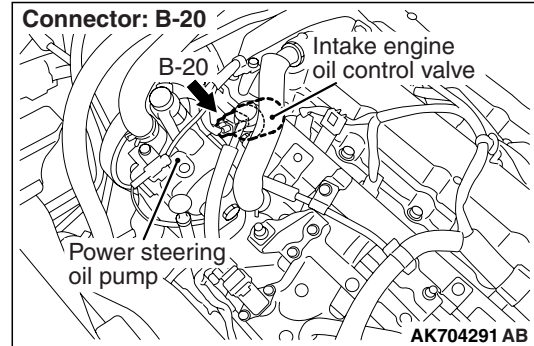
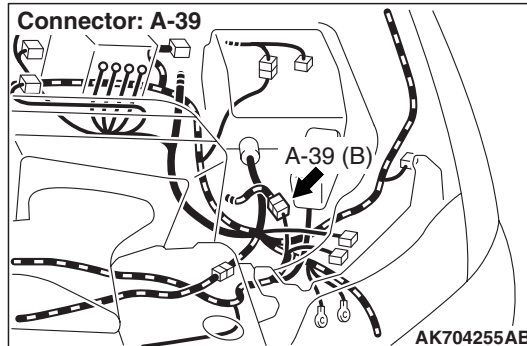
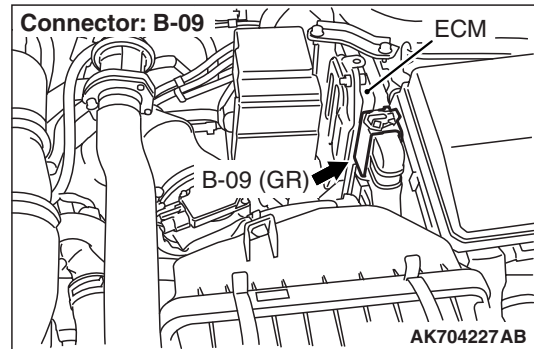
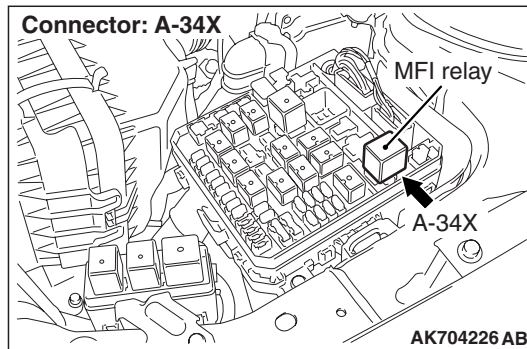
YES : The inspection is complete.

NO : Retry the troubleshooting.



INTAKE ENGINE OIL CONTROL VALVE CIRCUIT





CIRCUIT OPERATION

- The intake engine oil control valve power is supplied from the MFI relay (terminal No. 2).
- The ECM controls ground intake engine oil control valve by turning the power transistor in the ECM "ON" and "OFF".

TECHNICAL DESCRIPTION

- The intake engine oil control valve change the phase angle to operate the MIVEC system in the advance or retard side in accordance with the signals from the ECM.

DESCRIPTIONS OF MONITOR METHODS

Intake engine oil control valve circuit voltage is less than specified value.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

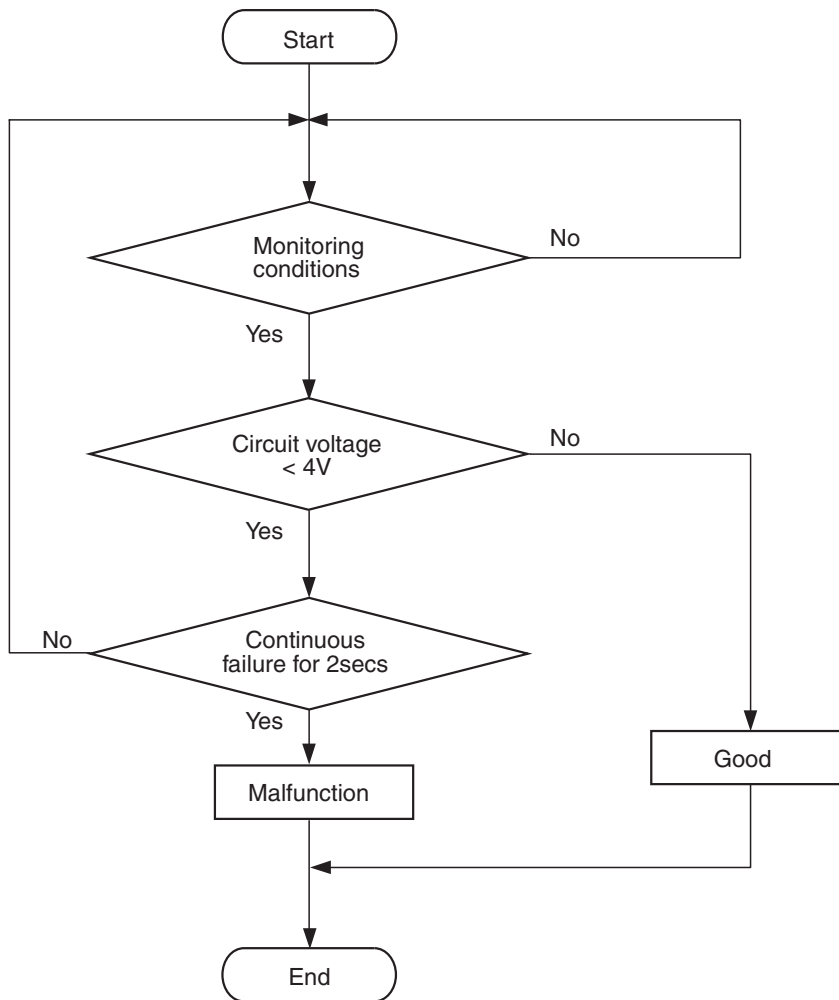
- Not applicable

Sensor (The sensor below is determined to be normal)

- Camshaft position sensor
- Crankshaft position sensor
- Engine coolant temperature sensor
- Throttle position sensor
- Engine oil control valve

DTC SET CONDITIONS

Logic Flow Chart



AK604350

Check Conditions

- Ignition switch is "ON" position.
- Intake engine oil control valve is "OFF".
- Battery positive voltage is between 10 and 16.5 volts.
- 0.1 second has elapsed after the above mentioned conditions have been met.

Judgement Criterion

- The ECM terminal voltage of intake engine oil control valve circuit is less than 4 volts for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- V.V.T.phase angle (intake) becomes most retarded angle.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Intake engine oil control valve failed.
- Open or shorted intake engine oil control valve circuit, or harness damage, or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB992110: Power Plant ECU Check Harness

STEP 1. Check harness connector B-20 at the intake engine oil control valve for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 10.

STEP 2. Check the intake engine oil control valve.

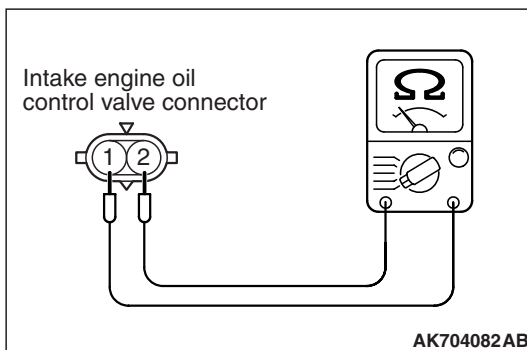
- (1) Disconnect the intake engine oil control valve connector B-20.
- (2) Measure the resistance between intake engine oil control valve side connector terminal No. 1 and No. 2.

Standard value: 6.9 –7.9 Ω [at 20° C (68° F)]

Q: Is the measured resistance between 6.9 and 7.9 Ω [at 20° C (68° F)]?

YES : Go to Step 3.

NO : Replace the intake engine oil control valve. Then go to Step 10.



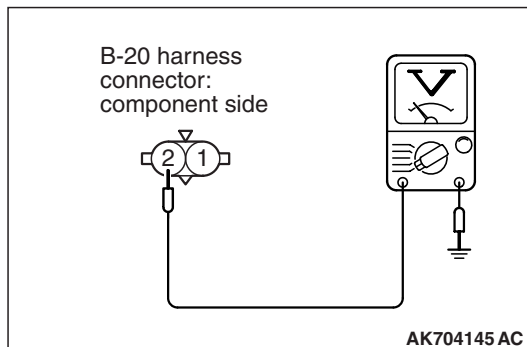
STEP 3. Measure the power supply voltage at intake engine oil control valve harness side connector B-20.

- (1) Disconnect the connector B-20 and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 2 and ground.
 - Voltage should be battery positive voltage.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 5.

NO : Go to Step 4.



STEP 4. Check harness connector A-34X at MFI relay for damage.

Q: Is the harness connector in good condition?

YES : Check harness connector A-39 at intermediate connector for damage, and repair or replace as required. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). If intermediate connector is in good condition, repair harness wire between MFI relay connector A-34X (terminal No. 2) and intake engine oil control valve connector B-20 (terminal No. 2) because of open circuit or short circuit to ground. Then go to Step 10.

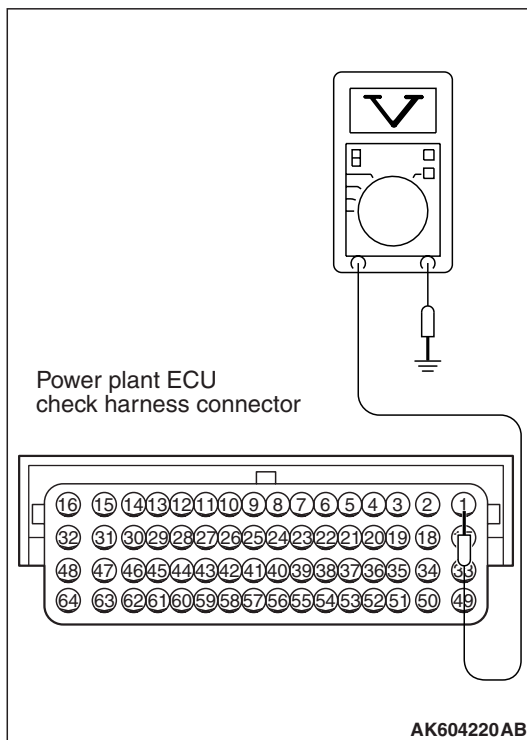
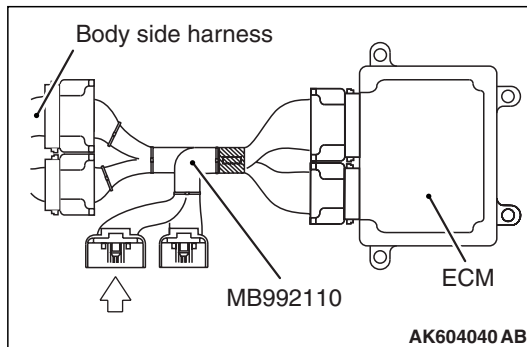
NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 10.

STEP 5. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 6.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 10.



STEP 6. Measure the power supply voltage at ECM connector B-09 by using power plant ECU check harness special tool MB992110.

(1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.

(2) Turn the ignition switch to the "ON" position.

(3) Measure the voltage between terminal No. 1 and ground.

- Voltage should be battery positive voltage.

(4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 7.

NO : Repair harness wire between intake engine oil control valve connector B-20 (terminal No. 1) and ECM connector B-09 (terminal No. 1) because of open circuit or short circuit to ground. Then go to Step 10.

STEP 7. Check for harness damage between MFI relay connector A-34X (terminal No. 2) and intake engine oil control valve connector B-20 (terminal No. 2).

NOTE: Check harness after checking intermediate connector A-39. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection P.00E-2. Then go to Step 10.

Q: Is the harness wire in good condition?

YES : Go to Step 8.

NO : Repair it. Then go to Step 10.

STEP 8. Check for harness damage between intake engine oil control valve connector B-20 (terminal No. 1) and ECM connector B-09 (terminal No. 1).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 10.

STEP 9. Check the trouble symptoms.

(1) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).

(2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1021 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 10.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 10. Test the OBD-II drive cycle.

(1) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).

(2) Check the diagnostic trouble code (DTC).

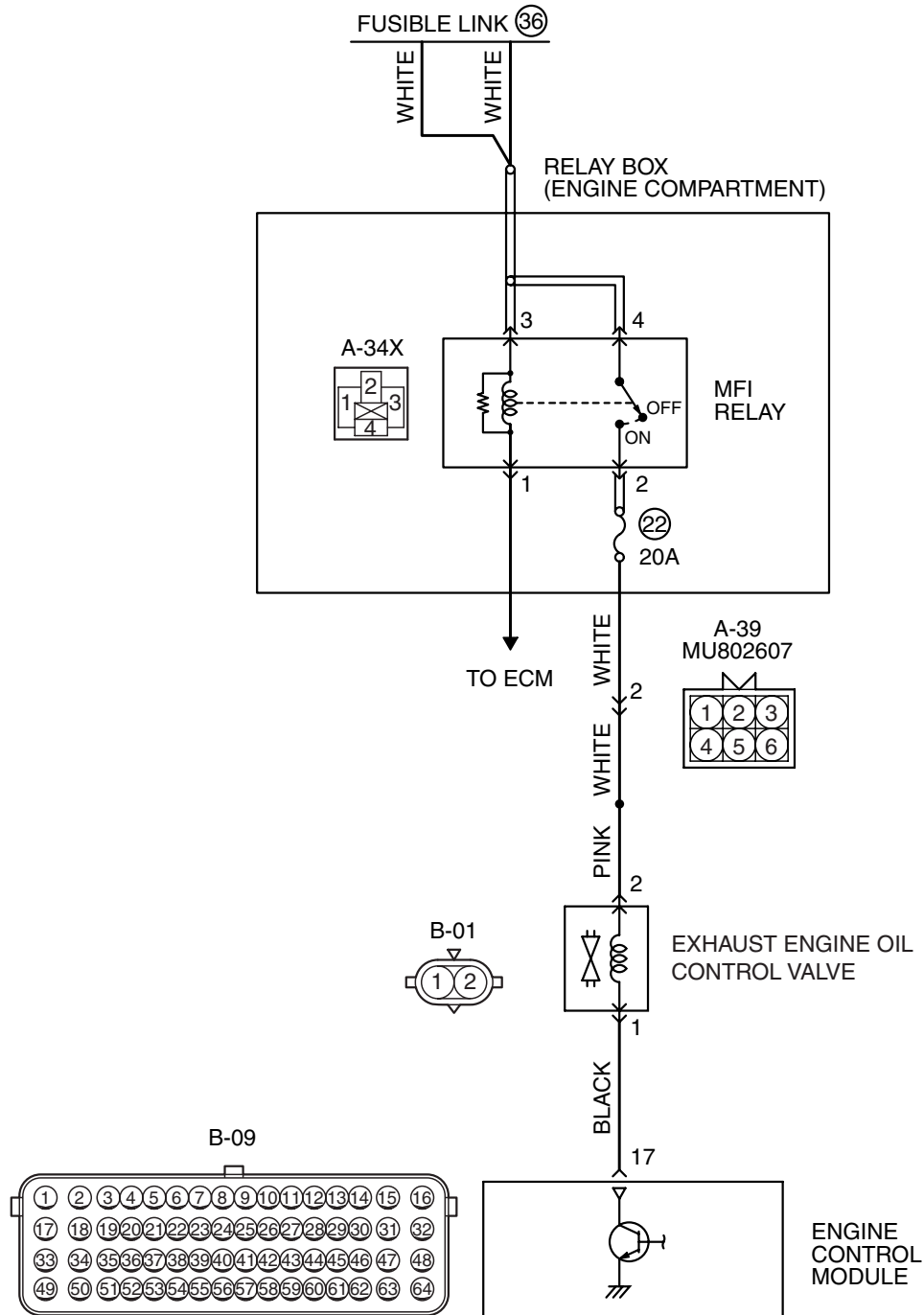
Q: Is DTC P1021 set?

YES : Retry the troubleshooting.

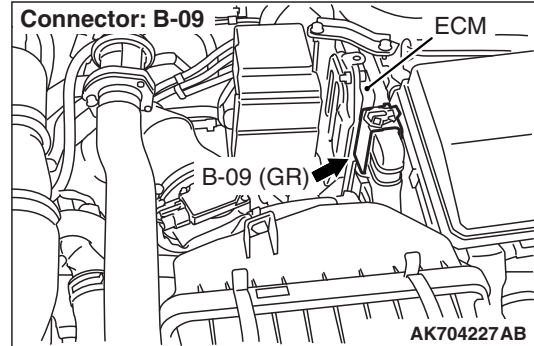
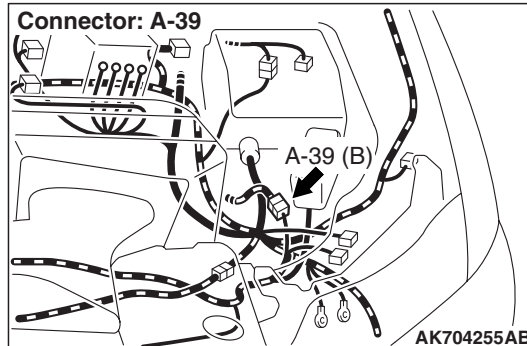
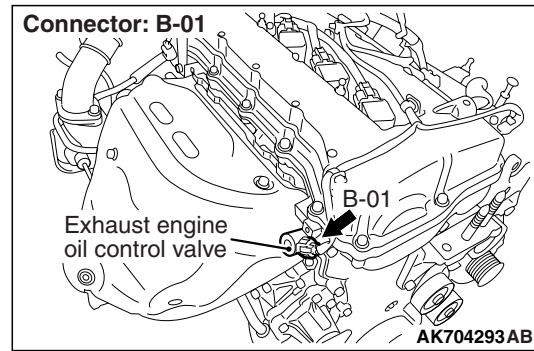
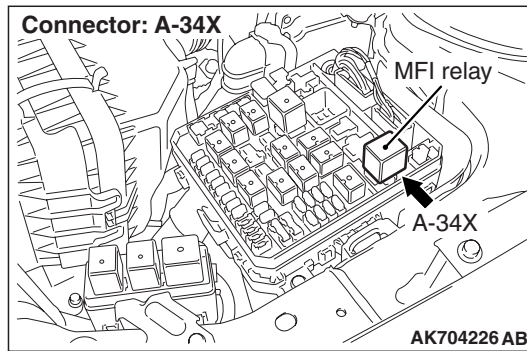
NO : The inspection is complete.

DTC P1025: Exhaust Engine Oil Control Valve Circuit

EXHAUST ENGINE OIL CONTROL VALVE CIRCUIT



AK704292 AB



CIRCUIT OPERATION

- The exhaust engine oil control valve power is supplied from the MFI relay (terminal No. 2).
- The ECM controls ground exhaust engine oil control valve by turning the power transistor in the ECM "ON" and "OFF".

TECHNICAL DESCRIPTION

- The exhaust engine oil control valve change the phase angle to operate the MIVEC system in the advance or retard side in accordance with the signals from the ECM.

DESCRIPTIONS OF MONITOR METHODS

Exhaust engine oil control valve circuit voltage is less than specified value.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

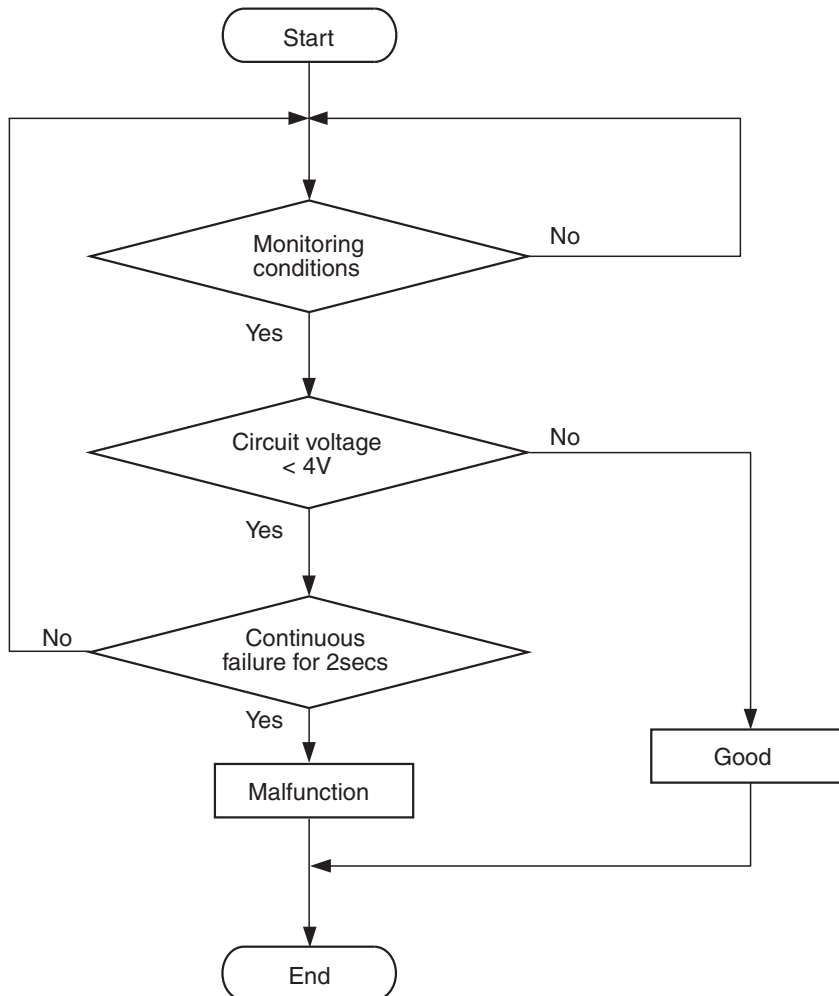
- Not applicable

Sensor (The sensor below is determined to be normal)

- Camshaft position sensor
- Crankshaft position sensor
- Engine coolant temperature sensor
- Throttle position sensor
- Engine oil control valve

DTC SET CONDITIONS

Logic Flow Chart



AK604350

Check Conditions

- Ignition switch is "ON" position.
- Exhaust engine oil control valve is "OFF".
- Battery positive voltage is between 10 and 16.5 volts.
- 0.1 second has elapsed after the above mentioned conditions have been met.

Judgement Criterion

- The ECM terminal voltage of exhaust engine oil control valve circuit is less than 4 volts for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- V.V.T.phase angle (exhaust) becomes most advanced angle.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

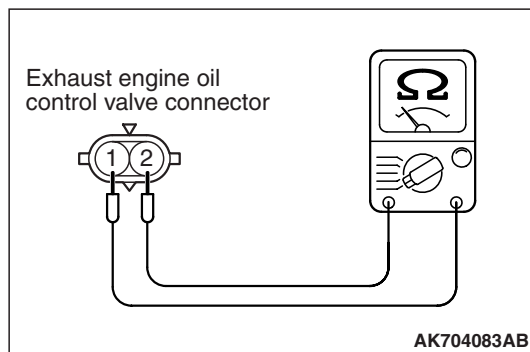
- Exhaust engine oil control valve failed.
- Open or shorted exhaust engine oil control valve circuit, or harness damage, or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

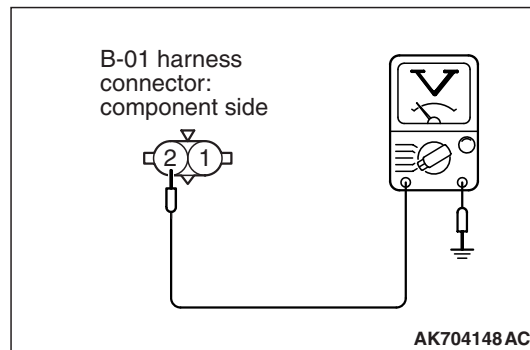
- MB992110:Power Plant ECU CCheck Harness

STEP 1. Check harness connector B-01 at the exhaust engine oil control valve for damage.**Q: Is the harness connector in good condition?****YES :** Go to Step 2.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 10.**STEP 2. Check the exhaust engine oil control valve.**

- (1) Disconnect the exhaust engine oil control valve connector B-01.
- (2) Measure the resistance between exhaust engine oil control valve side connector terminal No. 1 and No. 2.

Standard value: 6.9 –7.9 Ω [at 20° C (68° F)]**Q: Is the measured resistance between 6.9 and 7.9 Ω [at 20° C (68° F)]?****YES :** Go to Step 3.**NO :** Replace the exhaust engine oil control valve. Then go to Step 10.**STEP 3. Measure the power supply voltage at exhaust engine oil control valve harness side connector B-01.**

- (1) Disconnect the connector B-01 and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 2 and ground.
 - Voltage should be battery positive voltage.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?**YES :** Go to Step 5.**NO :** Go to Step 4.

STEP 4. Check harness connector A-34X at MFI relay for damage.

Q: Is the harness connector in good condition?

YES : Check harness connector A-39 at intermediate connector for damage, and repair or replace as required. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). If intermediate connector is in good condition, repair harness wire between MFI relay connector A-34X (terminal No. 2) and exhaust engine oil control valve connector B-01 (terminal No. 2) because of open circuit or short circuit to ground. Then go to Step 10.

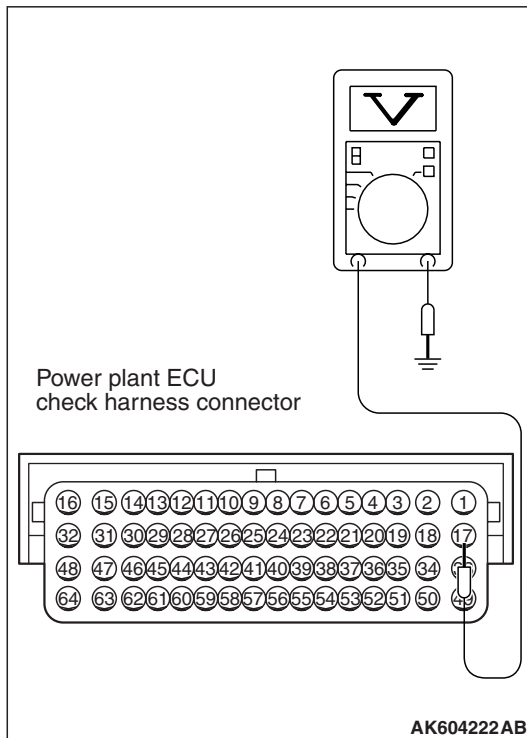
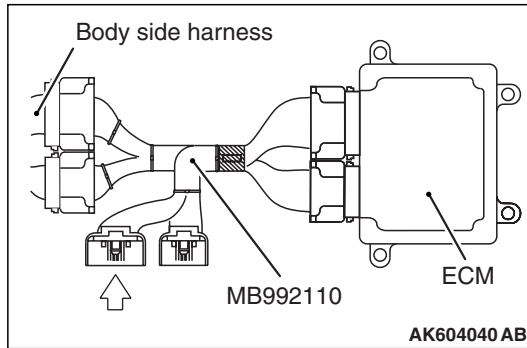
NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 10.

STEP 5. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 6.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 10.



STEP 6. Measure the power supply voltage at ECM connector B-09 by using power plant ECU check harness special tool MB992110.

(1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.

(2) Turn the ignition switch to the "ON" position.

(3) Measure the voltage between terminal No. 17 and ground.

- Voltage should be battery positive voltage.

(4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Go to Step 7.

NO : Repair harness wire between exhaust engine oil control valve connector B-01 (terminal No. 1) and ECM connector B-09 (terminal No. 17) because of open circuit or short circuit to ground. Then go to Step 10.

STEP 7. Check for harness damage between MFI relay connector A-34X (terminal No. 2) and exhaust engine oil control valve connector B-01 (terminal No. 2).

NOTE: Check harness after checking intermediate connector A-39. If intermediate connector is damaged, repair or replace it. Refer to GROUP 00E, Harness Connector Inspection P.00E-2. Then go to Step 10.

Q: Is the harness wire in good condition?

YES : Go to Step 8.

NO : Repair it. Then go to Step 10.

STEP 8. Check for harness damage between exhaust engine oil control valve connector B-01 (terminal No. 1) and ECM connector B-09 (terminal No. 17).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 10.

STEP 9. Check the trouble symptoms.

(1) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).

(2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1025 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 10.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 10. Test the OBD-II drive cycle.

(1) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).

(2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1025 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1231: Active Stability Control Plausibility

TECHNICAL DESCRIPTION

- Checks for an abnormal signal of active stability control (ASC) via the CAN communication.

DTC SET CONDITIONS

Check Condition

- Ignition switch is "ON" position.

Judgment Criterion

- A torque demand signal from the active stability control is not normal.

FAIL-SAFE AND BACKUP FUNCTION

- Torque requested by active stability control (ASC) is ignored.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Active stability control system failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

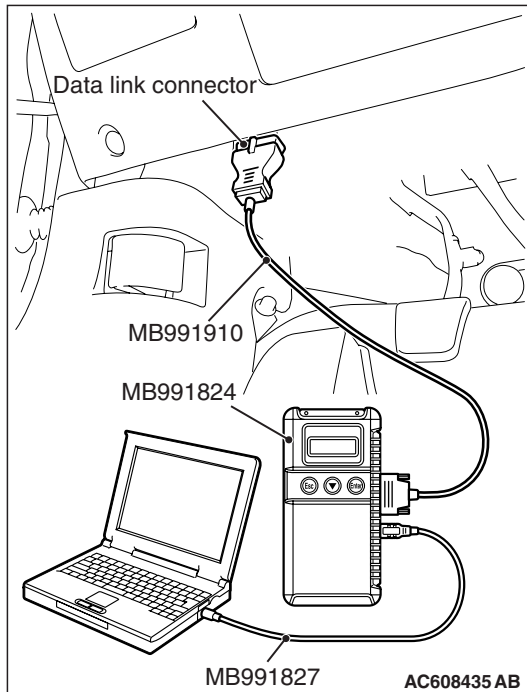
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the DTC P1231 set?

YES : Go to Step 2.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the ASC-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the ASC-DTC set?

YES : Refer to GROUP 35C, Active Stability Control System –Diagnostic Trouble Code Chart [P.35C-20](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

DTC P1232: Fail Safe System

TECHNICAL DESCRIPTION

- To judge whether fail-safe control can be performed, check that power supply to the throttle actuator control motor circuit can be stopped by turning the throttle actuator control motor relay to OFF position momentarily.

DTC SET CONDITIONS

Check Condition

- Ignition switch is "ON" position.

Judgment Criterion

- Power supply to the throttle actuator control motor cannot be shut down (though power supply is stopped).

FAIL-SAFE AND BACKUP FUNCTION

- Microcomputer is reset.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle actuator control motor relay circuit failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

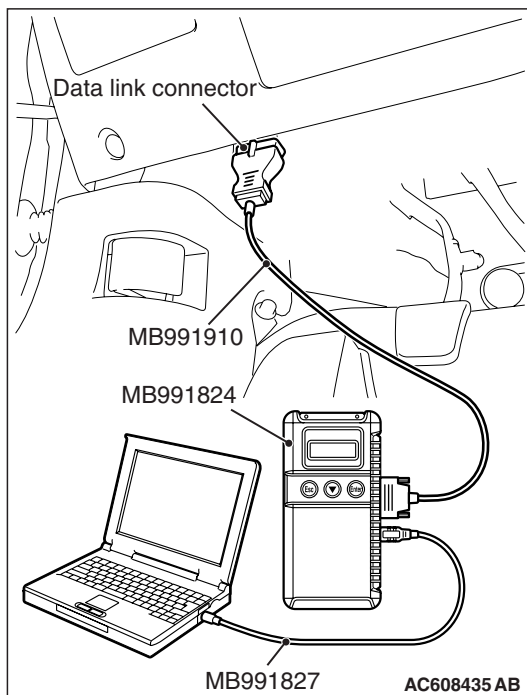
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the DTC P1232 set?

YES : Go to Step 2.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0657 set?

YES : Refer to DTC P0657 –Throttle actuator control motor relay circuit malfunction [P.13A-581](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

DTC P1233: Throttle Position Sensor (main) Plausibility

TECHNICAL DESCRIPTION

Compare the actual measurement of volumetric efficiency by a mass airflow sensor signal with volumetric efficiency estimated from a throttle position sensor (main) signal.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

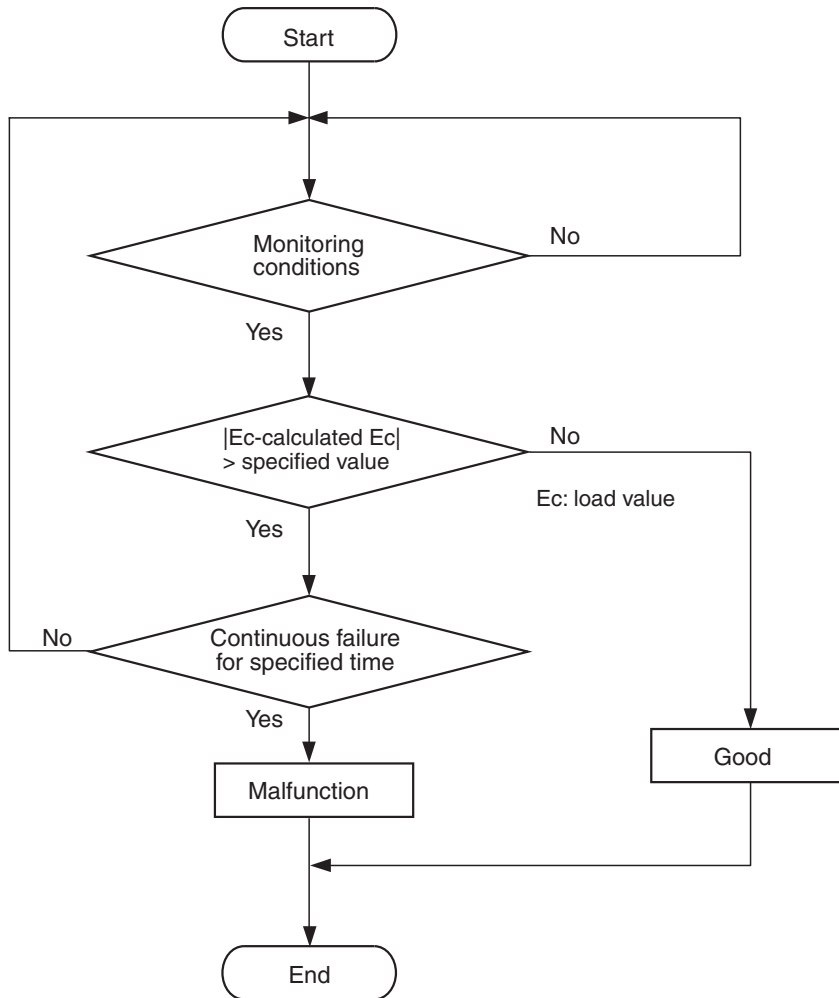
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604351

Check Conditions

- The difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (main) is 0 percent or more.
- The engine speed is between 750 and 3,000 r/min. Or, the throttle position sensor (main) output voltage is 3 volts or lower.

Judgment Criterion

- For 0.8 second, the difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (main) is 43 percent or more.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if throttle position sensor (sub) fails.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 16 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle position sensor (main) system failed.
- Intake system vacuum leak.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A
- MB991658: Test Harness

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

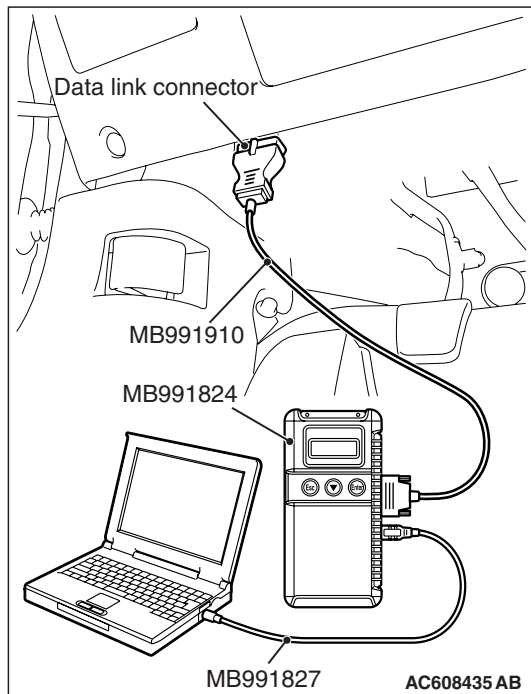
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1233 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list item 13: Throttle Position Sensor (main).

- (1) Turn the ignition switch to the "ON" position.
- (2) Detach the intake air hose at the throttle body.
- (3) Disconnect the connector of the throttle position sensor.
- (4) Use test harness special tool (MB991658) to connect only terminals No. 3, No. 4, No. 5, and No. 6.
- (5) Set scan tool MB991958 to the data reading mode for item 13, Throttle Position Sensor (main).
 - Output voltage should be between 0.3 and 0.7 volt when the throttle valve is fully closed with your finger.
 - Output voltage should be 4.0 volts or more when the throttle valve is fully open with your finger.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 3.

STEP 3. Check for intake system vacuum leak.

Q: Are there any abnormalities?

YES : Repair it. Then go to Step 5.

NO : Go to Step 4.

STEP 4. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1233 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 5.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 5. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1233 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1234: Throttle Position Sensor (Sub) Plausibility**TECHNICAL DESCRIPTION**

Compare the actual measurement of volumetric efficiency by a mass airflow sensor signal with volumetric efficiency estimated from a throttle position sensor (sub) signal.

MONITOR EXECUTION

Continuous

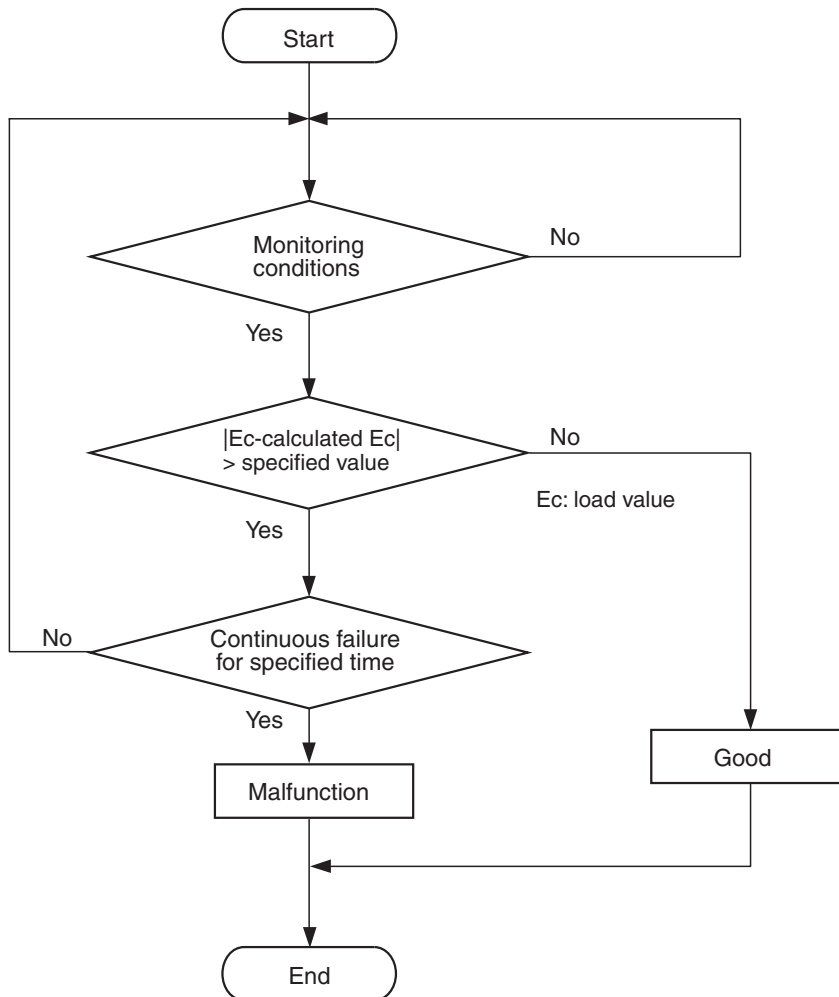
**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Logic Flow Chart**

AK604351

Check Conditions

- The difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (sub) is 0 percent or more.
- The engine speed is between 750 and 3,000 r/min. Or, the throttle position sensor (main) output voltage is 3 volts or lower.

Judgment Criterion

- For 0.8 second, the difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (sub) is 43 percent or more.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if throttle position sensor (main) fails.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 16 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle position sensor (sub) system failed.
- Intake system vacuum leak.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A
- MB991658: Test Harness

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

CAUTION

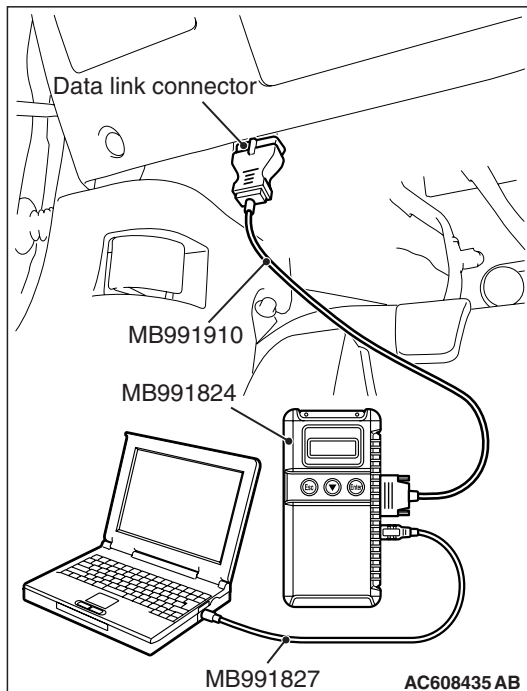
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1234 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list item 15: Throttle Position Sensor (sub).

- (1) Turn the ignition switch to the "ON" position.
- (2) Detach the intake air hose at the throttle body.
- (3) Disconnect the connector of the throttle position sensor.
- (4) Use test harness special tool (MB991658) to connect only terminals No. 3, No. 4, No. 5, and No. 6.
- (5) Set scan tool MB991958 to the data reading mode for item 15, Throttle Position Sensor (sub).
 - Output voltage should be 4.0 volts or more when the throttle valve is fully closed with your finger.
 - Output voltage should be 1.0 volt or less when the throttle valve is fully open with your finger.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 3.

STEP 3. Check for intake system vacuum leak.**Q: Are there any abnormalities?**

YES : Repair it. Then go to Step 5.

NO : Go to Step 4.

STEP 4. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1234 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 5.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 5. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1234 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1235: Mass Airflow Sensor Plausibility

TECHNICAL DESCRIPTION

Compare the actual measurement of volumetric efficiency by a mass airflow sensor signal with volumetric efficiency estimated from a throttle position sensor (main or sub) signal.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

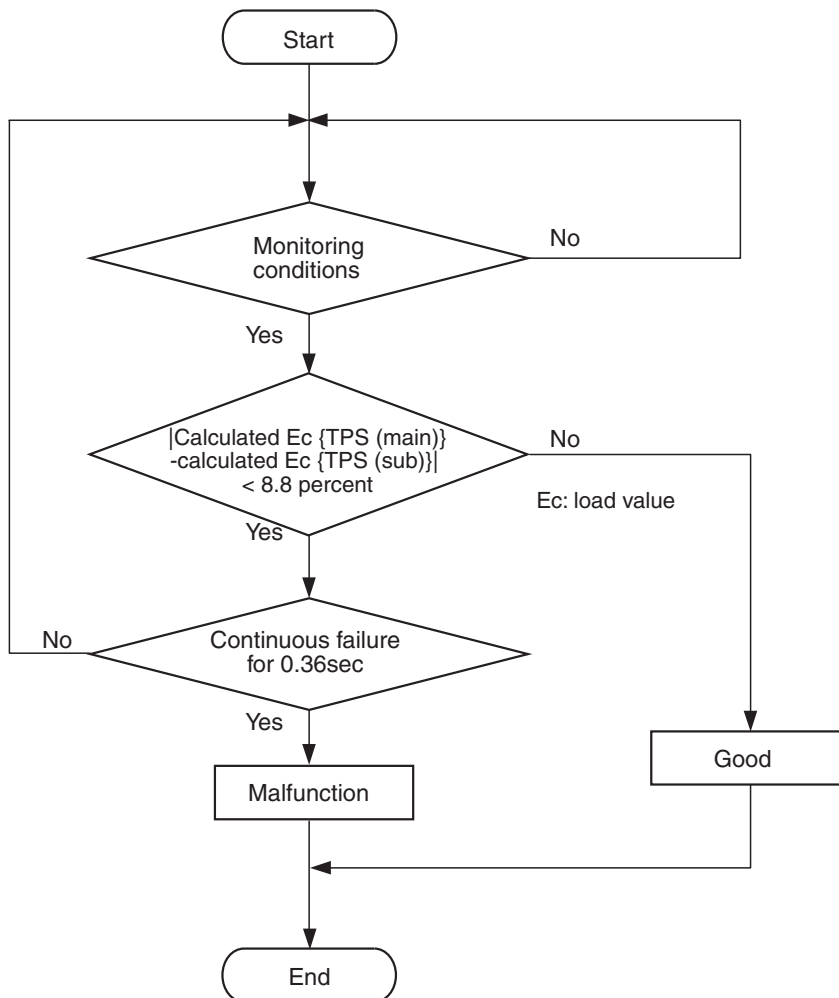
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604352

Check Conditions

- The plausibility error of the throttle position sensor (main) is detected.
- The plausibility error of the throttle position sensor (sub) is detected.

Judgment Criterion

- For 0.76 second, the difference between the volumetric efficiency estimated by the throttle position sensor (main) and the volumetric efficiency estimated by the throttle position sensor (sub) is 8.8 percent or less.

FAIL-SAFE AND BACKUP FUNCTION

- Fuel control is carried out using throttle opening degree and engine speed in accordance with pre-set map.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 16 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Mass airflow sensor system failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

CAUTION

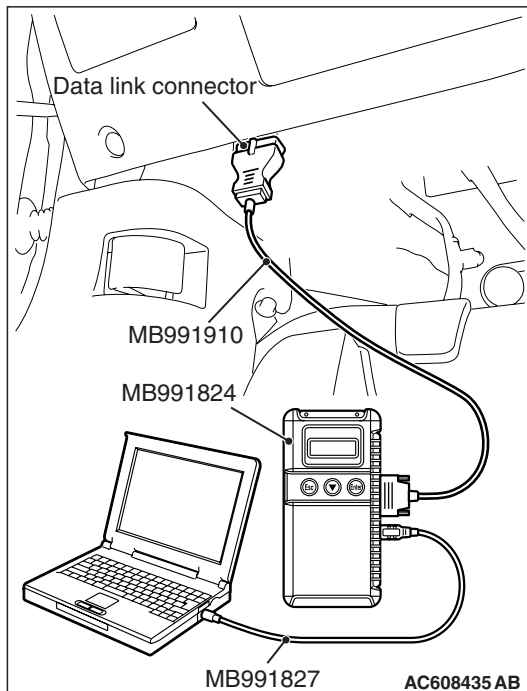
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- Connect scan tool MB991958 to the data link connector.
- Turn the ignition switch to the "ON" position.
- Read the DTC.
- Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1235 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list item 10: Mass Airflow Sensor.

- (1) Start the engine and run at idle.
- (2) Set scan tool MB991958 to the data reading mode for item 10, Mass Airflow Sensor.
- (3) Warm up the engine to normal operating temperature: 80° C to 95° C (176° F to 203° F).
 - The standard value during idling should be between 1,350 and 1,670 millivolts.
 - When the engine is revved, the mass airflow rate should increase according to the increase in engine speed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : Go to Step 3.

NO : Refer to, DTC P0101 –Mass Airflow Circuit Range/Performance Problem [P.13A-107](#), DTC P0102 –Mass Airflow Circuit Low Input [P.13A-113](#), DTC P0103 –Mass Airflow Circuit High Input [P.13A-120](#).

STEP 3. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1235 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 4. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1235 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1236: A/D Converter

TECHNICAL DESCRIPTION

Monitors whether input voltage from the accelerator pedal position sensor (sub) is normally converted into a digital signal in the ECM.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS**(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Check Condition**

- Ignition switch is "ON" position.

Judgment Criterion

- When the input voltage from the accelerator pedal position sensor (sub) is periodically 0 V for 0.45 second, the digital value of the input voltage indicates 0.2 V or more.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

- None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

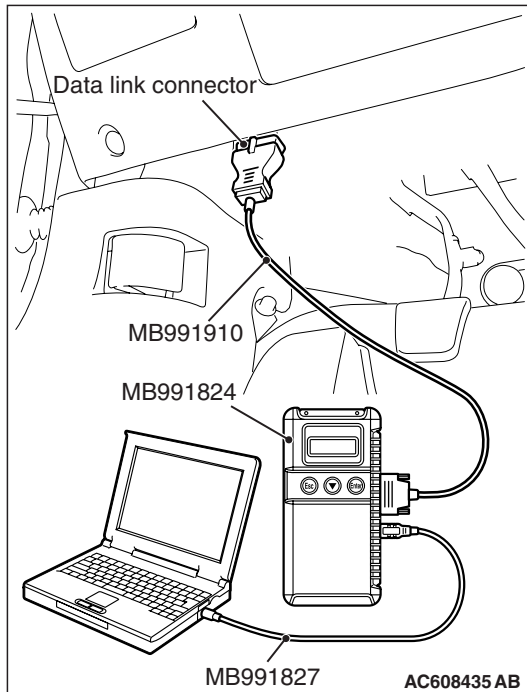
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1236 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P1237: Accelerator Pedal Position Sensor Plausibility

TECHNICAL DESCRIPTION

Compare the output value of the accelerator pedal position sensor (main) with that of the accelerator pedal position sensor (sub).

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

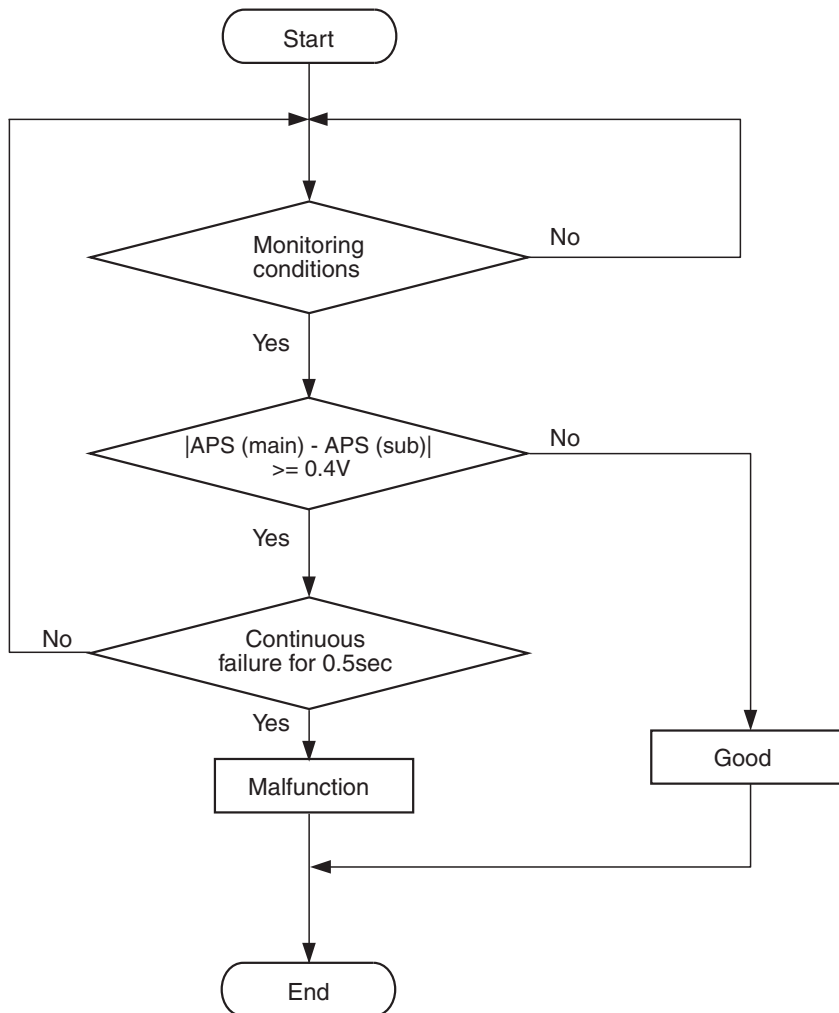
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604353

Check Condition

- Change of accelerator pedal position sensor (sub) output voltage per 40 milliseconds is lower than 0.06 volt.
- Range/performance error of accelerator pedal position sensor (main and sub) circuit is not detected.

Judgment Criterion

- Voltage obtained with the formula given below is 0.4 volt or higher for 0.5 second: accelerator pedal position sensor (main) output voltage – accelerator pedal position sensor (sub) output voltage.

NOTE: The accelerator pedal position sensor voltage used for the judgement is converted into the accelerator pedal position sensor voltage for the internal processing by the ECM.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

- None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Accelerator pedal position sensor failed.

- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

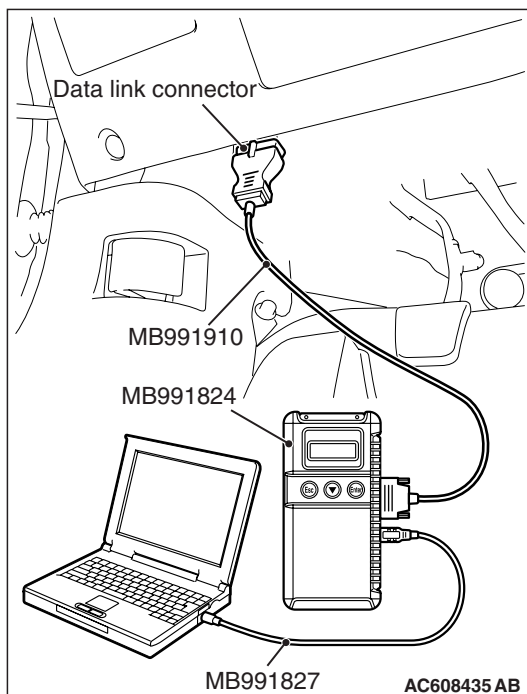
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1237 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list.

- (1) Turn the ignition switch to the "ON" position.
- (2) Check the following item in the data list. Refer to Data List Reference Table [P.13A-852](#).
 - a. Item 11: Accelerator pedal position sensor (main)
 - b. Item 12: Accelerator pedal position sensor (sub)
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : Go to Step 3.

NO : Repair or replace it. Then go to Step 4.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1237 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.
- NO :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 4. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1237 set?

- YES :** Retry the troubleshooting.
- NO :** The inspection is complete.

DTC P1238: Mass Airflow Sensor Plausibility (Torque Monitor)

TECHNICAL DESCRIPTION

Compare the actual measurement of volumetric efficiency by a mass airflow sensor signal with volumetric efficiency estimated from a throttle position sensor (main or sub) signal.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

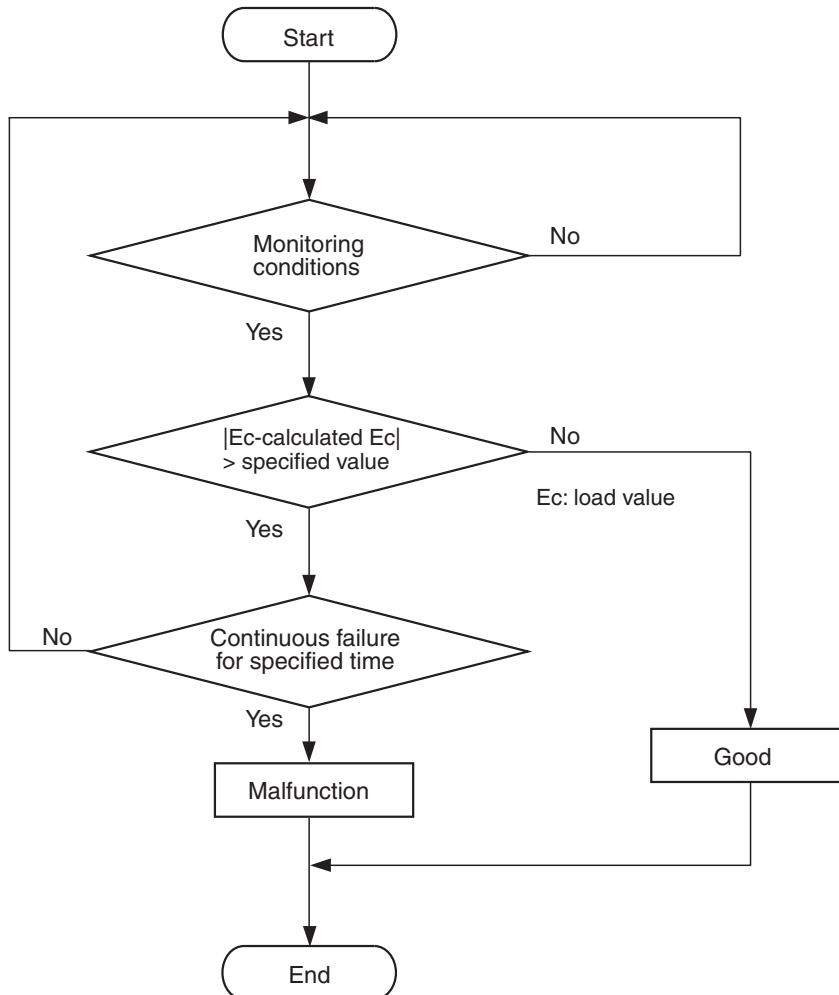
Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS



AK604351

Check Conditions

- The difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (main) is 0 percent or more.
- Engine speed is 750 r/min or higher.

Judgment Criterion

- For 0.9 second, the difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (main) is 45 percent or more.

Check Conditions

- The difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (sub) is 0 percent or more.
- Engine speed is 750 r/min or higher.

Judgment Criterion

- For 0.9 second, the difference between the actual volumetric efficiency and the volumetric efficiency estimated by the throttle position sensor (sub) is 45 percent or more.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle –
Pattern 16 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Mass airflow sensor system failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

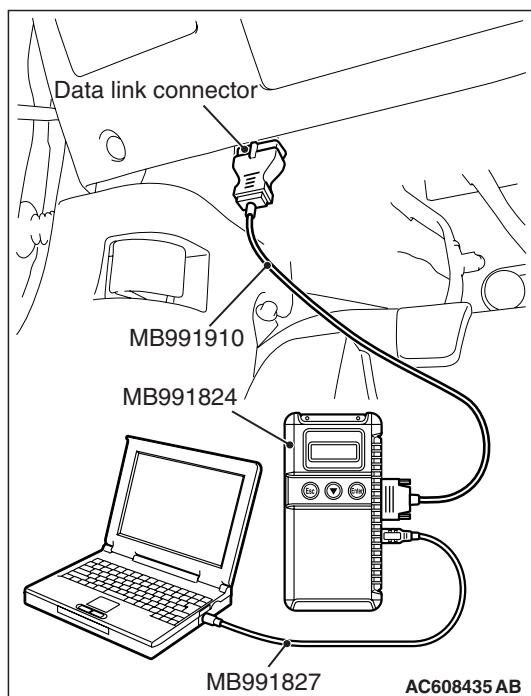
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1238 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list item 10: Mass Airflow Sensor.

- (1) Start the engine and run at idle.
- (2) Set scan tool MB991958 to the data reading mode for item 10, Mass Airflow Sensor.
- (3) Warm up the engine to normal operating temperature: 80° C to 95° C (176° F to 203° F).
 - The standard value during idling should be between 1,350 and 1,670 millivolts.
 - When the engine is revved, the mass airflow rate should increase according to the increase in engine speed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : Go to Step 3.

NO : Refer to, DTC P0101 –Mass Airflow Circuit Range/Performance Problem [P.13A-107](#), DTC P0102 –Mass Airflow Circuit Low Input [P.13A-113](#), DTC P0103 –Mass Airflow Circuit High Input [P.13A-120](#).

STEP 3. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1238 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 4. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1238 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1239: Engine RPM Plausibility**TECHNICAL DESCRIPTION**

Compares the actual engine speed calculated from the crankshaft position sensor signal cycle with the estimated engine speed calculated from the number of the crankshaft position sensor signal pulses after a lapse of the specified time.

MONITOR EXECUTION

Continuous

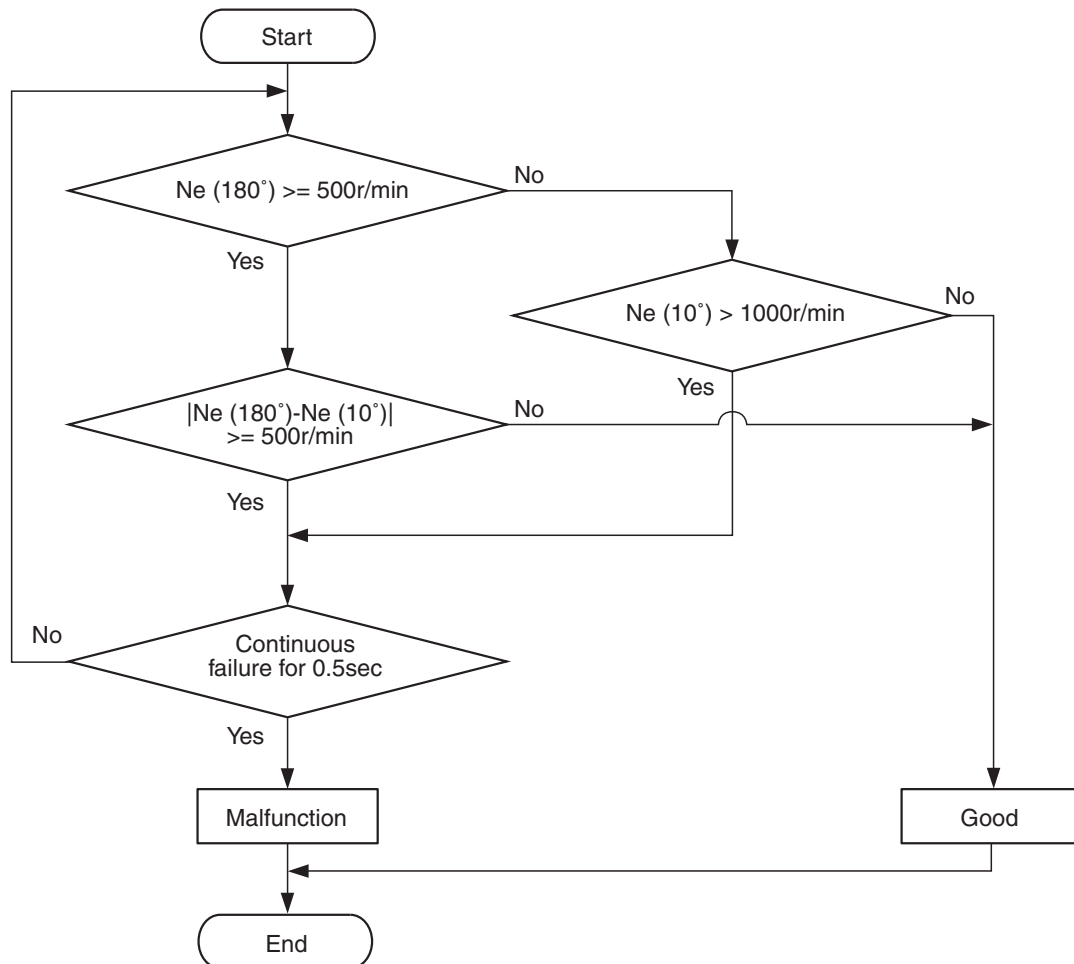
**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Logic Flow Chart**

AK604355

Check Condition

- The engine speed monitored with a 180 degree-cycle pulse is 500 r/min or more.

Judgment Criterion

- The difference between the engine speed monitored with a 180 degree-cycle pulse and the engine speed monitored with a 10 degree-cycle pulse is 500 r/min or more.

Check Condition

- The engine speed monitored with a 180 degree-cycle pulse is 500 r/min or less.

Judgment Criterion

- The engine speed monitored with a 10 degree-cycle pulse is 1,000 r/min or more.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Crankshaft position sensor system failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

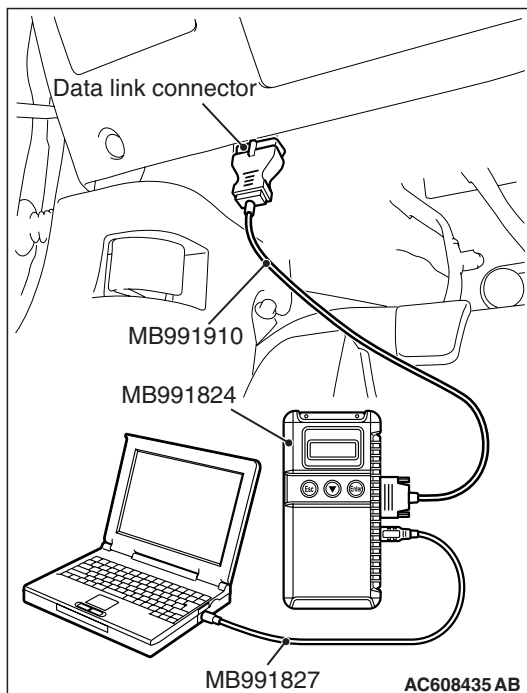
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1239 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1239 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.
- NO :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1239 set?

- YES :** Retry the troubleshooting.
- NO :** The inspection is complete.

DTC P1240: Ignition Angle

TECHNICAL DESCRIPTION

Checks for an abnormal ignition timing retard angle demand signal from the active stability control (ASC).

DTC SET CONDITIONS**Check Condition**

- Ignition switch is "ON" position.

Judgment Criterion

- The ignition timing retard angle demand signal is not normal.

FAIL-SAFE AND BACKUP FUNCTION

- Ignition retard is not carried out.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

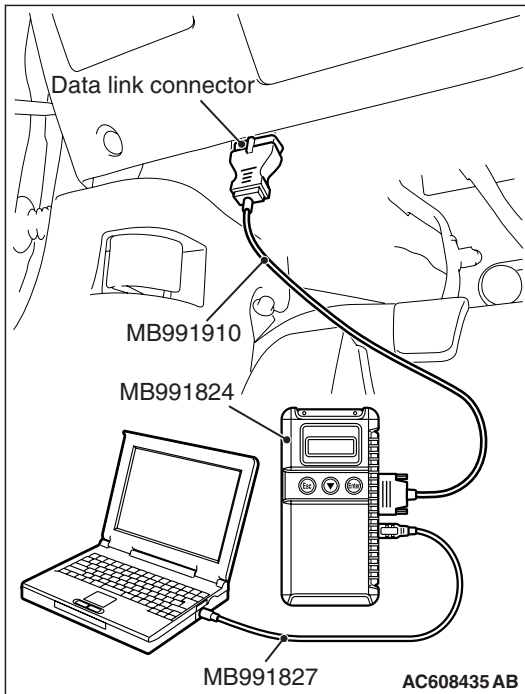
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

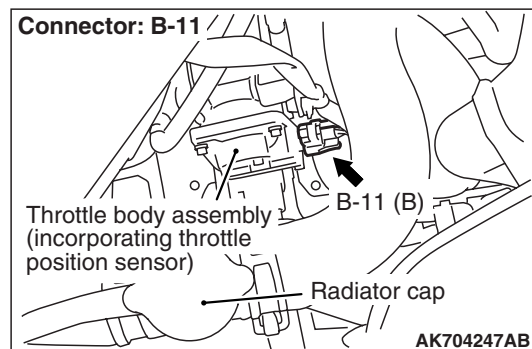
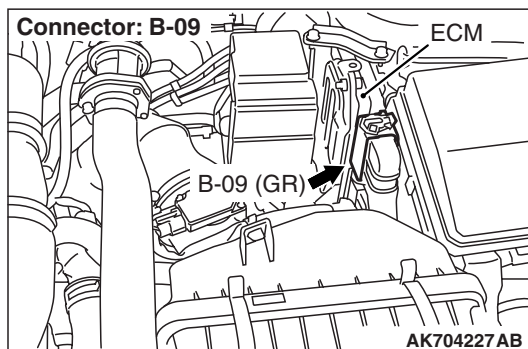
Q: Is the ASC-DTC set?

YES : Refer to GROUP 35C, Active Stability Control System –Diagnostic Trouble Code Chart [P.35C-20](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).



DTC P1241: Torque Monitor



TECHNICAL DESCRIPTION

Compares the actual torque signal computed from a mass air flow sensor signal with the driver demand torque signal computed from an accelerator pedal position sensor signal.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

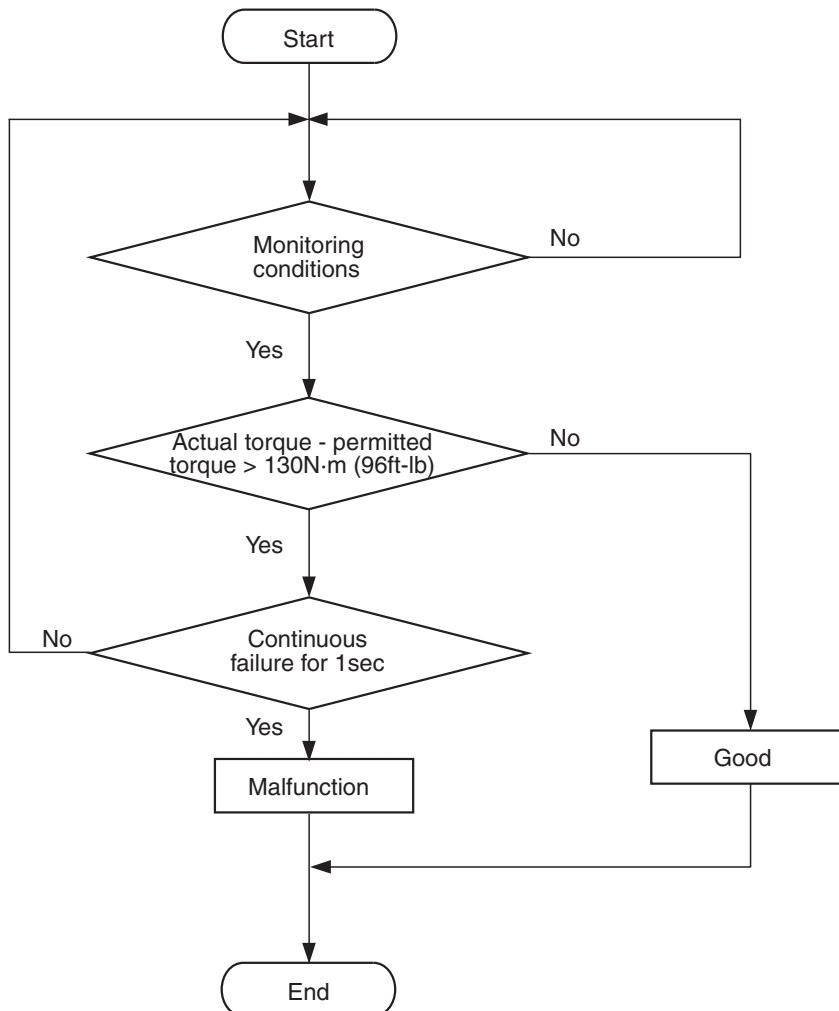
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK704414

Check Conditions

- Engine speed is 500 r/min or higher.
- Volumetric efficiency is 16 percent or higher.

Judgement Criterion

- The actual torque signal has continued to be more than the requested torque signal by 130 N·m (96 ft-lb) or more for 1 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 16 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle actuator control motor failed.
- Connector damage.
- Harness damage.
- Vacuum leak.
- ECM failed

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

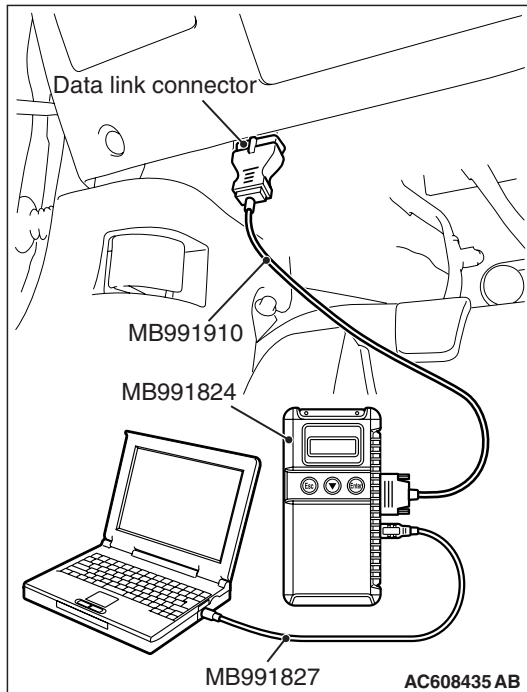
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1241 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Using scan tool MB991958, check data list.

- (1) Turn the ignition switch to the "ON" position.
- (2) Check the following items in the data list. Refer to Data List Reference Table [P.13A-852](#).
 - a. Item 10: Mass Airflow Sensor.
 - b. Item 11: Accelerator Pedal Position Sensor (main).
 - c. Item 12: Accelerator Pedal Position Sensor (sub).
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Are they operating properly?

YES : Go to Step 3.

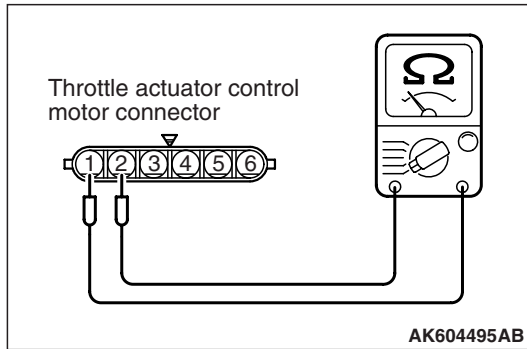
NO : Repair or replace it. Then go to Step 9.

STEP 3. Check harness connector B-11 at throttle actuator control motor and harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 4.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.

**STEP 4. Check the throttle actuator control motor.**

- (1) Disconnect the connector B-11.
- (2) Measure the resistance between throttle actuator control motor side connector terminal No. 1 and No. 2.

Standard value: 0.3 – 80 Ω [at 20° C (68° C)]

Q: Is the measured resistance between 0.3 and 80 Ω [at 20° C (68° C)]?

YES : Go to Step 5.

NO : Replace the throttle body assembly. Then go Step 9.

STEP 5. Check for harness damage between throttle actuator control motor connector B-11 (terminal No. 1) and PCM connector B-09 (terminal No. 15).

Q: Is the harness wire in good condition?

YES : Go to Step 6.

NO : Repair it. Then go to Step 9.

STEP 6. Check for harness damage between throttle actuator control motor connector B-11 (terminal No. 2) and PCM connector B-09 (terminal No. 16).

Q: Is the harness wire in good condition?

YES : Go to Step 7.

NO : Repair it. Then go to Step 9.

STEP 7. Check for intake system vacuum leak.

Q: Are there any abnormalities?

YES : Repair it. Then go to Step 9.

NO : Go to Step 8.

STEP 8. Check the trouble symptoms.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).

- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1241 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 9.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 9. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 16 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1241 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1242: Fail Safe Control Monitor

TECHNICAL DESCRIPTION

Monitors the engine speed during fail-safe control.

DTC SET CONDITIONS

Check Condition

- During fail safe control.

Judgment Criterion

- The engine speed is higher than assumed.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

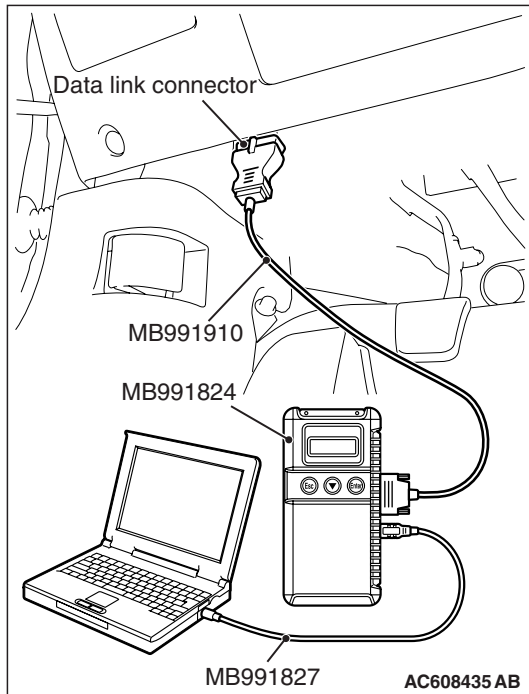
TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A



STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1242 set?

YES : Refer to, Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2 .

STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1242 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

DTC P1243: Inquiry/Response Error

TECHNICAL DESCRIPTION

Monitors the computation function of the ECM.

DTC SET CONDITIONS

Check Condition

- Ignition switch is "ON" position.

Judgment Criterion

- ECM can not calculate input data.

FAIL-SAFE AND BACKUP FUNCTION

- Microcomputer is reset.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

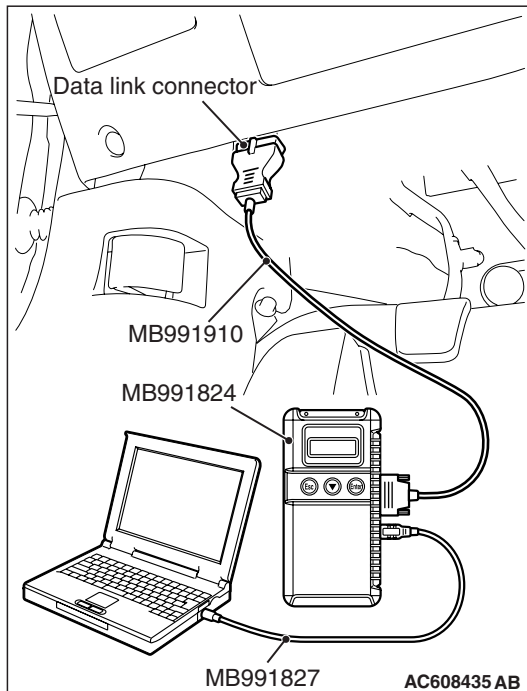
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1243 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P1244: RAM Test For All Area**TECHNICAL DESCRIPTION**

Monitors the computation function of the ECM periodically.

DTC SET CONDITIONS**Check Condition**

- Ignition switch is "ON" position.

Judgment Criterion

- All RAM data of ECM are defect.

FAIL-SAFE AND BACKUP FUNCTION

- Microcomputer is reset.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

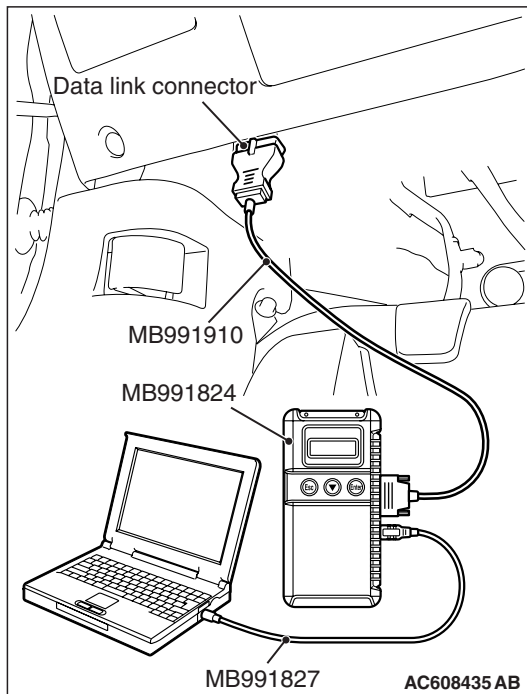
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- Connect scan tool MB991958 to the data link connector.
- Turn the ignition switch to the "ON" position.
- Read the DTC.
- Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1244 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P1245: Cycle RAM Test (Engine)

TECHNICAL DESCRIPTION

Monitors the computation function of the ECM periodically.

DTC SET CONDITIONS

Check Condition

- Ignition switch is "ON" position.

Judgment Criterion

- RAM data (engine) is defect.

FAIL-SAFE AND BACKUP FUNCTION

- Microcomputer is reset.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

CAUTION

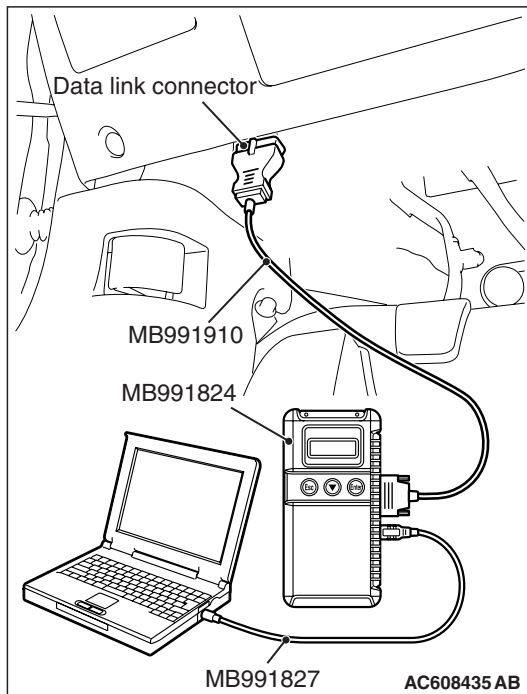
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- Connect scan tool MB991958 to the data link connector.
- Turn the ignition switch to the "ON" position.
- Read the DTC.
- Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1245 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P1247: TC-SST Plausibility <TC-SST>**TECHNICAL DESCRIPTION**

Checks for an abnormal signal of TCM.

DTC SET CONDITIONS**Check Condition**

- Ignition switch is "ON" position.

Judgment Criterion

- A torque demand signal from TCM is not normal.

FAIL-SAFE AND BACKUP FUNCTION

- Torque requested by TC-SST-ECU is ignored.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- TC-SST system failed.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

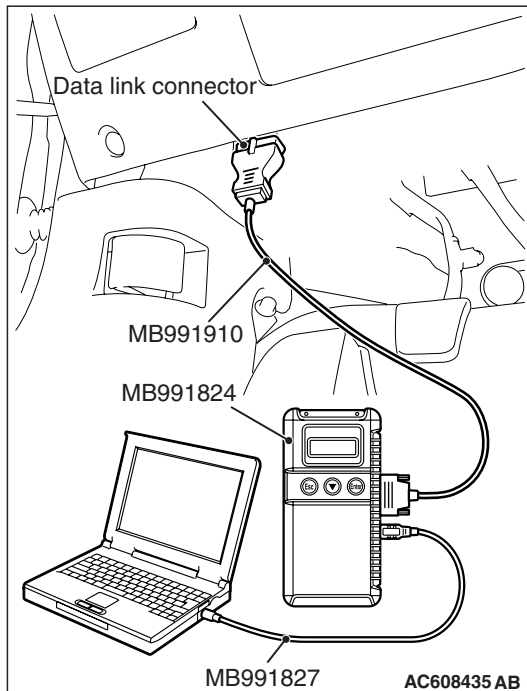
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the TC-SST-DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the TC-SST-DTC set?

YES : Refer to GROUP 22C, Diagnosis <TC-SST> – Diagnostic Trouble Code Chart [P.22C-10](#).

NO : Go to Step 2 .



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1247 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).
- NO :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

DTC P1320: Ignition Timing Retard Insufficient

TECHNICAL DESCRIPTION

- After starting the engine in the cold state, the ECM retards the ignition timing to early activate the catalyst by letting out the high temperature exhaust gases.
- The ECM monitors whether the retard ignition timing control is normal or not.

DESCRIPTIONS OF MONITOR METHODS

After starting the engine in the cold state, the retard ignition timing control is abnormal, which is caused by a decrease in idle speed and so on.

MONITOR EXECUTION

Once per driving cycle

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

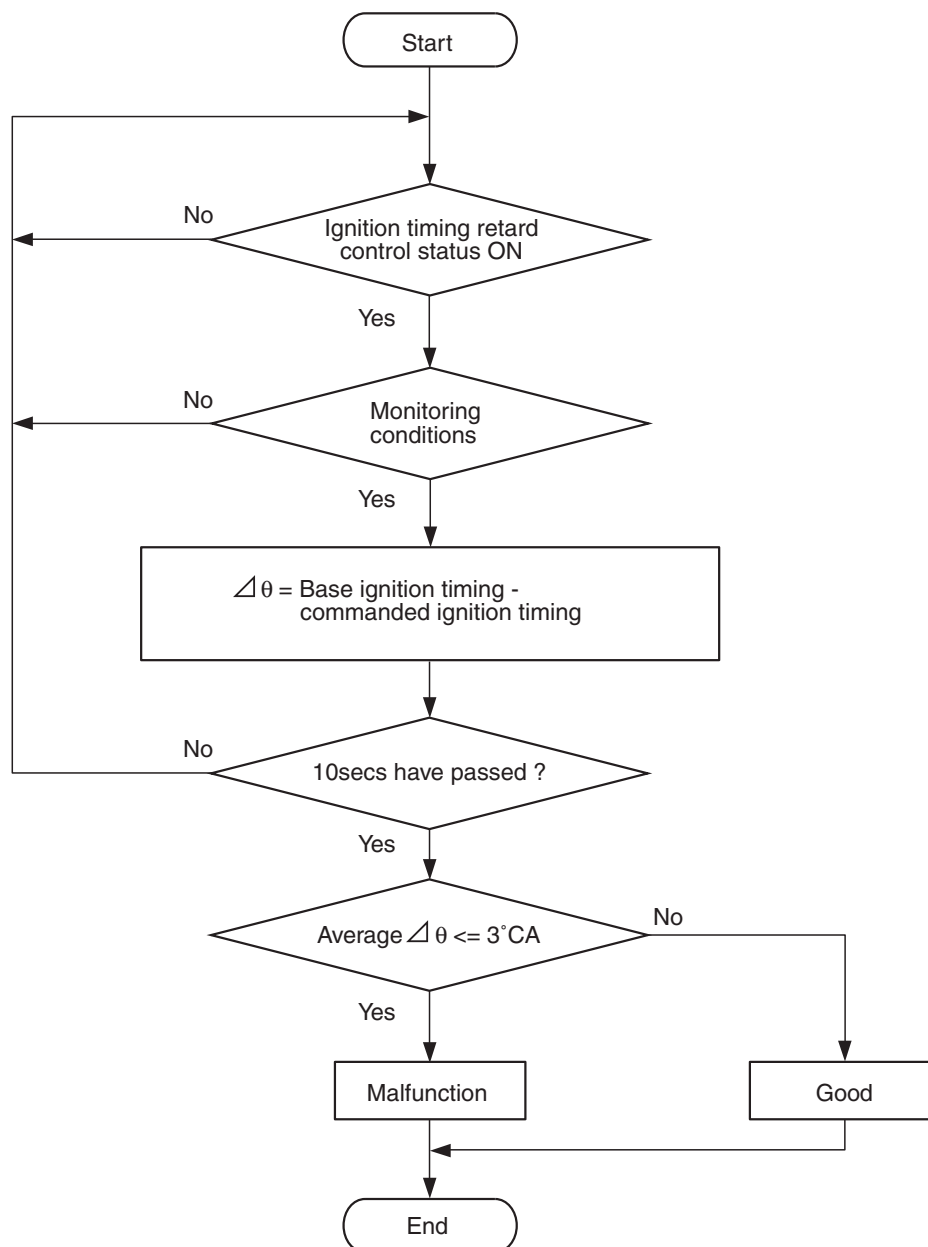
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK704294

Check Conditions

- Under the ignition timing retard control.
- Engine coolant temperature is between 7° C (45° F) and 36° C (97° F).

Judgement Criteria

- For 10 seconds, the difference between the basic ignition timing and the target/specified ignition timing is 3° CA or less on average during the retard control.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 19 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- The thermostat is faulty.

- ECM failed.

DIAGNOSIS

Required Special Tools

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check the throttle body. (throttle valve area)

Q: Is the throttle valve area dirty?

- YES :** Perform cleaning. Refer to Throttle body (throttle valve area) cleaning [P.13A-897](#). Then go to Step 3.
- NO :** Repair it. Then go to Step 2.

STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

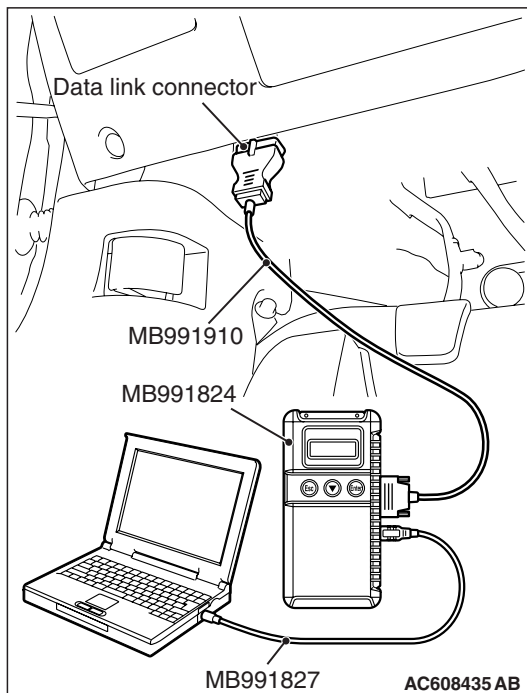
⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1320 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.
- NO :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 3. Test the OBD-II drive cycle.

- (1) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 19 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1320 set?

- YES :** Retry the troubleshooting.
- NO :** The inspection is complete.

DTC P1506: Idle Control System RPM Lower Than Expected at Low Temperature

TECHNICAL DESCRIPTION

- The amount of air taken in during idling is regulated by the opening and closing of the throttle valve.
- The ECM checks the difference between the actual engine speed and the target engine speed.

DESCRIPTIONS OF MONITOR METHODS

Difference between actual and target idle speed is over the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

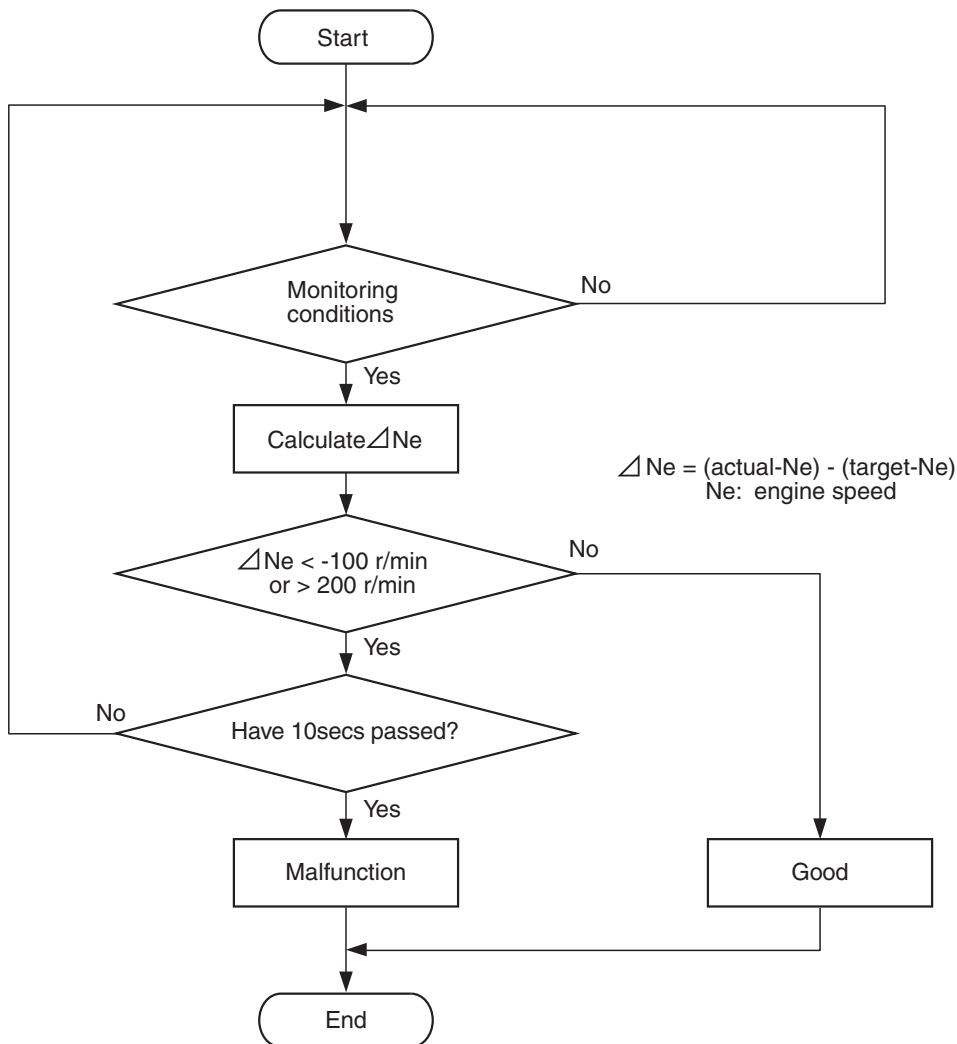
- Misfire monitor
- Fuel system monitor
- Vehicle speed signal monitor

Sensor (The sensor below is determined to be normal)

- Mass airflow sensor
- Engine coolant temperature sensor
- Intake air temperature sensor
- Barometric pressure sensor
- Throttle position sensor

DTC SET CONDITIONS

Logic Flow Chart



AK704413

Check Conditions

- Under the closed loop idle speed control.
- The engine coolant temperature is between 7° C (45° F) and 41° C (106° F).
- Battery positive voltage is higher than 10 volts.
- Power steering pressure switch: OFF.
- Volumetric efficiency is lower than 70 percent.
- Barometric pressure is higher than 76 kPa (22.4 in.Hg).
- Intake air temperature is higher than -10° C (14° F).
- More than 3 seconds have elapsed from the previous monitoring.
- Target air flow rate is 24 L/sec or more.

Judgment Criterion

- The actual idle speed is more than 100 r/min lower than the target idle speed for 10 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 19 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle valve area is dirty.
- Throttle body assembly failed.
- ECM failed.

DIAGNOSIS**Required Special Tools**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

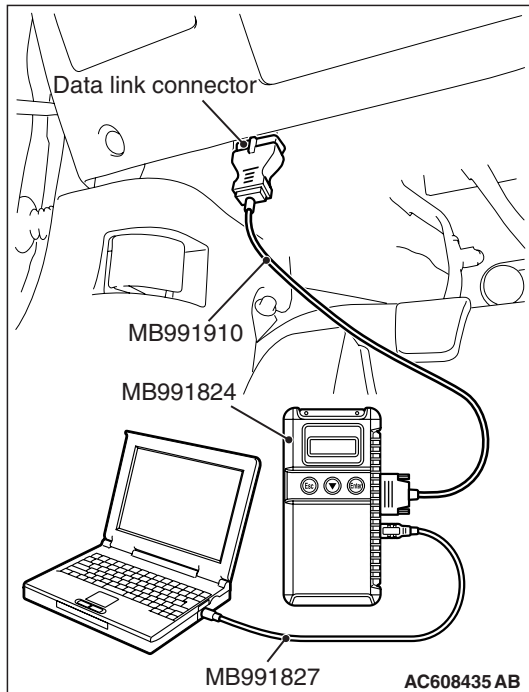
⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1506 set?

- YES** : Refer to Diagnostic Trouble Code Chart [P.13A-44](#).
NO : Go to Step 2.



STEP 2. Check the throttle body. (throttle valve area)

Q: Is the throttle valve area dirty?

- YES** : Perform cleaning. Refer to Throttle body (throttle valve area) cleaning [P.13A-897](#). Then go to Step 4.
NO : Go to Step 3.

STEP 3. Replace the throttle body assembly.

- (1) Replace the throttle body assembly.
- (2) Carry out test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 19 [P.13A-9](#).
- (3) Check the diagnostic trouble code (DTC).

Q: Is DTC P1506 set?

- YES** : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.
NO : The inspection is complete.

STEP 4. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 19 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1506 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1507: Idle Control System RPM Higher Than Expected at Low Temperature

TECHNICAL DESCRIPTION

- The amount of air taken in during idling is regulated by the opening and closing of the throttle valve.
- The ECM checks the difference between the actual engine speed and the target engine speed.

DESCRIPTIONS OF MONITOR METHODS

Difference between actual and target idle speed is over the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

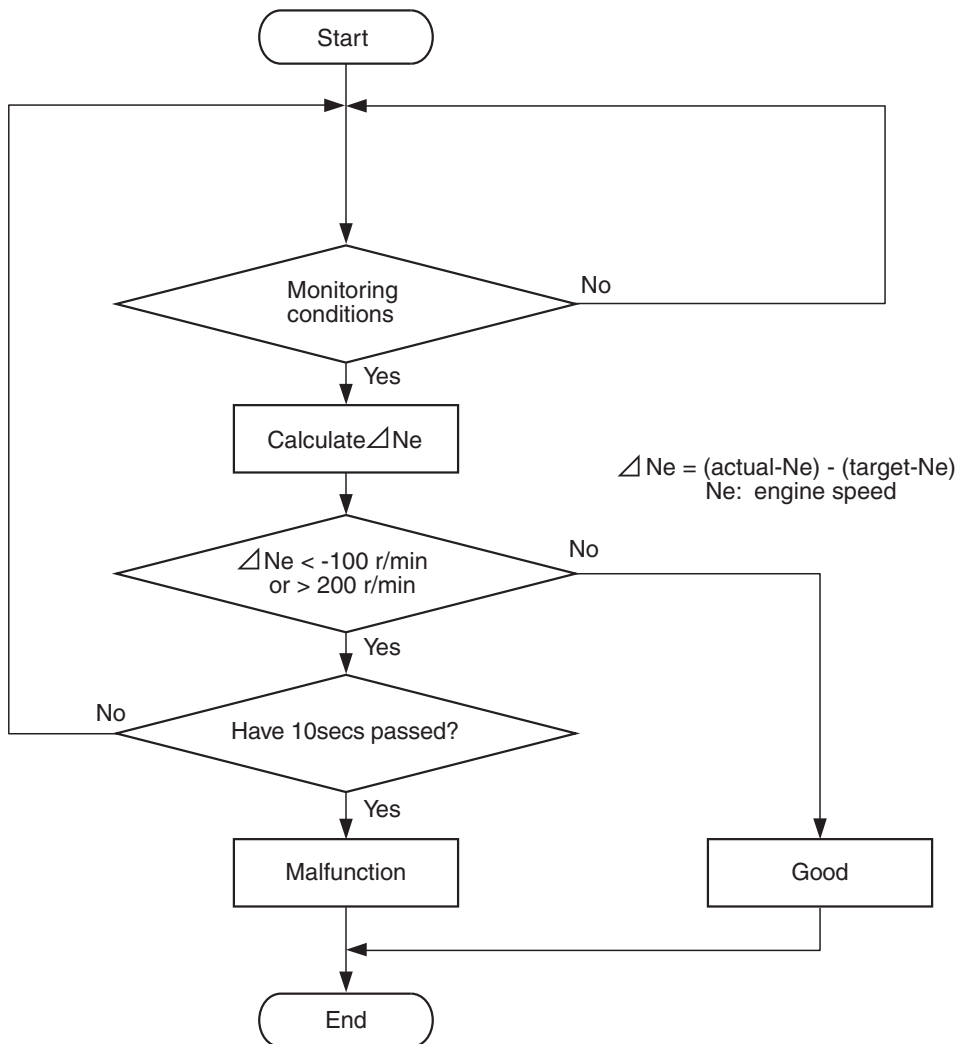
- Misfire monitor
- Fuel system monitor
- Vehicle speed signal monitor

Sensor (The sensor below is determined to be normal)

- Mass airflow sensor
- Engine coolant temperature sensor
- Intake air temperature sensor
- Barometric pressure sensor
- Throttle position sensor

DTC SET CONDITIONS

Logic Flow Chart



AK704413

Check Conditions

- Under the closed loop idle speed control.
- Engine coolant temperature is between 7° C (45° F) and 41° C (106° F).
- Battery positive voltage is higher than 10 volts.
- Barometric pressure is higher than 76 kPa (22.4 in.Hg).
- Intake air temperature is higher than -10° C (14° F).
- More than 3 seconds have elapsed from the previous monitoring.
- Target air flow rate is 0 L/sec.

Judgement Criterion

- Actual idle speed has continued to be higher than the target idle speed by 200 r/min (300 r/min*) or more for 10 seconds.

*: Specs in parentheses are applicable if the maximum air temperature during the previous operation was more than 45° C (113° F).

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 19 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Intake system vacuum leak.
- ECM failed.
- Throttle body assembly failed.

DIAGNOSIS

Required Special Tools

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

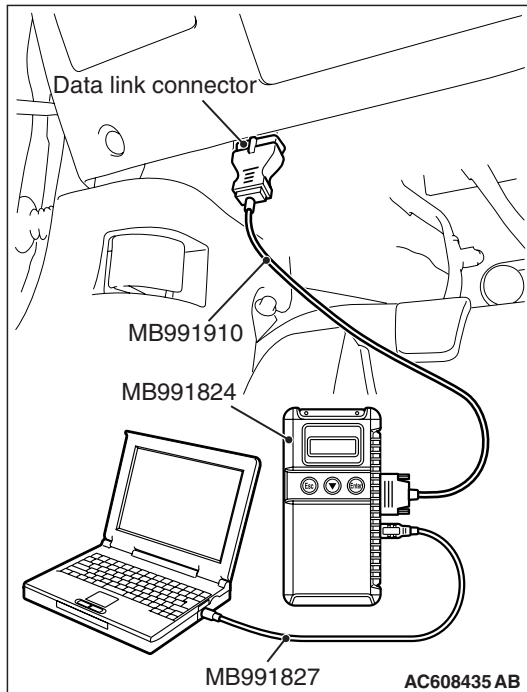
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the diagnostic trouble code other than P1507 set?

YES : Refer to Diagnostic Trouble Code Chart [P.13A-44](#).

NO : Go to Step 2.



STEP 2. Check the intake system vacuum leak.

Q: Are there any abnormalities?

YES : Repair or replace it. Then go to Step 4.

NO : Go to Step 3.

STEP 3. Replace the throttle body assembly.

- (1) Replace the throttle body assembly.
- (2) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 19 [P.13A-9](#).
- (3) Check the diagnostic trouble code (DTC).

Q: Is DTC P1507 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.

NO : The inspection is complete.

STEP 4. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 19 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P1507 set?**YES** : Retry the troubleshooting.**NO** : The inspection is complete.

DTC P1590: TCM to ECM Communication Error in Torque Reduction Request

**TECHNICAL DESCRIPTIONS OF
MONITOR METHODS**

The ECM checks whether the errors in data communications sent by the TCM exist or not.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

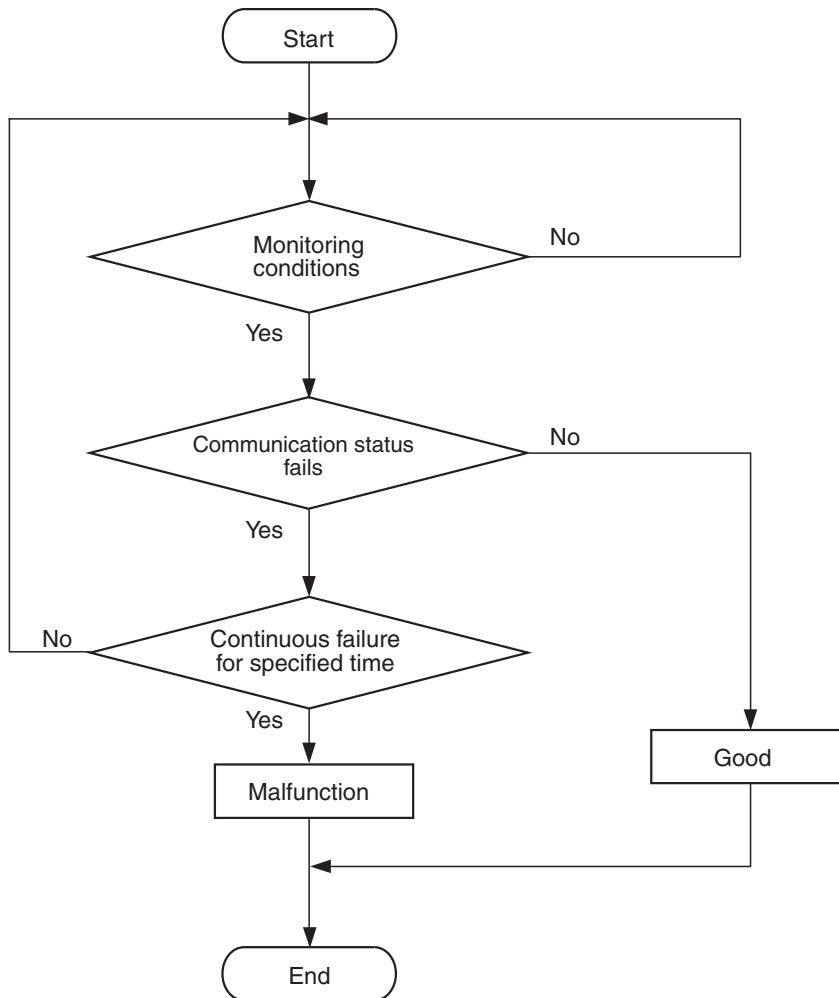
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604369

Check Condition

- Ignition switch is "ON" position.

Judgment Criterion

- ECM detects an error in communication between ECM and TCM.

FAIL-SAFE AND BACKUP FUNCTION

- Engine output is restricted.

OBD-II DRIVE CYCLE PATTERN

- None.

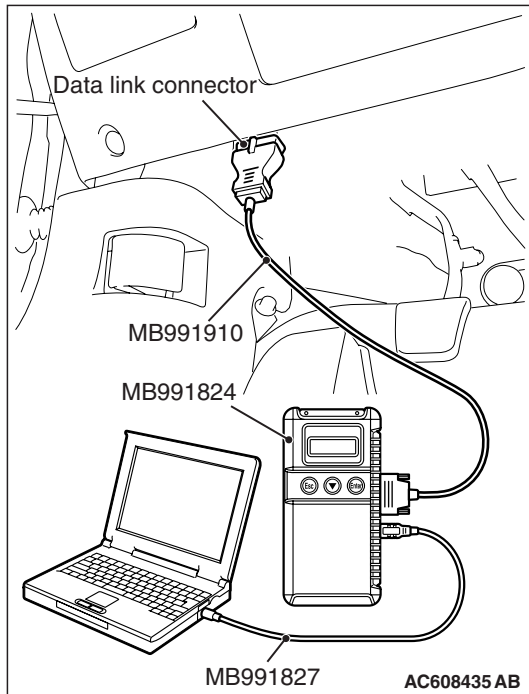
TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A



STEP 1. Using scan tool MB991958, diagnose CAN bus line.

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2 .

NO : Repair the CAN bus line. Refer to GROUP 54C, Can Bus Diagnostics Table [P.54C-14](#).

STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

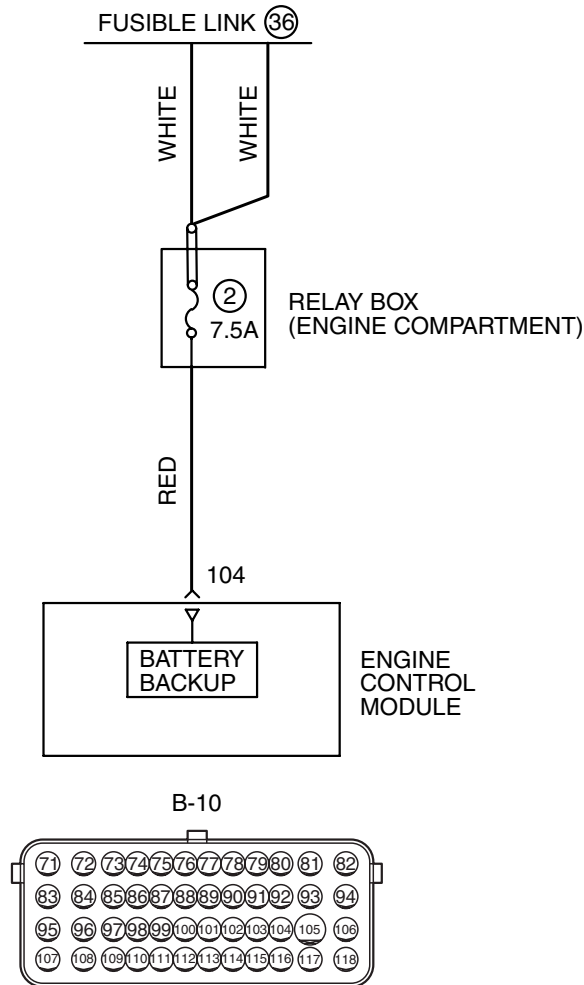
Q: Is DTC P1590 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

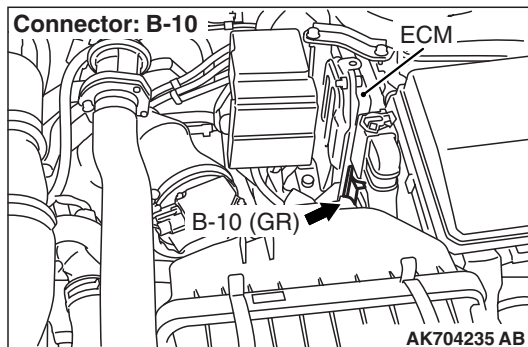
NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

DTC P1603: Battery Backup Circuit Malfunction

BATTERY BACKUP CIRCUIT



AK604256AD



TECHNICAL DESCRIPTION

- The ECM checks the open circuit of battery backup line.

NOTE: When the system detects an open circuit in the battery backup line, it makes 1 failure judgment of other diagnostic trouble codes (DTCs).

DESCRIPTIONS OF MONITOR METHODS

Battery backup line voltage is under specified value.

MONITOR EXECUTION

Continuous

Sensor (The sensor below is determined to be normal)

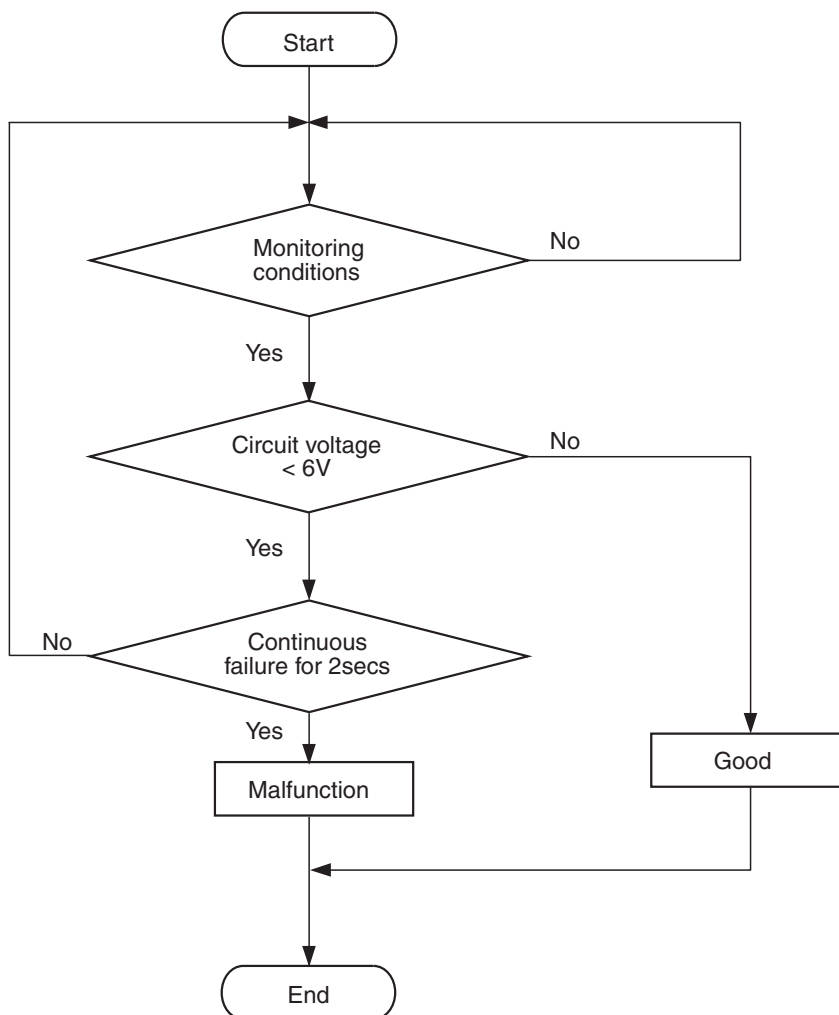
- Not applicable

MONITOR EXECUTION CONDITIONS

(Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

DTC SET CONDITIONS**Logic Flow Chart**

AK604357

Check Conditions

- Engine starting sequence was completed.

- Battery positive voltage is higher than 10 volts.

Judgement Criterion

- Battery backup line voltage has continued to be 6 volts or lower for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- All diagnosis codes are stored once failure judgment is completed.

OBD-II DRIVE CYCLE PATTERN

- None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Open or shorted battery backup line, harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A
- MB992110: Power Plant ECU Check Harness

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

CAUTION

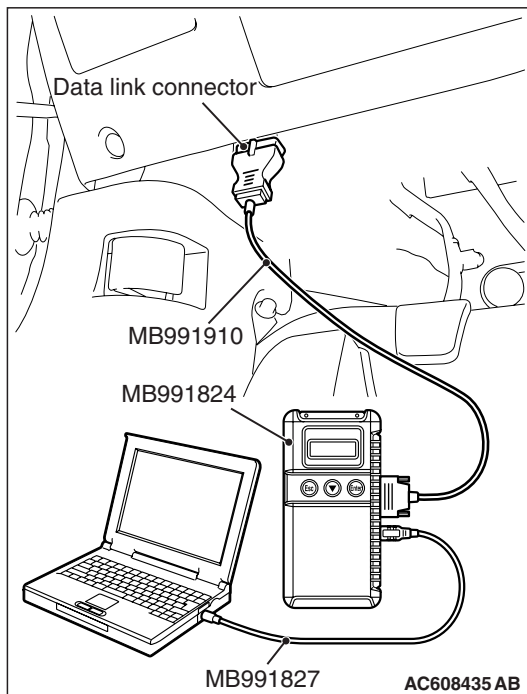
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTC.
- (4) Start the engine and run it at idle.
- (5) Read the DTC.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1603 set?

YES : Go to Step 2.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

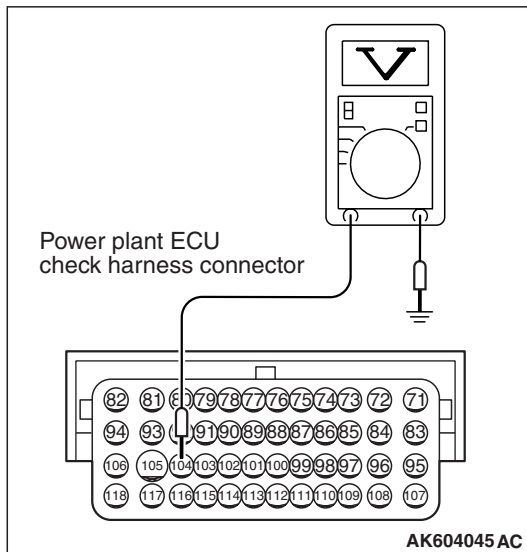
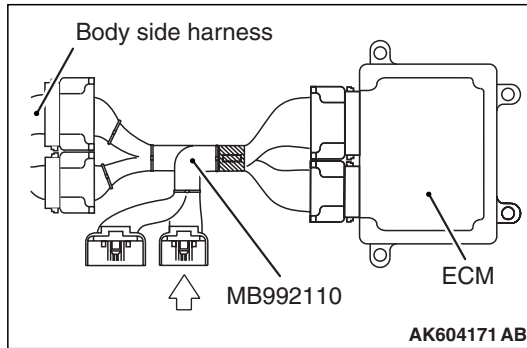


STEP 2. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 5.



STEP 3. Measure the backup power supply voltage at ECM connector B-10 by using power plant ECU check harness special tool MB992110.

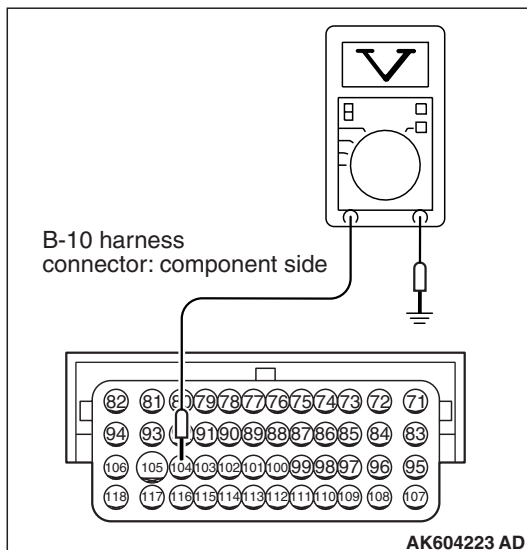
- (1) Disconnect all ECM connectors. Connect the power plant ECU check harness special tool MB992110 between the separated connectors.

- (2) Measure the voltage between terminal No. 104 and ground.
 - Voltage should be battery positive voltage.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 5.

NO : Go to Step 4.



STEP 4. Measure the backup power supply voltage at ECM harness side connector B-10.

- (1) Disconnect the ECM connector B-10 and measure at the harness side.
- (2) Measure the voltage between terminal No. 104 and ground.
 - Voltage should be battery positive voltage.

Q: Is battery positive voltage (approximately 12 volts) present?

YES : Repair harness wire between battery and ECM connector B-10 (terminal No. 104) because of harness damage. Then go to Step 5.

NO : Repair harness wire between battery and ECM connector B-10 (terminal No. 104) because of open circuit or short circuit to ground. Then go to Step 5.

STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1603 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P1676: Variant Coding

TECHNICAL DESCRIPTION

- The ECM performs proper engine control based on vehicle information entered.
- A vehicle identification number (VIN) is entered at a plant before shipment. This diagnosis code occurs when no vehicle information is entered during ECM replacement.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

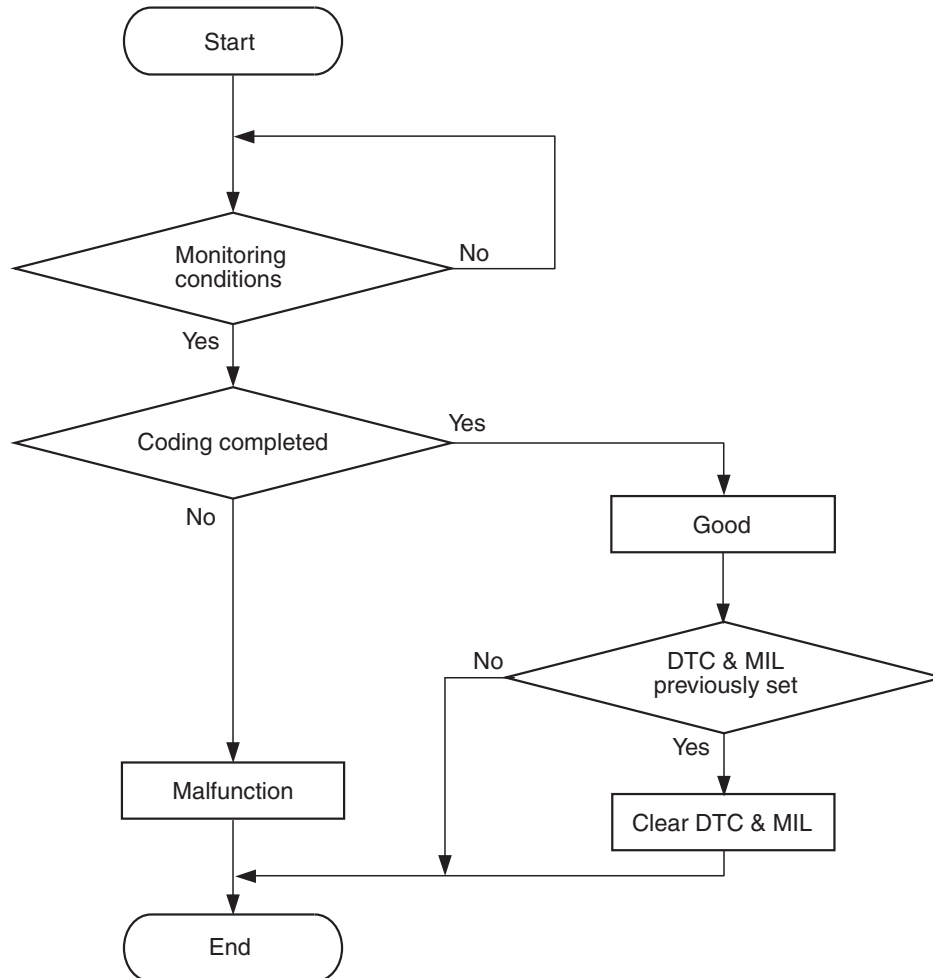
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604358

Check Condition

- Ignition switch is "ON" position.

Judgment Criterion

- Vehicle information is not entered into the ECM.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

- None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

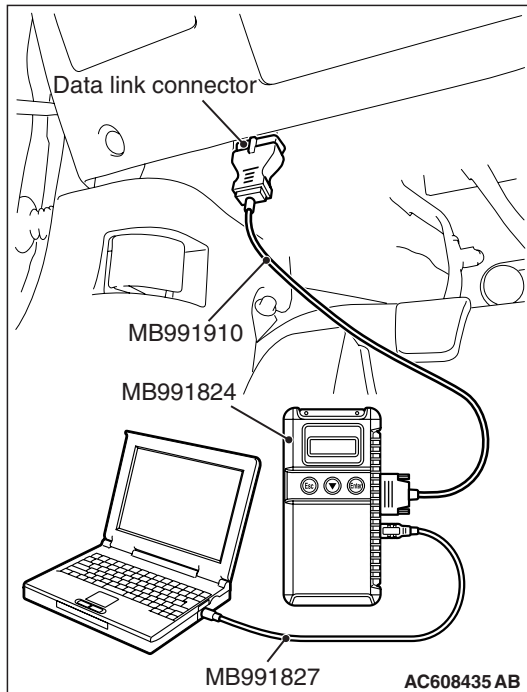
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P1676 set?

YES : Replace the ECM with one into which vehicle information is correctly entered.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P2066: Fuel Level Sensor (sub) Circuit Range/Performance

TECHNICAL DESCRIPTION

- The fuel level sensor converts the rest of the fuel to a voltage and sends it to the combination meter.
- The combination meter sends the data regarding the rest of the fuel to the ECM.
- The ECM checks whether this data is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Detect malfunction if change of fuel level sensor output voltage is small when sum of fuel injection is large.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

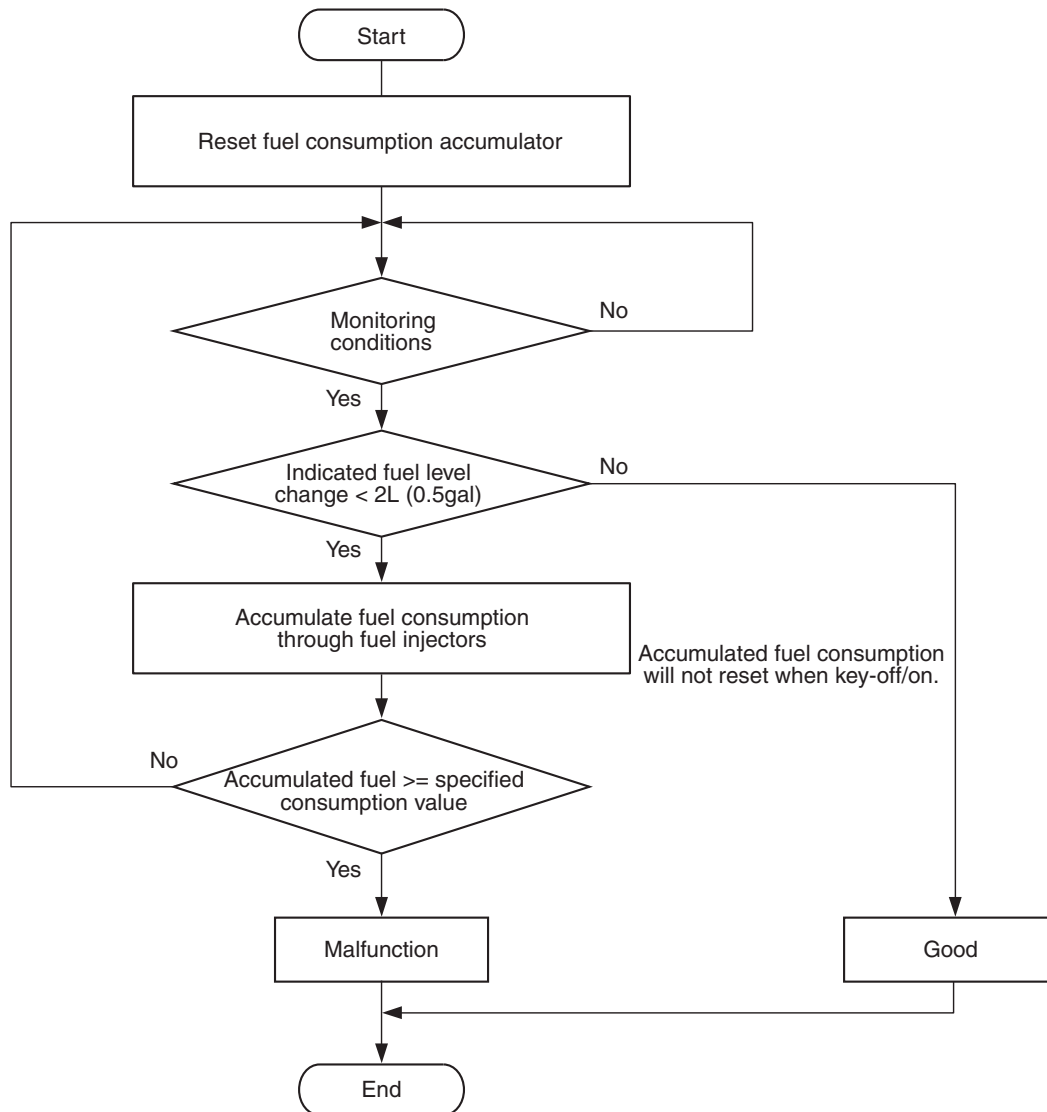
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604341

Check Condition, Judgement Criterion

- When the fuel consumption calculated from the operation time of the injector amounts to 30 liters (7.9 gal), the diversity of the amount of fuel in tank calculated from the fuel level sensor is 2 liters (0.5 gal) or less.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

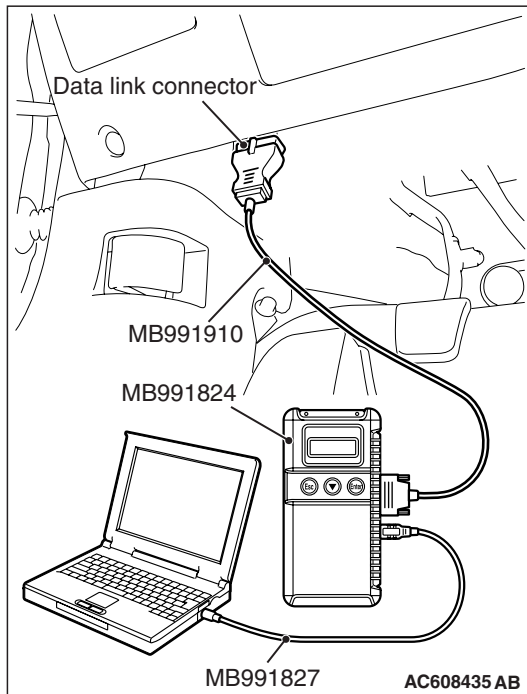
TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Fuel pump module or fuel level sensor (sub) failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A



STEP 1. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Read the DTC.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P0461 set?

YES : Go to Step 2.

NO : Go to Step 4.

STEP 2. Check fuel gauge.

Q: Is the fuel gauge functioning?

YES : Go to Step 3.

NO : Refer to GROUP 54A, Combination Meter –Trouble Symptom Chart [P.54A-59](#).

STEP 3. Check the trouble symptoms.

Check that the fuel gauge operates correctly.

Q: Does the fuel gauge operates correctly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, Diagnosis –ID Codes Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, Diagnosis –ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 6.

STEP 4. Check the fuel level sensor (sub).

Check to see if the fuel level sensor is normal. Refer to GROUP 54A, Combination Meter –On-vehicle Service –Fuel Gauge Unit Check [P.54A-91](#).

Q: Is the fuel level sensor (sub) normal?

YES : Go to Step 5.

NO : Replace the fuel level sensor (sub). Then go to Step 6.

STEP 5. Check the trouble symptoms.

Check that the fuel gauge operates correctly.

Q: Does the fuel gauge operates correctly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, Diagnosis –ID Codes Registration Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, Diagnosis –ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 6.

STEP 6. Check the trouble symptoms.

Check that the fuel gauge operates correctly.

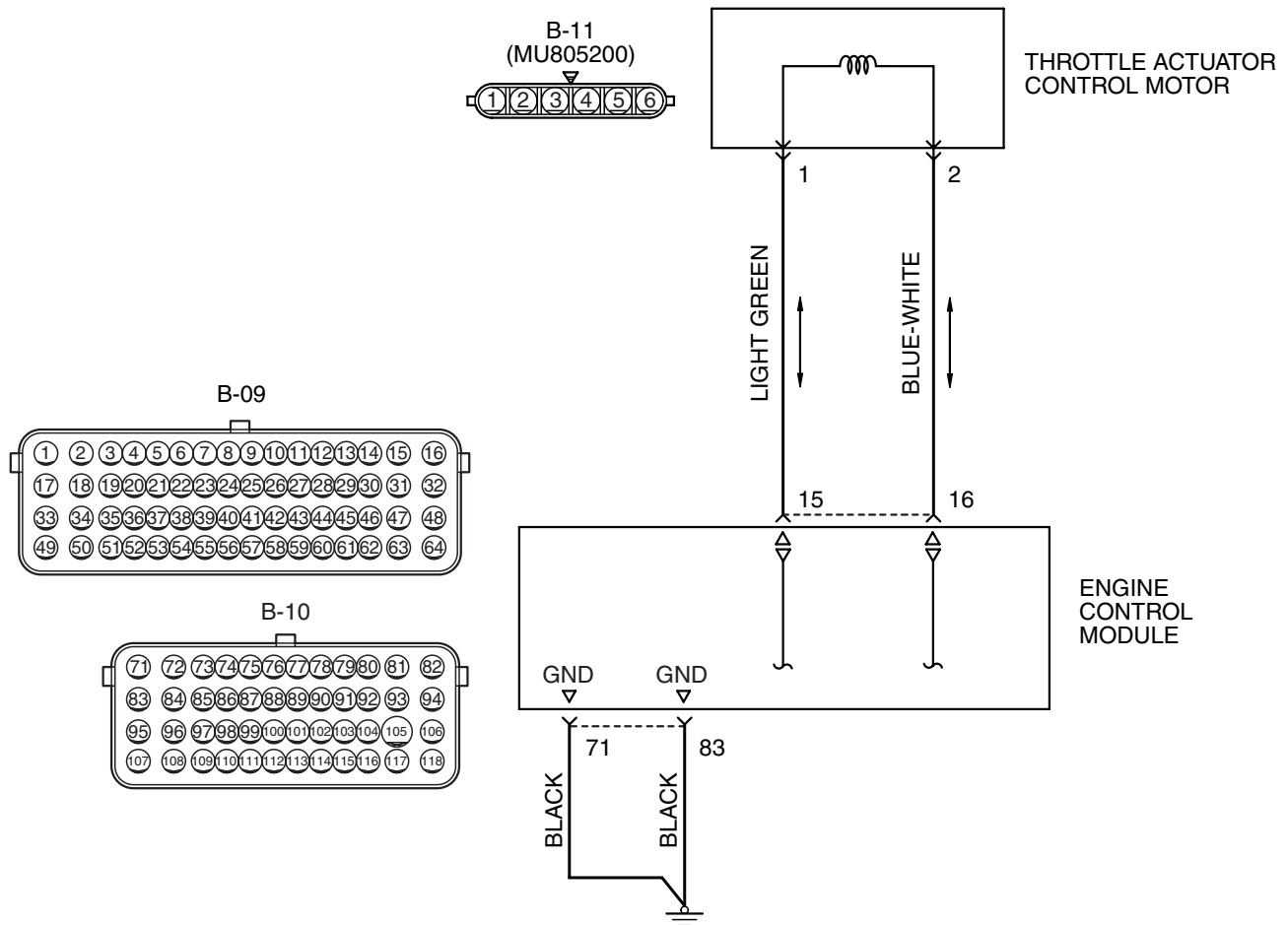
Q: Does the fuel gauge operates correctly?

YES : The inspection is complete.

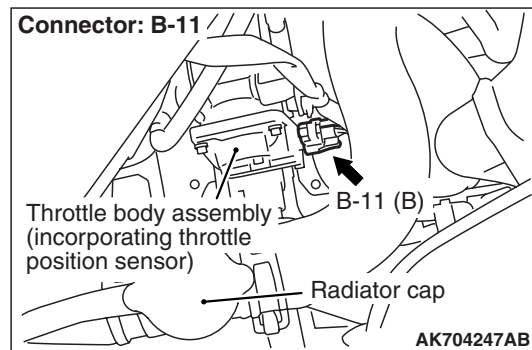
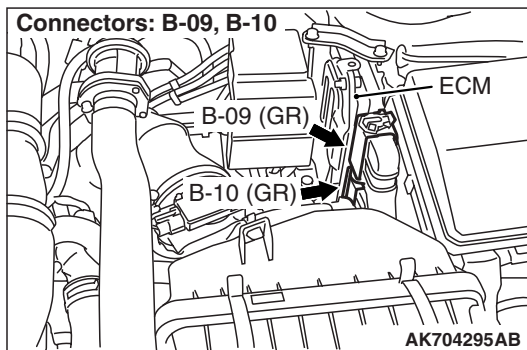
NO : Retry the troubleshooting.

DTC P2100: Throttle Actuator Control Motor Circuit (open)

THROTTLE ACTUATOR CONTROL MOTOR CIRCUIT



AK604296AD



CIRCUIT OPERATION

- Controls the current that is applied from the ECM (terminals No. 15, No. 16) to the throttle actuator control motor (terminals No. 1, No. 2).

TECHNICAL DESCRIPTION

- ECM varies the direction and the amperage of the current that is applied to the throttle actuator control motor in order to control the opening of the throttle valve.

DESCRIPTIONS OF MONITOR METHODS

The voltage difference between the throttle position sensor output voltage and the throttle position sensor output at the throttle actuator control motor relay off is smaller than the specified value.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Check Conditions**

- Battery positive voltage is higher than 8.3 volts.
- The output voltage of the throttle position sensor (main) minus the proposed output voltage becomes 0.1 volt or more.
- The output voltage of the throttle position sensor (sub) minus the proposed output voltage becomes 0.1 volt or more.
- The drive duty of the throttle actuator control motor is 100 percent or more.

- Except while during cranking.
- or
- Battery positive voltage is higher than 8.3 volts.
 - The proposed output voltage minus the output voltage of the throttle position sensor (main) becomes 0.1 volt or more.
 - The proposed output voltage minus the output voltage of the throttle position sensor (sub) becomes 0.1 volt or more.
 - The drive duty of the throttle actuator control motor is 100 percent or more.
 - Except while during cranking.

Judgement Criterion

- For 0.4 second, the difference between the output voltage of throttle position sensor (main) and the learning value of the middle-opened degree is 0.2 volt or less.
- or
- For 0.4 second, the difference between the output voltage of throttle position sensor (sub) and the learning value of the middle-opened degree is 0.2 volt or less.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle actuator control motor failed.
- Open throttle actuator control motor circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

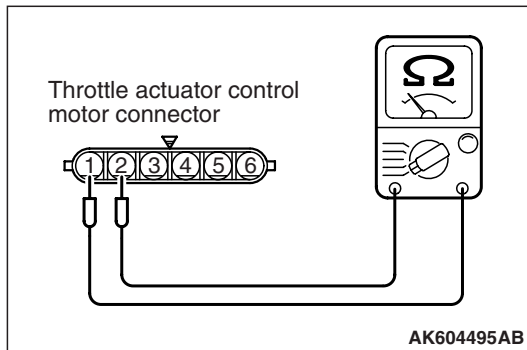
- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector B-11 at throttle actuator control motor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.



STEP 2. Check the throttle actuator control motor.

- (1) Disconnect the connector B-11.
- (2) Measure the resistance between throttle actuator control motor side connector terminal No. 1 and No. 2.

Standard value: 0.3 –80 Ω [at 20° C (68° F)]

Q: Is the measured resistance between 0.3 and 80 Ω [at 20° C (68° F)]?

YES : Go to Step 3.

NO : Replace the throttle body assembly. Then go to Step 9.

STEP 3. Check harness connector B-10 at ECM for damage

Q: Is the harness connector in good condition?

YES : Go to Step 4.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.

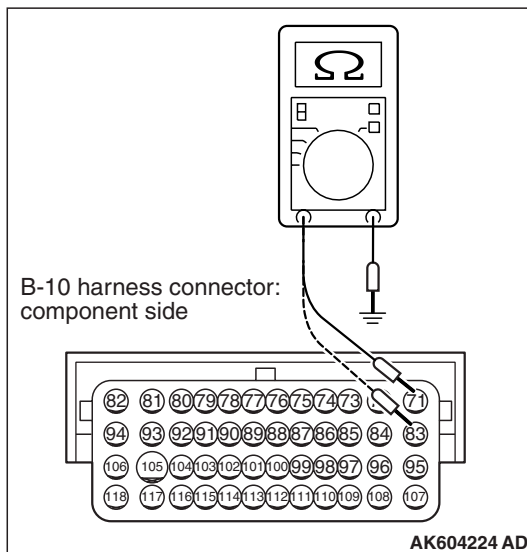
STEP 4. Check the continuity at ECM harness side connector B-10.

- (1) Disconnect the connector B-10 and measure at the harness side.
- (2) Measure the continuity between terminals No. 71, No. 83 and ground.
 - Continuity (2 ohms or less)

Q: Does continuity exist?

YES : Go to Step 5.

NO : Repair harness wire between ECM connector B-10 (terminals No. 71, No. 83) and ground because of open circuit or harness damage. Then go to Step 9.



STEP 5. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 6.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 9.

STEP 6. Check for open circuit and harness damage between throttle actuator control motor connector B-11 (terminal No. 1) and ECM connector B-09 (terminal No. 15).

Q: Is the harness wire in good condition?

YES : Go to Step 7.

NO : Repair it. Then go to Step 9.

STEP 7. Check for open circuit and harness damage between throttle actuator control motor connector B-11 (terminal No. 2) and ECM connector B-09 (terminal No. 16).

Q: Is the harness wire in good condition?

YES : Go to Step 8.

NO : Repair it. Then go to Step 9.

STEP 8. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

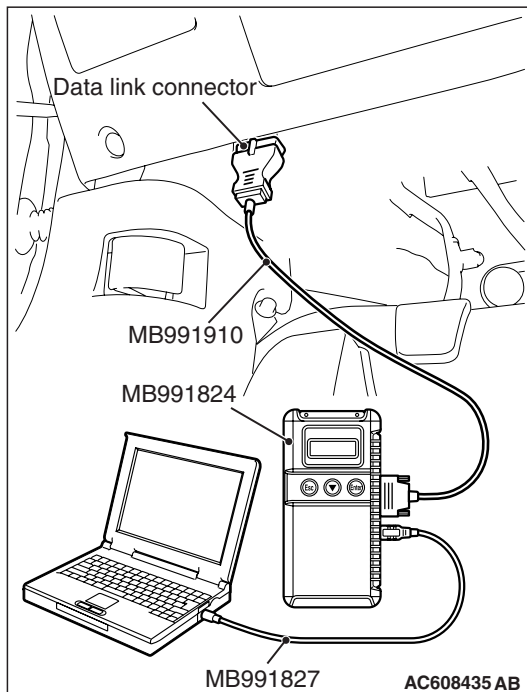
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2100 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 9.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 9. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

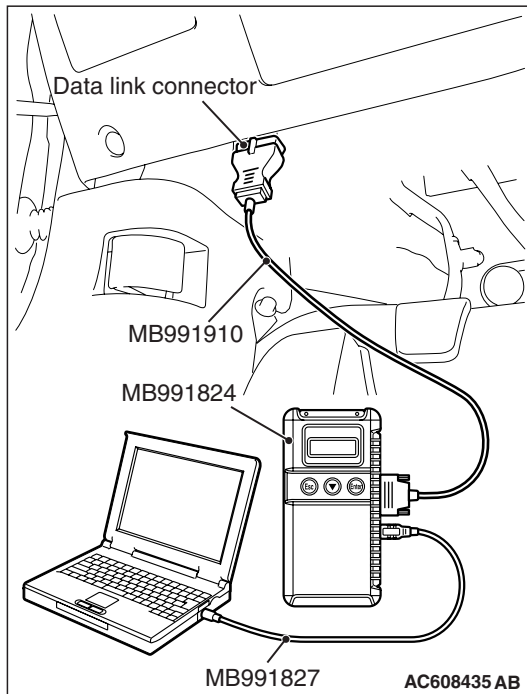
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2100 set?

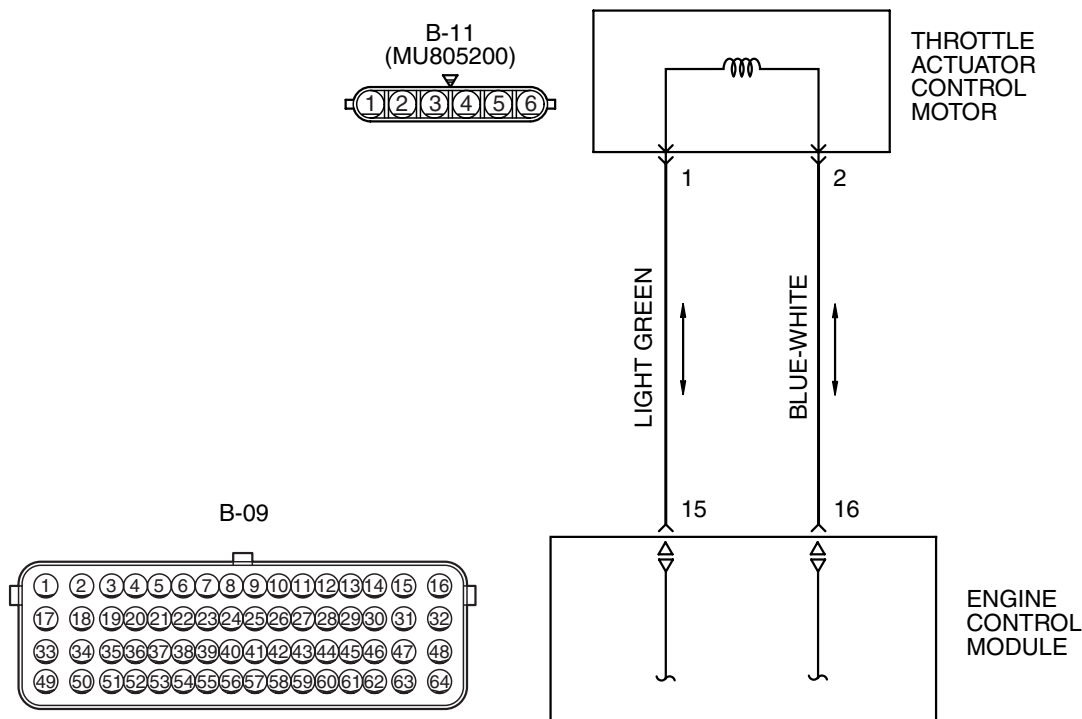
YES : Retry the troubleshooting.

NO : The inspection is complete.

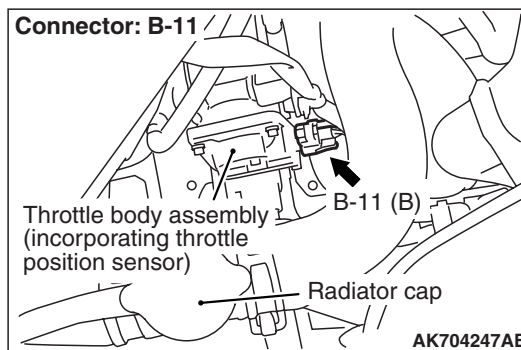
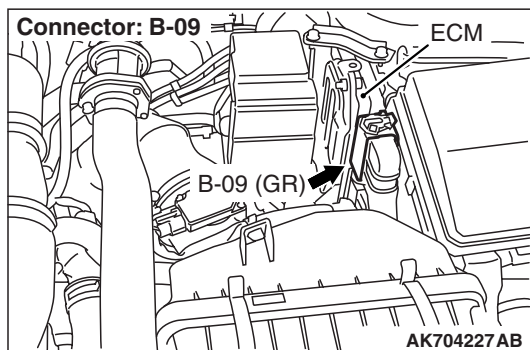


DTC P2101: Throttle Actuator Control Motor Magneto Malfunction

THROTTLE ACTUATOR CONTROL MOTOR CIRCUIT



AK604297AD



CIRCUIT OPERATION

- Controls the current that is applied from ECM (terminals No. 15, No. 16) to the throttle actuator control motor (terminals No. 1, No. 2)

TECHNICAL DESCRIPTION

- ECM check whether the throttle actuator control motor magneto has failed.

DESCRIPTIONS OF MONITOR METHODS

Throttle actuator control motor intelligent power device detects shorted-high/low and overheat of itself.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Check Condition

- Battery positive voltage is higher than 8.3 volts.

Judgement Criterion

- The coil current of the throttle actuator control motor is 8 ampere or more for 0.3 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Throttle actuator control motor failed.
- Shorted throttle actuator control motor circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector B-11 at throttle actuator control motor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 2.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7.

STEP 2. Check the throttle actuator control motor.

(1) Disconnect the connector B-11.

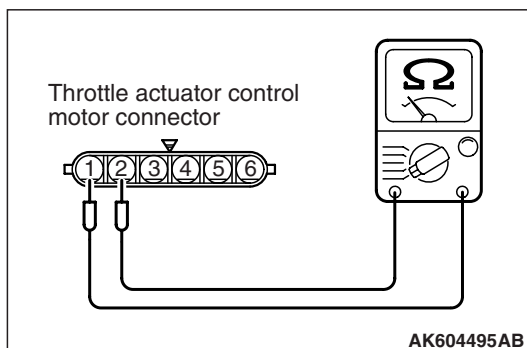
(2) Measure the resistance between throttle actuator control motor side connector terminal No. 1 and No. 2.

Standard value: 0.3 –80 Ω [at 20° C (68° F)]

Q: Is the measured resistance between 0.3 and 80 Ω [at 20° C (68° F)]?

YES : Go to Step 3.

NO : Replace the throttle body assembly. Then go to Step 7.



STEP 3. Check harness connector B-09 at ECM for damage**Q: Is the harness connector in good condition?****YES :** Go to Step 4.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 7.**STEP 4. Check for short circuit to ground between throttle actuator control motor connector B-11 (terminal No. 1) and ECM connector B-09 (terminal No. 15).****Q: Is the harness wire in good condition?****YES :** Go to Step 5.**NO :** Repair it. Then go to Step 7.**STEP 5. Check for short circuit to ground between throttle actuator control motor connector B-11 (terminal No. 2) and ECM connector B-09 (terminal No. 16).****Q: Is the harness wire in good condition?****YES :** Go to Step 6.**NO :** Repair it. Then go to Step 7.**STEP 6. Using scan tool MB991958, read the diagnostic trouble code (DTC).****⚠ CAUTION**

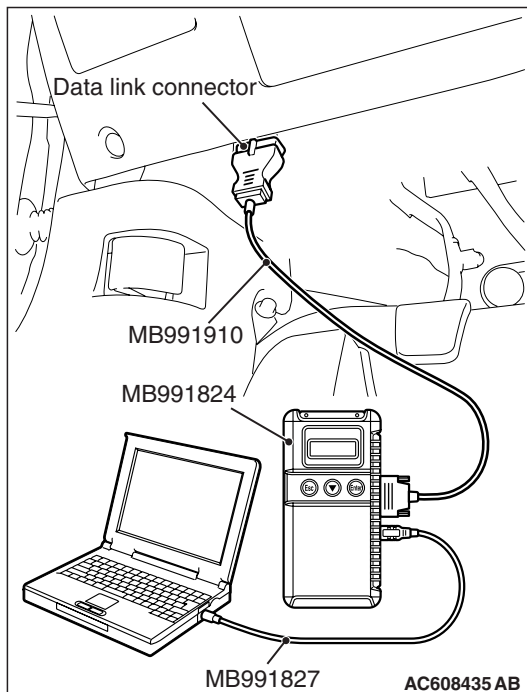
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2101 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 7.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 7. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

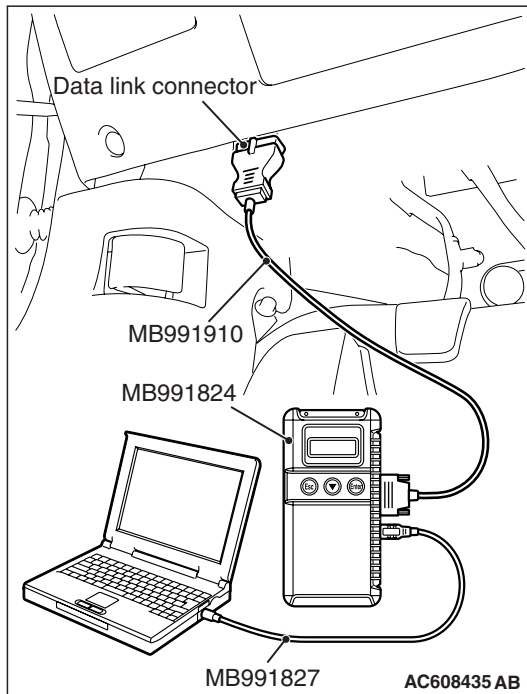
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2101 set?

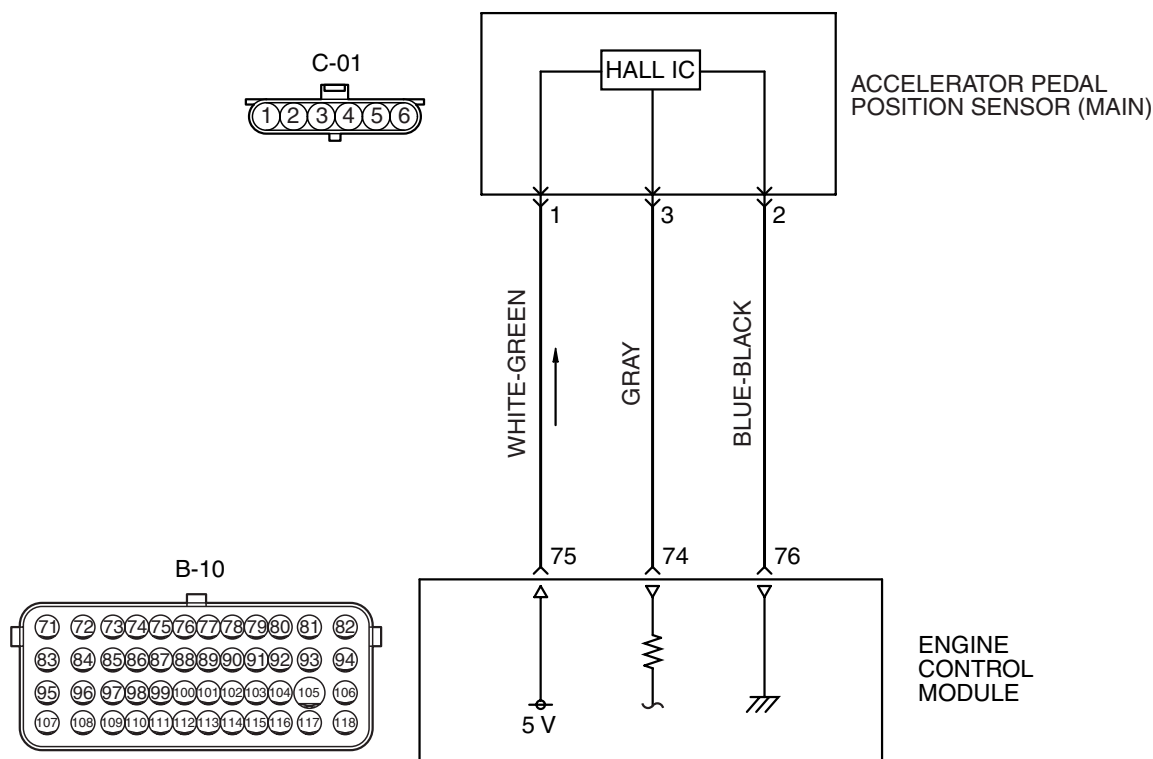
YES : Retry the troubleshooting.

NO : The inspection is complete.

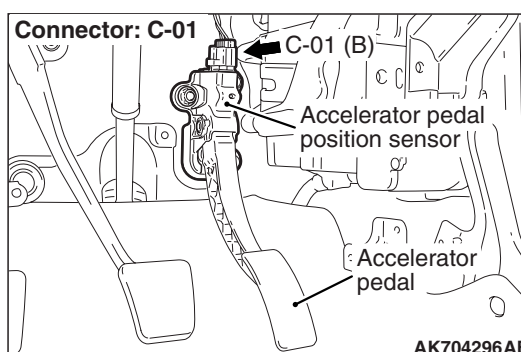
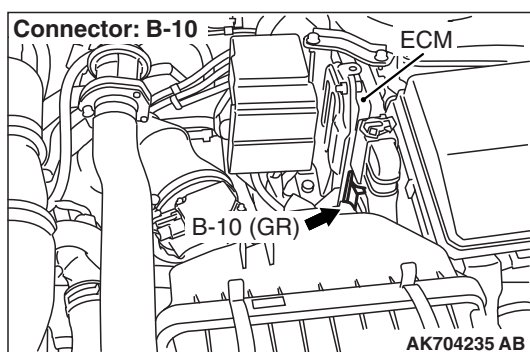


DTC P2122: Accelerator Pedal Position Sensor (main) Circuit Low Input

ACCELERATOR PEDAL POSITION SENSOR (MAIN) CIRCUIT



AK604257AG



CIRCUIT OPERATION

- A 5-volt power supply is applied on the accelerator pedal position sensor (main) power terminal (terminal No. 1) from the ECM (terminal No. 75).
- The ground terminal (terminal No. 2) is grounded with ECM (terminal No. 76).
- A voltage that is according to the accelerator opening angle is sent to the ECM (terminal No. 74) from the accelerator pedal position sensor (main) output terminal (terminal No. 3).

TECHNICAL DESCRIPTION

- The accelerator pedal position sensor (main) outputs voltage which corresponds to the accelerator pedal depression.
- The ECM checks whether the voltage is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Accelerator pedal position sensor (main) output voltage is out of specified range.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

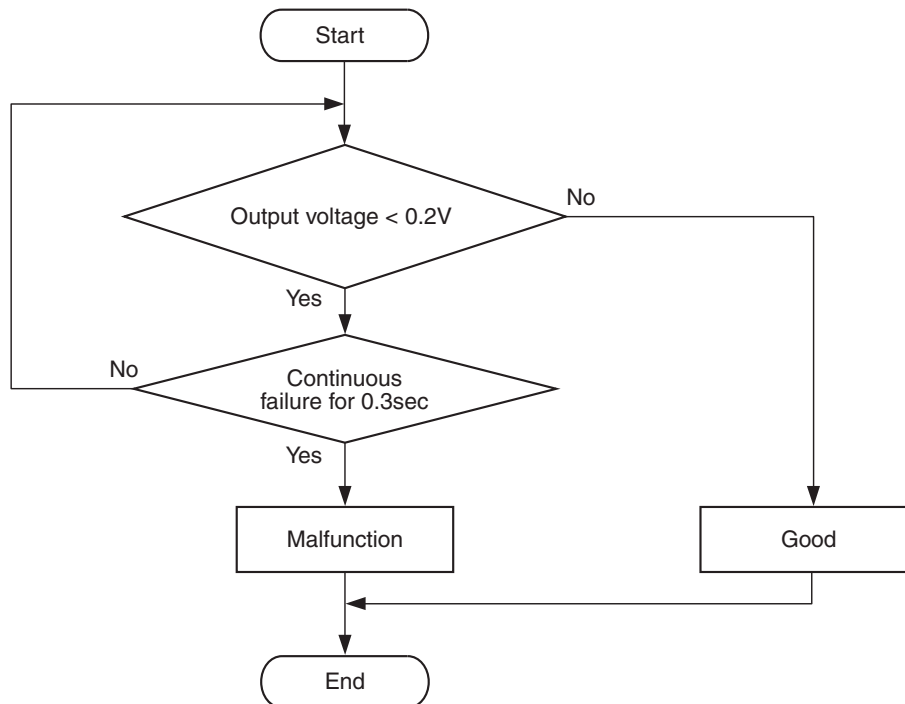
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604359

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- Accelerator pedal position sensor (main) output voltage is 0.2 volt or less for 0.3 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if accelerator pedal position sensor (sub) fails.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Accelerator pedal position sensor failed.
- Open or shorted accelerator pedal position sensor (main) circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, check data list item 11: Accelerator Pedal Position Sensor (main).

⚠ CAUTION

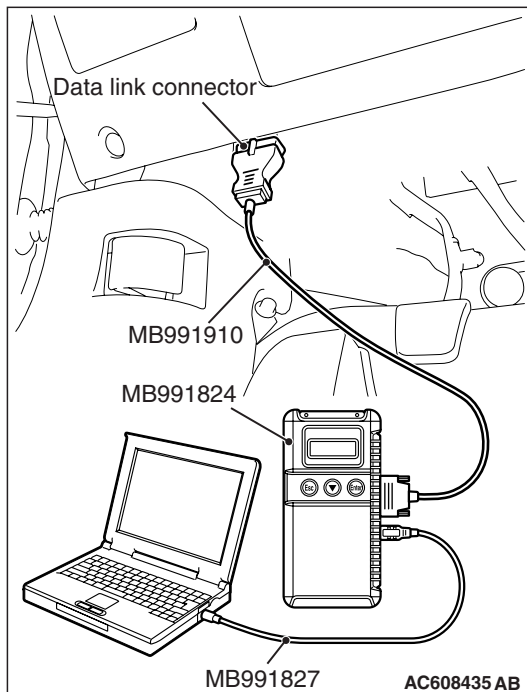
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958 to the data reading mode for item 11, Accelerator Pedal Position Sensor (main).
 - Output voltage is between 0.9 and 1.1 volts when foot is released from accelerator pedal.
 - Output voltage is 4.0 volts or higher when accelerator pedal is fully depressed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2.

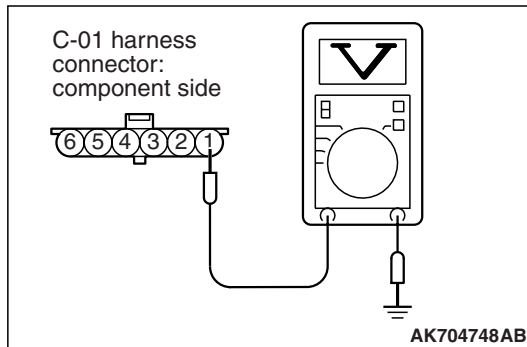


STEP 2. Check harness connector C-01 at accelerator pedal position sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.



STEP 3. Measure the sensor supply voltage at accelerator pedal position sensor harness side connector C-01.

- (1) Disconnect the connector C-01 and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 1 and ground.
 - Voltage should be between 4.9 and 5.1 volts.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage between 4.9 and 5.1 volts?

YES : Go to Step 7.

NO : Go to Step 4.

STEP 4. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 5.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.

STEP 5. Check for open circuit and short circuit to ground between accelerator pedal position sensor connector C-01 (terminal No. 1) and ECM connector B-10 (terminal No. 75).

Q: Is the harness wire in good condition?

YES : Go to Step 6.

NO : Repair it. Then go to Step 11.

STEP 6. Using scan tool MB991958, check data list item 11: Accelerator Pedal Position Sensor (main).

- (1) Turn the ignition switch to the "ON" position.
- (2) Set scan tool MB991958 to the data reading mode for item 11, Accelerator Pedal Position Sensor (main).
 - Output voltage is between 0.9 and 1.1 volts when foot is released from accelerator pedal.
 - Output voltage is 4.0 volts or higher when accelerator pedal is fully depressed.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 11.

STEP 7. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 8.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.

STEP 8. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 1) and ECM connector B-10 (terminal No. 75).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 11.

STEP 9. Check for open circuit, short circuit to ground and harness damage between accelerator pedal position sensor connector C-01 (terminal No. 3) and ECM connector B-10 (terminal No. 74).

Q: Is the harness wire in good condition?

YES : Go to Step 10.

NO : Repair it. Then go to Step 11.

STEP 10. Replace the accelerator pedal position sensor.

- (1) Replace the accelerator pedal position sensor.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2122 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 11.

NO : The inspection is complete.

STEP 11. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

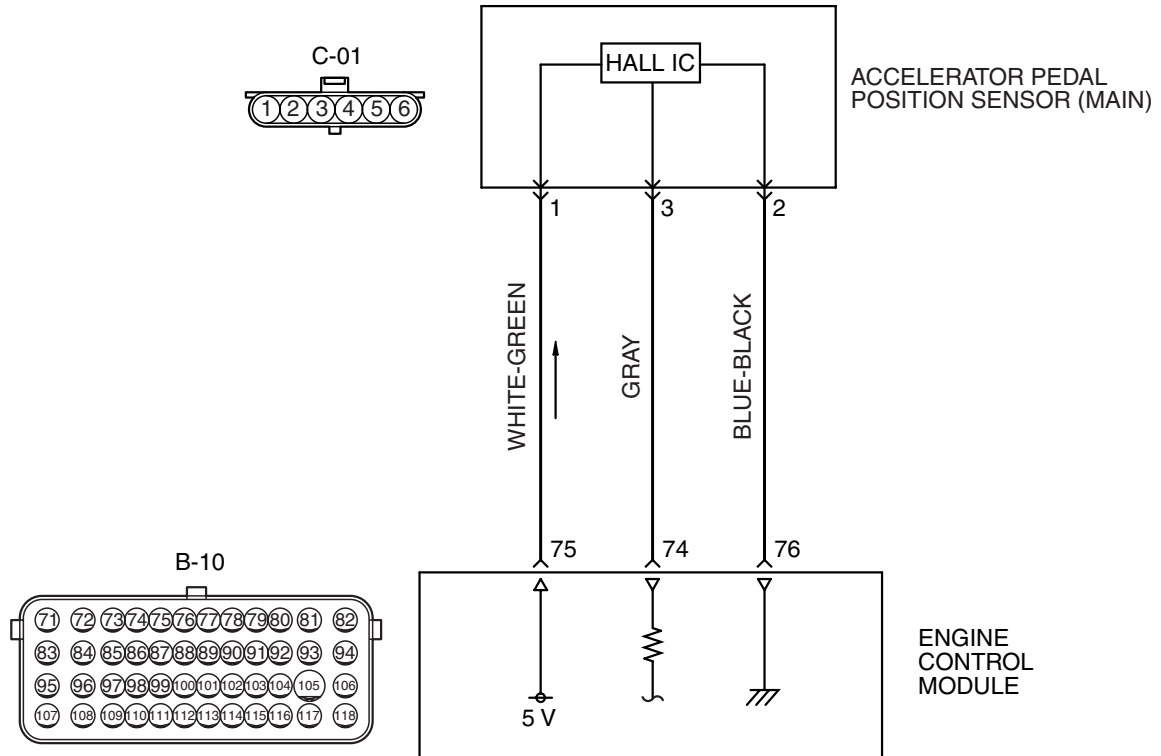
Q: Is DTC P2122 set?

YES : Retry the troubleshooting.

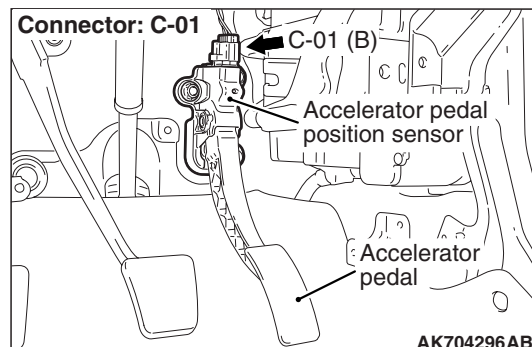
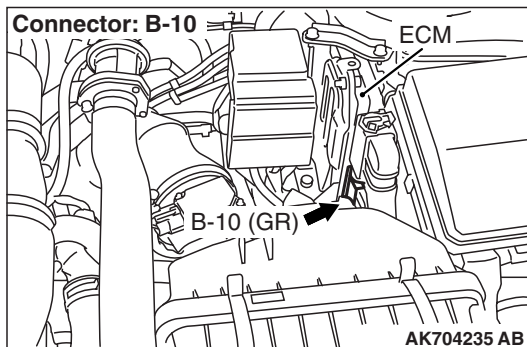
NO : The inspection is complete.

DTC P2123: Accelerator Pedal Position Sensor (main) Circuit High Input

ACCELERATOR PEDAL POSITION SENSOR (MAIN) CIRCUIT



AK604257AG



CIRCUIT OPERATION

- A 5-volt power supply is applied on the accelerator pedal position sensor (main) power terminal (terminal No. 1) from the ECM (terminal No. 75).
- The ground terminal (terminal No. 2) is grounded with ECM (terminal No. 76).
- A voltage that is according to the accelerator opening angle is sent to the ECM (terminal No. 74) from the accelerator pedal position sensor (main) output terminal (terminal No. 3).

TECHNICAL DESCRIPTION

- The accelerator pedal position sensor (main) outputs voltage which corresponds to the accelerator pedal depression.
- The ECM checks whether the voltage is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Accelerator pedal position sensor (main) output voltage is out of specified range.

MONITOR EXECUTION

Continuous

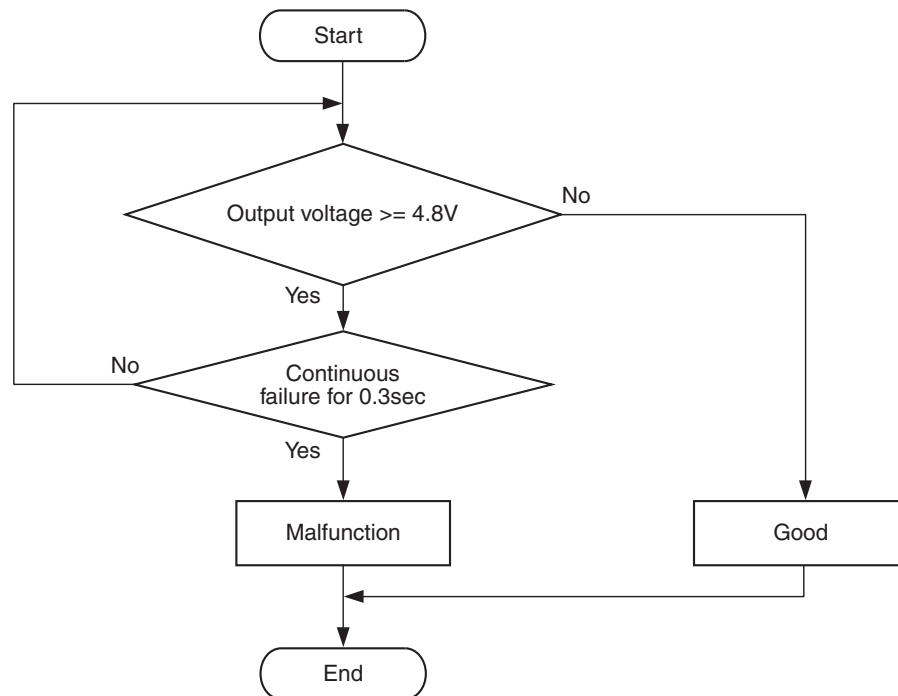
**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Logic Flow Chart**

AK604360

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- Accelerator pedal position sensor (main) output voltage should be 4.8 volts or higher for 0.3 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if accelerator pedal position sensor (sub) fails.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Accelerator pedal position sensor failed.
- Open accelerator pedal position sensor (main) circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, check data list item 11: Accelerator Pedal Position Sensor (main).

CAUTION

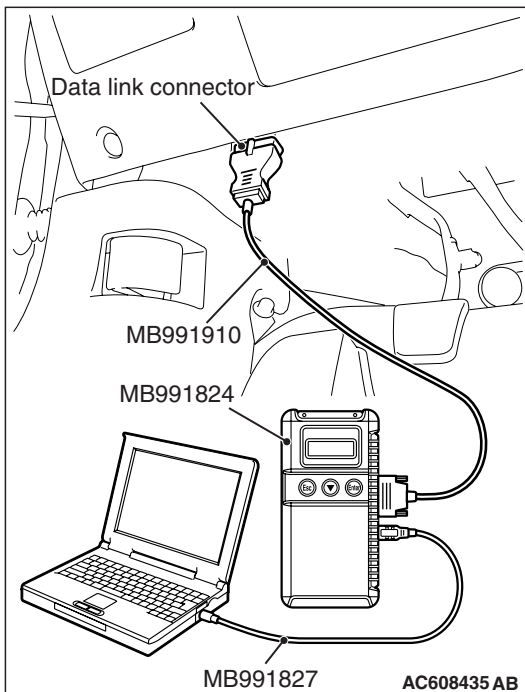
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958 to the data reading mode for item 11, Accelerator Pedal Position Sensor (main).
 - Output voltage is between 0.9 and 1.1 volts when foot is released from accelerator pedal.
 - Output voltage is 4.0 volts or higher when accelerator pedal is fully depressed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2.

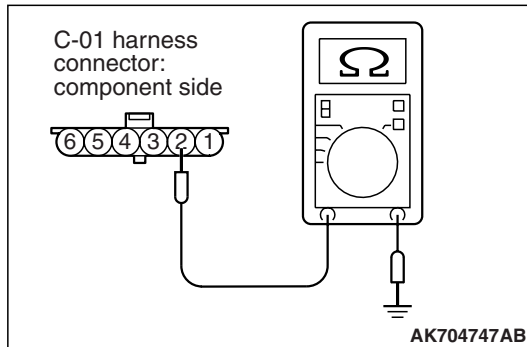


STEP 2. Check harness connector C-01 at accelerator pedal position sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 8.

**STEP 3. Check the continuity at accelerator pedal position sensor harness side connector C-01.**

- (1) Disconnect the connector C-01 and measure at the harness side.
- (2) Measure the continuity between terminal No. 2 and ground.
 - Continuity (2 ohms or less)

Q: Does continuity exist?**YES :** Go to Step 7.**NO :** Go to Step 4.**STEP 4. Check harness connector B-10 at ECM for damage.****Q: Is the harness connector in good condition?****YES :** Go to Step 5.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 8.**STEP 5. Check for open circuit and harness damage between accelerator pedal position sensor connector C-01 (terminal No. 2) and ECM connector B-10 (terminal No. 76).****Q: Is the harness wire in good condition?****YES :** Go to Step 6.**NO :** Repair it. Then go to Step 8.**STEP 6. Using scan tool MB991958, check data list item 11: Accelerator Pedal Position Sensor (main).**

- (1) Turn the ignition switch to the "ON" position.
- (2) Set scan tool MB991958 to the data reading mode for item 11, Accelerator Pedal Position Sensor (main).
 - Output voltage is between 0.9 and 1.1 volts when foot is released from accelerator pedal.
 - Output voltage is 4.0 volts or higher when accelerator pedal is fully depressed.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?**YES :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).**NO :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 8.

STEP 7. Replace the accelerator pedal position sensor.

- (1) Replace the accelerator pedal position sensor.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2123 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 8.

NO : The inspection is complete.

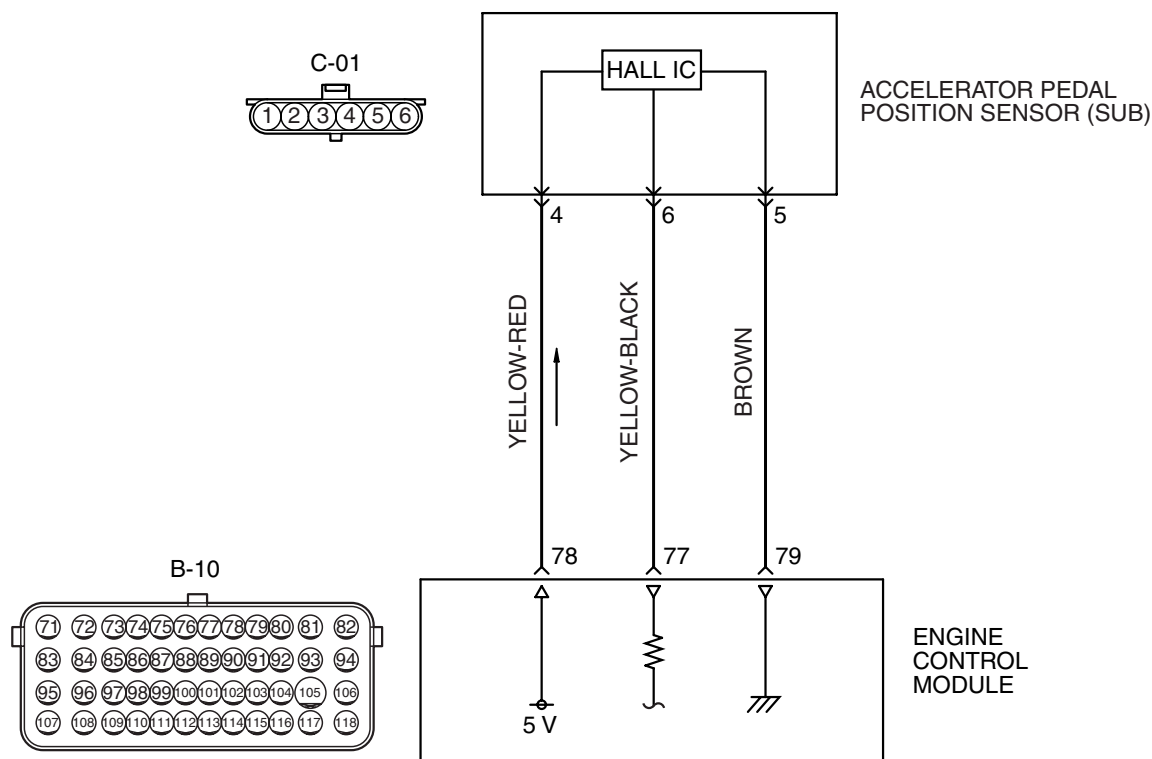
STEP 8. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

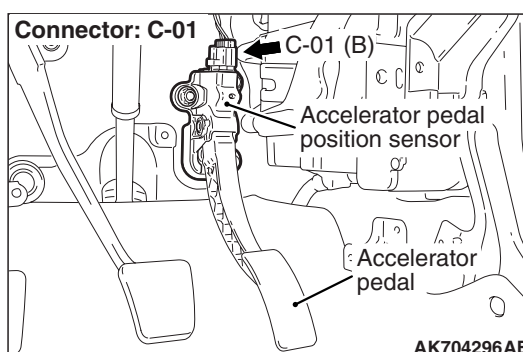
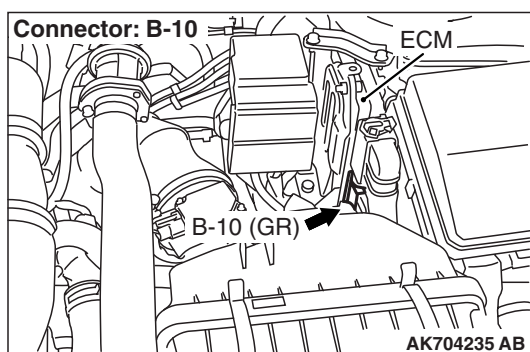
Q: Is DTC P2123 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P2127: Accelerator Pedal Position Sensor (sub) Circuit Low Input**ACCELERATOR PEDAL POSITION SENSOR (SUB) CIRCUIT**

AK604257AH

**CIRCUIT OPERATION**

- A 5-volt power supply is applied on the accelerator pedal position sensor (sub) power terminal (terminal No. 4) from the ECM (terminal No. 78).
- A voltage that is according to the accelerator opening angle is sent to the ECM (terminal No. 77) from the accelerator pedal position sensor (sub) output terminal (terminal No. 6).
- The ground terminal (terminal No. 5) is grounded with ECM (terminal No. 79).

TECHNICAL DESCRIPTION

- The accelerator pedal position sensor (sub) outputs voltage which corresponds to the accelerator pedal depression.
- The ECM checks whether the voltage is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Accelerator pedal position sensor (sub) output voltage is out of specified range.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

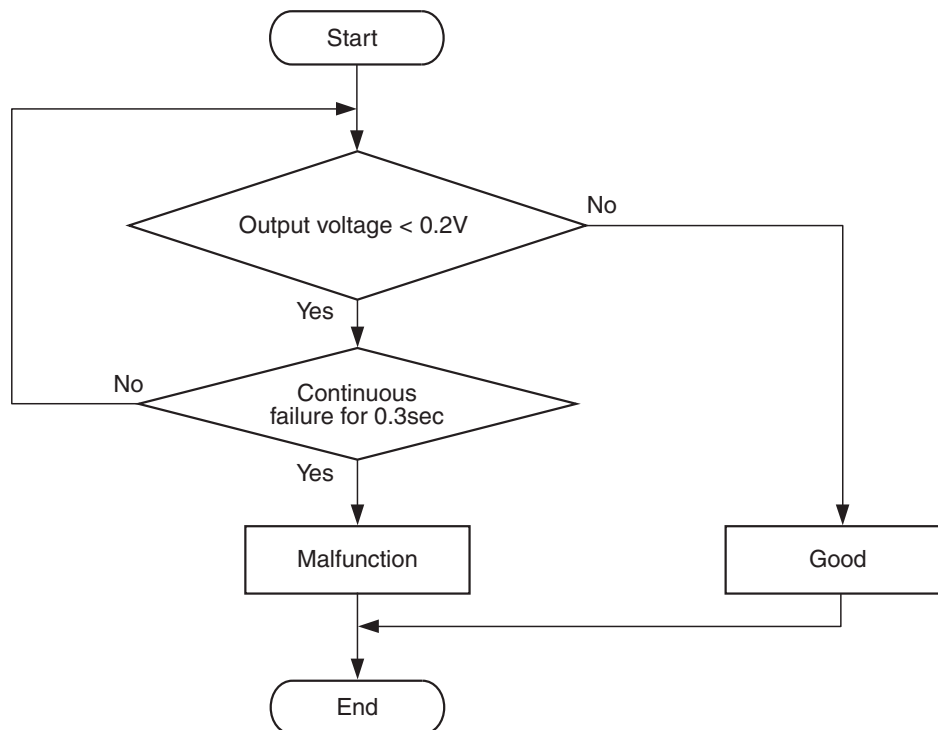
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604359

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- Accelerator pedal position sensor (sub) output voltage is 0.2 volt or less for 0.3 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if accelerator pedal position sensor (main) fails.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Accelerator pedal position sensor failed.
- Open or shorted accelerator pedal position sensor (sub) circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, check data list item 12: Accelerator Pedal Position Sensor (sub).

⚠ CAUTION

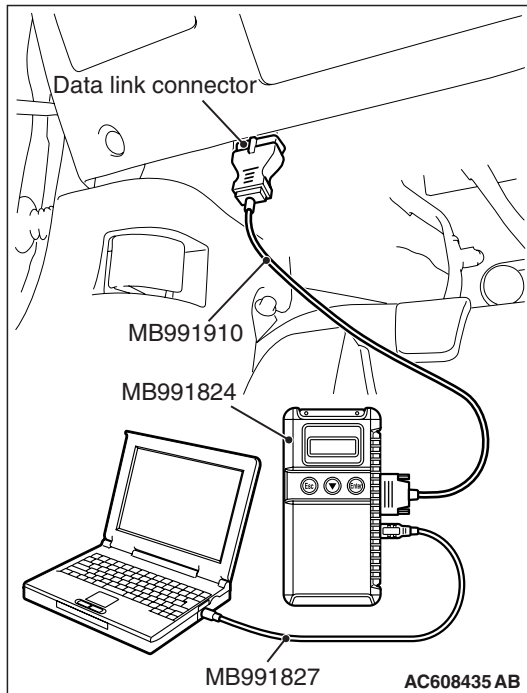
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958 to the data reading mode for item 12, Accelerator Pedal Position Sensor (sub).
 - Output voltage is between 0.4 and 0.6 volt when foot is released from accelerator pedal.
 - Output voltage is 2.0 volts or higher when accelerator pedal is fully depressed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2.

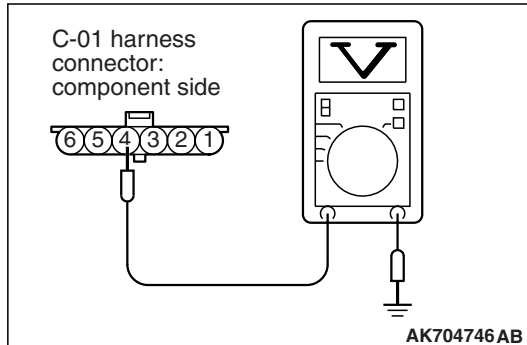


STEP 2. Check harness connector C-01 at accelerator pedal position sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.



STEP 3. Measure the sensor supply voltage at accelerator pedal position sensor harness side connector C-01.

- (1) Disconnect the connector C-01 and measure at the harness side.
- (2) Turn the ignition switch to the "ON" position.
- (3) Measure the voltage between terminal No. 4 and ground.
 - Voltage should be between 4.9 and 5.1 volts.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the measured voltage between 4.9 and 5.1 volts?

YES : Go to Step 7.

NO : Go to Step 4.

STEP 4. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 5.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.

STEP 5. Check for open circuit and short circuit to ground between accelerator pedal position sensor connector C-01 (terminal No. 4) and ECM connector B-10 (terminal No. 78).

Q: Is the harness wire in good condition?

YES : Go to Step 6.

NO : Repair it. Then go to Step 11.

STEP 6. Using scan tool MB991958, check data list item 12: Accelerator Pedal Position Sensor (sub).

- (1) Turn the ignition switch to the "ON" position.
- (2) Set scan tool MB991958 to the data reading mode for item 12, Accelerator Pedal Position Sensor (sub).
 - Output voltage is between 0.4 and 0.6 volt when foot is released from accelerator pedal.
 - Output voltage is 2.0 volts or higher when accelerator pedal is fully depressed.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

NO : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 11.

STEP 7. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 8.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.

STEP 8. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 4) and ECM connector B-10 (terminal No. 78).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 11.

STEP 9. Check for open circuit, short circuit to ground and harness damage between accelerator pedal position sensor connector C-01 (terminal No. 6) and ECM connector B-10 (terminal No. 77)

Q: Is the harness wire in good condition?

YES : Go to Step 10.

NO : Repair it. Then go to Step 11.

STEP 10. Replace the accelerator pedal position sensor.

- (1) Replace the accelerator pedal position sensor.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2127 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 11.

NO : The inspection is complete.

STEP 11. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

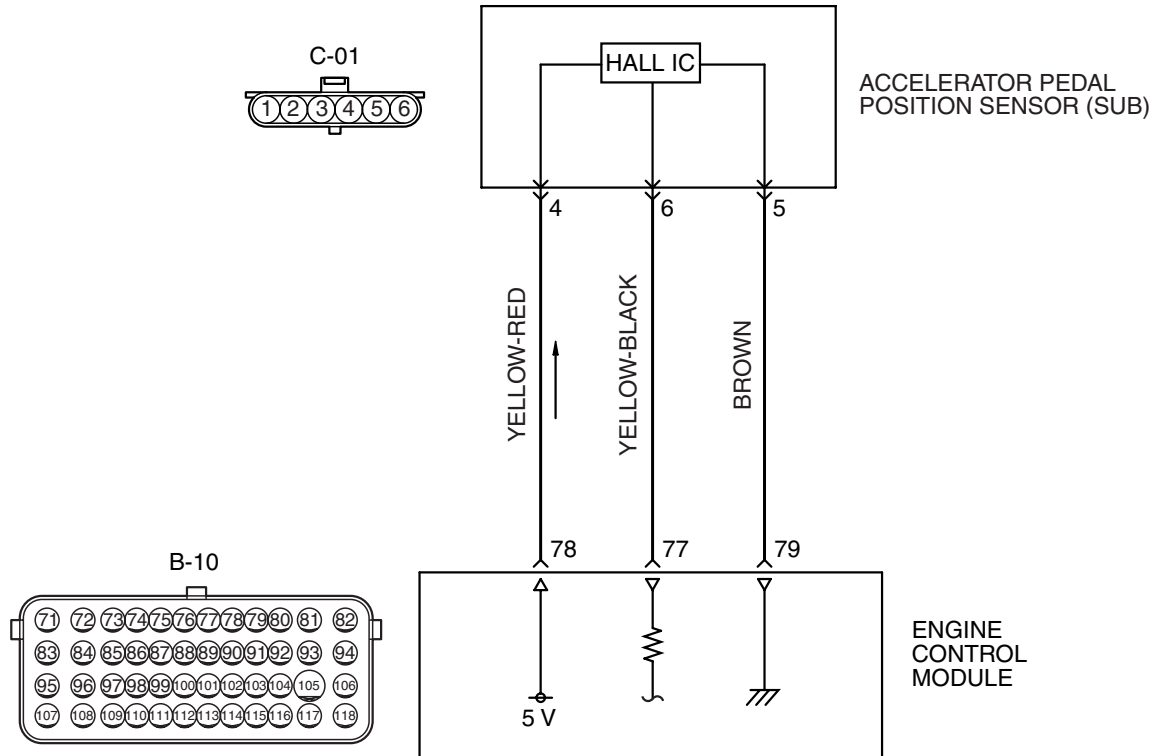
Q: Is DTC P2127 set?

YES : Retry the troubleshooting.

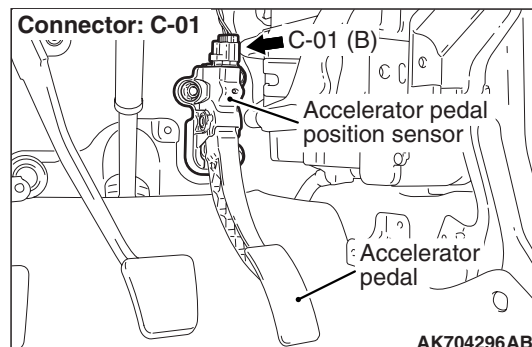
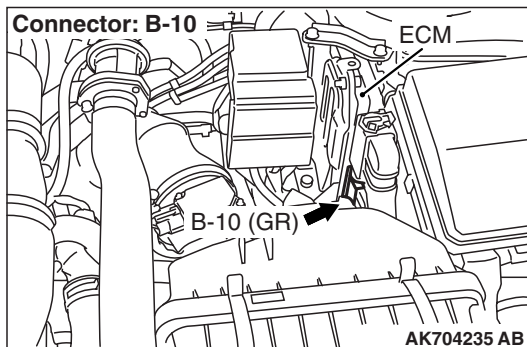
NO : The inspection is complete.

DTC P2128: Accelerator Pedal Position Sensor (sub) Circuit High Input

ACCELERATOR PEDAL POSITION SENSOR (SUB) CIRCUIT



AK604257AH



CIRCUIT OPERATION

- A 5-volt power supply is applied on the accelerator pedal position sensor (sub) power terminal (terminal No. 4) from the ECM (terminal No. 78).
- The ground terminal (terminal No. 5) is grounded with ECM (terminal No. 79).
- A voltage that is according to the accelerator opening angle is sent to the ECM (terminal No. 77) from the accelerator pedal position sensor (sub) output terminal (terminal No. 6).

TECHNICAL DESCRIPTION

- The accelerator pedal position sensor (sub) outputs voltage which corresponds to the accelerator pedal depression.
- The ECM checks whether the voltage is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Accelerator pedal position sensor (sub) output voltage is out of specified range.

MONITOR EXECUTION

Continuous

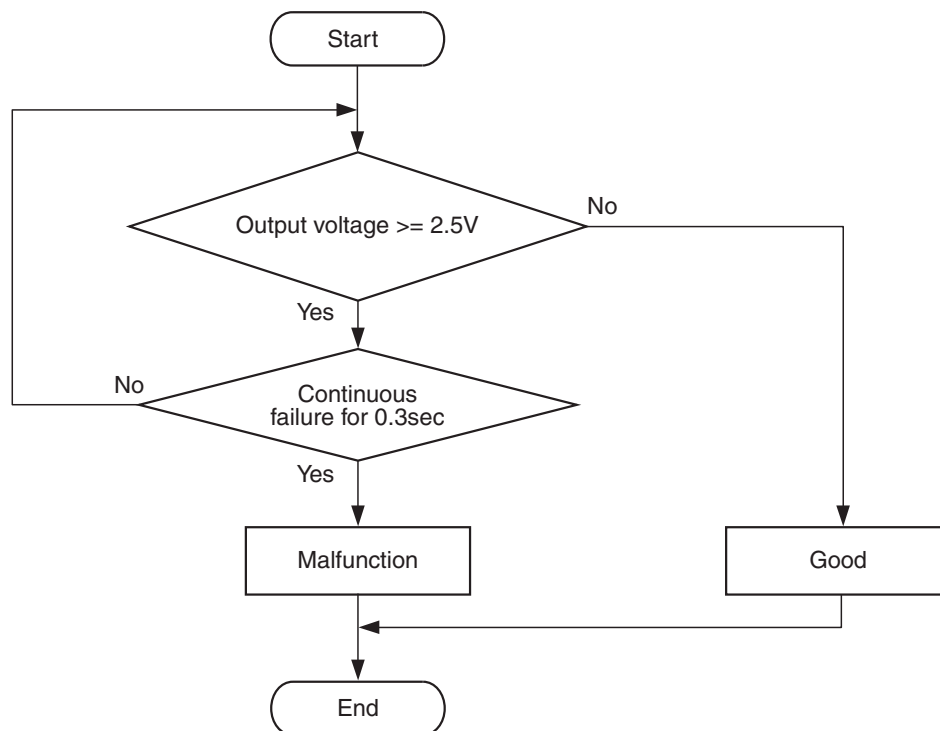
**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS**Logic Flow Chart**

AK604340

Check Condition

- Ignition switch is "ON" position.

Judgement Criterion

- Accelerator pedal position sensor (sub) output voltage is 2.5 volts or higher for 0.3 second.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if accelerator pedal position sensor (main) fails.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Accelerator pedal position sensor failed.
- Open accelerator pedal position sensor (sub) circuit, harness damage or connector damage.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, check data list item 12: Accelerator Pedal Position Sensor (sub).

CAUTION

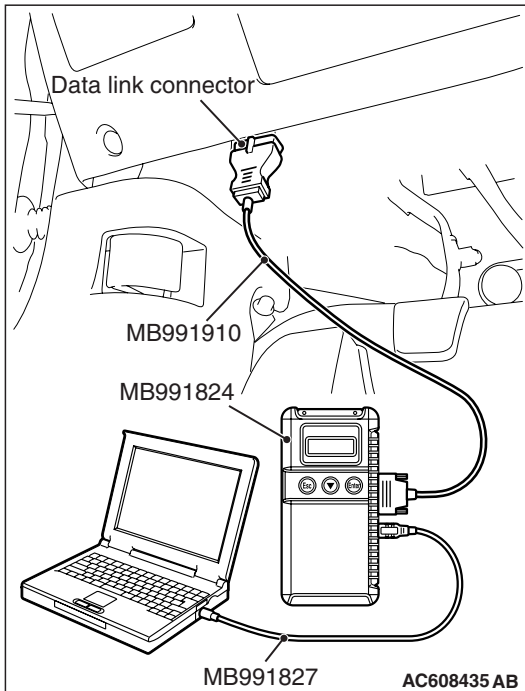
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958 to the data reading mode for item 12, Accelerator Pedal Position Sensor (sub).
 - Output voltage is between 0.4 and 0.6 volt when foot is released from accelerator pedal.
 - Output voltage is 2.0 volts or higher when accelerator pedal is fully depressed.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?

YES : It can be assumed that this malfunction is intermittent.
Refer to GROUP 00, How to Use
Troubleshooting/Inspection Service Points –How to
Cope with Intermittent Malfunctions [P.00-15](#).

NO : Go to Step 2.

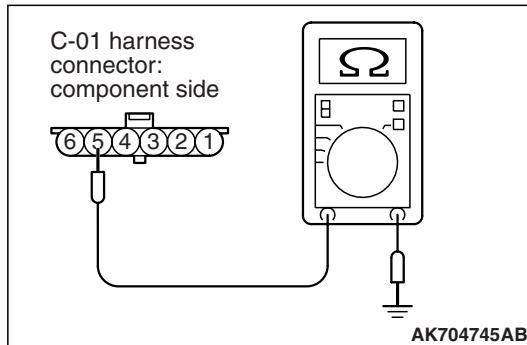


STEP 2. Check harness connector C-01 at accelerator pedal position sensor for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 3.

NO : Repair or replace it. Refer to GROUP 00E, Harness
Connector Inspection [P.00E-2](#). Then go to Step 8.

**STEP 3. Check the continuity at accelerator pedal position sensor harness side connector C-01.**

- (1) Disconnect the connector C-01 and measure at the harness side.
- (2) Measure the continuity between terminal No. 5 and ground.
 - Continuity (2 ohms or less)

Q: Does continuity exist?**YES :** Go to Step 7.**NO :** Go to Step 4.**STEP 4. Check harness connector B-10 at ECM for damage.****Q: Is the harness connector in good condition?****YES :** Go to Step 5.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 8.**STEP 5. Check for open circuit and harness damage between accelerator pedal position sensor connector C-01 (terminal No. 5) and ECM connector B-10 (terminal No. 79).****Q: Is the harness wire in good condition?****YES :** Go to Step 6.**NO :** Repair it. Then go to Step 8.**STEP 6. Using scan tool MB991958, check data list item 12: Accelerator Pedal Position Sensor (sub).**

- (1) Turn the ignition switch to the "ON" position.
- (2) Set scan tool MB991958 to the data reading mode for item 12, Accelerator Pedal Position Sensor (sub).
 - Output voltage is between 0.4 and 0.6 volt when foot is released from accelerator pedal.
 - Output voltage is 2.0 volts or higher when accelerator pedal is fully depressed.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the sensor operating properly?**YES :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).**NO :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 8.

STEP 7. Replace the accelerator pedal position sensor.

- (1) Replace the accelerator pedal position sensor.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2128 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 8.

NO : The inspection is complete.

STEP 8. Using scan tool MB991958, read the diagnostic trouble code (DTC).

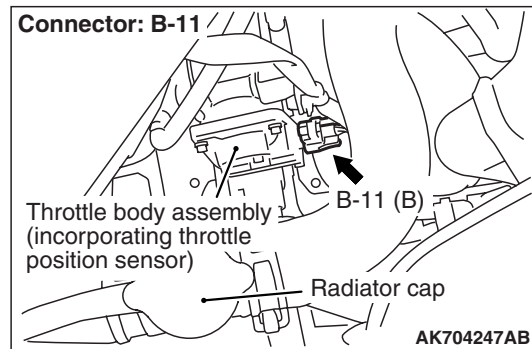
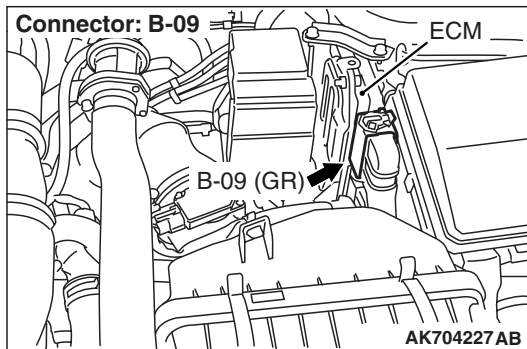
- (1) Turn the ignition switch to the "ON" position.
- (2) After the DTC has been deleted, read the DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2128 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC P2135: Throttle Position Sensor (main and sub) Range/Performance Problem



THROTTLE POSITION SENSOR (main and sub) RANGE/PERFORMANCE PROBLEM CIRCUIT

- Refer to DTC P0122 –Throttle Position Sensor (main) Circuit Low Input [P.13A-181](#).
- Refer to DTC P0222 –Throttle Position Sensor (sub) Circuit Low Input [P.13A-317](#).

CIRCUIT OPERATION

- Refer to DTC P0122 –Throttle Position Sensor (main) Circuit Low Input [P.13A-181](#).
- Refer to DTC P0222 –Throttle Position Sensor (sub) Circuit Low Input [P.13A-317](#).

TECHNICAL DESCRIPTION

- ECM checks the throttle position sensor output signal characteristics for abnormal conditions.

DESCRIPTIONS OF MONITOR METHODS

Detect malfunction if the relation between throttle position sensor (main) and throttle position sensor (sub) is wrong.

MONITOR EXECUTION

Continuous

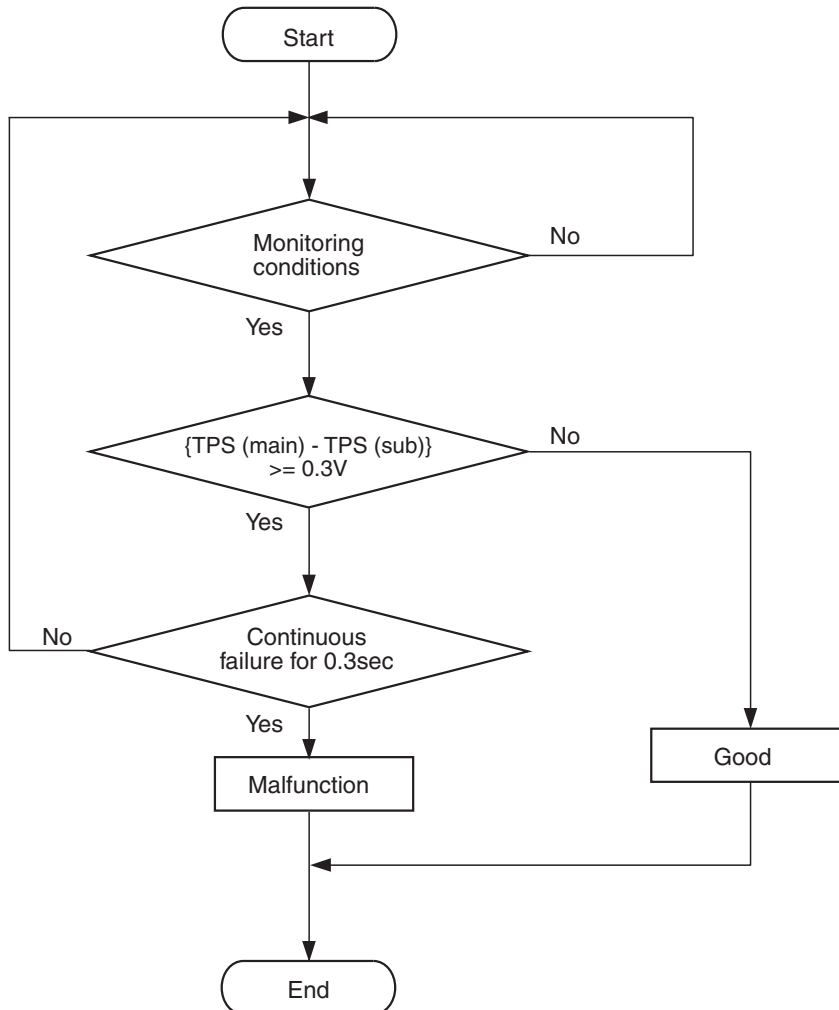
MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS <Range/Performance problem-relation between main and sub>**Logic Flow Chart**

AK604361

Check Conditions

- The target output voltage of the throttle position sensor is 0.9 volt or less.
- Throttle position sensor (main) output voltage is between 0.2 and 4.8 volts.
- Throttle position sensor (sub) output voltage is between 0.2 and 4.8 volts.
- Battery positive voltage is higher than 8.3 volts.

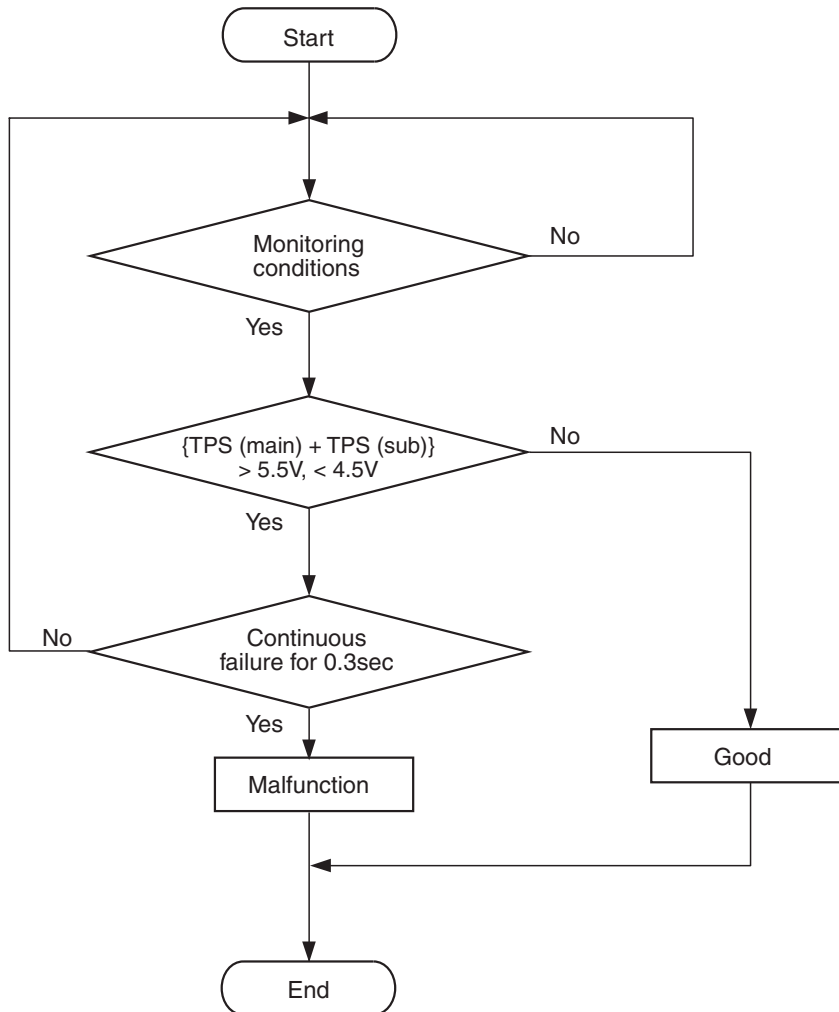
Judgement Criterion

- Voltage obtained with the formula given below is 0.3 volt or higher for 0.3 second: [throttle position sensor (main) voltage – throttle position sensor (sub) voltage]

NOTE: The throttle position sensor voltage used for the judgement is converted into the throttle position sensor voltage for the internal processing by the ECM.

DTC SET CONDITIONS <Range/Performance problem-relation between main and sub>

Logic Flow Chart



AK604362

Check Conditions

- Throttle position sensor (main) output voltage is between 0.2 and 4.8 volts.
- Throttle position sensor (sub) output voltage is between 0.2 and 4.8 volts.

Judgement Criterion

- For 0.3 second, the sum of the output voltage of the throttle position sensors (main) and (sub) is 4.5 V or less, or 5.5 V or more.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree position is in default position.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

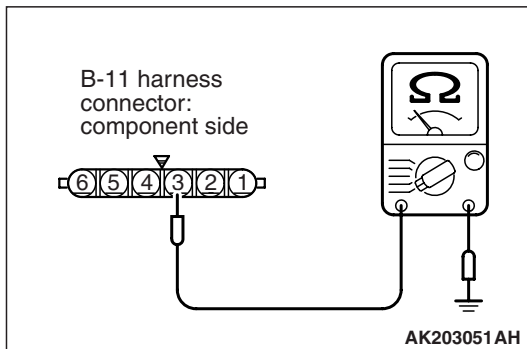
- Throttle position sensor failed.
- Harness damage in throttle position sensor circuit or connector damage.
- ECM failed.

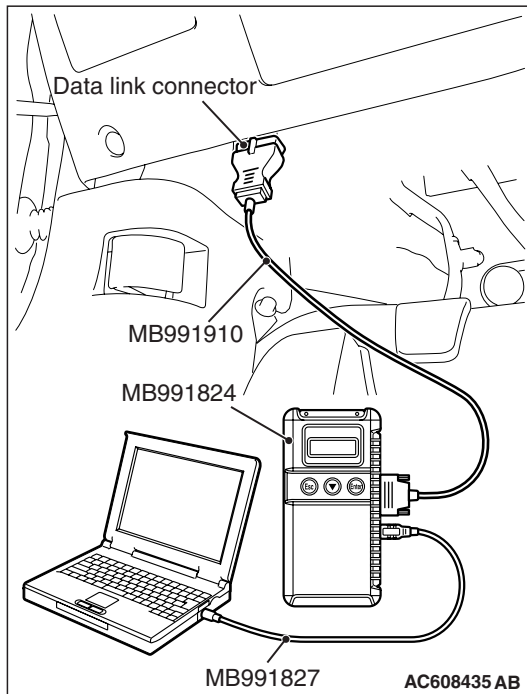
DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector B-11 at throttle position sensor for damage.**Q: Is the harness connector in good condition?****YES :** Go to Step 2.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.**STEP 2. Check the continuity at throttle position sensor harness side connector B-11.**

- (1) Disconnect the connector B-11 and measure at the harness side.
- (2) Measure the continuity between terminal No. 3 and ground.
 - Continuity (2 ohms or less)

Q: Does continuity exist?**YES :** Go to Step 6.**NO :** Go to Step 3.**STEP 3. Check harness connector B-09 at ECM for damage.****Q: Is the harness connector in good condition?****YES :** Go to Step 4.**NO :** Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.**STEP 4. Check for harness damage between throttle position sensor connector B-11 (terminal No. 3) and ECM connector B-09 (terminal No. 13).****Q: Is the harness wire in good condition?****YES :** Go to Step 5.**NO :** Repair it. Then go to Step 11.



STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTC.
- (4) Depress the accelerator pedal fully for a few seconds.
- (5) Check the DTC.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2135 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 11.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 6. Check harness connector B-09 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 7.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 11.

STEP 7. Check for harness damage between throttle position sensor connector B-11 (terminal No. 5) and ECM connector B-09 (terminal No. 12).

Q: Is the harness wire in good condition?

YES : Go to Step 8.

NO : Repair it. Then go to Step 11.

STEP 8. Check for harness damage between throttle position sensor connector B-11 (terminal No. 4) and ECM connector B-09 (terminal No. 10).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 11.

STEP 9. Check for harness damage between throttle position sensor connector B-11 (terminal No. 6) and ECM connector B-09 (terminal No. 11).

Q: Is the harness wire in good condition?

YES : Go to Step 10.

NO : Repair it. Then go to Step 11.

STEP 10. Replace the throttle body assembly.

- (1) Replace the throttle body assembly.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTC.
- (4) Depress the throttle fully for a few seconds.
- (5) Check the DTC.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2135 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 11.

NO : The inspection is complete.

STEP 11. Using scan tool MB991958, read the diagnostic trouble code (DTC).

CAUTION

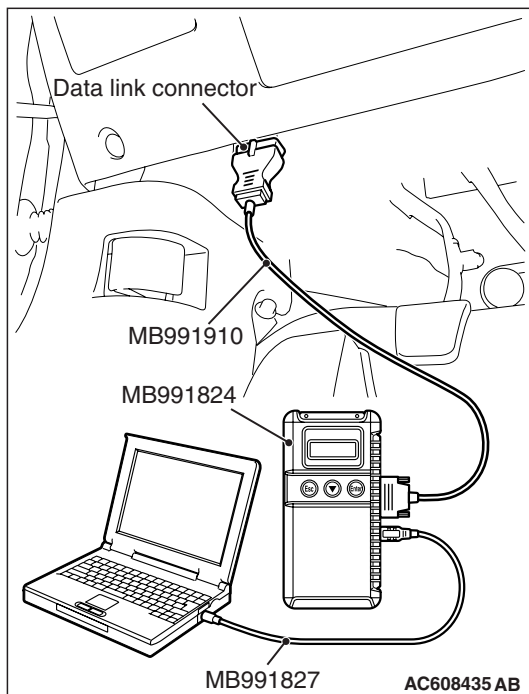
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Erase the DTC.
- (4) Depress the throttle fully for a few seconds.
- (5) Check the DTC.
- (6) Turn the ignition switch to the "LOCK" (OFF) position.

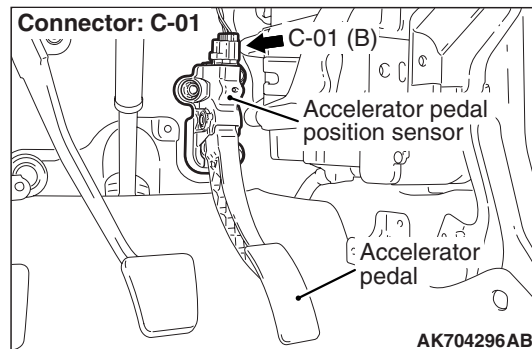
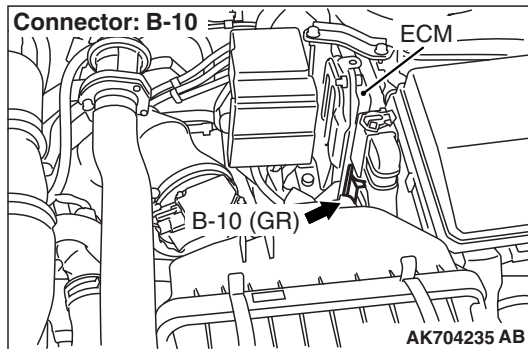
Q: Is DTC P2135 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.



DTC P2138: Accelerator Pedal Position Sensor (main and sub) Range/Performance Problem



ACCELERATOR PEDAL POSITION SENSOR (main and sub) RANGE/PERFORMANCE PROBLEM CIRCUIT

- Refer to DTC P2122 –Accelerator Pedal Position Sensor (main) Circuit Low Input [P.13A-672](#).
- Refer to DTC P2127 –Accelerator Pedal Position Sensor (sub) Circuit Low Input [P.13A-682](#).

CIRCUIT OPERATION

- Refer to DTC P2122 –Accelerator Pedal Position Sensor (main) Circuit Low Input [P.13A-672](#).
- Refer to DTC P2127 –Accelerator Pedal Position Sensor (sub) Circuit Low Input [P.13A-682](#).

TECHNICAL DESCRIPTION

- ECM checks the accelerator pedal position sensor output signal characteristics for abnormal conditions.

DESCRIPTIONS OF MONITOR METHODS

Detect malfunction if the relation between accelerator pedal position sensor (main) and accelerator pedal position sensor (sub) is wrong.

MONITOR EXECUTION

Continuous

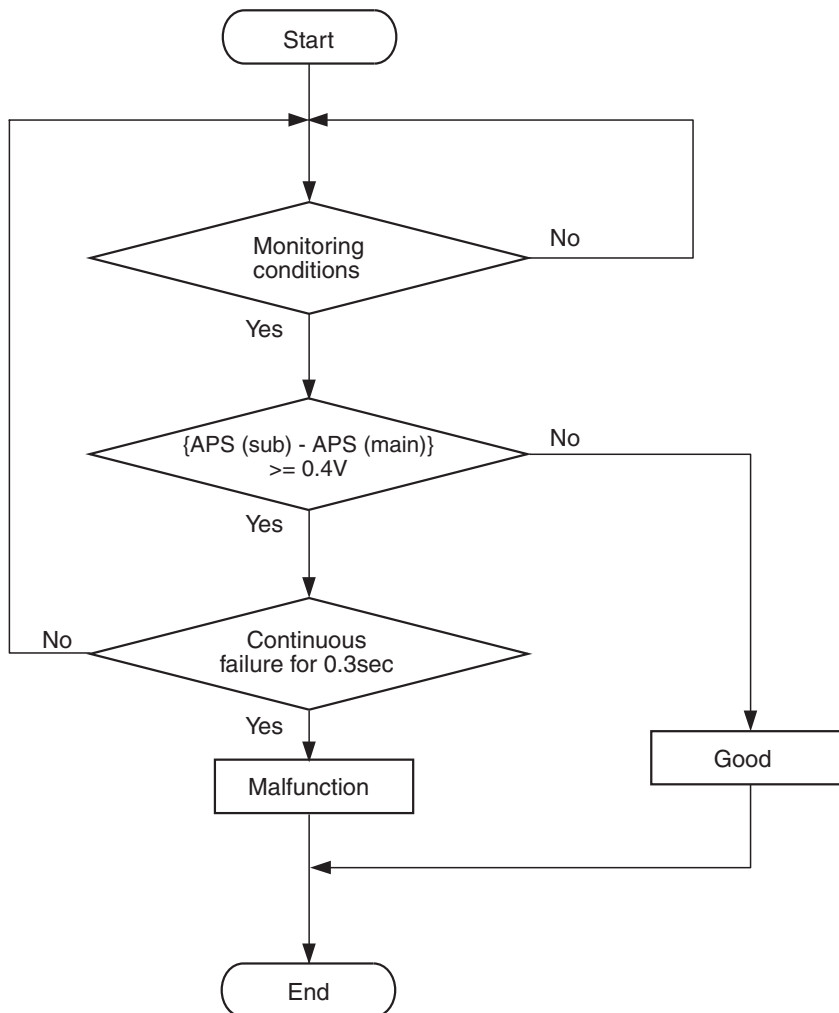
MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS <Range/Performance problem -relation between main and sub>**Logic Flow Chart**

AK604363

Check Conditions

- Change of accelerator pedal position sensor (sub) output voltage per 25 milliseconds is lower than 0.06 volt.

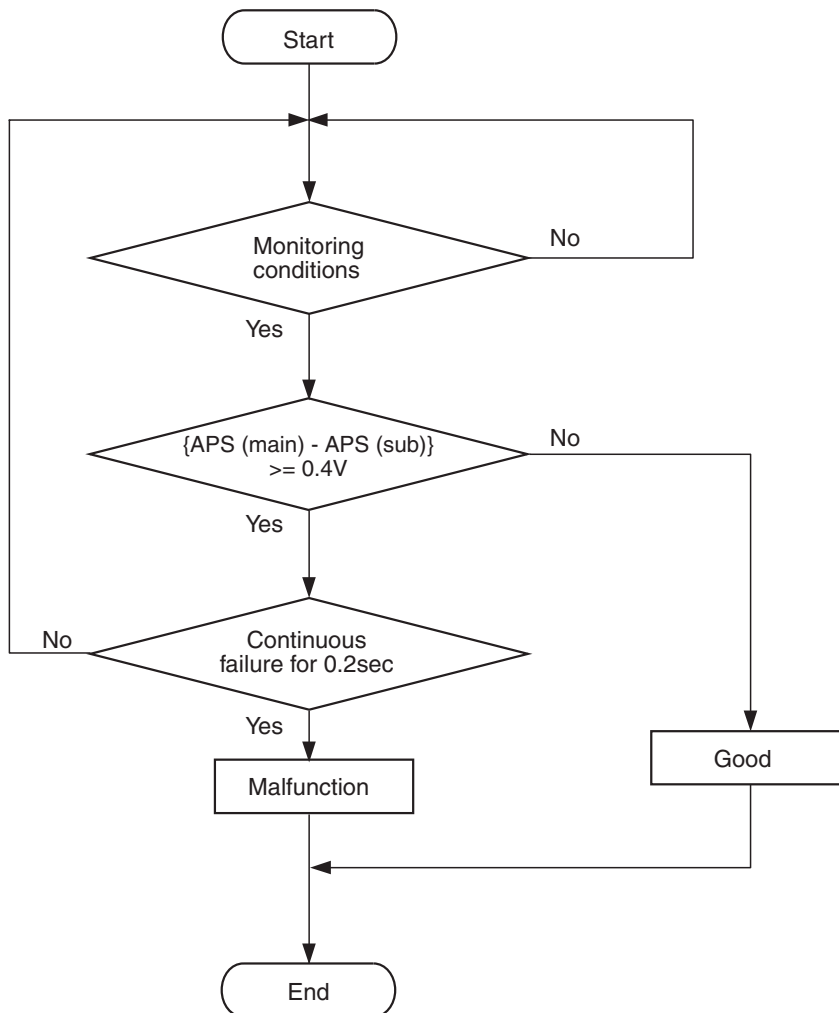
Judgement Criterion

- Voltage obtained with the formula given below is 0.4 volt or higher for 0.3 second: [accelerator pedal position sensor (sub) voltage – accelerator pedal position sensor (main) voltage].

NOTE: The accelerator pedal position sensor voltage used for the judgement is converted into the accelerator pedal position sensor voltage for the internal processing by the ECM.

DTC SET CONDITIONS <Range/Performance problem -relation between main and sub>

Logic Flow Chart



AK604364

Check Conditions

- Change of accelerator pedal position sensor (sub) output voltage per 25 milliseconds is lower than 0.06 volt.

Judgement Criterion

- Voltage obtained with the formula given below is 0.4 volt or higher for 0.2 second: [accelerator pedal position sensor (main) voltage – accelerator pedal position sensor (sub) voltage].

NOTE: The accelerator pedal position sensor voltage used for the judgement is converted into the accelerator pedal position sensor voltage for the internal processing by the ECM.

FAIL-SAFE AND BACKUP FUNCTION

- Throttle opening degree is restricted.
- Throttle opening degree position is in default position if accelerator pedal position sensor (sub) fails.

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

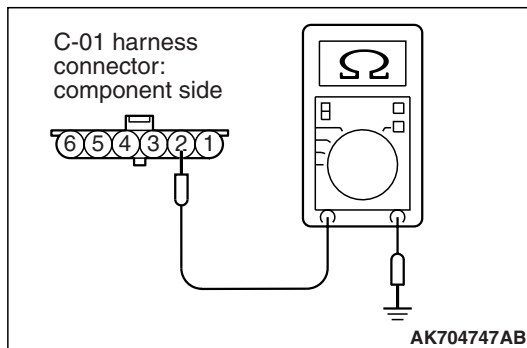
- Accelerator pedal position sensor failed.
- Harness damage in accelerator pedal position sensor circuit or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Check harness connector C-01 at accelerator pedal position sensor for damage.**Q: Is the harness connector in good condition?****YES** : Go to Step 2.**NO** : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 16.**STEP 2. Check the continuity at accelerator pedal position sensor harness side connector C-01.**

- (1) Disconnect the connector C-01 and measure at the harness side.
- (2) Measure the continuity between terminal No. 2 and ground.
 - Continuity (2 ohms or less)

Q: Does continuity exist?**YES** : Go to Step 6.**NO** : Go to Step 3.**STEP 3. Check harness connector B-10 at ECM for damage.****Q: Is the harness connector in good condition?****YES** : Go to Step 4.**NO** : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 16.**STEP 4. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 2) and ECM connector B-10 (terminal No. 76).****Q: Is the harness wire in good condition?****YES** : Go to Step 5.**NO** : Repair it. Then go to Step 16.

STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

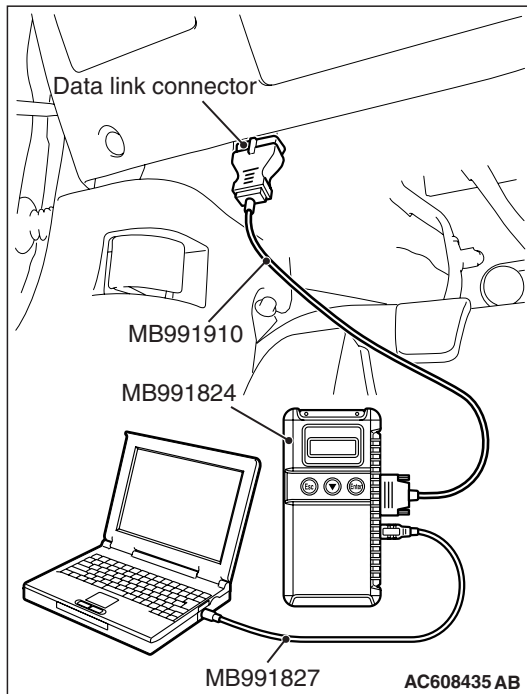
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2138 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 16.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



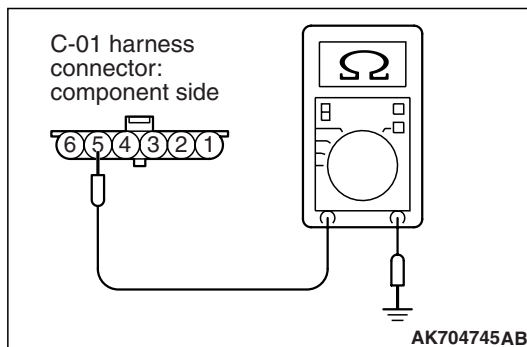
STEP 6. Check the continuity at accelerator pedal position sensor harness side connector C-01.

- (1) Disconnect the connector C-01 and measure at the harness side.
- (2) Measure the continuity between terminal No. 5 and ground.
 - Continuity (2 ohms or less)

Q: Does continuity exist?

YES : Go to Step 10.

NO : Go to Step 7.



STEP 7. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 8.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 16.

STEP 8. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 5) and ECM connector B-10 (terminal No. 79).

Q: Is the harness wire in good condition?

YES : Go to Step 9.

NO : Repair it. Then go to Step 16.

STEP 9. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

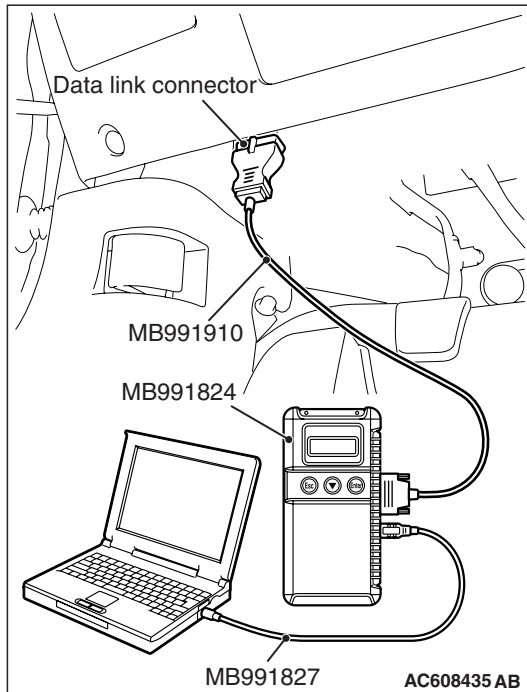
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2138 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 16.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



STEP 10. Check harness connector B-10 at ECM for damage.

Q: Is the harness connector in good condition?

YES : Go to Step 11.

NO : Repair or replace it. Refer to GROUP 00E, Harness Connector Inspection [P.00E-2](#). Then go to Step 16.

STEP 11. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 1) and ECM connector B-10 (terminal No. 75).

Q: Is the harness wire in good condition?

YES : Go to Step 12.

NO : Repair it. Then go to Step 16.

STEP 12. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 4) and ECM connector B-10 (terminal No. 78).

Q: Is the harness wire in good condition?

YES : Go to Step 13.

NO : Repair it. Then go to Step 16.

STEP 13. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 3) and ECM connector B-10 (terminal No. 74).

Q: Is the harness wire in good condition?

YES : Go to Step 14.

NO : Repair it. Then go to Step 16.

STEP 14. Check for harness damage between accelerator pedal position sensor connector C-01 (terminal No. 6) and ECM connector B-10 (terminal No. 77).

Q: Is the harness wire in good condition?

YES : Go to Step 15.

NO : Repair it. Then go to Step 16.

STEP 15. Replace the accelerator pedal position sensor.

- (1) Replace the accelerator pedal position sensor.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC P2138 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 16.

NO : The inspection is complete.

STEP 16. Using scan tool MB991958, read the diagnostic trouble code (DTC).

⚠ CAUTION

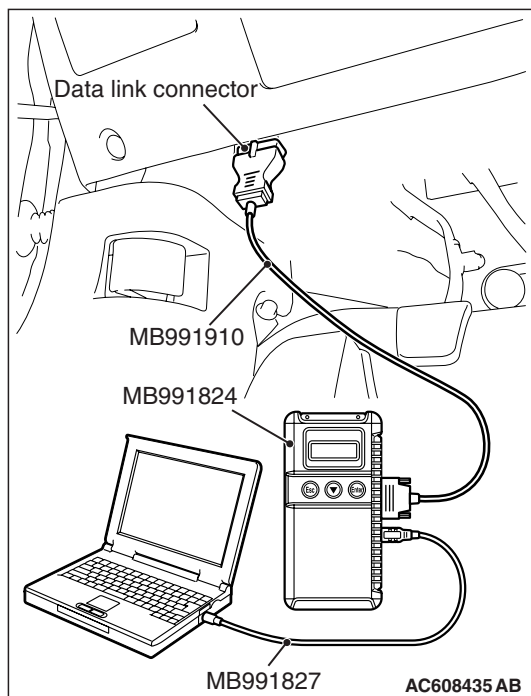
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) After the DTC has been deleted, read the DTC again.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

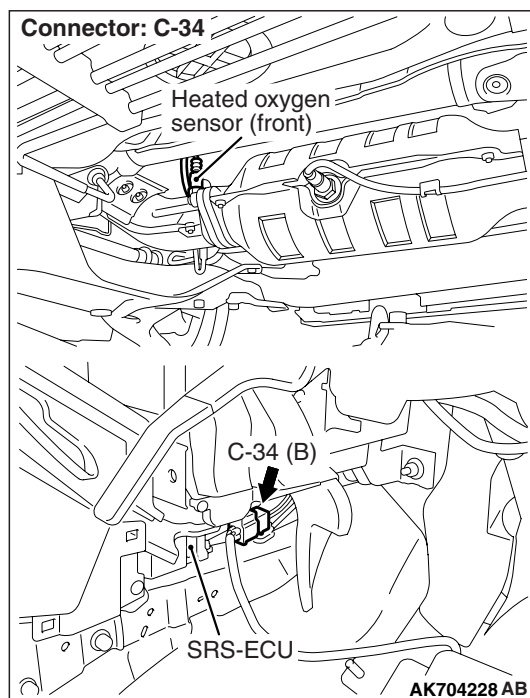
Q: Is DTC P2138 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.



DTC P2195: Heated Oxygen Sensor (front) Inactive



TECHNICAL DESCRIPTION

- The ECM effects air/fuel ratio feedback control in accordance with the signals from the heated oxygen sensor (front).
- If the heated oxygen sensor (front) has deteriorated, corrections will be made by the heated oxygen sensor (rear).
- DTC P2195 becomes stored in memory if a failure is detected in the heated oxygen sensor (front).

DESCRIPTIONS OF MONITOR METHODS

Heated oxygen sensor (front) output voltage does not cross lean/rich criteria (about 0.5 volt) within specified period.

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

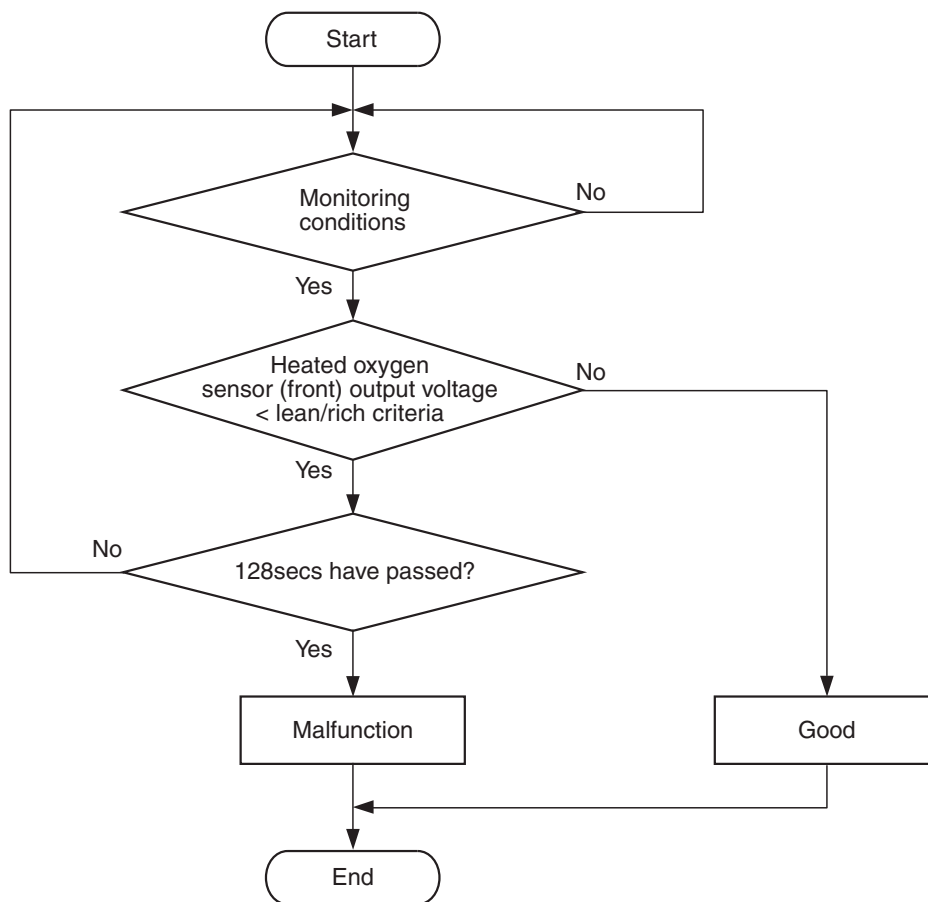
- Heated oxygen sensor (front) heater monitor
- Misfire monitor

Sensor (The sensor below is determined to be normal)

- Engine coolant temperature sensor
- Intake air temperature sensor

DTC SET CONDITIONS

Logic Flow Chart



AK604365

Check Conditions

- More than 20 seconds have passed since the engine starting sequence was completed.

- Engine coolant temperature is higher than 7° C (45° F).
- Under the closed loop air/fuel ratio control.
- Intake air temperature is more than -10° C (14° F).

Judgement Criterion

- Heated oxygen sensor (front) output voltage is less than lean/rich criteria (about 0.5 volt) for 128 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 22 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Heated oxygen sensor (front) deteriorated.
- ECM failed.

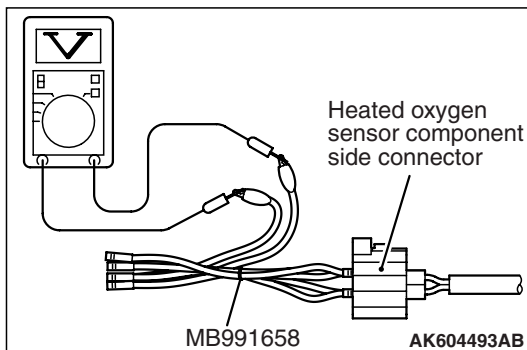
DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A
- MB991658: Test Harness

STEP 1. Check the heated oxygen sensor (front).

- (1) Disconnect the heated oxygen sensor (front) connector C-34 and connect test harness special tool MB991658 to the connector on the heated oxygen sensor (front) side.
- (2) Warm up the engine until engine coolant temperature reaches 80° C (176° F) or higher.
- (3) Rev the engine for 5 minutes or more with the engine speed of 2,000 r/min.
- (4) Connect a digital voltage meter between terminal No. 3 and terminal No. 4.
- (5) While repeatedly revving the engine, measure the heated oxygen sensor (front) output voltage.

Standard value: 0.6 –1.0 V

**CAUTION**

- Be very careful when connecting the jumper wires; incorrect connection can damage the heated oxygen sensor.
- Be careful the heater can be damaged if a voltage beyond 8 volts is applied to the heated oxygen sensor heater.

NOTE: If the sufficiently high temperature [of approximate 400° C (752° F) or more] is not reached although the heated oxygen sensor is normal, the output voltage would be possibly low although the rich air/fuel ratio. Therefore, if the output voltage is low, use a jumper wire to connect the terminal No. 1 and the terminal No. 2 of the heated oxygen sensor with the positive terminal and the negative terminal of 8 volts power supply respectively, then check again.

Q: Is the measured voltage between 0.6 and 1.0 volt?

YES : Go to Step 2.

NO : Replace the heated oxygen sensor (front). Then go to Step 3.

STEP 2. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 22 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P2195 set?

- YES :** Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.
- NO :** It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Test the OBD-II drive cycle.

- (1) Carry out a test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 22 [P.13A-9](#).
- (2) Check the diagnostic trouble code (DTC).

Q: Is DTC P2195 set?

- YES :** Retry the troubleshooting.
- NO :** The inspection is complete.

DTC P2228: Barometric Pressure Circuit Low Input

TECHNICAL DESCRIPTION

- The barometric pressure sensor outputs a voltage which corresponds to the barometric pressure.
- The ECM checks whether this voltage is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Barometric pressure sensor output voltage is out of reasonable barometric pressure range.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

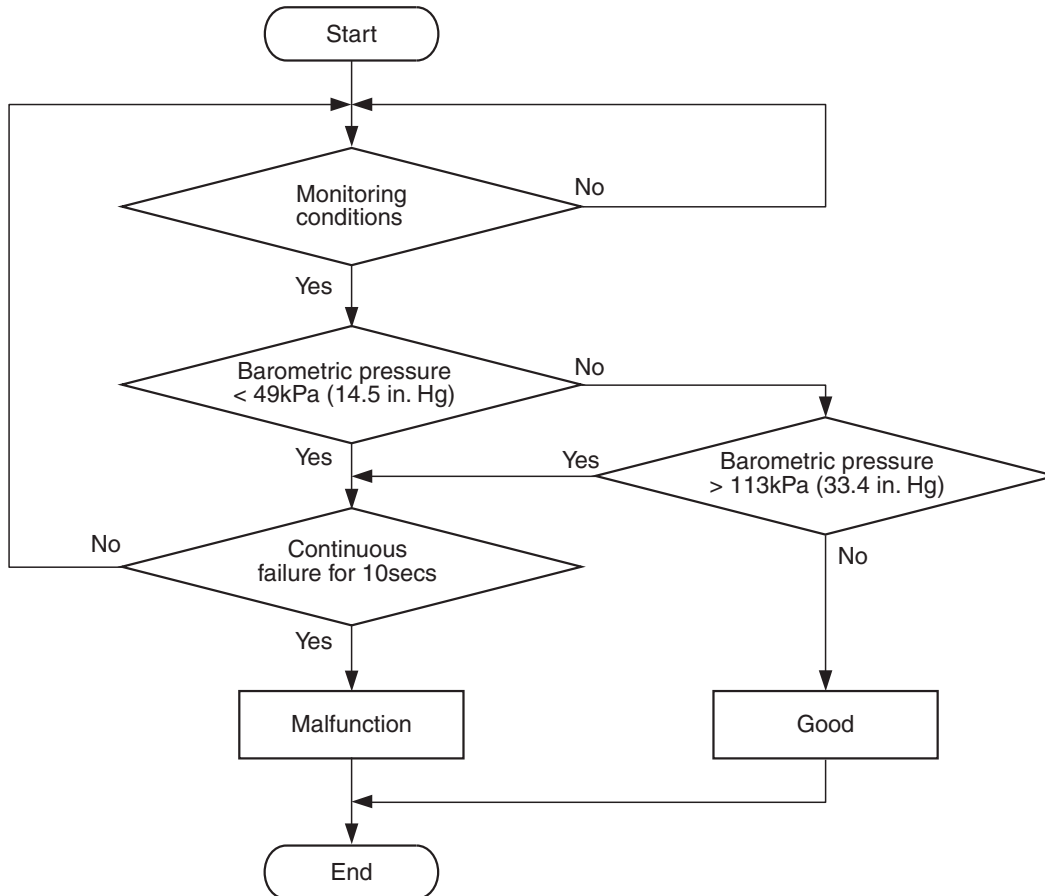
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK704299

Check Conditions

- 2 seconds or more have passed since the engine starting sequence was completed.
- Battery positive voltage is 8 volts or higher.

Judgement Criterion

- Barometric pressure sensor output signal has continued to be 49 kPa (14.5 in.Hg) or lower (approximately 15,000 ft above sea level) for 10 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- Control as if the barometric pressure is 101kPa.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Test the OBD-II drive cycle.

⚠ CAUTION

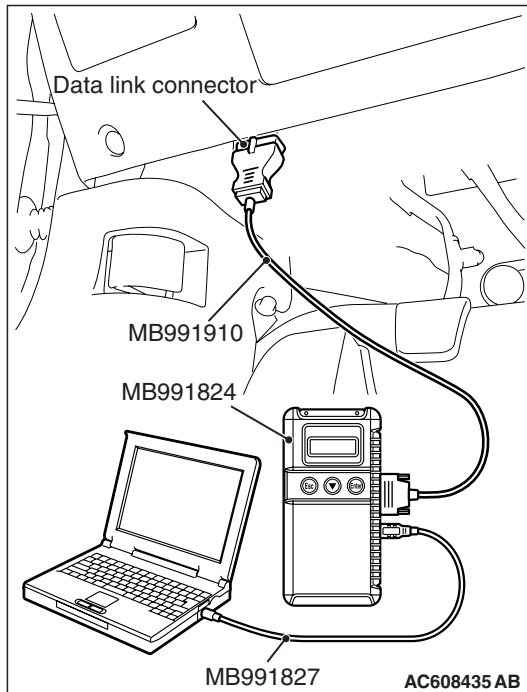
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Carry out the test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (4) Check the diagnostic trouble code (DTC).

Q: Is DTC P2228 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P2229: Barometric Pressure Circuit High Input

TECHNICAL DESCRIPTION

- The barometric pressure sensor outputs a voltage which corresponds to the barometric pressure.
- The ECM checks whether this voltage is within a specified range.

DESCRIPTIONS OF MONITOR METHODS

Barometric pressure sensor output voltage is out of reasonable barometric pressure range.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

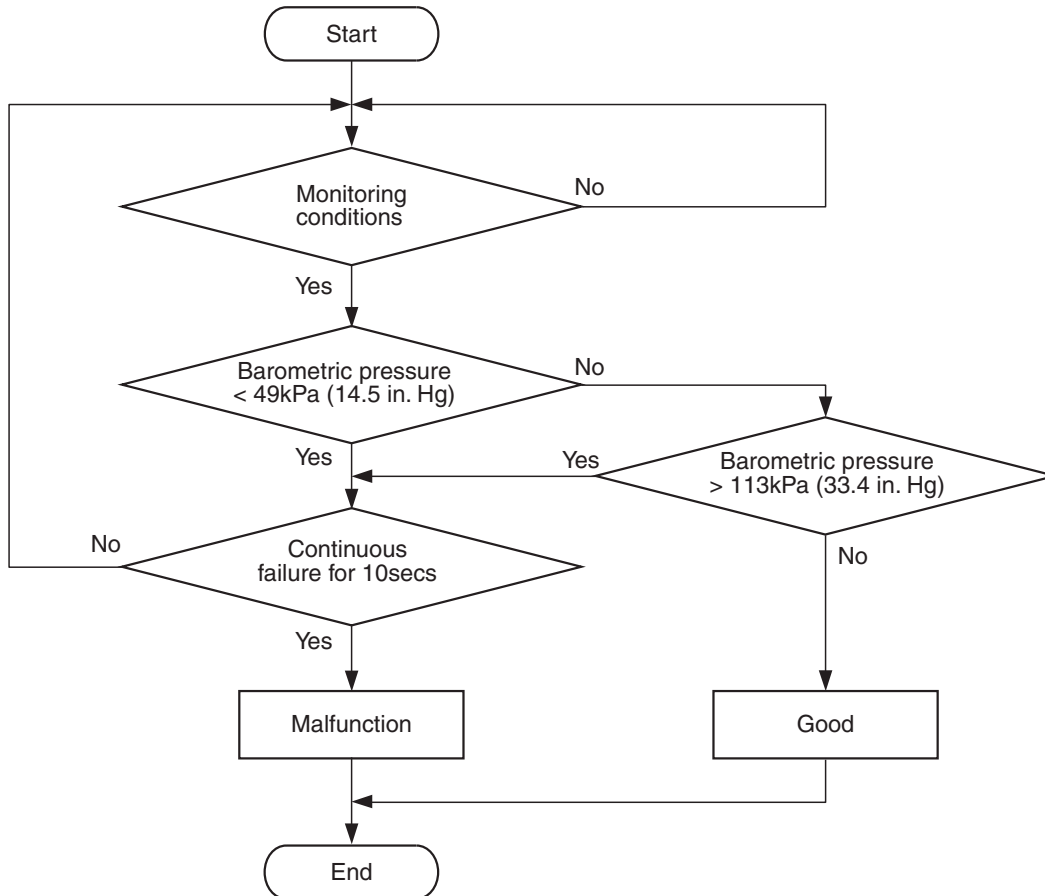
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK704299

Check Conditions

- 2 seconds or more have passed since the engine starting sequence was completed.
- Battery positive voltage is 8 volts or higher.

Judgement Criterion

- Barometric pressure sensor output signal has continued to be 113 kPa (33.4 in.Hg) or higher (approximately 4,000 ft below sea level) for 10 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- Control as if the barometric pressure is 101kPa.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnostic Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Test the OBD-II drive cycle.

⚠ CAUTION

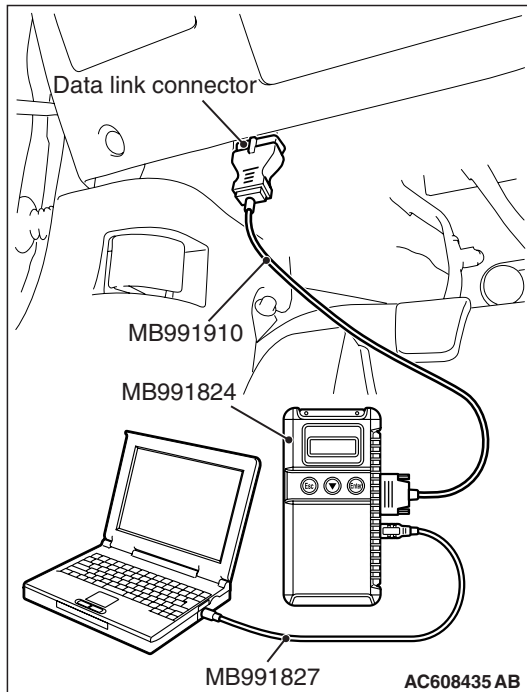
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Carry out the test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (4) Check the diagnostic trouble code (DTC).

Q: Is DTC P2229 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P2252: Heated Oxygen Sensor Offset Circuit Low Voltage

TECHNICAL DESCRIPTION

- The ECM applies an offset voltage of 0.5 volt to the heated oxygen sensor.
- The ECM checks for heated oxygen sensor offset voltage.

DESCRIPTIONS OF MONITOR METHODS

Heated oxygen sensor offset voltage is under specified range.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

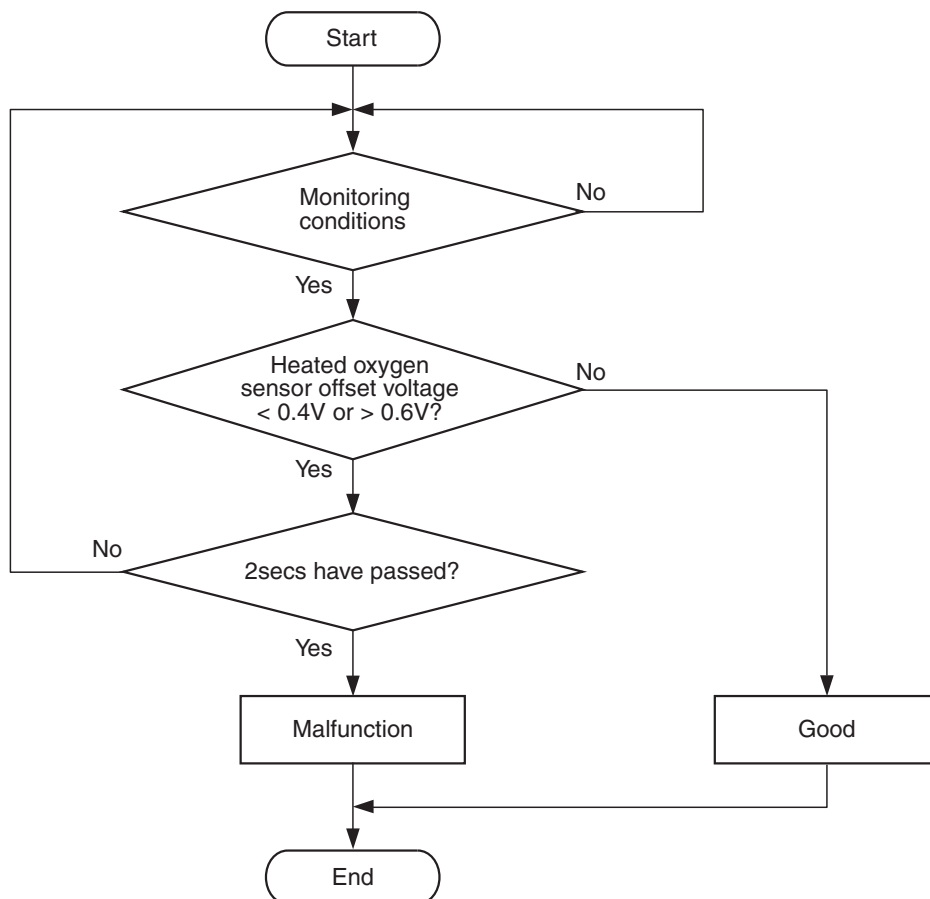
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITION

Logic Flow Chart



AK704300

Check Condition

- 2 seconds or more have passed since the engine starting sequence was completed.

Judgement Criterion

- Heated oxygen sensor offset voltage has continued to be 0.4 volt or lower for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnosis Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Test the OBD-II drive cycle.

⚠ CAUTION

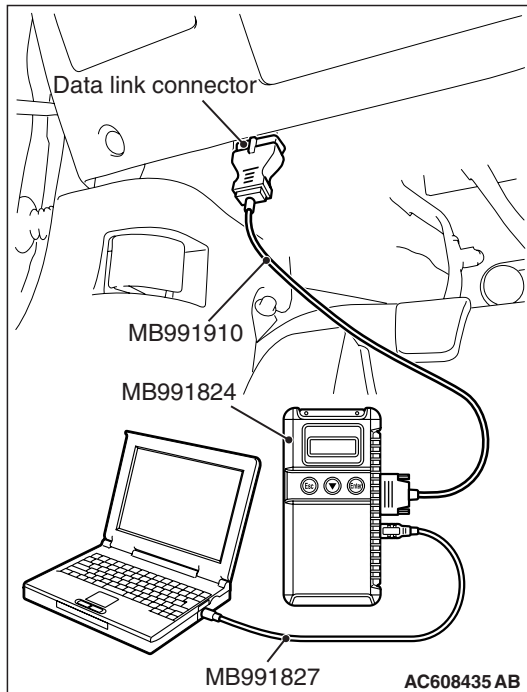
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Carry out the test drive with the drive cycle pattern. Refer to Diagnostic Function –OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (4) Check the diagnostic trouble code (DTC).

Q: Is DTC P2252 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P2253: Heated Oxygen Sensor Offset Circuit High Voltage

TECHNICAL DESCRIPTION

- The ECM applies an offset voltage of 0.5 volt to the heated oxygen sensor
- The ECM checks for heated oxygen sensor offset voltage.

DESCRIPTIONS OF MONITOR METHODS

Heated oxygen sensor offset voltage is over specified range.

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

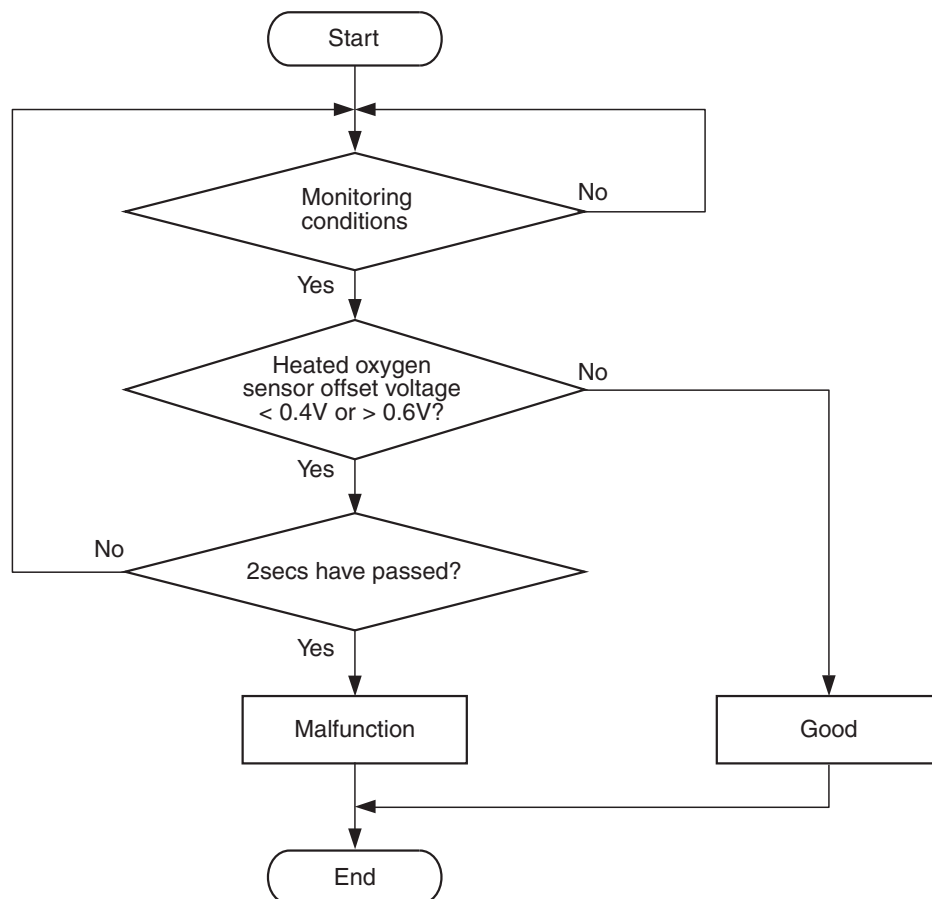
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITION

Logic Flow Chart



AK704300

Check Condition

- 2 seconds or more have passed since the engine starting sequence was completed.

Judgement Criterion

- Heated oxygen sensor offset voltage has continued to be 0.6 volt or higher for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- Does not control air-fuel ratio closed loop.

OBD-II DRIVE CYCLE PATTERN

Refer to Diagnosis Function –OBD-II Drive Cycle – Pattern 23 [P.13A-9](#).

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Test the OBD-II drive cycle.

⚠ CAUTION

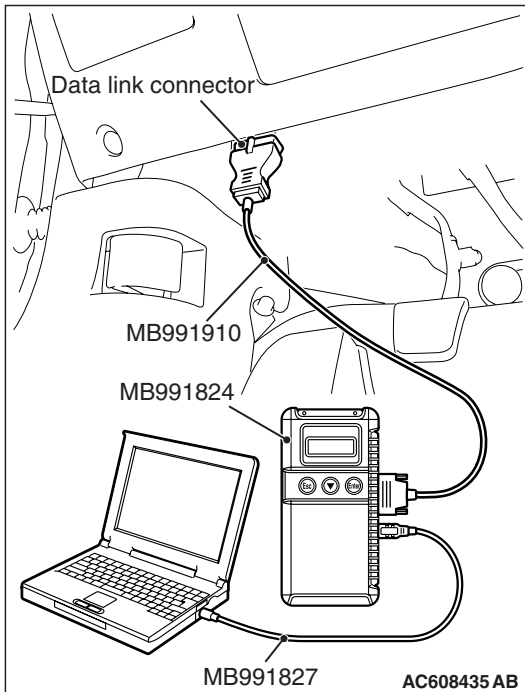
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Carry out the test drive with the drive cycle pattern. Refer to Diagnostic Function OBD-II Drive Cycle –Pattern 23 [P.13A-9](#).
- (4) Check the diagnostic trouble code (DTC).

Q: Is DTC P2253 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).



DTC P2263: Intake Charge System Malfunction

TECHNICAL DESCRIPTION

- The ECM indicates or monitors a minimum prescribed value of volumetric efficiency within the boost operation range, which is determined by the signals from the throttle position sensor and the engine speed.

DTC SET CONDITIONS

Check Conditions

- Engine speed is between 3,000 and 5,000 r/min.
- Throttle position sensor (main) output voltage is higher than 2 volts.

Judgment Criterion

- Volumetric efficiency is lower than 100 percent for 2 seconds.

FAIL-SAFE AND BACKUP FUNCTION

- Fuel is cut in engine overboost condition.

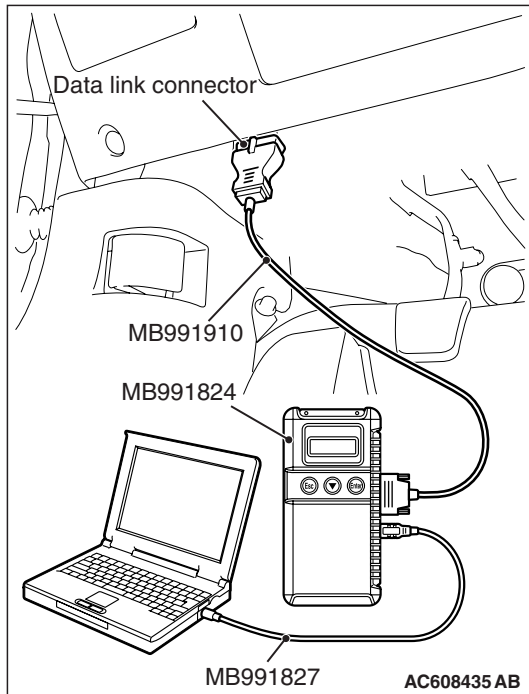
TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- Turbocharger wastegate actuator failed.
- Turbocharger wastegate regulating valve failed.
- Charging pressure control system failed.
- Intake charge pressure leak.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan Tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A



STEP 1. Using scan tool MB991958, check actuator test item 20: Turbocharger Wastegate Solenoid.

⚠ CAUTION

To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Set scan tool MB991958 to the actuator test mode for item 20, turbocharger wastegate solenoid.
 - An operation sound should be heard and vibration should be felt when the turbocharger wastegate solenoid is operated.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the solenoid operating properly?

YES : Go to Step 2.

NO : Refer to DTC P0243 –Turbocharger Wastegate Solenoid 1 Circuit [P.13A-330](#), DTC P0247 – Turbocharger Wastegate Solenoid 2 Circuit [P.13A-337](#).

STEP 2. Check the intake charge pressure.

Refer to GROUP 15, On-vehicle Service –Intake Charge Pressure Check. [P.15-5](#)

Q: Are there any abnormalities?

YES : Repair or replace it. Then go to Step 5.

NO : Go to Step 3.

STEP 3. Check the intake charge pressure control system.

Refer to GROUP 15, On-vehicle Service –Intake Charge Pressure Control System Check. [P.15-7](#)

Q: Are there any abnormalities?

YES : Repair or replace it. Then go to Step 5.

NO : Go to Step 4.

STEP 4. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Set scan tool MB991958 to the data reading mode for item 13, Throttle Position Sensor (main).

 CAUTION

During a test drive, the data list must be read by someone other than the driver.

- (3) Accelerate the vehicle within the engine speed range of 3,000 to 5,000 r/min while a minimum of 2 volts is being output by the throttle position sensor (main).
- (4) After completing the test drive, read the DTC. Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the DTC P2263 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID code Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 5.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/ Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Set scan tool MB991958 to the data reading mode for item 13, Throttle Position Sensor (main).

 CAUTION

During a test drive, the data list must be read by someone other than the driver.

- (3) Accelerate the vehicle within the engine speed range of 3,000 to 5,000 r/min while a minimum of 2 volts is being output by the throttle position sensor (main).
- (4) After completing the test drive, read the DTC. Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the DTC P2263 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC U0001: Bus Off**⚠ CAUTION**

- If the ECM output the DTC U0001, make sure to diagnose the CAN bus line.
- Before replacing the ECU, make sure that the communication circuit is operating normally.

DTC SET CONDITIONS**Check Conditions**

- Always

Judgement Criteria

- Bus off error detected

FAIL-SAFE AND BACKUP FUNCTION

- None

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- ECM failed.

DIAGNOSIS**Required Special Tools:**

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, diagnose CAN bus line.**⚠ CAUTION**

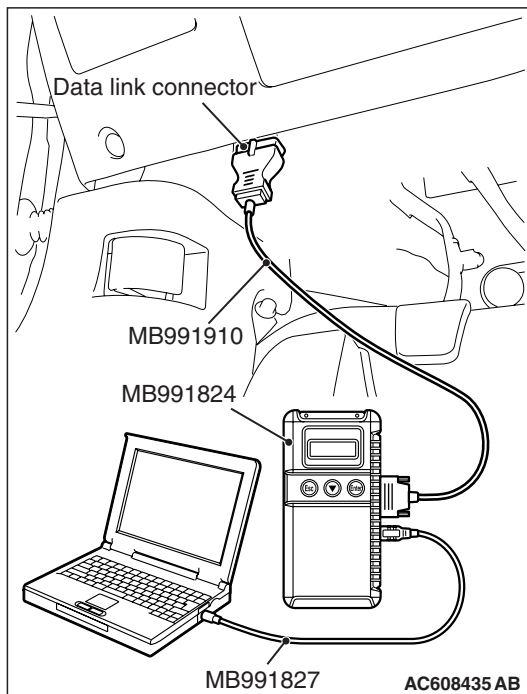
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2.

NO : Repair the CAN bus line. Refer to GROUP 54C, Can bus Diagnostics Table [P.54C-14](#). Then go to Step 3.



STEP 2. Recheck for diagnostic trouble code.

Check again if the DTC is set.

- (1) Erase the DTC.
- (2) Turn the ignition switch to the "ON" position.
- (3) Check if the DTC is set.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0001 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 3.

NO : It can be assumed that this malfunction is intermittent. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 3. Recheck for diagnostic trouble code.

Check again if the DTC is set.

- (1) Erase the DTC.
- (2) Turn the ignition switch to the "ON" position.
- (3) Check if DTC is set.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0001 set?

YES : Replay the troubleshooting.

NO : The inspection is complete.

DTC U0101: TC-SST-ECU Time-out

** CAUTION**

- If the ECM output the DTC U0101, make sure to diagnose the CAN bus line.
- Before replacing the ECU, make sure that the communication circuit is operating normally.

DESCRIPTIONS OF MONITOR METHODS

There is no data from TC-SST-ECU for the specified time. (TC-SST-ECU also detect communication error with ECM.)

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

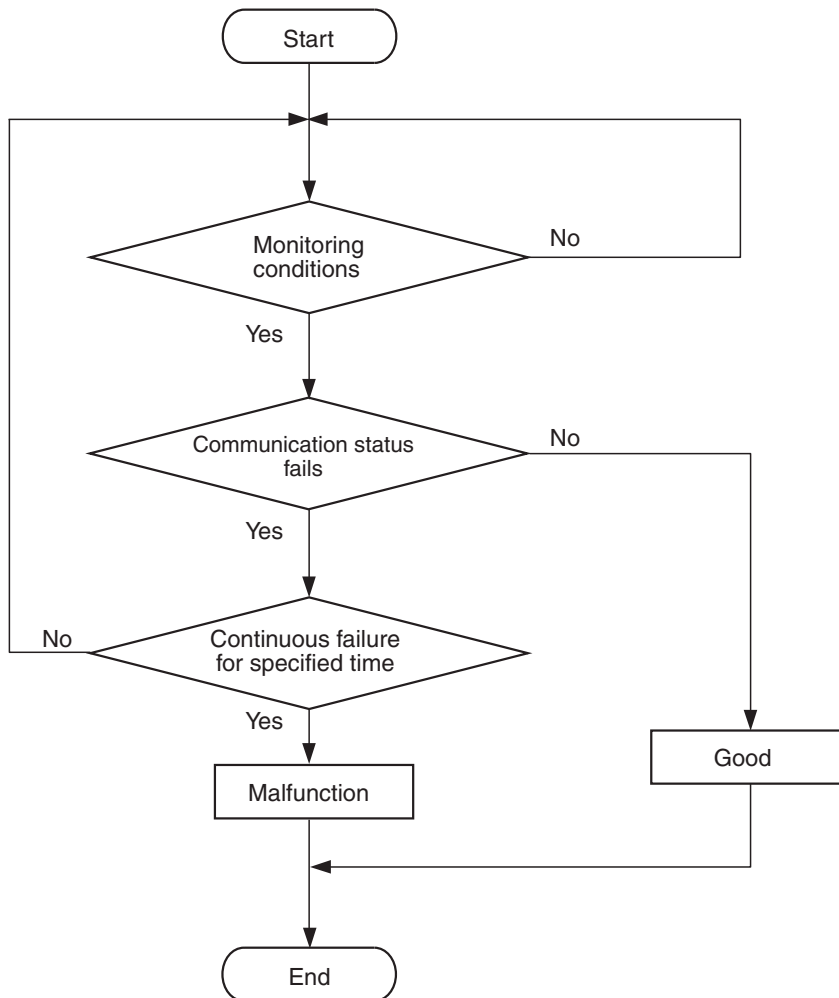
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604369

Check Condition

- Battery positive voltage is between 10 and 16.5 volts.

Judgement Criterion

- Unable to receive TC-SST-ECU signals through the CAN bus line for 4 seconds.

Past Trouble

- Proceed to troubleshoot based on a harness or connector damage on the CAN bus line between the ECM and TC-SST-ECU, and a failure in the TC-SST-ECU power supply system. Refer to

COMMENT**Current Trouble**

- Some of the possible causes are a harness or connector damage between the ECM and the TC-SST-ECU on the CAN bus line, a failure in the TC-SST-ECU power supply system, a failure in the TC-SST-ECU, or a failure in the ECM.

"How to cope with past trouble" (Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Treat Past Trouble [P.00-16](#)).

NOTE: If a malfunction occurred in the past, a failure cannot be discovered through the M.U.T.-III CAN bus diagnosis even if there might be a problem with the CAN bus. In this case, refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points – How to Cope with Intermittent Malfunctions P.00-15. Furthermore, it is possible to narrow down the areas of the possible failures from the DTCs that are output by the ECUs, which are communicating on the CAN bus (Refer to GROUP 54C, CAN bus line Diagnostics Flow P.54C-9).

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- TC-SST-ECU failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, diagnose CAN bus line.

CAUTION

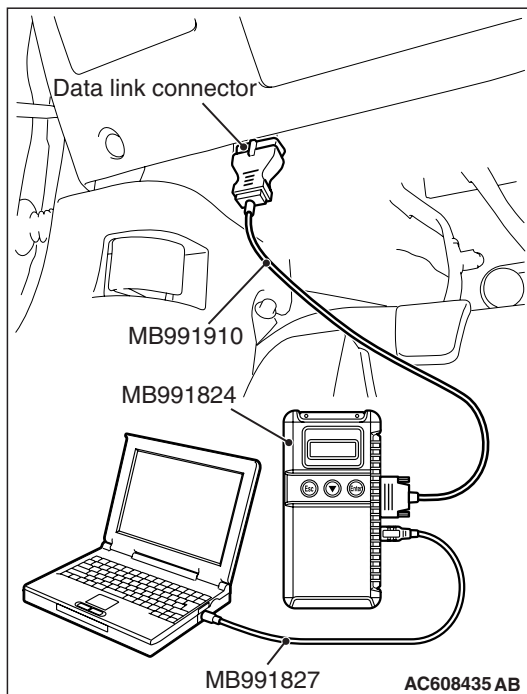
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2.

NO : Repair the CAN bus line. Refer to GROUP 54C, Can Bus Diagnostics Table P.54C-14. Then go to Step 6.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the TC-SST-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the TC-SST-DTC set?

YES : Refer to GROUP 22C, Diagnosis <TC-SST> – Diagnostic Trouble Code Chart [P.22C-10](#).

NO : Go to Step 3.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the S-AWC-DTC, Shift lever-DTC, ASC-DTC and ETACS-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0101 set?

YES : Go to Step 4.

NO : Go to Step 5.

STEP 4. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0101 set?

YES : Replace the TC-SST-ECU. Then go to Step 6.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and TC-SST-ECU. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0101 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 6.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and TC-SST-ECU. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 6. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0101 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC U0121: ASC-ECU Time-out

 CAUTION

- If the ECM output the DTC U0121, make sure to diagnose the CAN bus line.
- Before replacing the ECU, make sure that the communication circuit is operating normally.

DESCRIPTIONS OF MONITOR METHODS

There is no data from ASC-ECU for the specified time. (ASC-ECU also detect communication error with ECM.)

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

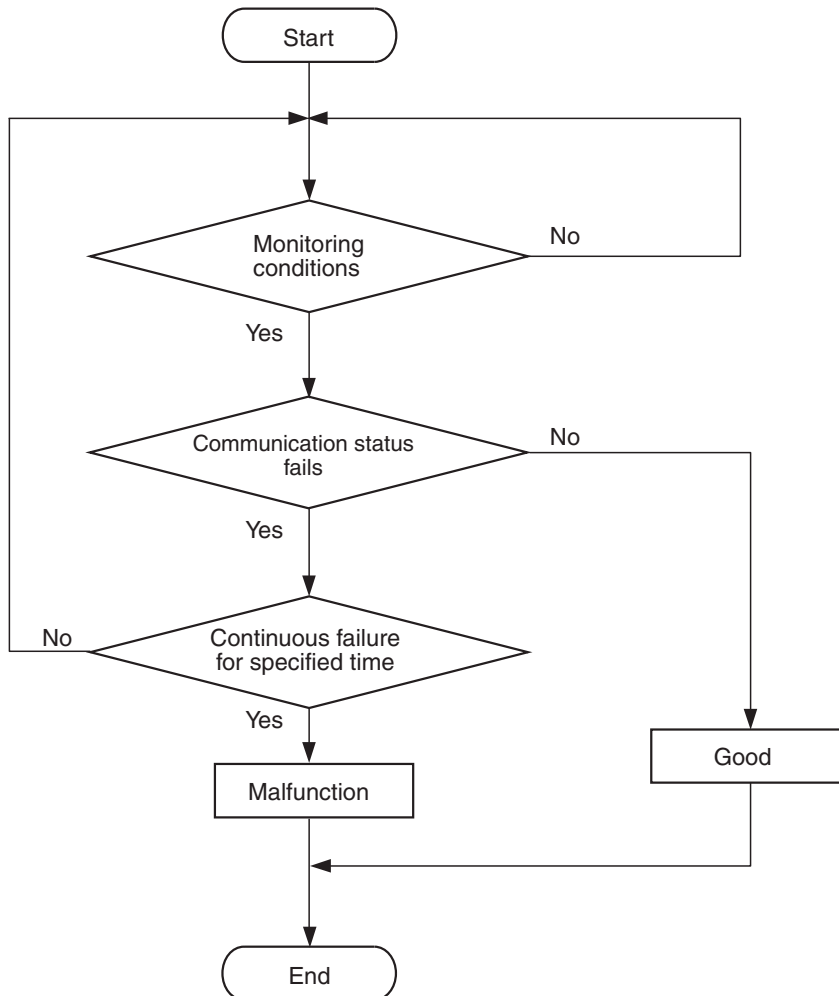
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604369

Check Condition

- Battery positive voltage is between 10 and 16.5 volts.

Judgement Criterion

- Unable to receive ASC-ECU signals through the CAN bus line for 4 seconds.

COMMENT**Current Trouble**

- Some of the possible causes are a harness or connector damage between the ECM and the ASC-ECU on the CAN bus line, a failure in the ASC-ECU power supply system, a failure in the ASC-ECU, or a failure in the ECM.

Past Trouble

- Proceed to troubleshoot based on a harness or connector damage on the CAN bus line between the ECM and ASC-ECU, and a failure in the ASC power supply system. Refer to "How to cope with past trouble" (Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Treat Past Trouble [P.00-16](#)).

NOTE: If a malfunction occurred in the past, a failure cannot be discovered through the MB991958 CAN bus diagnosis even if there might be a problem with the CAN bus. In this case, refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points – How to Cope with Intermittent Malfunctions [P.00-15](#). Furthermore, it is possible to narrow down the areas of the possible failures from the DTCs that are output by the ECUs, which are communicating on the CAN bus (Refer to GROUP 54C, CAN bus line Diagnostics Flow [P.54C-9](#)).

FAIL-SAFE AND BACKUP FUNCTION

- None

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- ASC-ECU failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, diagnose CAN bus line.

CAUTION

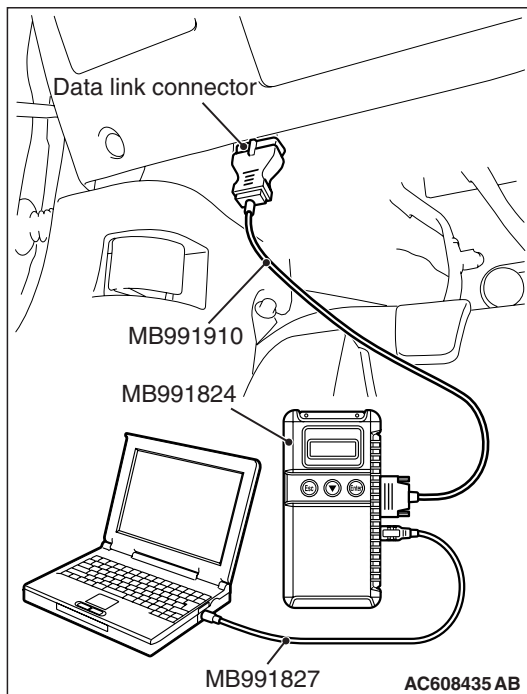
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2.

NO : Repair the CAN bus line. Refer to GROUP 54C, Can Bus Diagnostics Table [P.54C-14](#). Then go to Step 6.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the ASC-ECU.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the ASC-DTC set?

YES : Refer to GROUP 35C, Active Stability Control System –Diagnostic Trouble Code Chart [P.35C-20](#).

NO : Go to Step 3.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the S-AWC-DTC, TC-SST-DTC, Shift lever-DTC and ETACS-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0121 set?

YES : Go to Step 4.

NO : Go to Step 5.

STEP 4. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0121 set?

YES : Replace the ASC-ECU. Then go to Step 6.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and ASC-ECU. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0121 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 6.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and ASC-ECU. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 6. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0121 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC U0141: ETACS-ECU Time-out

CAUTION

- If the ECM output the DTC U0141, make sure to diagnose the CAN bus line.
- Before replacing the ECU, make sure that the communication circuit is operating normally.

DESCRIPTIONS OF MONITOR METHODS

There is no data from ETACS-ECU for the specified time. (ETACS-ECU also detect communication error with ECM.)

MONITOR EXECUTION

Continuous

MONITOR EXECUTION CONDITIONS (Other monitor and Sensor)

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

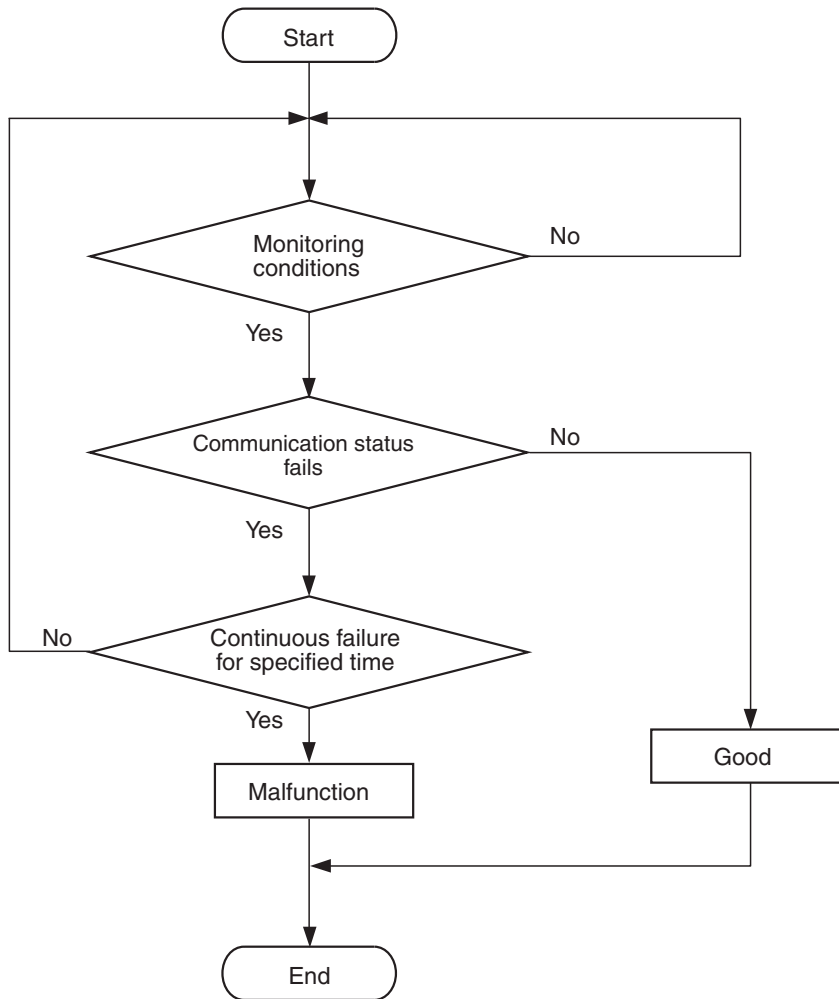
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604369

Check Condition

- Battery positive voltage is between 10 and 16.5 volts.

Judgement Criterion

- Unable to receive ETACS-ECU signals through the CAN bus line for 4 seconds.

Past Trouble

- Proceed to troubleshoot based on a harness or connector damage on the CAN bus line between the ECM and ETACS-ECU, and a failure in the ETACS power supply system. Refer to "How to cope with past trouble" (Refer to GROUP 00,

COMMENT**Current Trouble**

- Some of the possible causes are a harness or connector damage between the ECM and the ETACS-ECU on the CAN bus line, a failure in the ETACS-ECU power supply system, a failure in the ETACS-ECU, or a failure in the ECM.

How to Use Troubleshooting/Inspection Service Points –How to Treat Past Trouble [P.00-16](#)).

NOTE: If a malfunction occurred in the past, a failure cannot be discovered through the MB991958 CAN bus diagnosis even if there might be a problem with the CAN bus. In this case, refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points – How to Cope with Intermittent Malfunctions P.00-15. Furthermore, it is possible to narrow down the areas of the possible failures from the DTCs that are output by the ECUs, which are communicating on the CAN bus (Refer to GROUP 54C, CAN bus line Diagnostics Flow P.54C-9).

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- ETACS-ECU failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, diagnose CAN bus line.

CAUTION

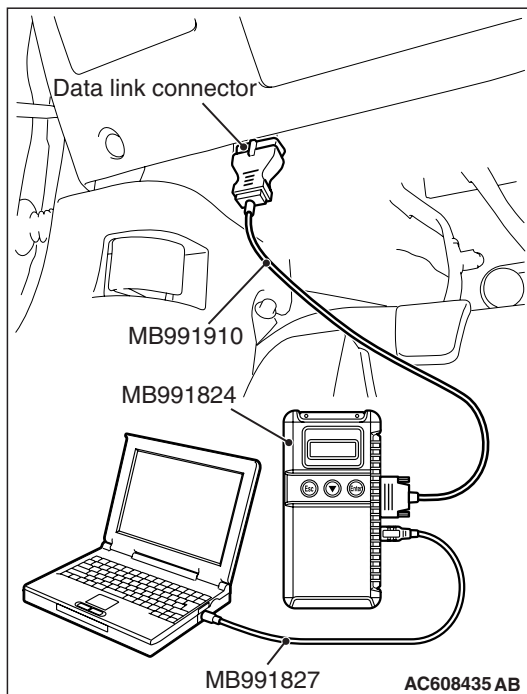
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2.

NO : Repair the CAN bus line. Refer to GROUP 54C, Can Bus Diagnostics Table P.54C-14. Then go to Step 6.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the ETACS-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the ETACS-DTC set?

YES : Refer to GROUP 54A, ETACS –Diagnostic Trouble Code Chart [P.54A-582](#).

NO : Go to Step 3.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the S-AWC-DTC, TC-SST-DTC, Shift lever-DTC, ASC-DTC, KOS-DTC or WCM-DTC, SRS-DTC, Occupant classification-DTC, Combination meter-DTC, Radio and CD player or CD changer-DTC, CAN box unit-DTC, Hands-free module-DTC, Satellite radio tuner-DTC and A/C-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0141 set?

YES : Go to Step 4.

NO : Go to Step 5.

STEP 4. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0141 set?

YES : Replace the ETACS-ECU. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 6.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and ETACS-ECU. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 5. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0141 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 6.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and ETACS-ECU. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 6. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0141 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.

DTC U0167: Immobilizer Communication Error

 CAUTION

- If the ECM output the DTC U0167, make sure to diagnose the CAN bus line.
- Before replacing the ECU, make sure that the communication circuit is operating normally.

DTC SET CONDITIONS

Check Conditions

- Ignition switch is in "ON" position.
- Battery positive voltage is 8 volts or higher.

Judgement Criterion

- Unable to receive KOS-ECU/WCM (immobilizer-ECU) signals through the CAN bus line.

COMMENT

Current Trouble

- Some of the possible causes are a harness or connector damage between the ECM and the KOS-ECU/WCM (immobilizer-ECU) on the CAN bus line, a failure in the KOS-ECU/WCM (immobilizer-ECU) power supply system, a failure in the KOS-ECU/WCM (immobilizer-ECU), or a failure in the ECM.

Past Trouble

- Proceed to troubleshoot based on a harness or connector damage on the CAN bus line between the ECM and KOS-ECU/WCM (immobilizer-ECU), and a failure in the KOS-ECU/WCM (immobilizer-ECU) power supply system. Refer to "How to cope with past trouble" (Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Treat Past Trouble [P.00-16](#)).

NOTE: If a malfunction occurred in the past, a failure cannot be discovered through the MB991958 CAN bus diagnosis even if there might be a problem with the CAN bus. In this case, refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points – How to Cope with Intermittent Malfunctions P.00-15. Furthermore, it is possible to narrow down the areas of the possible failures from the DTCs that are output by the ECUs, which are communicating on the CAN bus (Refer to GROUP 54C, CAN bus line Diagnostics Flow P.54C-9).

FAIL-SAFE AND BACKUP FUNCTION

- Engine start is prohibited.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- KOS-ECU/WCM (immobilizer-ECU) failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, diagnose CAN bus line.

CAUTION

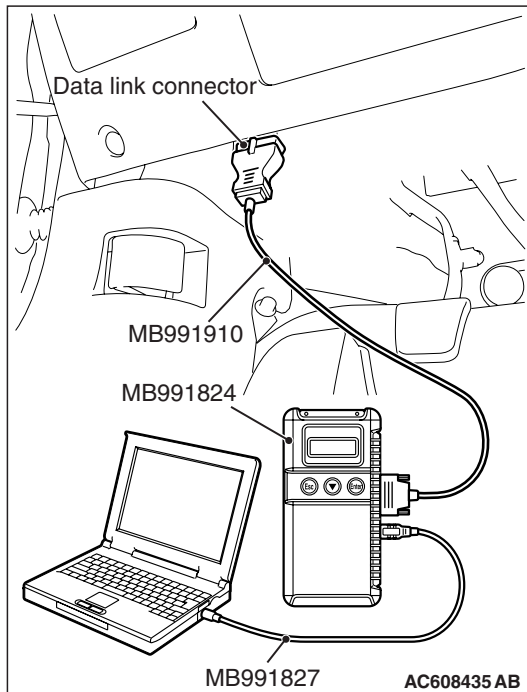
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2.

NO : Repair the CAN bus line. Refer to GROUP 54C, Can Bus Diagnostics Table P.54C-14. Then go to Step 3.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the immobilizer-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the immobilizer-DTC set?

YES : Refer to Refer to GROUP 42B, Keyless Operation System –Diagnosis Trouble Code Chart <Vehicles with KOS> [P.42B-20](#) or GROUP 42C, Wireless Control Module –Diagnosis Trouble Code Chart <Vehicles with WCM> [P.42C-14](#).

NO : Go to Step 3.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U0167 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#).

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and KOS-ECU/WCM (immobilizer-ECU). Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

DTC U1180: Combination Meter Time-out

 CAUTION

- If the ECM output the DTC U1180, make sure to diagnose the CAN bus line.
- Before replacing the ECU, make sure that the communication circuit is operating normally.

DESCRIPTIONS OF MONITOR METHODS

There is no data from combination meter for the specified time. (Meter also detect communication error with ECM.)

MONITOR EXECUTION

Continuous

**MONITOR EXECUTION CONDITIONS
(Other monitor and Sensor)**

Other Monitor (There is no temporary DTC stored in memory for the item monitored below)

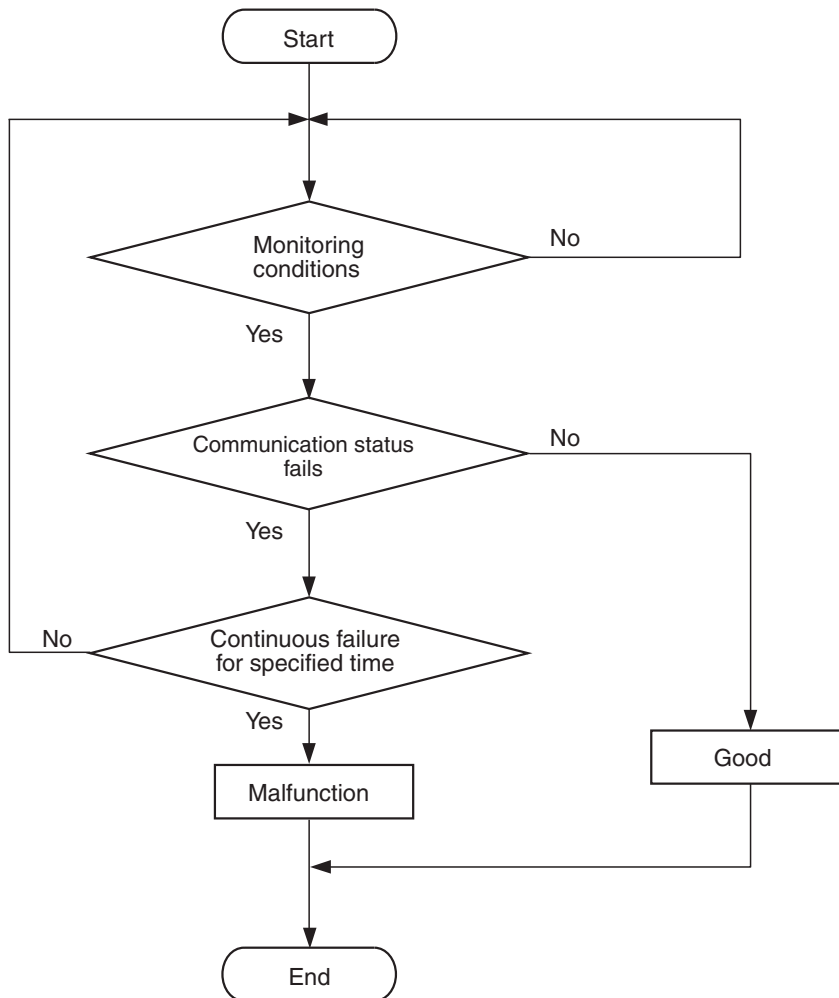
- Not applicable

Sensor (The sensor below is determined to be normal)

- Not applicable

DTC SET CONDITIONS

Logic Flow Chart



AK604369

Check Condition

- Battery positive voltage is between 10 and 16.5 volts.

Judgement Criterion

- Unable to receive combination meter signals through the CAN bus line for 4 seconds.

COMMENT**Current Trouble**

- Some of the possible causes are a harness or connector damage between the ECM and the combination meter on the CAN bus line, a failure in the combination meter power supply system, a failure in the combination meter, or a failure in the ECM.

Past Trouble

- Proceed to troubleshoot based on a harness or connector damage on the CAN bus line between the ECM and combination meter, and a failure in the combination meter power supply system. Refer to "How to cope with past trouble" (Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Treat Past Trouble [P.00-16](#)).

NOTE: If a malfunction occurred in the past, a failure cannot be discovered through the M.U.T.-III CAN bus diagnosis even if there might be a problem with the CAN bus. In this case, refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points – How to Cope with Intermittent Malfunctions [P.00-15](#).

Furthermore, it is possible to narrow down the areas of the possible failures from the DTCs that are output by the ECUs, which are communicating on the CAN bus (Refer to GROUP 54C, CAN bus line Diagnostics Flow [P.54C-9](#)).

FAIL-SAFE AND BACKUP FUNCTION

- None

OBD-II DRIVE CYCLE PATTERN

None.

TROUBLESHOOTING HINTS (The most likely causes for this code to be set are:)

- CAN line harness damage or connector damage.
- Combination meter failed.
- ECM failed.

DIAGNOSIS

Required Special Tools:

- MB991958: Scan tool (M.U.T.-III Sub Assembly)
 - MB991824: V.C.I.
 - MB991827: USB Cable
 - MB991910: Main Harness A

STEP 1. Using scan tool MB991958, diagnose CAN bus line.

CAUTION

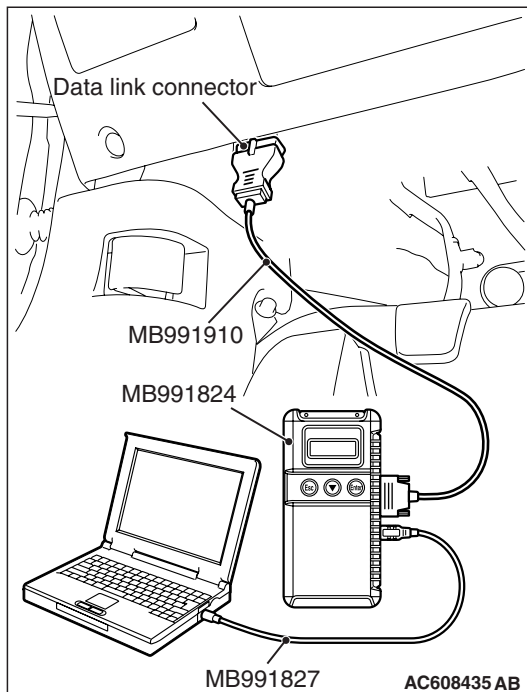
To prevent damage to scan tool MB991958, always turn the ignition switch to the "LOCK" (OFF) position before connecting or disconnecting scan tool MB991958.

- (1) Connect scan tool MB991958 to the data link connector.
- (2) Turn the ignition switch to the "ON" position.
- (3) Diagnose CAN bus line.
- (4) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the CAN bus line normal?

YES : Go to Step 2.

NO : Repair the CAN bus line. Refer to GROUP 54C, Can Bus Diagnostics Table [P.54C-14](#). Then go to Step 4.



STEP 2. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) Read the combination meter-DTC.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is the combination meter-DTC set?

YES : Refer to GROUP 54A, Chassis Electrical – Combination Meter Assembly –Diagnostic Trouble Code Chart [P.54A-28](#).

NO : Go to Step 3.

STEP 3. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U1180 set?

YES : Replace the ECM. When the ECM is replaced, register the ID code. Refer to GROUP 42B, ID Code Registration Necessity Judgment Table <Vehicles with KOS> [P.42B-12](#) or GROUP 42C, ID Codes Registration Judgment Table <Vehicles with WCM> [P.42C-9](#). Then go to Step 4.

NO : It can be assumed that this malfunction is intermittent of CAN bus line between ECM and combination meter. Refer to GROUP 00, How to Use Troubleshooting/Inspection Service Points –How to Cope with Intermittent Malfunctions [P.00-15](#).

STEP 4. Using scan tool MB991958, read the diagnostic trouble code (DTC).

- (1) Turn the ignition switch to the "ON" position.
- (2) After the MFI-DTC has been deleted, read the MFI-DTC again.
- (3) Turn the ignition switch to the "LOCK" (OFF) position.

Q: Is DTC U1180 set?

YES : Retry the troubleshooting.

NO : The inspection is complete.